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• **SIASUN 新松**

• General Operating
• Manual for SIASUN
• Robot

• Technical Information for Industrial Robot
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Thank you for choosing SIASUN Industrial Robot.

Read through all applicable users' manuals and safety instructions carefully and make sure that you understand the corresponding contents before use or operate the robot.

The specifications, data and information herein may be updated or changed by us without prior notice at any time to reflect the upgrade of product.

We do not guarantee that all descriptions herein are accurate or correct. We do not assume any responsibilities or liabilities to any direct or indirect consequences caused by any omissions or errors in this Manual.

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Safety notes

Please read and understand the content set out in this section before using a robot.

Before using a SIASUN industrial robot, the safety measures of operators and systems must be fully considered. The user shall fully understand its specifications via this Manual.

Before use (installation, transfer, maintenance, inspection/repair), please read and fully master this Manual and other ancillary information, and shall not start to use until familiarizing all equipment knowledge, safety knowledge and notes. However, SIASUN can't warrant that users will not suffer any harm even if they follow all the safety information given in this Manual.

1 Definition of user

The user is defined as follows.

-- Operator

Perform power ON/OFF operation of robot.

Start the robot program from operation panel.

- Programmer

Perform the operation of robot.

Conduct robot teaching in the safe area.

- Repair staff

Perform the operation of robot.

Conduct robot teaching in the safe area.

Conduct robot maintenance (repair, adjustment, replacement) job.

Operator is not allowed to operate in the safe area.

Programmer and maintenance staff are allowed to operate in the safe area.

Jobs in the safe area include: handling, setting, teaching, adjustment, repair, etc.

The specialized training in robot must be received before operating in the safe area.
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In the operation, editing and maintenance of robot, the operator, the programmer and the repair staff must pay attention to safety, and shall at least wear the following articles for operation.

- Work clothes suitable for the job content
- Safety shoes
- Safety helmet

- Protective glasses

2 Safety signs

The obvious tips where safety matters are necessary to be prompted will be made in this Manual, so as to remind designers or operators to ensure the safety of themselves, colleagues and equipment. We use "Danger", "Warning", "Attention" and "Note" to describe these safety precautions. Users must read and understand the content described in these safety precautions before using robot, and must also strictly follow them.

Signs	Definition
 DANGER	An emergency dangerous situation that may result in the death or serious personal injury of the operator if the operation is incorrect.
 Warnings	A dangerous situation that may result in the death or serious personal injury of the operator if the operation is incorrect.
 Note	A situation that may result in equipment damage or staff injury if the operation is incorrect.
Note	Supplement content in addition to warnings and attentions.

3 Safety notes for users

- (1) Please arrange the safe area before the robot is installed. The safety fence and safety warning signs must be arranged around the robot system. A safety door must be installed, and the staff must open the safety door to enter the fence. The safety fence must completely close the robot, so that the staff can't enter the safety fence via other places except for the safety door, preventing the entry of unrelated people and preventing robot from injuring people.
- (2) The peripheral devices are all required to be connected to the ground wire in accordance with relevant standards.
- (3) The peripheral devices are designed outside the motion range of robot as much as possible.
- (4) Safety locks shall be installed at the power supply according to the actual situation on site, so that the robot can only be connected to the power supply by specific staff.
- (5) Do not use the robot in flammable, explosive and radioactive environments.
- (6) Do not climb on the robot.
- (7) During the operation of the robot system, even if the robot is in or seems to be in a status of stopping motion, it may be that the robot is waiting for a signal to start motion or other signals, namely the robot or peripheral devices are in a status of imminent motion. In such a status where the robot can start at any time, the robot system shall be considered as being in motion. To ensure the safety of the staff, an indicator light or a sound can be installed to let the staff know that the robot system is in operation.
- (8) If peripheral devices of the robot are required to be commissioned alone, please

disconnect the power of the robot before performing the operation.

(9) After connecting the safety signal, stop operation signal and other signals that communicate with the peripheral devices of robot, please confirm whether these signals can effectively control the stop of robot (external emergency stop, safety door and other signals), because incorrect connection may cause serious personal injury.

(10) Please do not wear gloves to operate the teach pendant.

(11) Please regularly backup the parameters, variables, job and other information of robot system to prevent accidental data loss.

(12) In the initial operation after the installation of robot, please operate the robot at a low speed, and gradually increase the motion speed, confirming whether there is any abnormal status in robot.

(13) Before operating the robot, please confirm whether there are operators inside the safety fence. Check whether there are still potential dangers in the robot system, and please operate the robot after all the dangers have been eliminated.

(14) When handling and installing the robot, please follow the correct methods instructed by Siasun to avoid the overturning or falling of the robot or controller due to incorrect handling or installation methods, resulting in serious injuries to staff or equipment damage.

(15) There shall be no hanging objects above the robot, preventing it from falling to damage the robot and other equipment.

(16) It is not allowed to lean against the electric control cabinet or touch the button at will so as to prevent unexpected actions of the robot, causing personal injury or equipment damage.

(17) When disassembling the robot, attention shall be paid to the parts that may fall from the robot to injure people.

(18) It is not allowed to move any axis of the robot, otherwise it may cause personal injury and equipment damage.

(19) Protective measures shall be taken when wiring and piping between the electric control cabinet and the robot/peripheral devices, for example, the pipe, wire or cable shall be passed through the pit or a protective cover shall be added to cover it, so as to avoid the same being damaged by people or being crushed by forklifts.

(20) The action range and load conditions set by the software shall not exceed the specified value in the product specification table, because improper setting may cause staff injury or machine damage.



DANGER

Under no circumstances it is allowed to stand under the robot arm, so as to prevent an accident caused by abnormal motion of the robot or by someone else connecting to enable it.

■ Safety of operation outside the safety fence

When starting and stopping the robot integrated devices daily, the power of the robot system can be switched on/off outside the safety fence, and the operation of starting and stopping the robot system can be performed via the teach pendant or the human-machine interface of robot system.

- (1) When the robot is not required to be operated, the power shall be off, or the emergency stop button shall be pressed.
- (2) When designing the robot system, an external emergency stop button shall be set, through which the motion of robot can be stopped emergently. Please set the emergency stop button within the range of the operator's reach.
- (3) Before starting the robot system, please observe whether there is an unexpected entry of the staff in the safety fence, and the safety fence and safety door with a safety lock must be set.

Table (a) represents the jobs outside the safety fence. Each robot operator can perform the jobs marked [○] in this table.

Table (a) Jobs outside the safety fence

Job content	Operator	Programmer/teaching operator	Repair/maintenance staff
Control device power ON/OFF	○	○	○
Selection of operation mode (T1, T2)		○	○
Selection of remote/local mode		○	○
Select the program with the teach pendant		○	○
Select the program with the peripheral devices		○	○
Start the program with the operation panel	○	○	○
Start the program with the teach pendant		○	○
Reset alarm with the operation panel		○	○
Reset alarm with the teach pendant		○	○
Set by the teach pendant data		○	○
Taught by the teach pendant		○	○
Emergency stop by the operation panel	○	○	○
Emergency stop by the teach pendant	○	○	○
Repair of the operation panel			○
Repair of the teach pendant			○

■ Safety during programming and teaching

Personnel performing robot teaching/programming operations are strictly prohibited from entering the robot's work envelope and must maintain heightened awareness of personal

safety at all times.

- (1) When teaching the robot, please make sure that there is no danger or abnormal status in the robot or the surrounding integrated devices.
- (2) If it is necessary to enter the motion range of robot, please determine the status and position of the emergency stop button, the safety switch of robot and the integrated devices in advance.
- (3) The on-site teaching staff is obliged to ensure that there is no other staff within the motion range of robot.
- (4) On the teach pendant, SIASUN robot provides an emergency stop button and a key switch for selecting the execution mode, teaching mode and remote mode. In the case of automatic operation, the robot will stop operation when the safety door is opened.
 - a) In addition to the emergency stop switch, our teach pendant is also equipped with a three-way enable switch: DEADMAN.
 - b) Emergency stop button: as long as the emergency stop button is pressed, the robot will stop emergently.
 - c) DEADMAN: a three-way switch is set. When the switch is in the middle, the robot can move manually. Loosen or hold the switch, and the robot will stop in an emergency.
 - d) The teaching staff shall not make the robot move until DEADMAN is in the middle position, and the dangerous status shall be eliminated before the motion.
 - e) DEADMAN is enabled under the T1/T2 teaching mode.
- (5) When using the teach pendant or the human-machine interface to start the robot, it shall fully ensure that there is no staff and abnormalities within the motion range of robot before execution.
- (6) After teaching, commissioning and programming, please conduct the cycle of low-speed & single-step and low-speed & continuous-operation at least once, medium-speed & continuous operation once, normal- speed & continuous operation once. If there is no error, the robot can operate automatically.
- (7) When automatic operation is conducted, the staff shall work outside the safety fence.

■ Safety in repair and maintenance

To ensure the safety of repair staff, the full attention shall be paid to the following matters.

- (1) Do not enter the motion range of robot while it is in operation.
- (2) When repairing and maintaining the robot, power off the power supply of robot as far as possible and use the safety lock. If it is necessary to power on the power supply to repair and maintain the robot, please press the emergency stop button and make a warning sign.
- (3) When working in the robot safety fence, it is necessary to carefully understand whether there is a danger in the robot integrated system. Determine the status of the system, and work carefully.

- (4) When pneumatic systems or elements are to be maintained, the air pressure shall be reduced to 0 before operation.
- (5) Never operate the robot automatically when there are people in its motion range.
- (6) When maintaining the robot by several people in a narrow area, make sure not to block the escape access of others.
- (7) During repair job, another person is required to be near the emergency stop button and can press the emergency stop button at any time to ensure the safety of the staff.
- (8) When replacing robot components, please consult the SIASUN after-sales service. Do not judge the replacement by yourself, because this will lead to unexpected accidents, resulting in serious consequences.
- (9) When replacing robot components, the foreign matters are not allowed to enter the robot.
- (10) When inspecting/maintaining the inside of the control cabinet of robot, the power shall be off for more than 15 minutes to ensure that the driver is not in the status of high voltage charging, and the power-on light shall be off before proceeding.
- (11) When replacing components, please use the standard components provided by SIASUN, and do not choose the components from other manufacturers. Doing so will cause damage to the robot and is likely to cause fire.
- (12) When the maintenance of robot is completed, before starting the robot, it is necessary to make sure that there is no staff inside the safety fence and that there is no abnormality in the robot system as a whole.
- (13) When replacing the motor or brake, the crane shall be used to fix the robot, so that the robot will not fall due to the loss of the brake, and then the job can be conducted.
- (14) After the work is completed, the grease and lubricant left on the ground shall be cleaned up to prevent someone from slipping or causing a fire.
- (15) The servo motor, the inside of control cabinet, the reducer, the brake resistance and other element of robot will emit heat, so do not touch them in the heat status. If it is necessary to touch them please wear heat-resistant gloves and other safety protective tools.
- (16) When maintaining and repairing equipment, please do not stand on the robot to avoid danger. When working at height, make sure that the scaffolding is safe and a safety belt is worn.
- (17) When replacing the components disassembled during repair, they shall all be installed back to their original positions. If insufficient components or partially left components are found, they shall be rechecked for correct installation.
- (18) After the components have been replaced, please conduct a test run when the staff is outside the safety fence.

4 Safety of integrated devices and tools

- (1) The integrated devices are required to be equipped with sensors, limit switches and other devices to identify different states of peripheral devices through signals, ensuring that the robot stops motion when in danger. When the integrated devices or tool are in

fault, the robot is also required to stop motion at the same time.

(2) When the robot operates together with the peripheral integrated device, the interference range is required to be set, ensuring the operation safety of the robot and the integrated devices.

(3) The integrated devices of robot are required to work in an environment that will not be affected by oil, water, dust, etc.

(4) The cables inside the mechanical part of robot are required to follow the matters below to avoid unexpected hazards.

a) The cables inside the machinery shall use the assembled interface.

b) When installing the cables outside the machinery, the attention shall be paid to the impact on the robot motion.

c) When there are externally exposed cables in the machinery, please do not modify such externally exposed cables.

d) When installing peripheral devices on the robot, interference with the robot shall be avoided.

(5) Please do not use the emergency stop button when the robot is normally stopped, because doing so will damage the robot. Under normal circumstances, please stop the motion of robot first, and then use the emergency stop button to power off the robot.

5 Safety of robot body

■ Operation notes

(1) When the staff operates the robot, the motion speed of robot shall be adjusted to be slow enough, so that when danger occurs, the staff has enough time to react, avoiding the occurrence of accidents.

(2) When operating a robot, it is necessary to have a deep understanding on the results of each operation and the action of robot after each button-pressing, and then operate the robot.

■ Programming notes

(1) When the motion range of multiple robots overlaps, please arrange the interference range for the motion range interfered by the robot, ensuring that the robot will not damage due to colliding with each other in the motion process.

(2) The origin shall be set for the teaching program robot, and the start point and the end point of robot shall be at the origin of the program, so that the staff can directly observe the end of the program.

■ Notes on machinery

The motion range of robot shall be kept clean, and the use environment shall not be affected by oil, water, dust, etc.

■ Safety of end effector

The end effector shall be equipped with limit switches and sensors to make the control system know the position and status of the end effector via signals, and control the effector at the same time to prevent the occurrence of hazards.

■ Stop mode of robot

(1) Stop category 0 (STOP0)

Immediately disconnect the servo power supply of the robot, while the brake of each axis actuates, so that the robot stops in an instant. Since the action trajectory of each axis is not controlled after the servo power is disconnected, the robot will deviate from the predefined operation trajectory, which is an uncontrolled stop mode.

(2) Stop category 1 (STOP1)

The robot will basically perform deceleration in accordance with the predefined trajectory, while the controller and the safety system will disconnect the power supply of the servo in accordance with the preset time, which is a controlled stop mode.

(3) Stop category 2 (STOP2)

The robot will basically perform deceleration in accordance with the predefined trajectory. After stopping, the power supply of servo is not disconnected, and the motor of each axis is still in the power-on status.

6 Danger, warning, attention signs

(1) Electric shock warning sign



Figure (a) Electric shock warning sign

Content: There is danger of high voltage and electric shock where this sign is attached, and attention shall be paid.

(2) High temperature warning sign



Figure (b) High temperature warning sign

Content: There is heat where this sign is attached, and attention shall be paid. Wear protective equipment such as heat-resistant gloves when it is necessary to touch the equipment in a heat status.

(3) Hand pinch hazard warning sign



Figure (c) Hand pinch hazard warning sign

Content: Risk of hand entrapment by robotic components. Exercise caution and avoid these hazardous areas to prevent hand pinching injuries.

(4) No treading sign



Figure (d) No treading sign

Content: Do not put your feet on the robot or climb on the robot where this sign is attached. Treading will cause adverse effects on equipment, and may also cause injuries to operators.

(5) Robot injury warning sign



Figure (e) Robot injury warning sign

Content: There is a risk of being injured by the robot when operating within the motion range of robot where this sign is attached.

(6) No disassembly sign



Figure (f) No disassembly sign

Content: It is not allowed for user to disassemble where this sign is attached. Disassembly shall be conducted by professionals using professional tools.

(7) No releasing brake sign



Figure (g) No releasing brake sign

Content: It is not allowed to release brake where this sign is attached.

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1 Overview

This chapter mainly introduces the RC software, robotic arm and control cabinet of the robot.

SIASUN robot is mainly composed of these three parts.

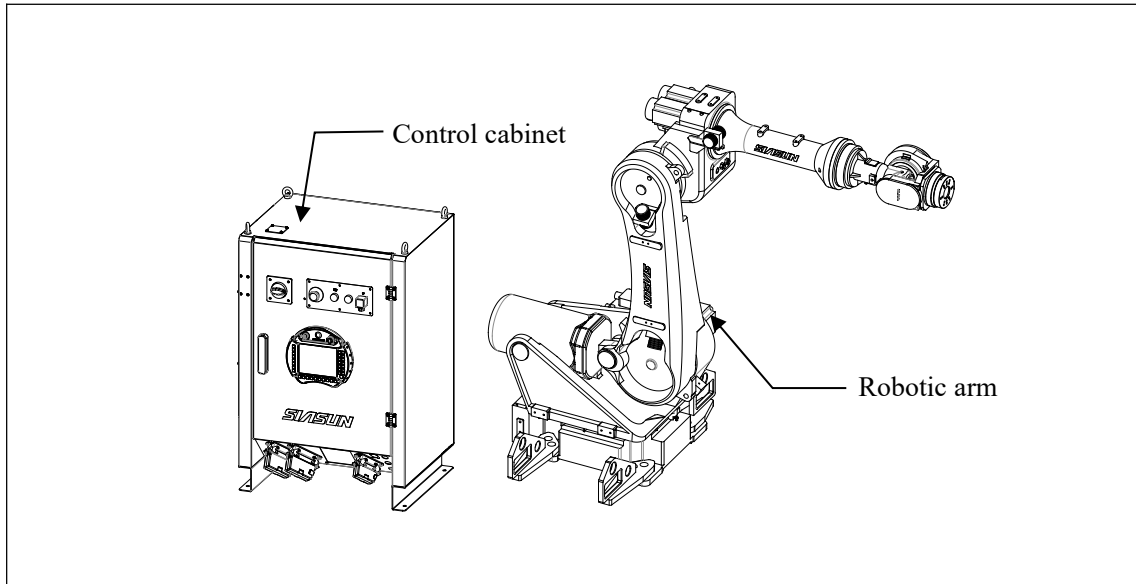


Figure 1- 1 Composition of an industrial robot

RC software:

The RC software specifically designed for robot control is embedded in the controller. Operators can control the robot to perform various operations required on site through interfaces such as buttons, menus, commands and icons on the teaching pendant. The RC software of the robot can control the peripheral devices of the robot, such as fixture, camera, safety device, etc., through communication or collect information from them. It can also control the input or output of peripheral devices, such as external sensors.

Robot:

The mechanical part of the robot body completes on-site operations by adding end effectors.

Control cabinet:

The control cabinet of the robot can provide power supply for the operation of the robot body.

The control cabinet of the robot is embedded with control software, which can control the teaching pendant and external devices.

External devices refer to the peripheral devices of the robot that can be controlled through remote I/O or bus communication, except the control cabinet. For example,

- 1) Control the operation of the robot from devices other than the control cabinet.
- 2) End effectors and sensors operated by the control cabinet through I/O or fieldbus.

1.1 Robot Operating Environment

If the robot is suspended for a long time under near-freezing conditions (close to 0°C) or during holidays/nighttime when temperatures fall below 0°C, alarms may be triggered upon restart due to large rotational resistance in drive parts. In such cases, it is recommended to operate at low speed for several minutes to warm up.

1.2 Robot Body

The robot body is mainly composed of the mechanical part of the arm and the axis driven by the servo. Usually, the industrial robot has six joints. The robot operates the end effector loaded on the wrist end flange.

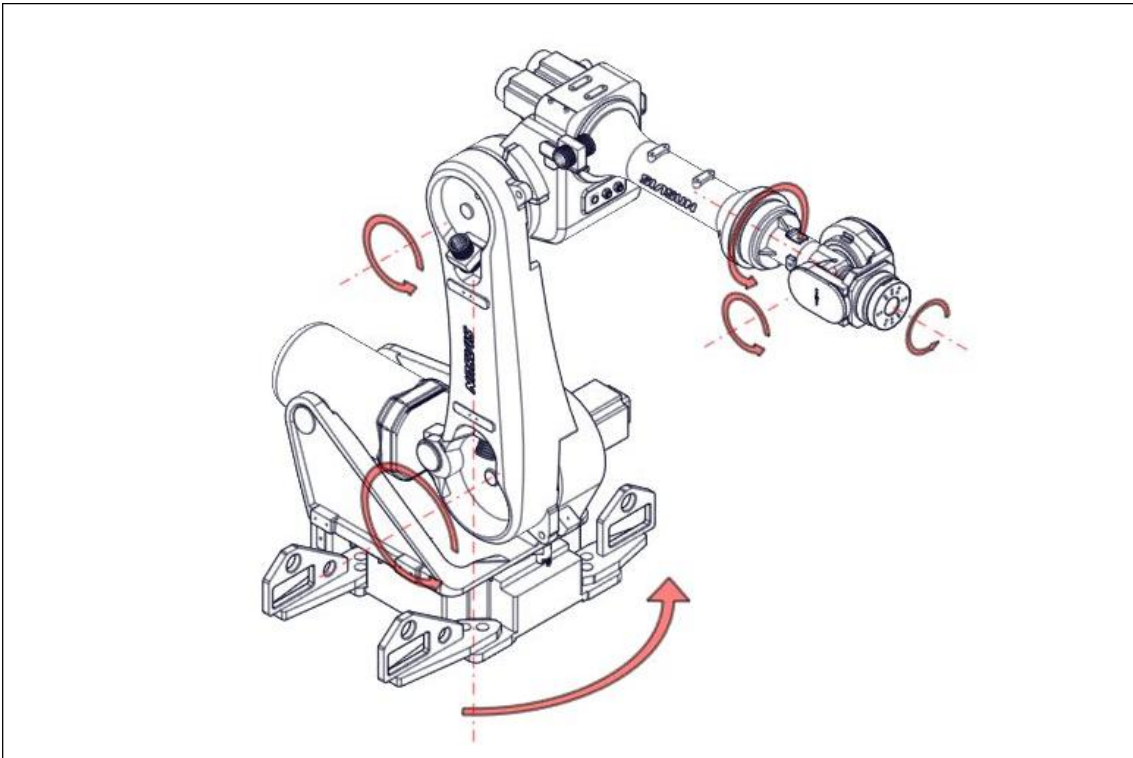


Figure 1-3 Directions of axes

1.3 Controller

The robot controller is composed of power supply, control unit, driver, user interface, IO interface, etc.

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To operate the controller, users need to use the teaching pendant.

The controller controls all servo motor drivers, including the external axis, through the CPU motherboard.

The settings, data and programs of the robot are stored in CF or TF card.

The controller can obtain communication with external devices by inputting and outputting signals through IO modules or fieldbuses.

Some controllers support the hot-swap function of the teaching pendant. For specific use methods, please refer to Electrical Operation and Maintenance Manual for SIASUN Robot.

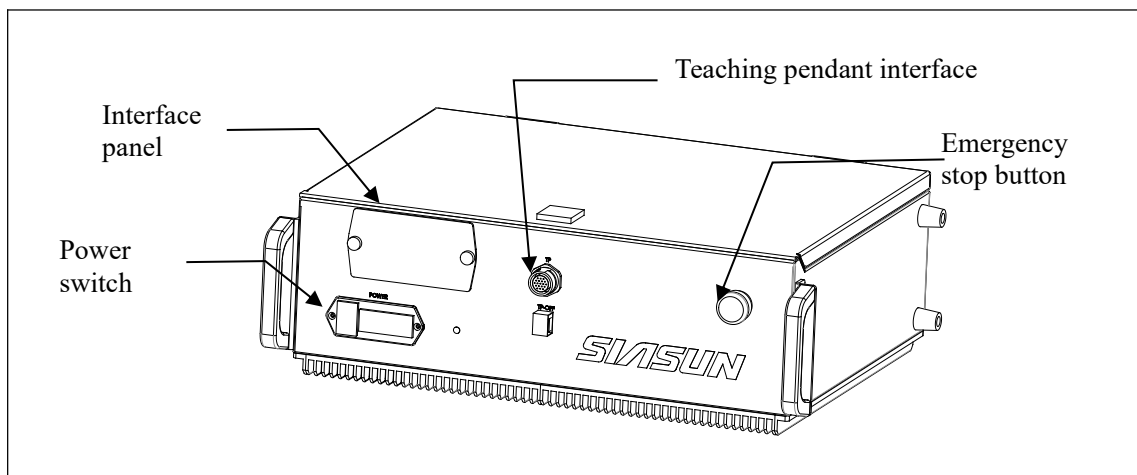


Figure 1-4 SRC C5 robot controller

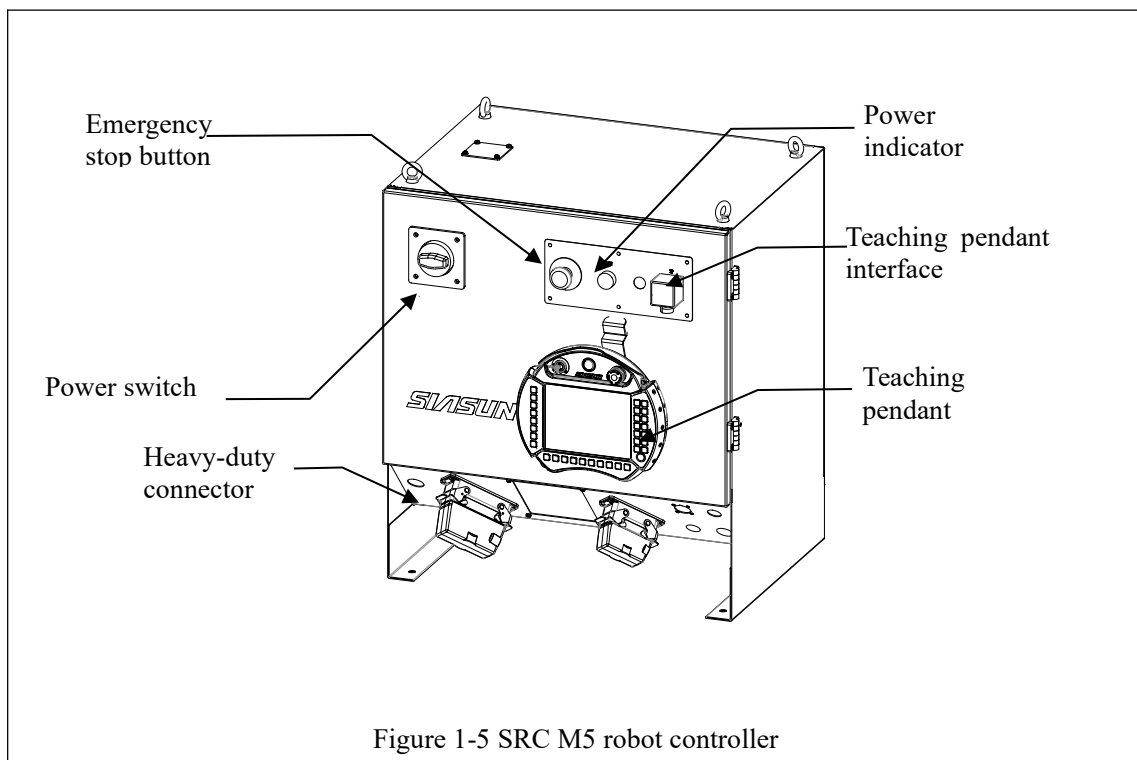


Figure 1-5 SRC M5 robot controller

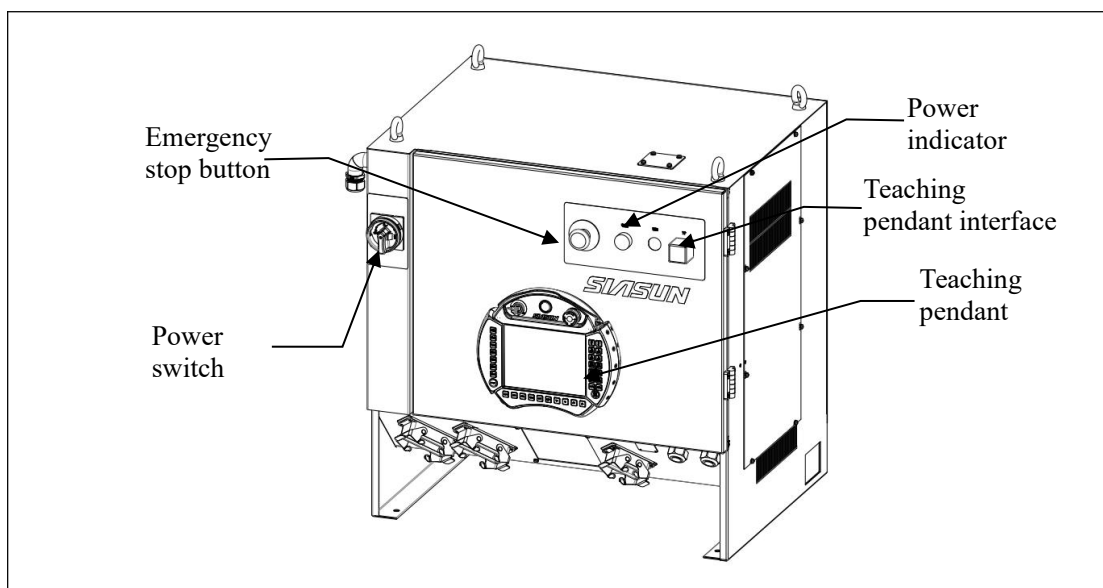


Figure 1-6 SRC E5 robot controller

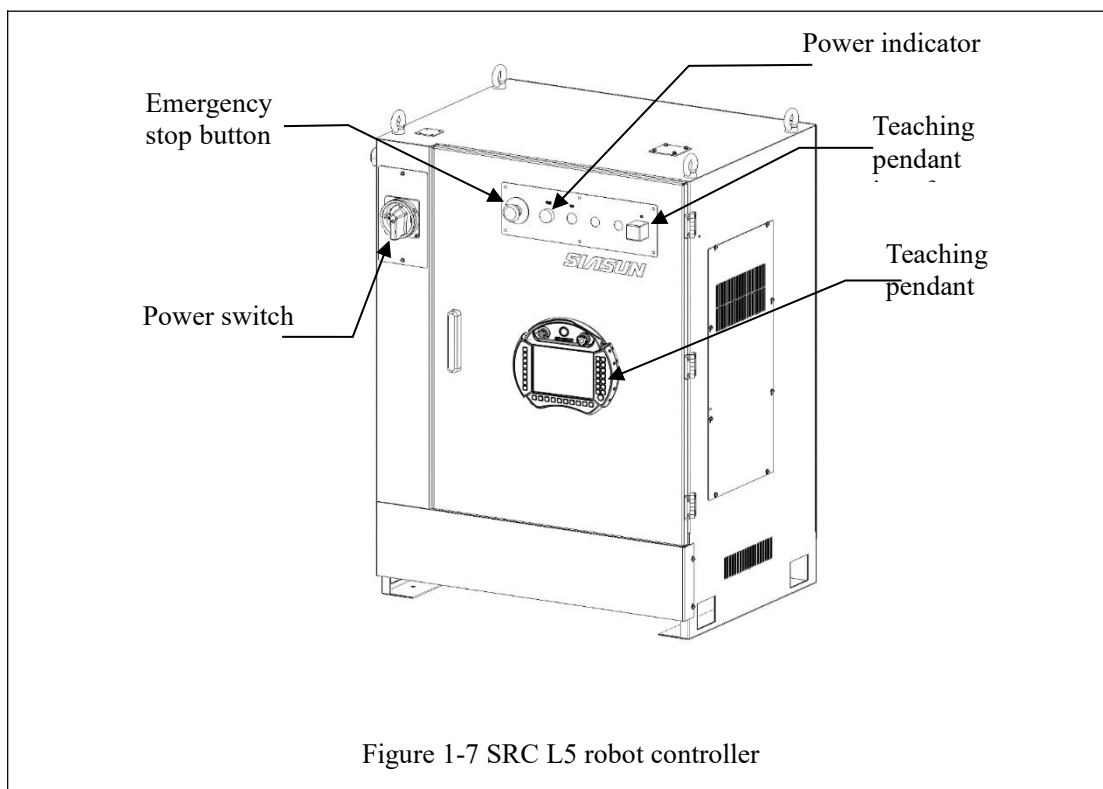


Figure 1-7 SRC L5 robot controller

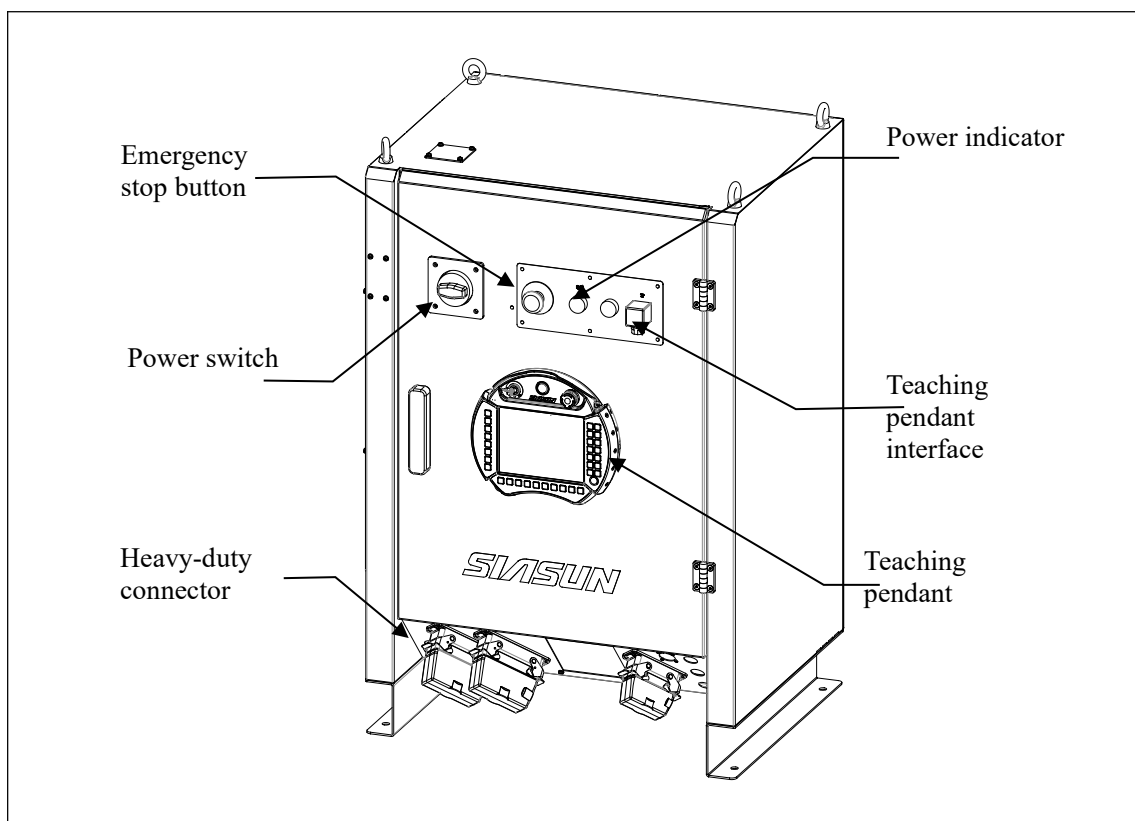


Figure 1-8 SRC G5 robot controller

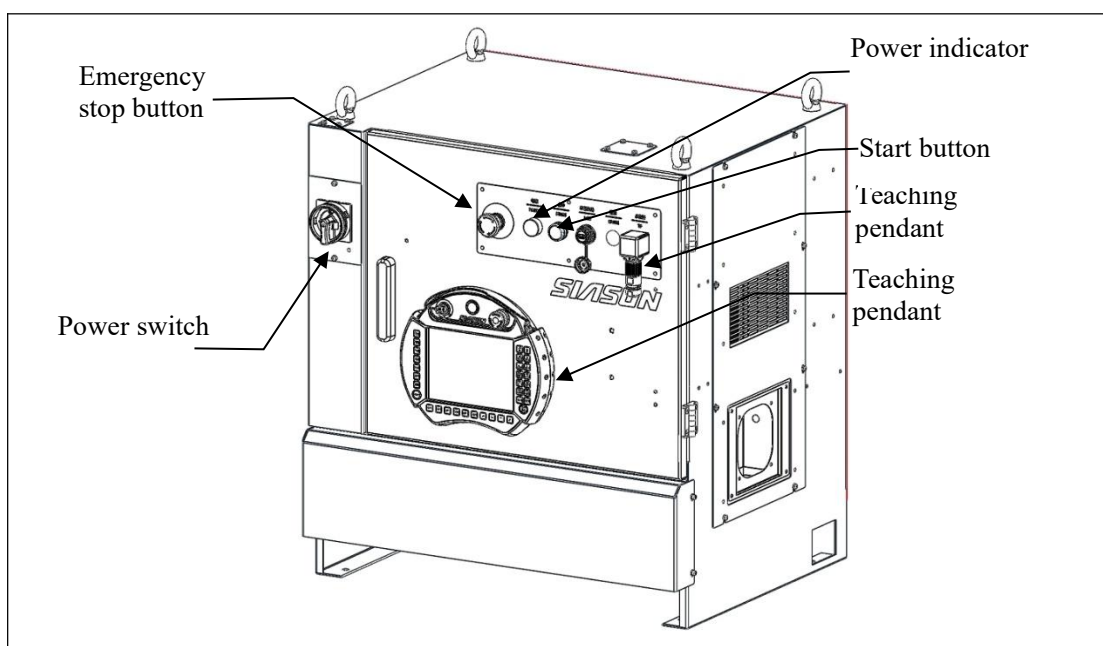


Figure 1-9 SRC M6 robot controller

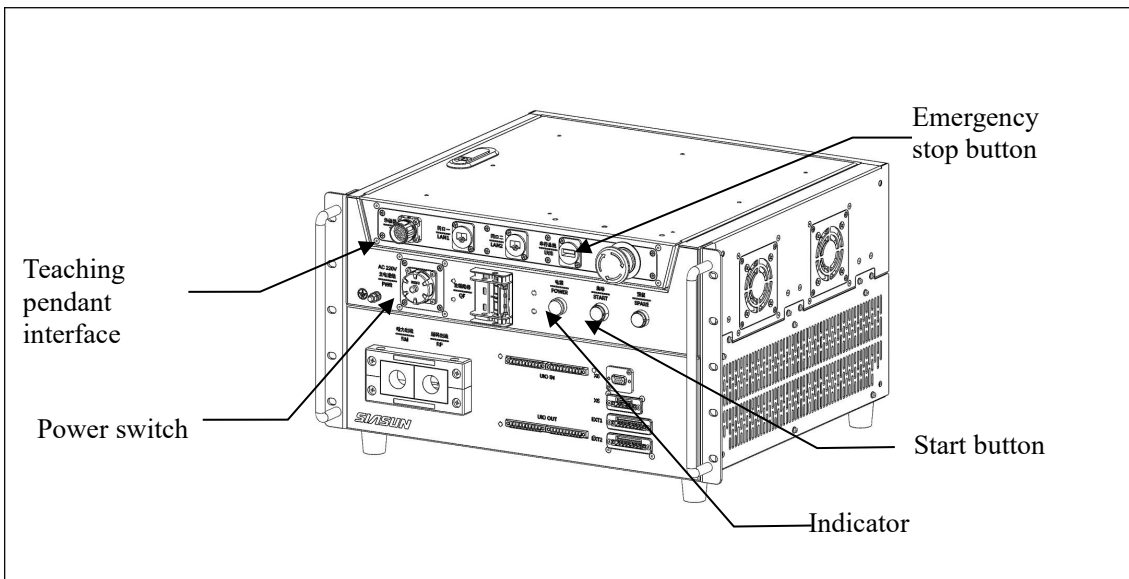


Figure 1-10 SRC C6 robot controller

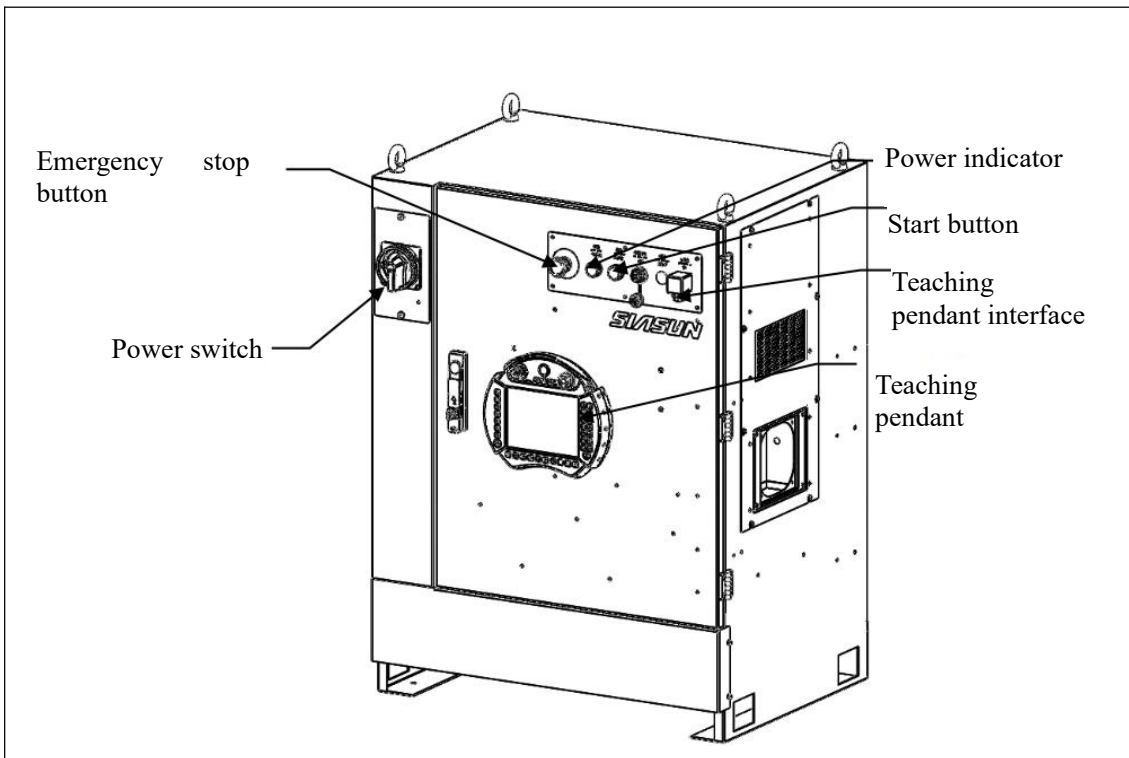


Figure 1-11 SRC L6 robot controller

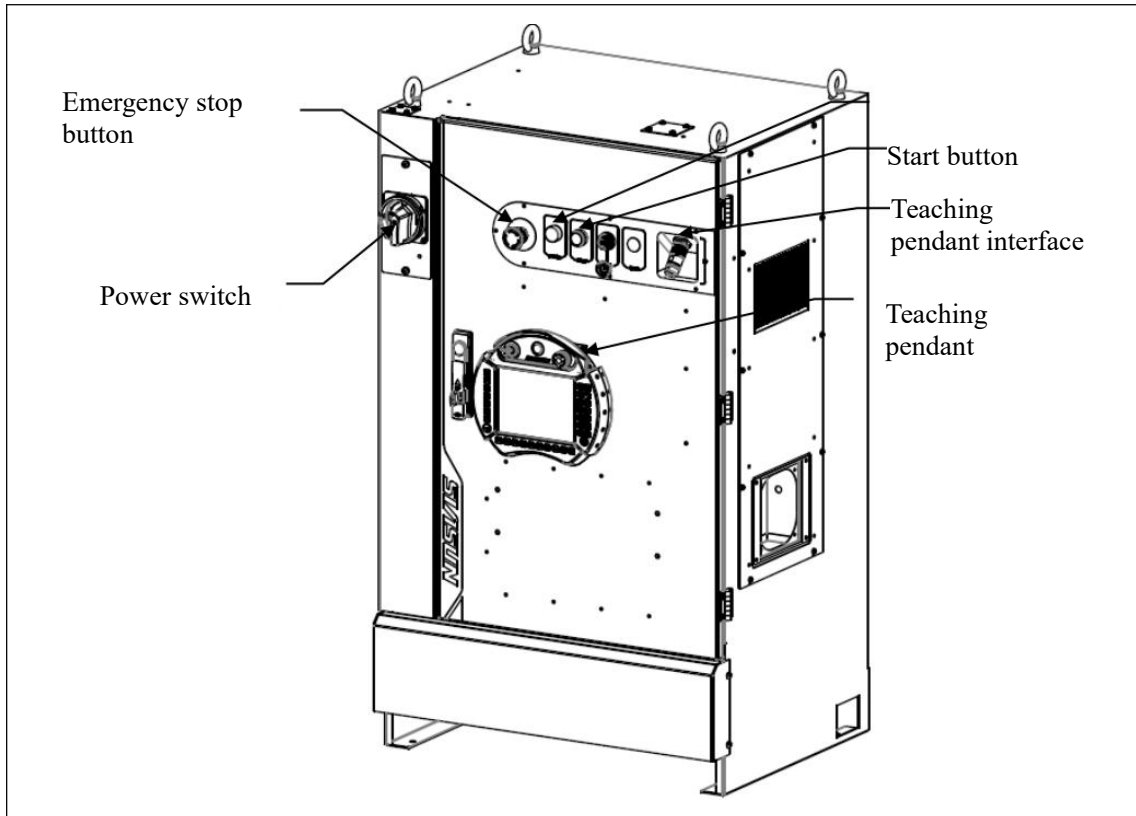


Figure 1-12 SRC ME6 robot controller

1.3.1 Teaching pendant hardware

The teaching pendant is the main interface device between the control software and the user, and can operate the robot. The teaching pendant is connected to the controller through cables.

The operations that can be performed by the teaching pendant include:

- 1) Manual robot motion.
- 2) Program creation and programming.
- 3) Program testing and execution.
- 4) Automatic program execution.
- 5) Confirmation of various statuses.

The teaching pendant consists of the following devices:

- 1) 8-inch LED touchscreen.
- 2) One LED indicator.
- 3) Function buttons.

Notes:

- 1) When touching multiple areas on the touchscreen simultaneously, the system may sometimes recognize positions different from the actually touched positions. Therefore, always touch only one position on the touchscreen.
- 2) The touchscreen may identify a position different from the one you touched due to a malfunction. Do not perform safety-related operations through the touchscreen.
- 3) Please operate the touchscreen with your fingers or a stylus for touchscreen operation. Operating with sharp end tools such as a pen used for taking notes may cause touchscreen malfunctions.

In addition, the teaching pendant is provided with three switches:

Table 1-1 Teaching Pendant Switches

Switch	Function
Mode selection switch	It can be switched among teaching mode, automatic operation mode and remote mode. Note: The teaching mode includes T1 and T2 modes, being T1 mode by default.
DEADMAN	The 3-position switch is active (allowing robot movement) only when set to the middle position. In either side position, it is inactive, and the robot will not move.
Emergency stop button	Regardless of the current state, the robot will immediately stop movement when this button is pressed.

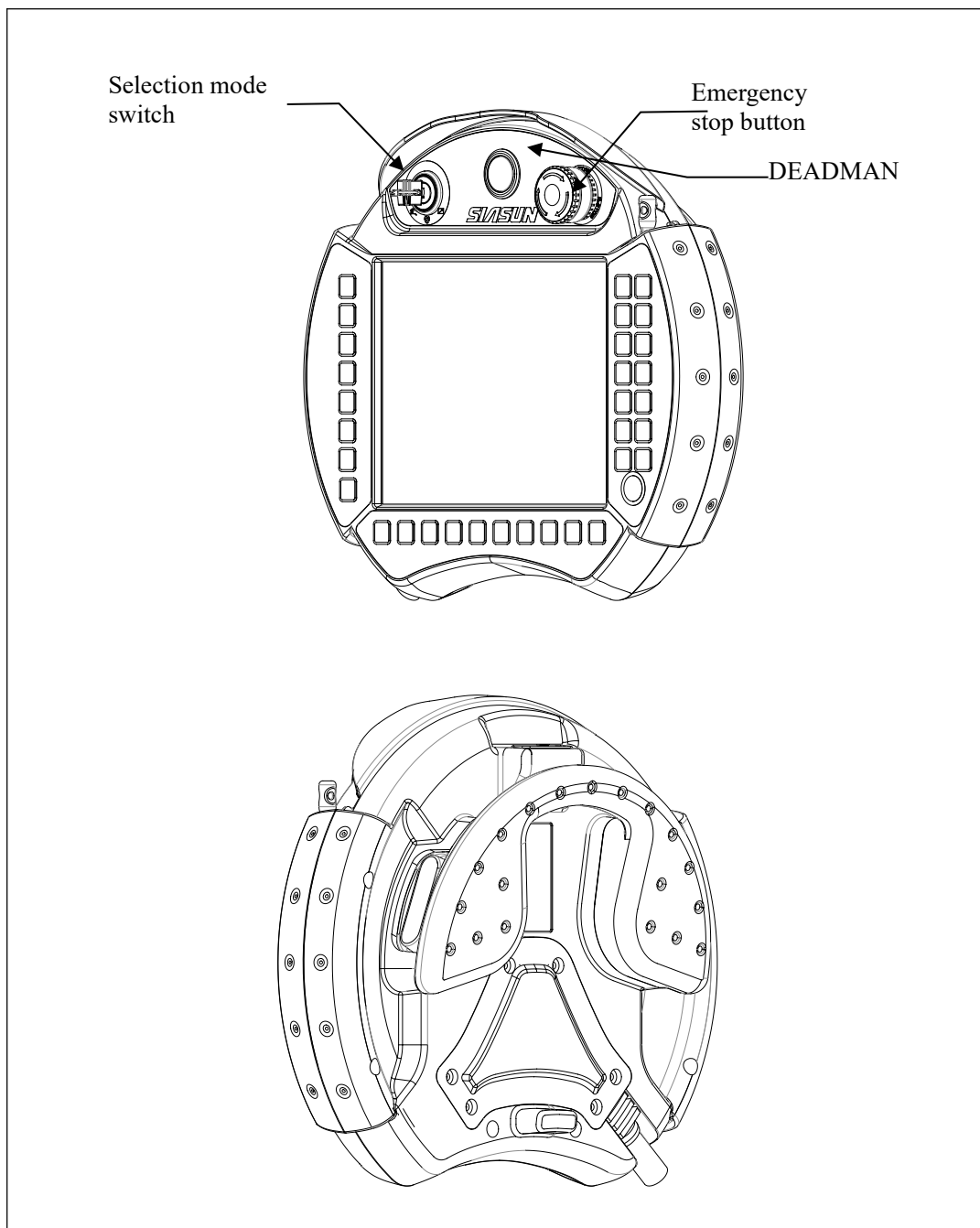


Figure 1-13 Teaching pendant switches

Teaching mode: Perform teaching work on the program, modify the edited jobs, and set various functions.

Execution mode: Automatically run the edited program.

Remote mode: The robot can be remotely and logically controlled through external input and output signals, such as servo power-on, calling the main job, start, pause and other operations.

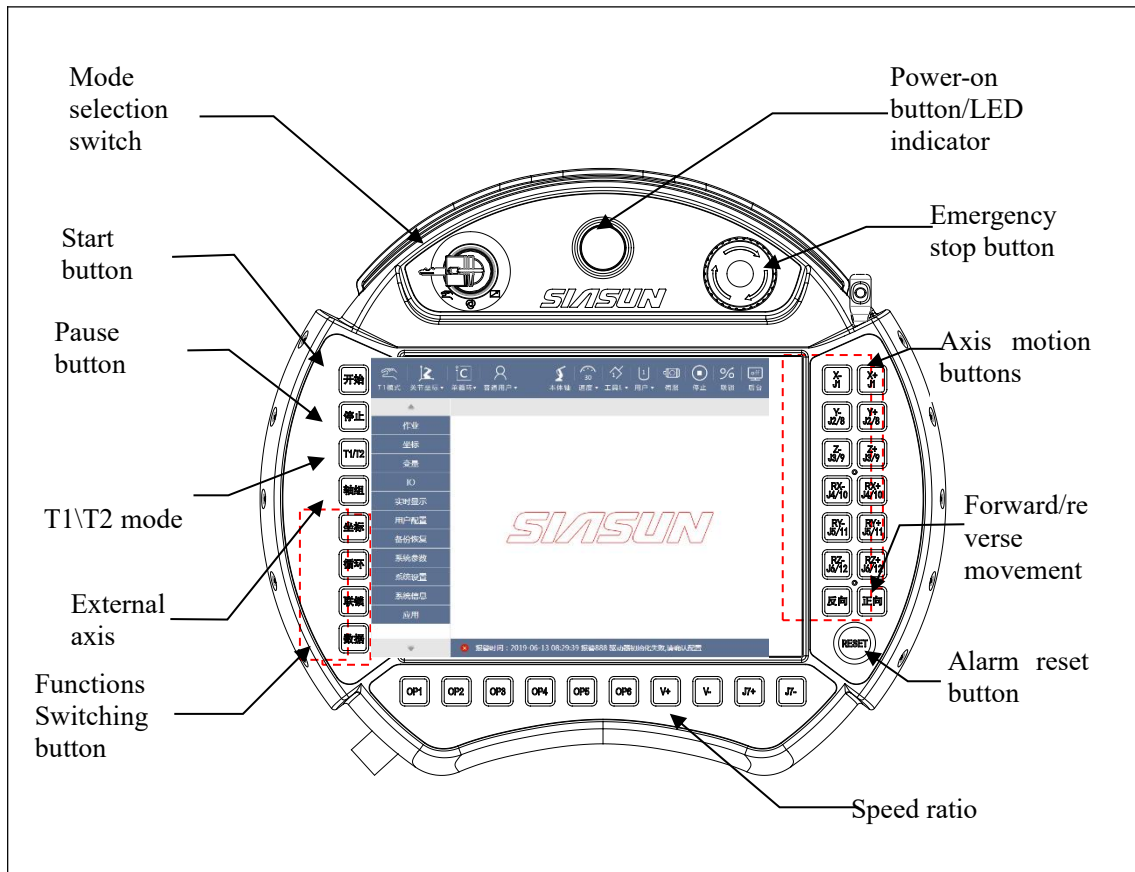


Figure 1-14 Teaching pendant button layout

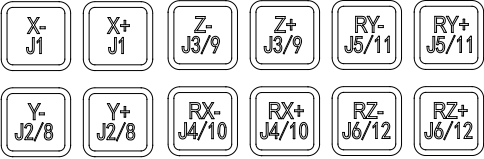
Teaching pendant buttons include:

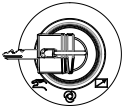
- 1) Buttons related to start and stop.
- 2) Buttons related to axis motion and frame motion.
- 3) Buttons related to execution.
- 4) Specific function buttons.

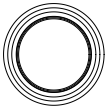
Table 1-2 Teaching Pendant Button Instructions

Button	Function
	The button for activating the selected program when the robot is in execution mode. The stop button can stop the running program in any mode.

	Switching between the T1 and T2 modes of the robot. Warning: In T2 mode, the movement speed of the robot can reach full speed. If not necessary, please do not use T2 mode to ensure the personal safety of the teaching engineer.
	Control switching between the robot body axis and the external axis. If the current type in use is the axis type, it can be displayed on the icon of the teaching pendant.
	Used for switching the frame in manual mode. The frame is switched among joint, rectangular, tool and user frames.
	In manual or automatic operation, switch among the three cycle modes of step, cycle and auto. Notes: Since the cycle mode is active both in manual and automatic operations, please confirm before manual or automatic operation.
	In manual operation, the robot can switch whether to execute or skip logical commands through the interlock key. Notes: Please confirm the interlock status before the forward movement to determine whether the logical

	command is executed during the forward movement.
	Display various data related to the operation of the robot, such as joint values, Rpy, code disk values, etc.
	Special function buttons, which are used when the robot requires a special software package.
	Change the operating speed of the robot.
	In the teaching mode, the axis or robot can be moved simultaneously in a single axis or multiple axes through the axis motion buttons. Only when the axis operation button and DEADMAN are pressed simultaneously, the robot moves. The robot moves according to the selected frame and speed ratio. Before the robot moves, please confirm the frame and speed.
	Control the movement of the robot's external axis 7.
	When Forward Movement and DEADMAN are pressed simultaneously, the robot will operate according to the motion points and commands taught by the program. Please confirm whether the interlock function works before movement. When the interlock function is enabled, the robot executes logical commands and

	motion commands. When the interlock function is inactive, the robot only executes motion commands. Meanwhile, please confirm the cycle mode of the robot and confirm whether the robot program is executed in step mode or continuously. The robot will move at the selected speed. Please determine the operating speed of the robot before moving, especially T2 mode. Warning: In T2 mode, the movement speed of the robot can reach full speed. If not necessary, please do not use T2 mode to ensure the personal safety of the teaching engineer.
	When Reverse Movement and DEADMAN are pressed simultaneously, the robot will reversely move according to the motion points taught by the program. The robot will move at the selected speed. Please determine the operating speed of the robot before moving, especially T2 mode.
	Reset the alarm of the robot. If the alarm is not released, the robot will continue to alarm until the alarm is released and reset.
	Teach\execute\remotely switch the robot mode. In the teaching mode, programming by the teaching pendant enables manual axis operations and forward and reverse movements of the program.

	The robot in the teaching mode cannot be started or started by an external start signal. In the execution mode, the robot can be started locally to automatically run the selected program. The execution mode cannot be initiated by an external start signal. In remote mode, the robot can be externally controlled through remote I/O signals. In remote mode, the start signal on the robot teaching pendant is invalid.
	Servo power-on button. Press this button in the execution mode, and the servo will be powered on. In manual mode, the servo is ready. In case of emergency stop or local alarm, the servo will be powered off. When the servo is powered on, the servo icon will display as powered. Power-on button. Meanwhile, this button is also an LED display light. When the program runs, the button is green. When the program is paused, the button is yellow. When the alarm fails to start, the button is red.

1.3.2 Teaching Pendant Window

The top of the teaching pendant window is the status bar, and the bottom thereof has a message area, which displays status information for various common robot functions.

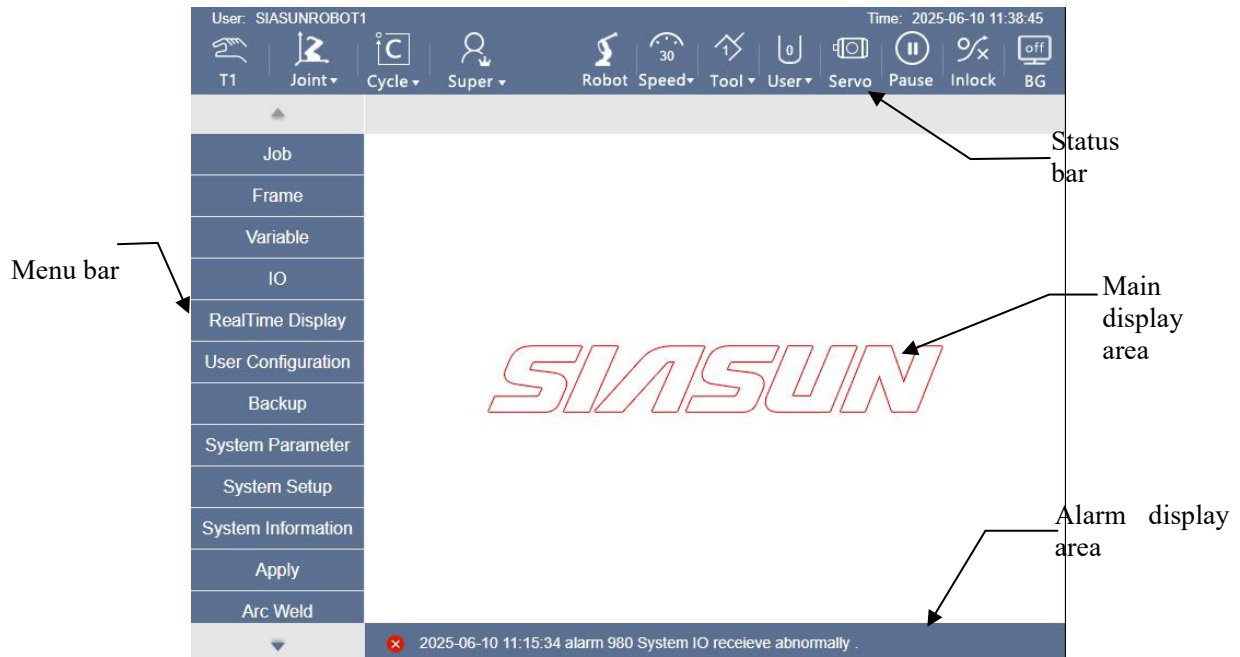


Figure 1-15 Teaching pendant window

Job Editing Interface

In the job teaching interface, the user can perform routine operations such as writing, copying, pasting, deleting, and modifying commands.

In addition, the user can scroll the page through the page up and down buttons beside.

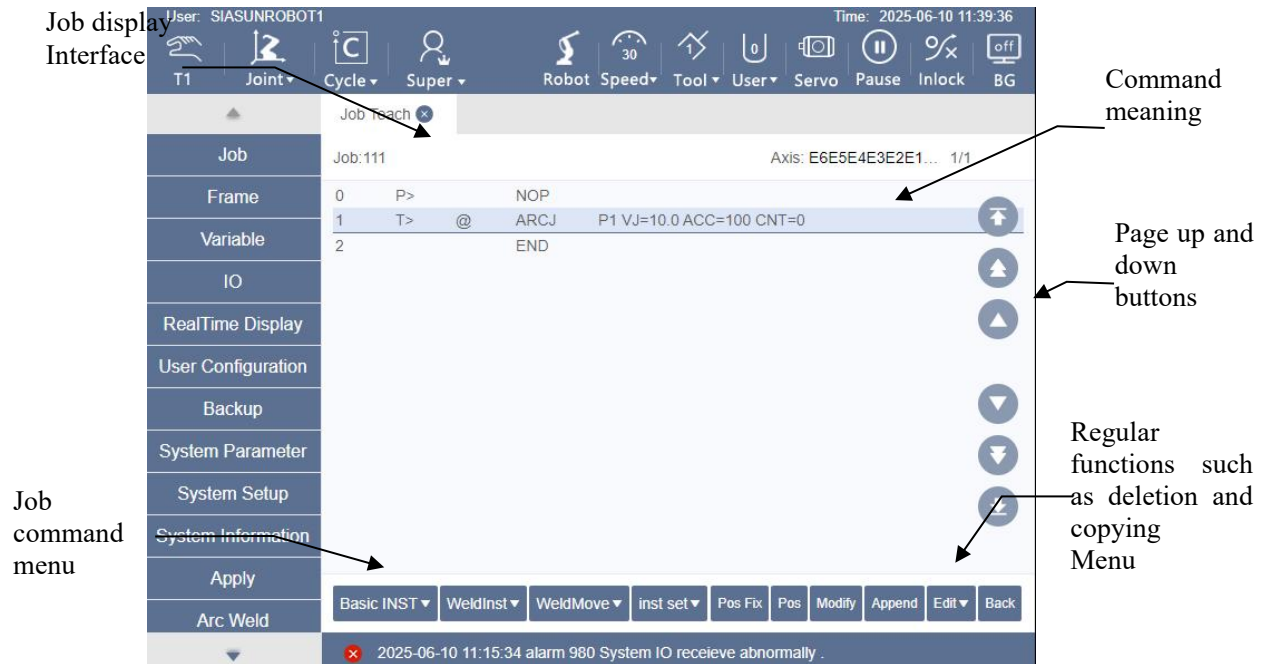


Figure 1-16 Teaching program editing window

Editing cursor and execution cursor:

General Operating Manual for SIASUN Robot

The cursor is divided into the editing cursor and the execution cursor.

The execution cursor records the command position of the job operation, and only changes along with the execution of the job command; the editing cursor records the command position after the user's editing operation.

When the jobs with the same name are opened consecutively, the positions of the last execution cursor and the editing cursor of these jobs can be automatically restored to prompt the user of the command execution status and command editing status of these jobs; as long as the consecutively opened jobs are not of the same name (i.e., different names), both the editing cursor and the execution cursor will be restored to the first line of the job, that is, the NOP line.

When the execution cursor coincides with the editing cursor, the job only displays the execution cursor, as shown in the figure; when the positions of the execution cursor and the editing cursor are not consistent, the editing cursor and the execution cursor on the job page will be observed, as shown in the figure.

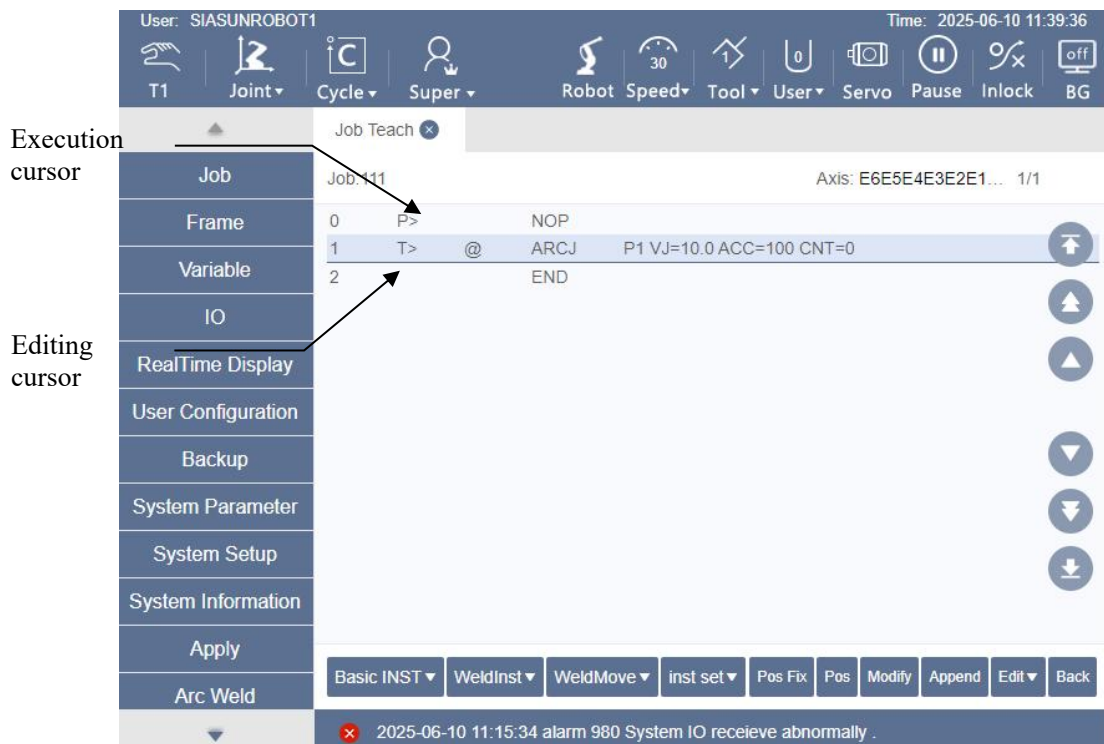


Figure 1-17 Editing cursor and execution cursor

When starting a job (including automatic operation, forward and reverse operations), the editing cursor and the execution cursor must be consistent; otherwise, the job cannot be started, and a pop-up message will appear and prompt the user to choose whether to start the job from the editing cursor or the execution cursor.

The execution cursor of the current job is consistent with the editing cursor. When the user

triggers the forward and reverse buttons at this time, the job can be started and executed immediately.

When the robot moves to this position, the icon @ will be displayed on this command, indicating that the actual position of the robot is the position indicated by this command.

The execution cursor of the current job is inconsistent with the editing cursor. When the user triggers the forward and reverse buttons at this time, the job is not started, and a pop-up prompt box appears, indicating "The editing cursor of the current job is inconsistent with the execution cursor. The job is prohibited from starting.". When "Confirm" is selected, the job will be started from the editing cursor; when "Cancel" is selected, the job will be started from the execution cursor. When "Confirm" is selected, the execution cursor will automatically move to the editing cursor, that is, the two cursors overlap; when "Cancel" is selected, the editing cursor needs to be manually moved to the execution cursor to ensure that the two cursors coincide. Regardless of the selection, the job can only be officially started from the editing cursor or the execution cursor after Forward and Reverse buttons are triggered again. Similarly, the same is true for starting a job in the execution mode.

Page up and down buttons:

There are three types of page up and down buttons on the right side of the editing interface, which can be used to turn pages for the job.

Table 1-3 Introduction to Page Up and Down Buttons

Button	Description
	The cursor moves to the first line and last line of the job.
	When paging up or down, the cursor position remains unchanged.
	The cursor moves up or down one line.

Status bar:

The status bar above the robot screen will display various contents or data related to the robot's status.

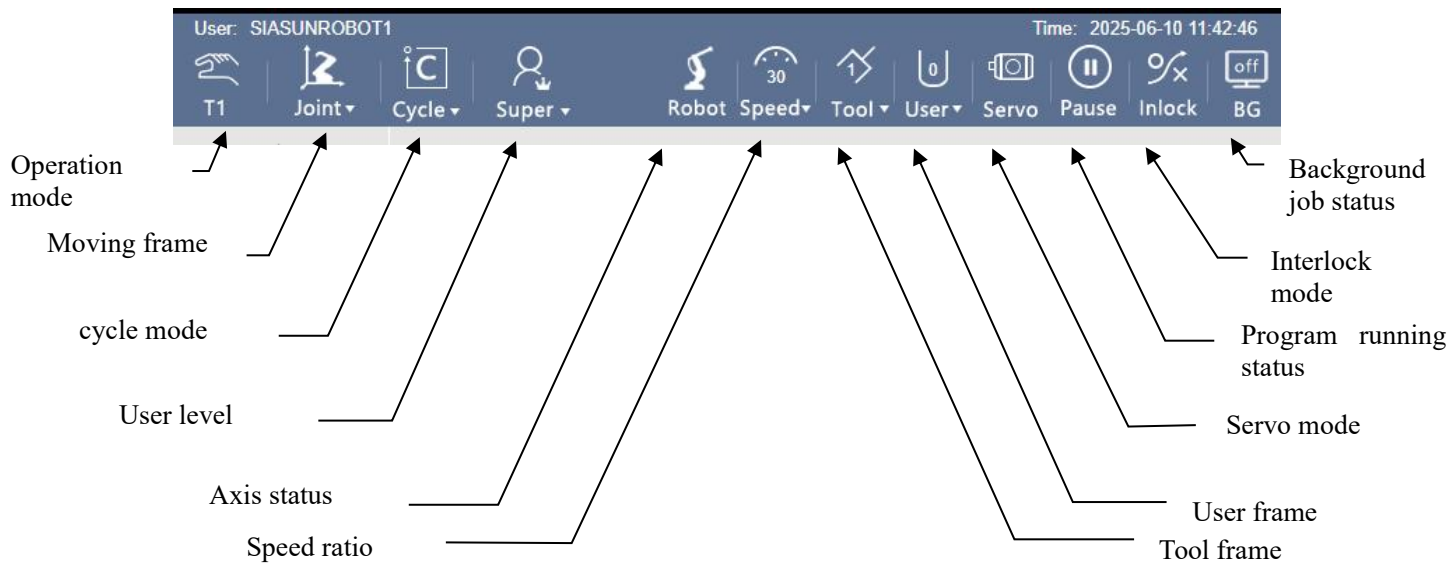
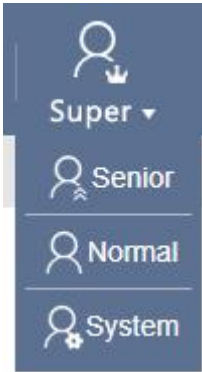


Table 1-4 Meanings of Status Bar

Icon	Meaning
	Status display icon of robot in current execution mode. They are respectively T1 / T2 mode, execution mode and remote mode. Warning: In T2 mode, the movement speed of the robot can reach full speed. If not necessary, please do not use T2 mode to ensure the personal safety of the teaching engineer.
	Used for switching the frame in manual mode. The frame is switched among axis, rectangular, tool and user frames.

	Program cycle mode: step, auto, and cycle. These three modes are valid in automatic or manual operation. Notes: Since the cycle mode is active both in manual and automatic operations, please confirm before manual or automatic operation.
	Different permissions can be granted for different user levels. The passwords are "00001111" for advanced user and "12341111" for superuser. System Users is designed for developers and is not intended for customer use. Note: When the input content is completed, press [Go] button in the lower right corner of the soft keyboard first before actually entering the content into the parameter box.
	When manually moving the robot and switching between the robot body axis and the external axis, the icons will change accordingly.
	Icon for speed ratio display. When the speed ratio changes, the corresponding value changes.
	Tool, external tool icon. When the file number of the tool frame changes, the number on the icon changes accordingly.

	Notes: The change to the tool frame number has an impact on the motion path of the robot program. Please confirm before manually or automatically executing the program.
	User frame icon. When the file number of the user frame changes, the number on the icon changes accordingly. Notes: The change to the user frame number, in some cases, such as offset command, has an impact on the motion path of the robot program. Please confirm before manually or automatically executing the program.
	Servo status icons, representing servo power-off, servo power-on, and servo enable.
	Program running status icons, representing program start, pause and stop.
	Background job status icons. ON indicates that the background is in the started state, and OFF indicates that the background is in the stopped state.
	Indicate whether the job executes logical commands in manual mode. When both are circles, it indicates that the logical command and the motion command are being executed in the forward movement. When there is a circle and a cross, it indicates that no logical

	command is executed in the forward movement, and only motion command is executed. Notes: Please confirm the interlock status before the forward movement to determine whether the logical command is executed during the forward movement.
--	--

Introduction to main menu bar:

The main menu is located on the left side of the interface and serves as the entry point for various robot operations.

Table 1-5 Introduction to Main Menu of Robot

Icon	Description
	Enter the job list interface, where the user can create new programs, open programs, and manage jobs. Meanwhile, the existing job list and job types can be seen.
	Operations, such as viewing, calibrating and setting various frames of the robot, can be performed.
	Operations, such as viewing and writing, can be performed on various types of variables.
	Operations, such as viewing input and output, manually outputting states, configuring group I/O, mapping remote I/O and setting up bus I/O, can be performed.
	Display data such as robot joint values, code disk values, pose values and IO status.

User Configuration	Enter the user configuration interface and configure various conditions, including the configuration of functions such as teaching, origin, external device communication, interference zone and interrupt operation.
Backup	Back up or restore the operation, parameters and other data of the robot.
System Parameter	The interface for configuring the system parameters of the robot.
System Setup	The system setup interface of the robot.
System Information	Display the system version, system library, memory and other information of the robot.
Apply	Various application menus.

Text input operation:

When the user clicks on the area where the user needs to enter letters or parameters on the touchscreen, a soft keyboard will pop up at the bottom of the screen. The required content can be entered through the soft keyboard.

Notes:

- 1) When the input content is completed, press [Go] button in the lower right corner of the soft keyboard first before actually entering the content into the parameter box.
- 2) The entered content is based on the content before the first trigger of [Go]

button. After [Go] button is triggered for the first time, the content subsequently edited will be invalid.

- 3) Before triggering [Go] button, please carefully confirm the content you have edited. If [Go] button has been triggered and the edited content requires changing, click [Cancel] button on the soft keyboard.

Alarm interface:

Click on the alarm bar at the bottom of the screen, the alarm interface will pop up. Alarm information can be displayed.

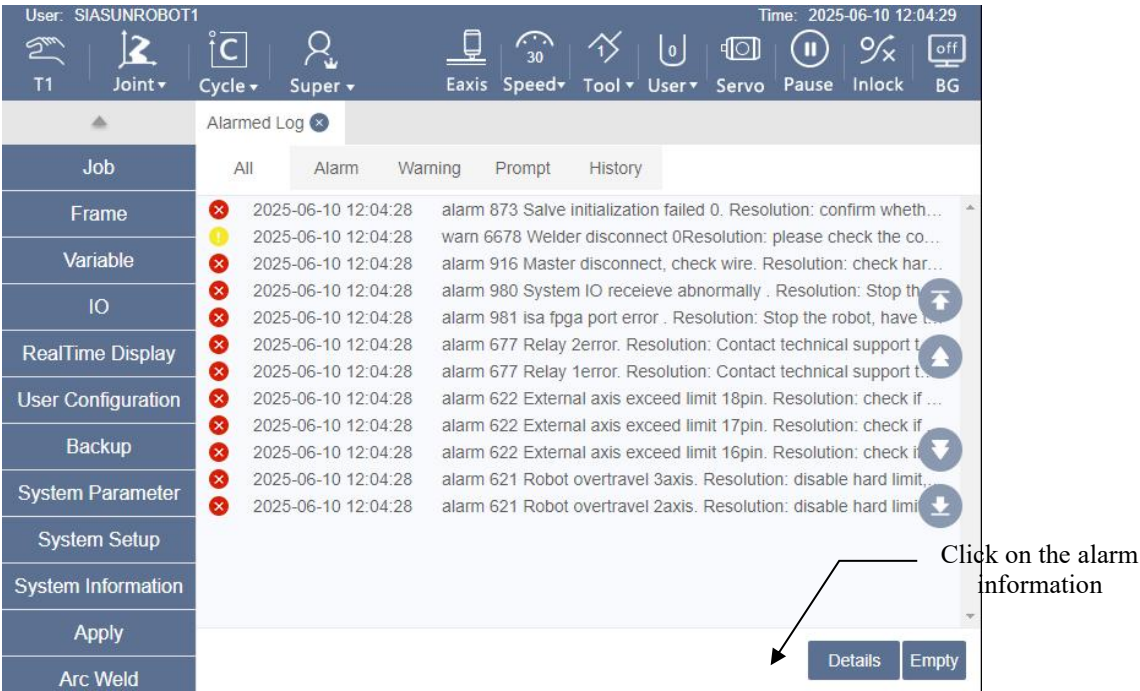


Figure 1-19 Alarm Interface

Table 1-6 Introduction to Alarm Interface Icons

Alarm prompt bar	Display alarm, warning and prompt information.
	Robot alarm, warning and prompt icons.

Menu tree structure:

The menu tree structure is shown in the following table.

Table 1-7 Main Menu List

First-level menu	main menu	Second-level menu	main menu	Third-level menu	main menu	User level
Job		LastJob				Regular user
		New job				Advanced user
		OpenJob				Regular user
		Copy				Advanced user
		Delete				Advanced user
		More		Job Property		Regular user
				Change		Advanced user
				Register/Cancel		Advanced user
Coordinate		User coordinates		Calibration		Advanced user
				External calibration		Advanced user
				Setting		Advanced user
				External setting		Advanced user
		Tool coordinate		5-point calibration		Advanced user
				20-point calibration		Advanced user
				Pose calibration		Advanced user
				Setting		Advanced user
		Workpiece coordinate		Calibration		Advanced user
				Setting		Advanced user
		External tool		Calibration		Advanced user
				Setting		Advanced user
		World coordinate		Setting		Advanced user
		Current coordinate				Regular user
Variable		User integer				Advanced user
		User float				Advanced user
		User byte				Advanced user
		Character variable	string			Advanced user

	Position variable		Advanced user
	System integer		Advanced user
	System float		Advanced user
IO	Manual operation IO	User IO input	Advanced user
		User IO output	Advanced user
	Group IO configuration		Advanced user
	Remote IO mapping	Input	Advanced user
		Output	Advanced user
	Bus IO mapping	Input IO mapping	Advanced user
		Output IO mapping	Advanced user
	Explosion mapping	Input	Advanced user
		Output	Advanced user
		Explosion-proof switch	Advanced user
		Waiting time	Advanced user
	Analog IO configuration	Input	Advanced user
		Output	Advanced user
	Custom group IO	Input	Advanced user
		Output	Advanced user
	Logic IO distribution	Input IO distribution	Advanced user
		Output IO distribution	Advanced user
Real-time display	Real-time display		Regular user
	User IO input		Regular user
	User IO output		Regular user
	System IO input		Regular user
	System IO output		Regular user
	Alarm log		Regular user
	Command information		Regular user

	Operation log		Regular user
User configuration	Teaching condition		Regular user
	External communication setting	DevNet	Advanced user
		Serial port	Advanced user
		EtherCAT	Advanced user
		Switch configuration	Advanced user
		LAN switch configuration	Advanced user
		LAN IP setting	Advanced user
		Modbus slave station configuration	Advanced user
		Modbus master station configuration	Advanced user
		EtherCAT NT151	Advanced user
	Home position record	Home position	Advanced user
		Range	Advanced user
		External axis	Advanced user
	Reservation	Reservation setting	Advanced user
		Reservation status	Regular user
	Interference area	Axis interference	Advanced user
		Space shape calibration	Advanced user
		Space interference property	Advanced user
	Robot orientation		Advanced user
	Username		Advanced user
	Set IP		Advanced user
	Language		Advanced user
	Collision protection	Teaching sensitivity	Advanced user
		Execution sensitivity	Advanced user
	Load parameter	Load setting	Advanced user

		Additional load	Advanced user
	Cycle mode		Superuser
	Vibration suppression parameter		Superuser
	Feedforward friction		Superuser
	Plan library		Superuser
	External axis speed control		Advanced user
	Custom alarm		Advanced user
	Interlock		Advanced user
	Alarm code output configuration		Advanced user
	User permission configuration	Change password	Advanced user
		Reset password	Regular user
		Permission logout	Advanced user
Backup recovery	Parameter		Advanced user
	Position variable		Advanced user
	Job upload		Advanced user
	Job download		Advanced user
	Uninstall USB		Advanced user
System parameters	Motion parameters	Straight line configuration	Superuser
		Speed	Superuser
		Reduction ratio	Superuser
		Code disk number	Superuser
	Control parameters	Soft limit	Advanced user
		Safety protection	Superuser
		Torque limit	Superuser
		Rod length compensation	Superuser
		Indefinite rotation	Advanced user

	Certificate	Remaining time	Advanced user
		Authorization	Superuser
		Certification setting	Superuser
System setting	Application configuration		Superuser
	Model configuration		Superuser
	Bus device		Superuser
	Bus attribute		Superuser
	Bus optimization		Superuser
	Zero position	Zero calibration	Superuser
		Zero position setting	Superuser
		Code disk clearing	Superuser
	Number of system axes		Superuser
	Execution switch		Advanced user
	Security setting		Superuser
	External axis	Axis configuration	Advanced user
		System configuration	Advanced user
		Variable 1 calibration	Advanced user
		Variable 2 calibration	Advanced user
		Variable 3 calibration	Advanced user
		Variable 1 setting	Advanced user
		Variable 2 setting	Advanced user
		Variable 3 setting	Advanced user
		Slide calibration	Advanced user
		Slide setting	Advanced user
		Transformation matrix	Advanced user
		Turntable setting	Advanced user
		Rotating shaft setting	Advanced user
		Share external axis	Advanced user

		External axis manual/automatic	Advanced user
	Cabinet configuration		Superuser
	File quantity extension		Superuser
System information	System version		Regular user
	System library		Superuser
	Memory		Regular user
	Upgrade		Superuser
	Uninstall		Superuser
	System time		Advanced user
	Restart		Regular user
Application			Superuser

1.3.3 Communication

The interfaces configured for bus communication include:

- 1) RS232, the communication interface is located at the COM interface of the motherboard;
- 2) DeviceNet, the MCB board is located at the X5 interface, and the SCB board is located at the P5 interface;
- 3) Profinet, the communication interface is located at the LAN1 interface of the motherboard;
- 4) Modbus TCP, the communication interface is located at the LAN5 interface of the motherboard;
- 5) EtherCAT, the communication interface is located at the LAN3 interface of the motherboard;

Notes:

The above introduction is about the meanings of M5 controller interfaces. When using other controllers, please refer to Electrical Operation and Maintenance Manual for SIASUN Robot.

1.3.4 I/O

The input and output signals of the robot, namely I/O signals, can transmit data between the robot control software and external devices through the user I/O signals.

I/O types include:

- 1) User I/O signal.
- 2) Group I/O signal.
- 3) Analog I/O.

User I/O is divided into hardware direct connection IO and bus communication IO, supporting a total of 4096-bit input and 4096-bit output.

Hardware direct connection IO has 8-bit input and 8-bit output, as well as 16-bit input and 16-bit output depending on different control cabinets. It can be expanded by adding modules, and the number of IOs varies with different hardware configurations. Among them, Huatai module supports up to 64-bit output and 64-bit input. Refer to Electrical Operation and Maintenance Manual for SIASUN Robot for details.

1.3.5 Remote I/O

Remote I/O is the signal for communication between the robot and the peripheral devices. These signals need to be defined as special signals and set through the software interface.

- 1) Call the main job.
- 2) Start or stop the program.
- 3) Cancel the alarm.
- 4) Interaction with other signals.

1.3.6 Robot Action

The movement of the robot moving the tool center point (TCP) from the current position to the target position is processed as a command, and in this way, the editing of the robot action is completed. The robot software will automatically process the relevant path, acceleration, speed and position information.

There are two ways to make a robot move: manual control through buttons and execution of actions through programs.

The robot movement can be manually controlled through the buttons on the teaching pendant.

The robot actions are determined by the selected frame and speed.

When a robot executes automatically according to a program, its movements are determined by the position data, motion type, speed and acceleration given in the program commands.

The action types include J (axis) motion, L (linear) motion, and C (arc) motion. During J motion, the TCP point moves irregularly between the two teaching points, and the motion path repeats the same path. During L motion, the TCP point moves along the straight line between the two teaching points. During C motion, the TCP point moves along the arc among three teaching points.

When in operation, the robot can turn on or off the continuous path, that is, it does not stop at the target point but slides rapidly in the vicinity of the target point.

1.3.7 Emergency stop button

The robot has the following emergency stop buttons:

- 1) Emergency stop buttons on the teaching pendant and the control cabinet.
- 2) External emergency stop buttons configured by the user.

When the emergency stop button is pressed, the robot will trigger the emergency stop state regardless of its status. The interface terminal of the external emergency stop button is located in the internal control unit or interface panel of the robot control cabinet.

2 Robot Settings

Before using the robot system, set functions.

2.1 Set frames

Frames are position frames defined on the robot or in space to determine its positions and poses. Frames include joint frame, rectangular frame, tool frame and user frame.

Joint frame:

The joint frame is the frame of the robot on each axis joint. It enables each axis of the robot to move independently.

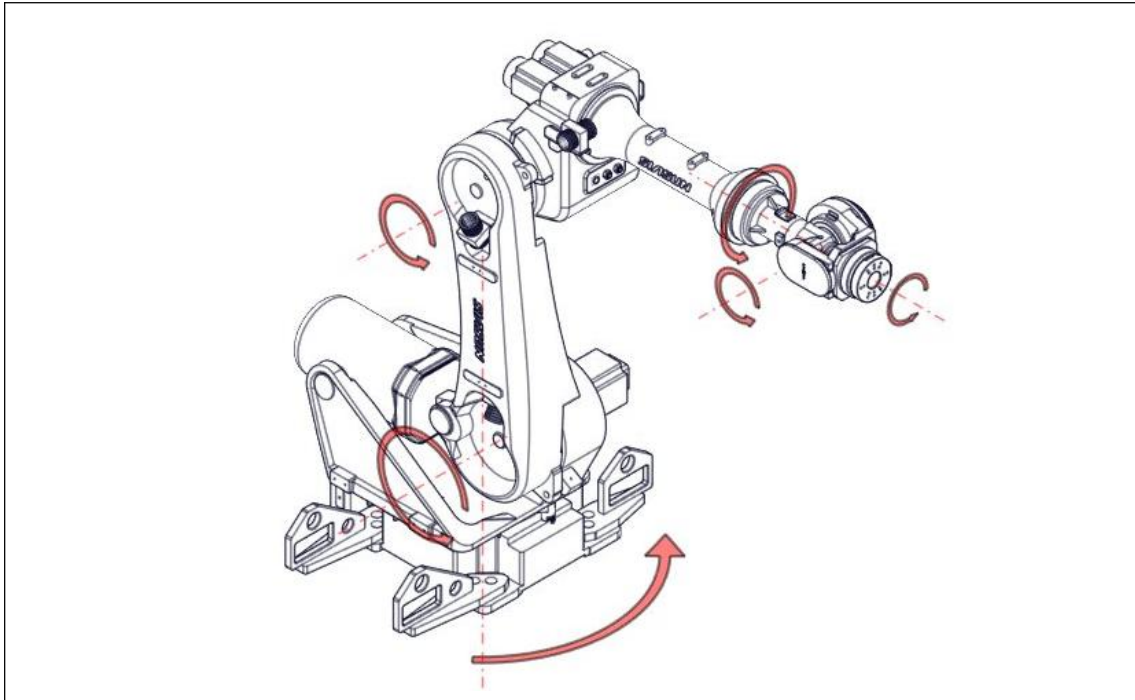


Figure 2 - 1 Joint frame

Three Cartesian frames:

SIASUN robot has three Cartesian frames: rectangular frame, tool frame and user frame. The position and pose of the robot are defined by the frames x, Y, and z from the origin of the rectangular frame in space to the origin of the Cartesian frame of the tool (TCP point), and the rotation angles R_x , R_y and R_z of the Cartesian frame on the tool side around the X-axis, Y-axis and Z-axis of the rectangular frame in space.

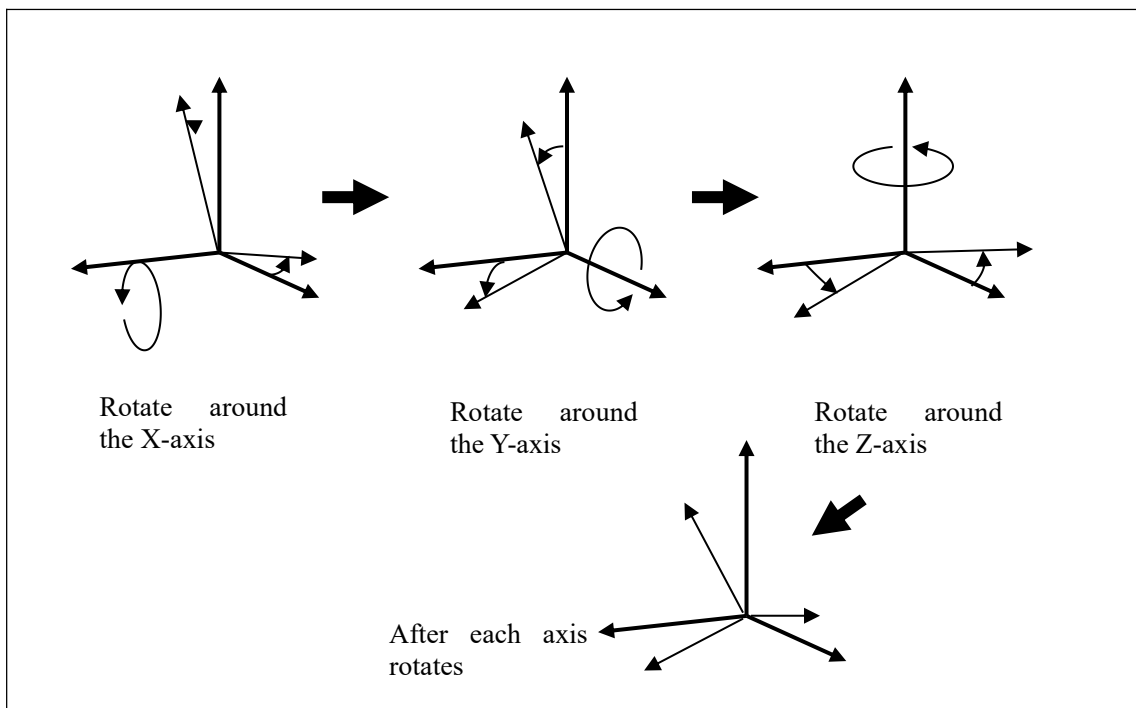


Figure 2- 2 Meanings of (Rx, Ry, Rz)

Rectangular frame:

Rectangular frame of SIASUN robot. The origin is defined at the intersection of the axial line of the axis 1 and the horizontal plane where the axis 2 lies.

The TCP of the robot moves parallel to the X\Y\Z-axis.

Tool frame:

The tool frame defines the position and pose of the TCP. The tool frame needs to be calibrated or set by the user. Before calibration, the tool frame is set to a 6-axis flange frame by default.

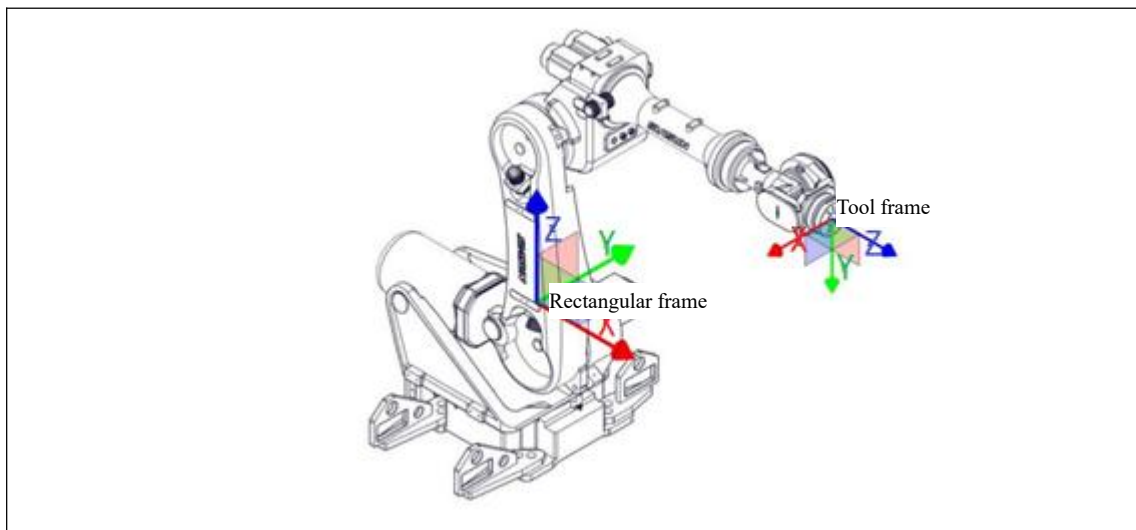


Figure 2- 3 Rectangular and tool frames

User frame:

The user frame is a Cartesian frame in which the user defines the space of each job. It is used for teaching and executing position storage, and executing position compensation commands, etc. The default user frame is located on the rectangular frame (the same as the rectangular frame).

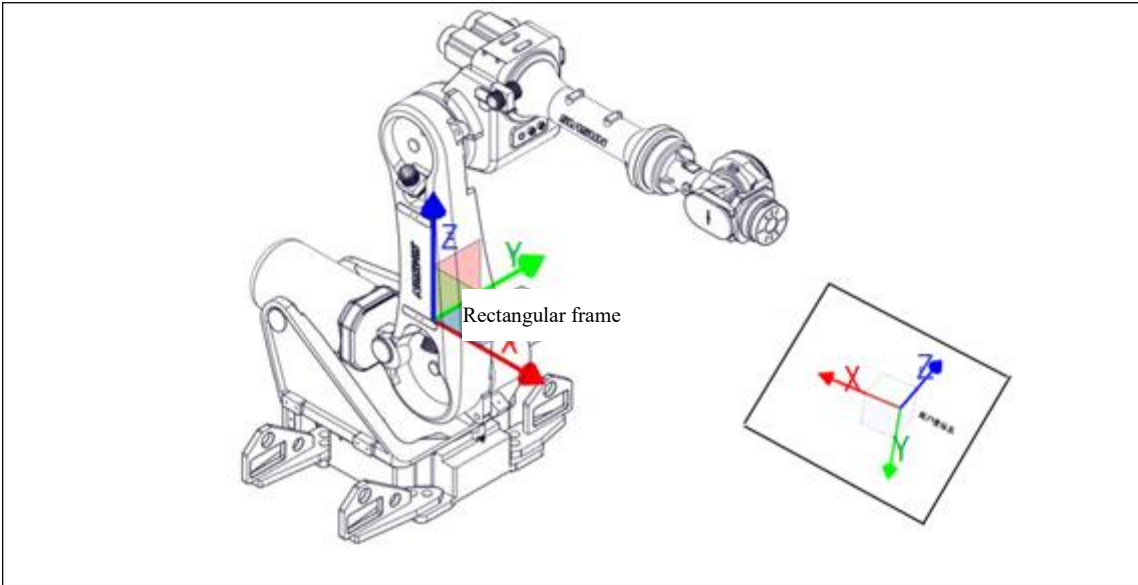


Figure 2-4 User frame

Warning:

If the teaching program changes the tool or user frame, each program point of the program must be re-taught and manually verified to avoid personal injury or equipment damage.

External tool frame:

When the tool is installed at a position other than the robot, the calibrated tool frame is called the external frame, and the calibration method is the same as that of the user frame.

2.1.1 Frames and Axis Motion Buttons

Joint frame:

The joint frame can be used to move each axis of the robot independently.

Table 2-1 Robot Motion in Joint Frame

Axis name	Axis operation buttons	Action
Axis 1	 	J1 axis rotates left and right.
Axis 2	 	J1 axis moves forward and backward.
Axis 3	 	J3 axis moves up and down.

Axis 4	 	J4 axis rotates.
Axis 5	 	J5 axis moves up and down.
Axis 6	 	J6 axis rotates.

Rectangular frame:

In the rectangular frame, the TCP of the robot will move parallel to the X/Y/Z-axis of the robot. Meanwhile, the robot can change its pose with the TCP point as the center while the TCP point remains unchanged.

Table 2-2 Robot Motion in Rectangular Frame

Axis name	Axis operation buttons	Action
X-axis	 	TCP moves parallel to the X-axis.
Y-axis	 	TCP moves parallel to the Y-axis.
Z-axis	 	TCP moves parallel to the Z-axis.
Rx-axis	 	TCP rotates along the X-axis and changes its pose.
Ry-axis	 	TCP rotates along the Y-axis and changes its pose.
Rz-axis	 	TCP rotates along the Z-axis and changes its pose.

Tool frame:

In the tool frame, the TCP of the robot will move parallel to the X/Y/Z-axis defined by the tool. Meanwhile, the robot can change its pose with the TCP point as the center while the TCP point remains unchanged.

Table 2-3 Robot Motion in Tool Frame

Axis name	Axis operation buttons	Action
X-axis	 	TCP moves parallel to the X-axis.
Y-axis	 	TCP moves parallel to the Y-axis.
Z-axis	 	TCP moves parallel to the Z-axis.
Rx-axis	 	TCP rotates along the X-axis and changes its pose.
Ry-axis	 	TCP rotates along the Y-axis and changes its pose.
Rz-axis	 	TCP rotates along the Z-axis and changes its pose.

Notes

The direction of each axis in the tool frame will move along with the change of the wrist end. However, the movement direction of the robot under the tool frame will be based on the tool itself and will not change with the position or pose of the robot.

User frame:

In the user frame, the TCP of the robot will move parallel to the X/Y/Z-axis defined in space. Meanwhile, the robot can change its pose with the TCP point as the center while the TCP point remains unchanged.

Table 2-4 Robot Motion in Rectangular Frame

Axis name	Axis operation buttons	Action
X-axis	 	TCP moves parallel to the X-axis.
Y-axis	 	TCP moves parallel to the Y-axis.
Z-axis	 	TCP moves parallel to the Z-axis.
Rx-axis	 	TCP rotates along the X-axis and changes its pose.
Ry-axis	 	TCP rotates along the Y-axis and changes its pose.
Rz-axis	 	TCP rotates along the Z-axis and changes its pose.

2.1.2 Set Tool Frame

Tool frame is a Cartesian frame representing the TCP and pose. Under normal circumstances, it takes TCP as the origin and the tool direction as the Z-axis. Before calibrating the tool frame, the default tool frame is the flange frame.

The tool frame is determined by the position values (x, y, z) of the tool center point (TCP) of the tool frame and the tool pose values (Rx, Ry, Rz).

The tool center point (TCP) is defined by the x, y and z distance values relative to the center of the 6-axis flange. The tool pose is determined by the rotation angles Rx, Ry, and Rz of the x-axis, y-axis and z-axis in the flange frame.

After the TCP is calibrated, the linear path is based on the position of TCP during teaching. Pose transformation using tool frame is also carried out according to the pose of the tool frame, that is, TCP rotates along the axis of the tool frame.

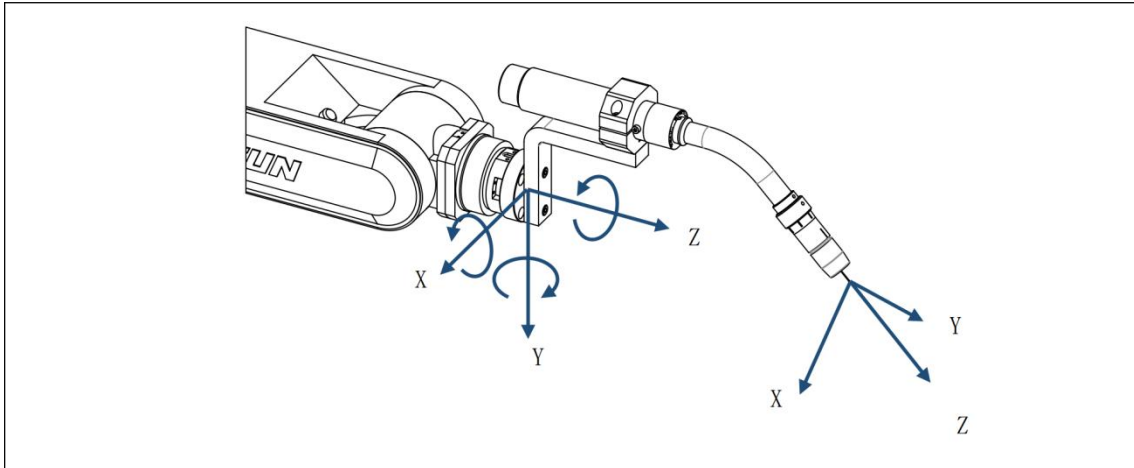


Figure 2-5 Tool frame

Notes:

When the axis motion buttons can be used for single-axis motion in the joint frame, the axis motion joint value does not change with the tool frame or the user frame. However, when the program runs automatically or manual forward movement is performed, the position and pose of the robot will be affected by both the tool and user frames.

The TCP of the frame is determined by the following method:

Tool frame setting.

There are two methods for calibrating the origin of tool frame: 5-point method and 20-point method.

The difference between the 20-point method and the 5-point method: The 20-point method changes the original zero position of the robot.

The similarity between the 20-point method and the 5-point method: They can both calibrate the origin of the robot tool frame. When using the same tool frame number for calibration, the later calibrated value will overwrite the earlier calibrated value.

20-point tool frame calibration:

The 20-point method can determine:

1. TCP, that is, the x, y and z values of the origin of tool frame relative to the flange center point.
2. Update the zero position of the robot.

Note: Before calibration, please confirm the zero position of the robot. The deviation of the zero position of each axis based on the label position should be less than 5°; otherwise, calibration failure will occur during the calculation after the 20-point

calibration. The maximum zero compensation is 5 degrees. The tip of the external calibration needle should be higher than the origin of the rectangular frame (the point where the first axis intersects the second axis).

Calibration steps of 20-point method:

- 1) Click [Frame] - [Tool Frame] - [20PtCalibration] in the main menu to enter the calibration interface. The tool number to be calibrated can be selected through the page up and down buttons.

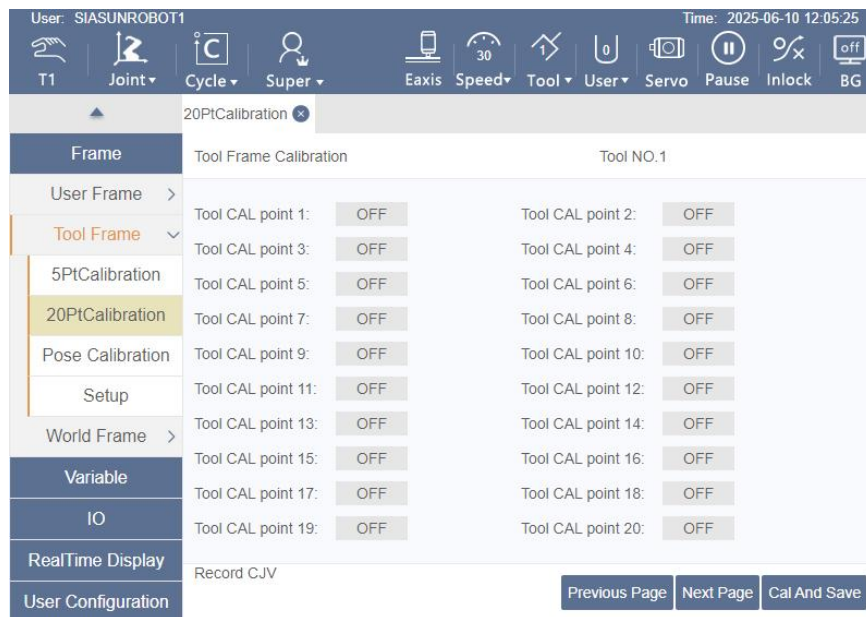


Figure 2-6 20-point tool calibration interface

- 2) The tool calibration requires the selection of a calibration needle and a fixed reference point (needle tip). As shown in Figure 2-20, the tool calibration needle is installed on the robot, and the calibration reference needle is fixed outside the robot to ensure that its position remains fixed during calibration.

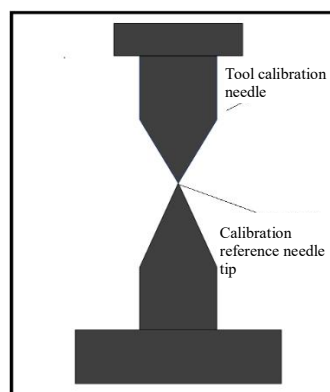


Figure 2-7 Tool calibration

- 3) Align the reference point in 20 different poses in sequence and click "OFF". When "OFF" changes to "ON", this point is successfully calibrated. Click "ON". When "ON" changes to "OFF", this point is cancelled. There are a total of 8 tool frames. The number is displayed on the tool number in the upper right corner of the interface.
- 4) For the selection of each point, the greater the difference among 20 points is, the better is. It is recommended to select 20 points and calibrate using the rectangular frame. At point 1, the tool needle's pose should be vertically downward. At points 2 to 4, move Rx+ and select one point every 30° to return to vertical. At points 5 to 7, move Rx- and select one point every 30° to return to vertical. At points 8 to 10, move Ry+ and select one point every 30° to return to vertical. At points 11 to 13, move Ry- and select one point every 30° and return to the vertical. At points 14 to 17, move Rz+ and select one point every 40° to return to the vertical. At points 18 to 20, move Rz- and select one point every 50°.

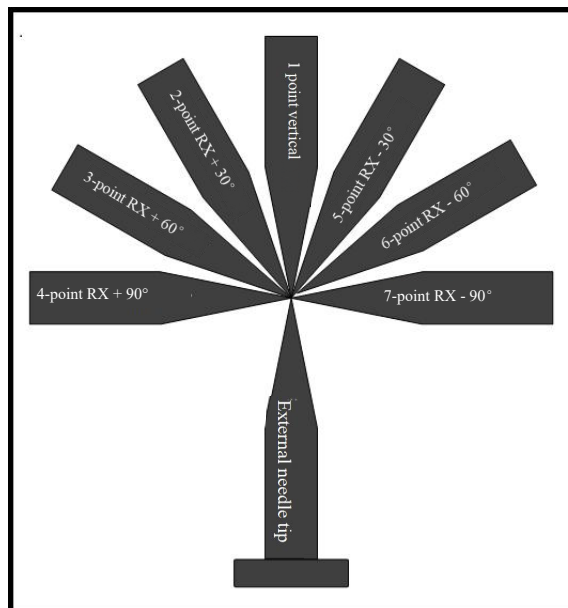


Figure 2- 8 Calibration diagram of points 1-7

- 5) Save the calibration results, release the enable button, and click [Cal and Save]. A prompt [Whether to modify zero position and calculate the tcp pose value] will pop up. Click Confirm, and a prompt of successful calibration will be given. The zero point update is completed.

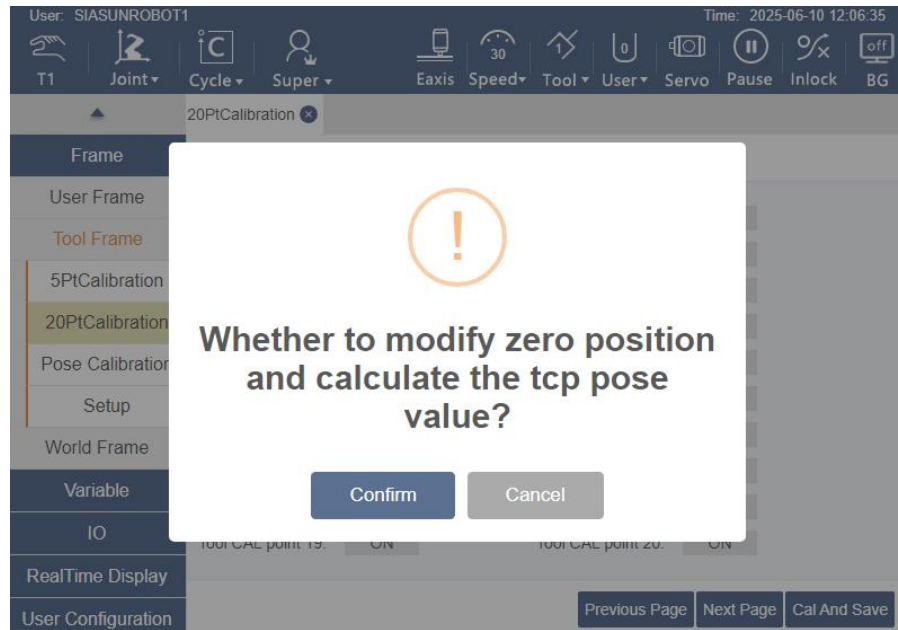


Figure 2-9 Modify zero position and TCP confirmation interface

- 6) The calibration results are displayed on the setting interface corresponding to the file number, i.e. the values of Px, Py, and Pz.

Notes

- 1) The calibration points are incomplete. Click [Cal and Save], then [Calibration data is incomplete, data cannot be calculated or saved] will pop up. Please supplement the uncalibrated points or recalibrate.
- 2) Click Cancel, then the calibration results will not be saved, the zero position will not be updated, and the calibration position and status will remain unchanged.
- 3) 20-point calibration will change the zero position. After saving, the zero position will be updated, thereby affecting the position points of the original program. Please confirm before saving.
- 4) When using 20-point calibration, the point position data in the previously calibrated file number will be cleared, changing from ON to OFF.
- 5) The calibration point status is saved and will remain as before when the page is reopened. After calibration and calculation are completed, ON will not change to OFF.

Calibration by 5-point method:

The 5-point method can determine the TCP, that is, the x, y and z values of the origin of tool frame relative to the flange center point. Use the manual motion robot to make the robot TCP

point to the same point in five different poses. The system will automatically calculate the TCP position value. During teaching, try to approach the target point with five different poses in different directions.

Notes:

The target point should be higher than the origin of the rectangular frame (the point where the first axis intersects the second axis).

After calibration by the 5-point method, only the TCP points (x, y, z) of the tool can be determined. At this time, the pose values (Rx, Ry, Rz) of the tool are the default (0,0,0).

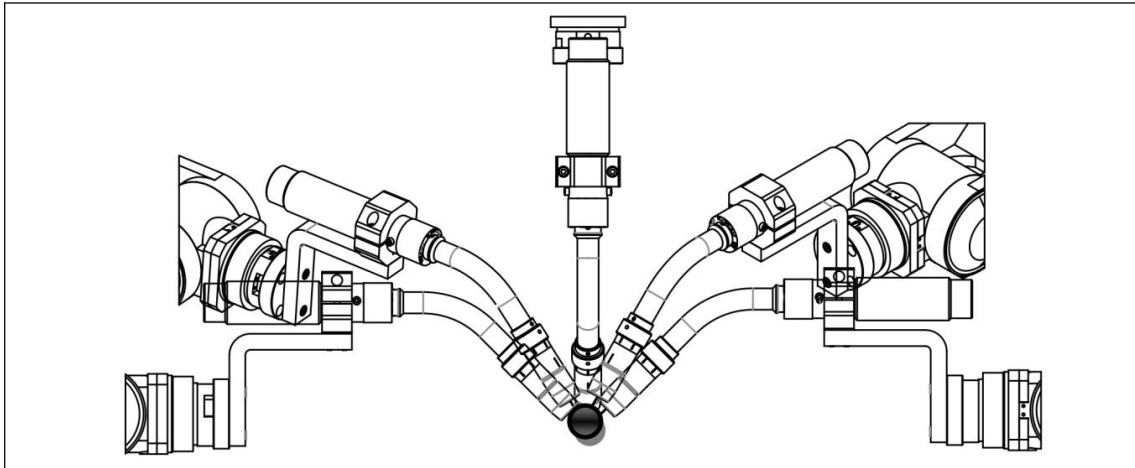


Figure 2- 10 Calibrate the tool TCP by the five-point method

Calibration steps of 5-point method:

- 1) Click [Frame] - [Tool Frame] - [Calibration] in the main menu to enter the tool frame calibration interface. The tool number to be calibrated can be selected through the page up and down buttons.

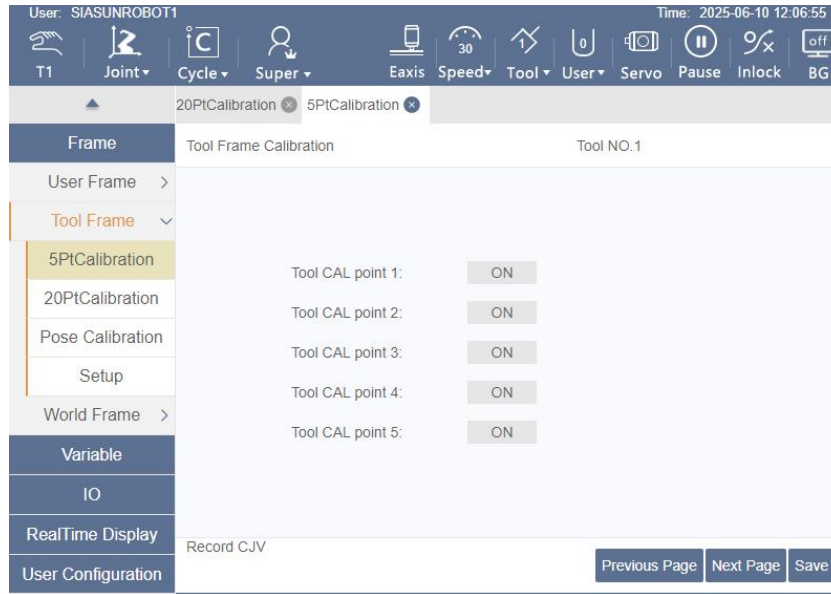


Figure 2- 11 Tool frame calibration interface

- 2) Approach the same target point in five different poses in sequence and click "OFF".
When "OFF" changes to "ON", this point is successfully calibrated. Press "Save" to end the calibration. Press "Next Page" to calibrate another tool frame. There are a total of 8 tool frames. The number is displayed on the tool number in the upper right corner of the interface.
- 3) The calibration results are displayed on the setting interface corresponding to the file number, i.e. the values of P_x , P_y , and P_z .

Tool pose calibration:

Use the 5-point method to calibrate the tool TCP, and then calibrate the pose. Calibrate the tool pose at any point in space, then calibrate the x-direction of the tool along the parallel direction of the X-axis, and finally calibrate the Z positive direction of the tool at a point on the XZ plane in the positive direction of the tool z-axis.

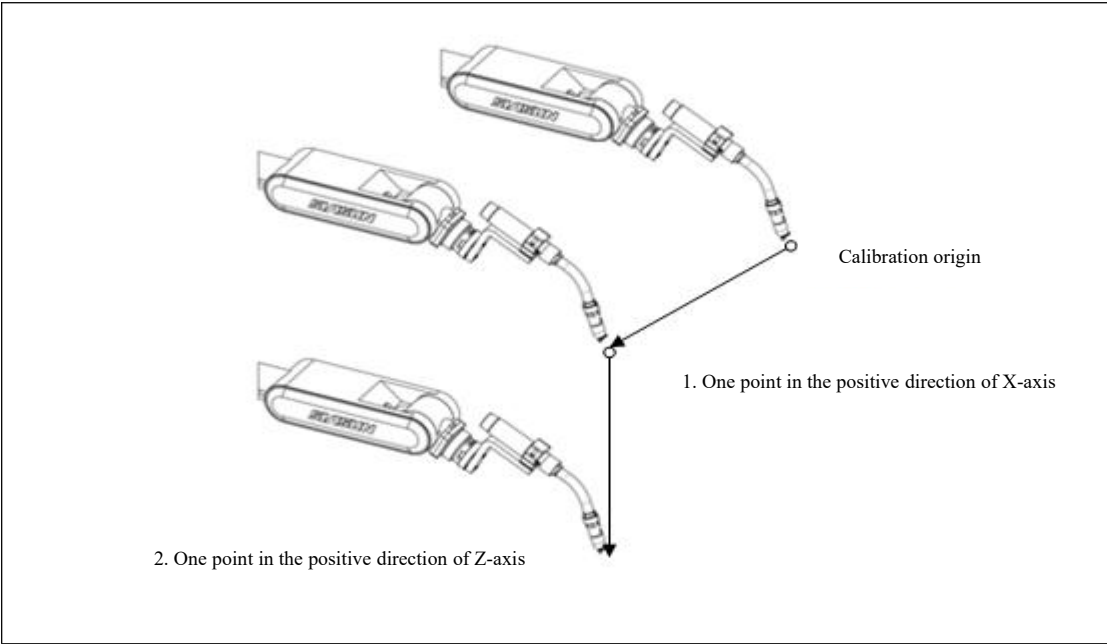


Figure 2- 12 Tool frame direction calibration

Pose calibration steps:

Notes:

If the calibrated frame number previously had a tool frame calibration, the pose value of this tool frame may not be (0,0,0).

Please go to Tool, reset them to (0,0,0), and then recalibrate the pose.

- 1) Click [Frame] - [Tool Frame] - [Calibration] on the main menu. Press "Previous Page" or "Next Page" to select the desired tool number.

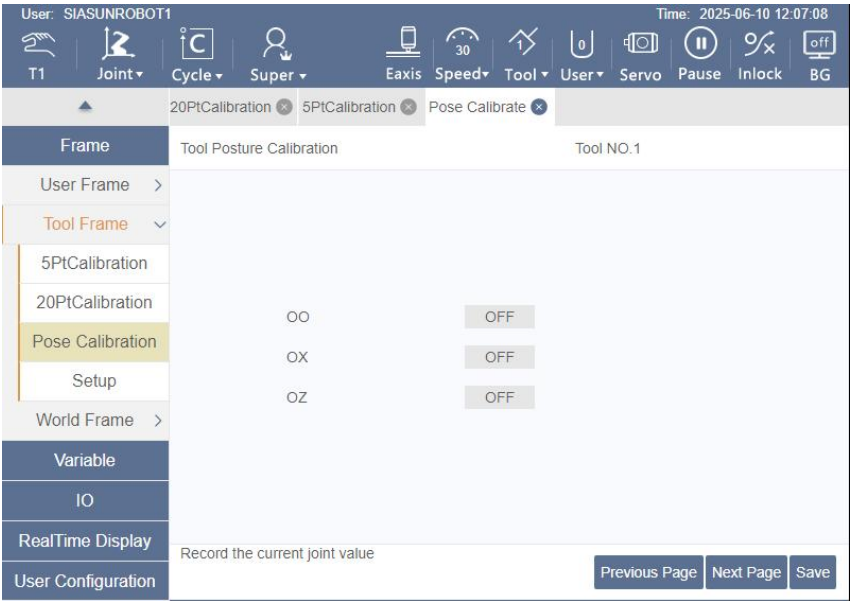


Figure 2- 13 Tool frame pose calibration interface

- 2) First, calibrate the selected points of TCP in space, and then calibrate a point in the positive direction of the X-axis of the tool frame and a point on the XZ plane in the positive direction of the z-axis. At the three points, click OFF/ON to calibrate.
- 3) After calibration is completed, press Save.
- 4) The calibration results are displayed on the setting interface corresponding to the file number, i.e. the values of Rx, Ry, and Rz.

Notes:

TCP and pose in the same frame need to be calibrated under the same tool number.

Setting steps for tool frame:

- 1) Click [Frame] - [Tool Frame] - [Calibration] on the main menu. The tool frame setting interface is displayed. Press Previous Page and Next Page to select the tool number desired to set.

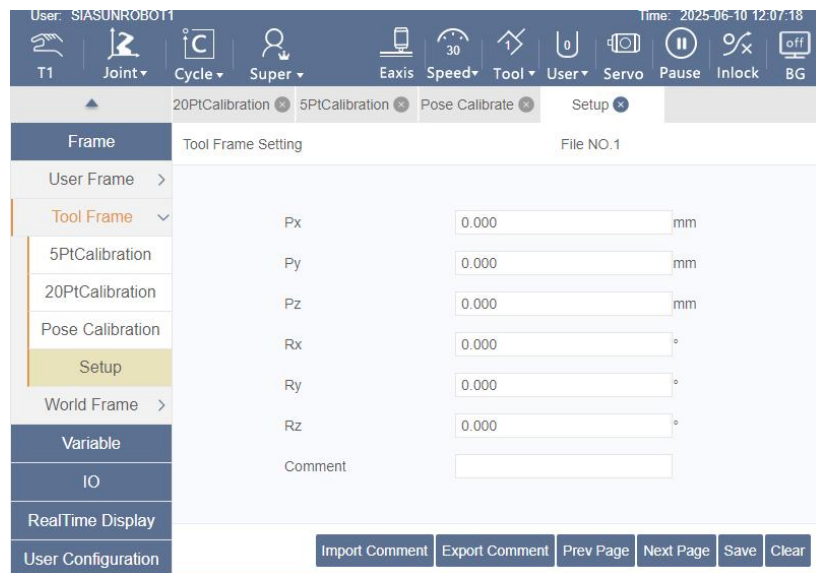


Figure 2- 14 Tool frame setting interface

- 2) Fill in the values X, Y, Z, Rx, Ry and Rz in sequence, and then click Save.

Comment: Comment can be added.

Import Comment: Comment from the USB flash drive can be imported into this page.

Export Comment: Comment from the USB flash drive can be exported to the USB flash drive.

Clear: The data of the current file number can be cleared.

Selection of tool frame in the job:

In the job, the tool frame number can be selected through commands. **SET TF#<frame**

number> is the tool frame selection command.

SET TF#<frame number> command can appear at the top of the job, as well as in the middle and at the end of the job. After the execution of this command, the current tool frame number of the system is changed, which not only affects the automatically executed jobs, but also requires the use of the newly set frame for teaching operations in the teaching mode.

Notes:

- 1) If the customer uses one tool frame, **SET TF#<frame number >** command can be absent in the job. If the customer uses multiple tool frames, it is recommended to add the **SET TF** command at the top of each job in order to avoid the confusion of the tool frames, so that the tool frame used by each command of each job is clearly defined during the execution of the job, avoiding the robot path error when executing the job due to the incorrect number of the current tool frame.
- 2) When multiple tool frames are used, it is necessary to determine that the current tool frame number is the one that the customer wants to use before adding and inserting commands.
- 3) When changing the tool frame number (the number after **SET TF#**) in the job, it is necessary to confirm whether all the points after this command use this tool frame file, and verify the forward motion for all the points that use the new tool frame number after this command. The specific verification method is as follows: Confirm or change the current tool frame number to the newly **SET** tool frame number, and then use the forward motion button to verify the teaching point after **SET TF#** command.

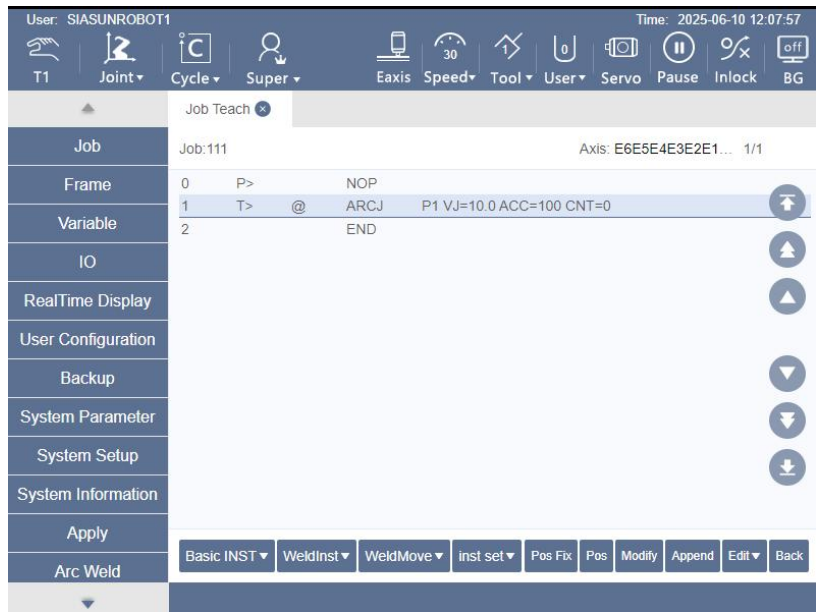


Figure 2- 15 Change to tool frame in the program

2.1.3 Set User Frame

The user frame is a Cartesian frame in which the user calibrates the space of each job. Among them, the user frame 0 is the world frame by default and cannot be modified.

The user frame is defined by the position (x, Y, Z) relative to the origin of the rectangular frame of the robot and the rotation angles (Rx, Ry, Rz) of the X-axis, Y-axis and Z-axis. The number of user frames ranges from 0 to 8 (expandable to 128), and can be switched as needed.

Notes:

When the axis motion buttons can be used for single-axis motion in the joint frame, the axis motion joint value does not change with the tool frame or the user frame. However, when the program runs automatically or manual forward movement is performed, the position and pose of the robot will be affected by both the tool and user frames.

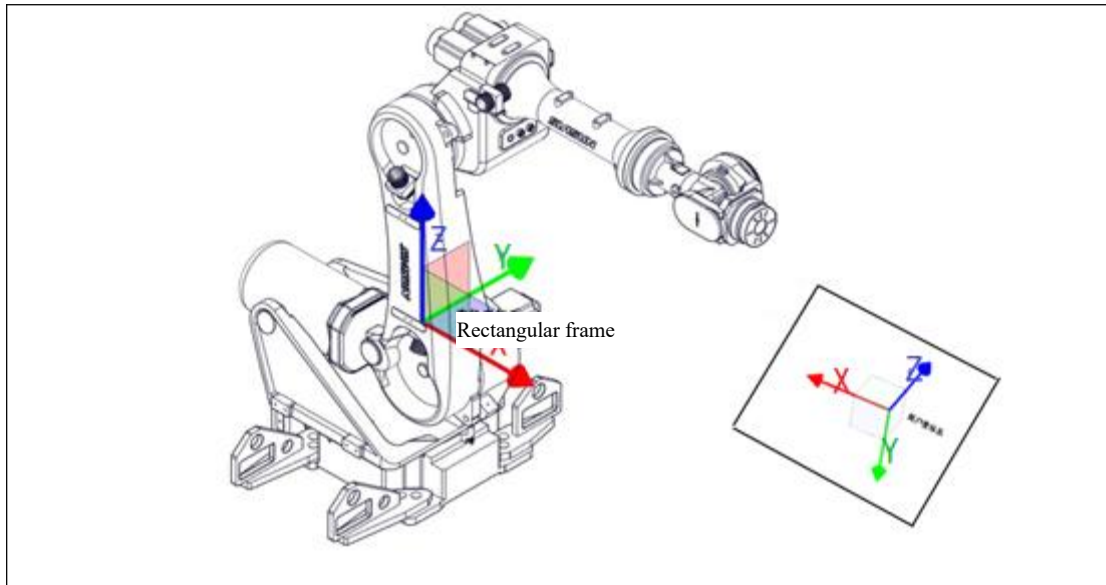


Figure 2- 16 User frame

3-point teaching:

Teach three points: the origin of space, the point in the X-axis direction, and the point on the XY plane.

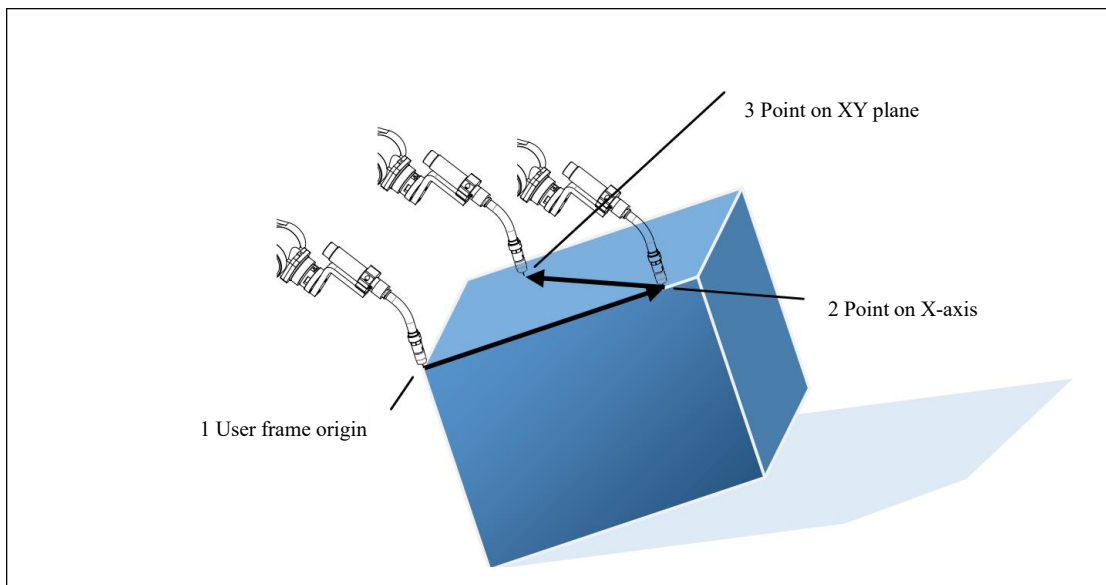


Figure 2- 17 Three-point teaching of user frame

Direct setting

Directly enter the position and pose relative to the user frame in the rectangular frame in the setting interface.

Calibration steps for user frame:

- 1) Click [Frame] - [User Frame] - [Calibration] on the main menu. Open the user frame calibration interface. Select the number of the user frame by pressing Previous Page and

Next Page.

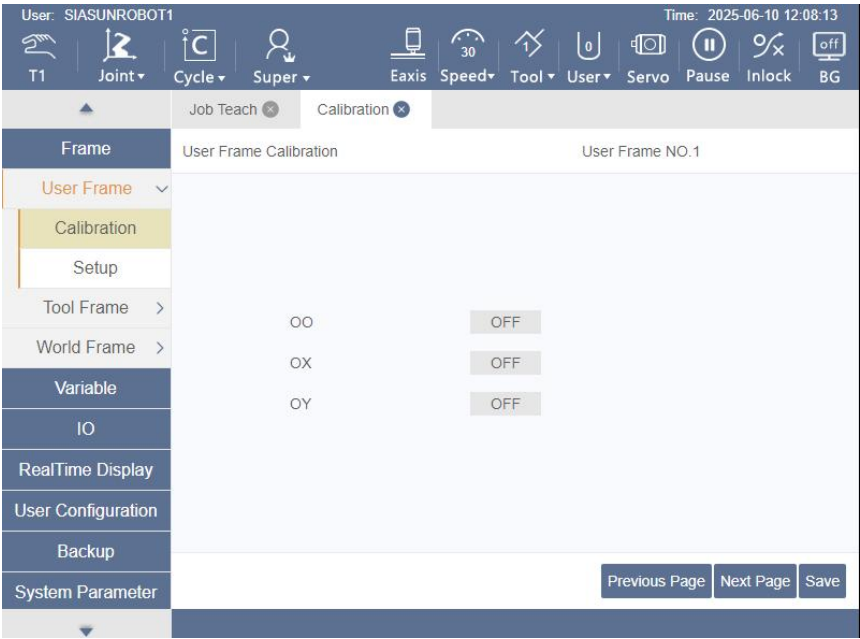


Figure 2- 18 User frame calibration interface

- 2) Move the robot to the origin of the user frame, the point on the X-axis and the point on the XY plane. Click OFF corresponding the position in sequence, and record the current position. After the three points are displayed as ON, click Save to complete the calibration.

Setup steps for user frame:

- 1) Click [Frame] - [User Frame] - [Setup] on the main menu. Open the user frame setup interface. Select the number of the user frame by pressing Previous Page and Next Page.

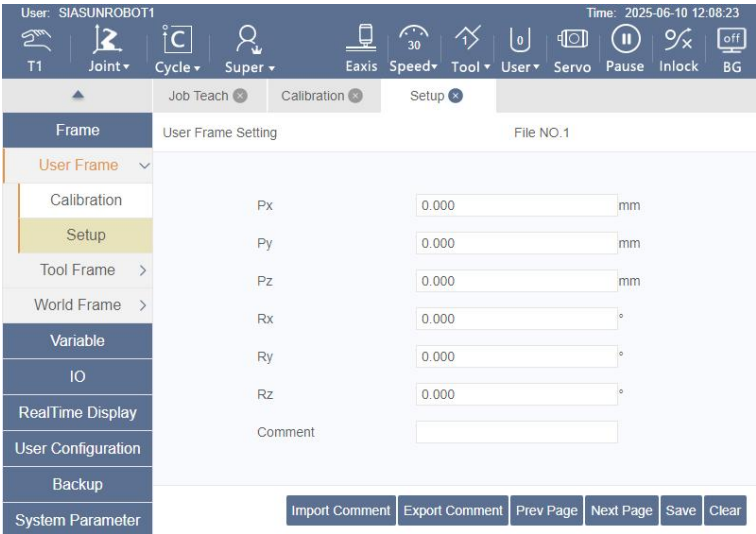


Figure 2- 19 User frame setup interface

- 2) Fill in the correct parameters at the corresponding positions, and click Save to set the user frame. Meanwhile, the calibrated user frame can also be viewed here.

Comment: Comment can be added.

Import Comment: Comment from the USB flash drive can be imported into this page.

Export Comment: Comment from the USB flash drive can be exported to the USB flash drive.

Selection of user frame number in the job:

In the job, the user frame number can be selected through commands. SET UF#<frame number> is the user frame selection command.

SET UF#<frame number> command can appear at the top of the job, as well as in the middle and at the end of the job.

After the execution of this command, the current user frame number of the system is changed, which not only affects the automatically executed jobs, but also requires the use of new user frame for the recording of axis operations or commands in the teaching mode.

Notes:

- 1) **If the customer uses one user frame, SET UF#<frame number > command can be absent in the job. If the customer uses multiple user frames, it is recommended to add the SET UF command at the top of each job in order to avoid the confusion of the user frames, so that the user frame used by each command of each job is clearly defined during the execution of the job, avoiding the robot path error when executing the job due to the incorrect number of the current user frame.**
- 2) **When multiple user frames are used, it is necessary to determine that the current user frame number is the one that the customer wants to use before adding and inserting commands.**
- 3) **When changing the user frame number (the number after SET UF#) in the job, it is necessary to confirm whether all the points after this command use this user frame file, and verify the forward motion for all the points that use the new user frame number after this command. The specific verification method is as follows: Confirm or change the current user frame number to the newly SET user frame number, and then use the forward motion button to verify the**

teaching point after SET UF# command.

2.1.4 Set External Tool Frame

Under normal circumstances, fixtures are installed on the flange of the robot, and then the robot performs operations such as picking and placing the workpiece. However, in some special cases, the tool needs to be installed on the outside of the robot, and the robot picks up the workpiece onto the external tool for processing. For example, in grinding, belt sanders are typically installed on the ground, separate from the robot. The robot picks up the workpiece onto the sanding belt for grinding.

To perform external tool calibration, the following steps are required:

- 1) To calibrate the robot tool frame, perform 5-point calibration (refer to Section 3.2.2 for detailed instructions on setting the tool frame).
- 2) The calibrated tool frame TCP point is required to calibrate the external tool frame.
- 3) Use the robot TCP point to align with the external TCP and calibrate the zero point of the external tool.
- 4) External tool calibration.

Calibration steps for external tool frame:

The same calibration interface as the user calibration method is used. Both the user frame calibration interface and the external TCP calibration interface use the user frame calibration interface.

The external tool calibration interface can be opened by following the path [Frame] - [User Frame] - [Calibration].

Switch by pressing Previous Page and Next Page.

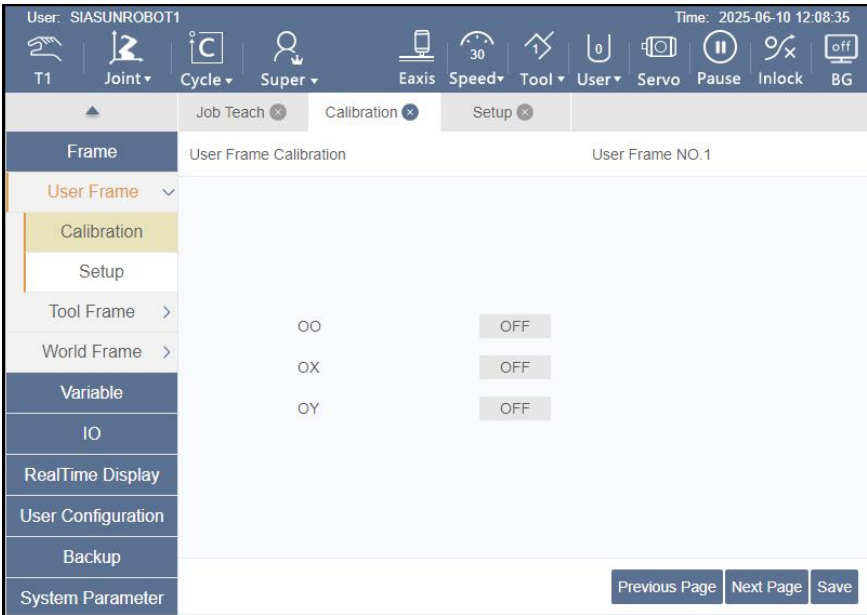


Figure 2- 20 External tool calibration page

Setup steps for external tool frame:

The external tool setup interface can be opened by following the path [Frame] - [User Frame] - [Setup].

Enter the external tool value settings directly on the external tool setup page.

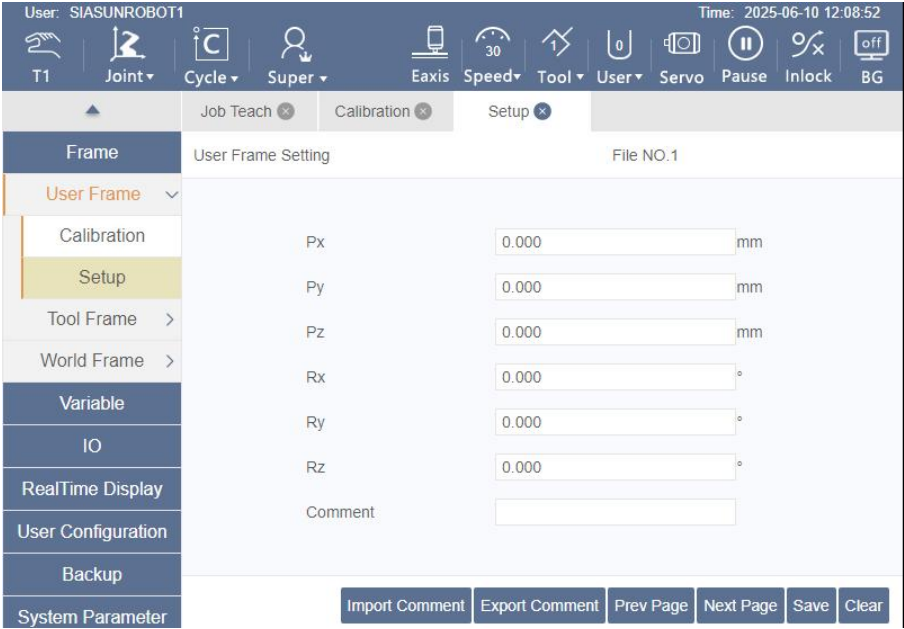


Figure 2- 21 External tool setup page

If an external TCP is used after calibration is completed, the external frame needs to be switched by clicking the status bar. After switching to the external tool frame, the icon in the status bar will change from User to UserE. Select in this dialog box to switch the external tool frame. This

window can switch between the user frame and the external TCP frame.

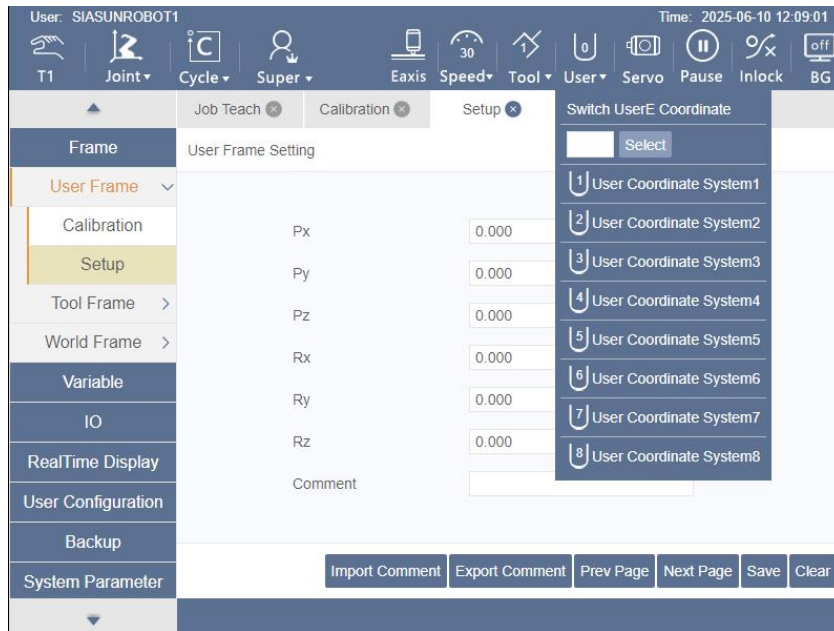


Figure 2- 22 External tool switching screen

Robot teaching operation:

The frame needs to be switched to the user frame.

When switching to the external tool: Press the X/Y/Z button, and the TCP of the robot moves in the X/Y/Z direction of the external TCP;

Press the Rx/Ry/Rz button, and the TCP of the robot rotates around the origin of the external TCP;

Commands used by external TCP:

Command format:

MOVL P2 VL=1000 ACC=100 CNT=0 ET

P2: Position variable number, which saves and records the position information of the target point.

VL: Running speed of current path.

ACC: Acceleration percentage of current path.

CNT: Whether the end point of the current path is continuous with the next point.

ET: The presence of ET indicates that the external tool is valid. The external tool number used is confirmed based on the user frame number in the position information.

Automatic operation of robot:

In automatic operation, the motion commands with ET are planned according to the user

frame number in the current P.

2.1.5 Set World Frame

Setup steps for world frame:

The world frame interface can be opened by following the path [Frame] - [World Frame] - [Setup].

Enter the world frame values directly on the world frame setup page for setting. After setting the world frame, the robot will move in the direction of the newly set world frame in the rectangular frame.

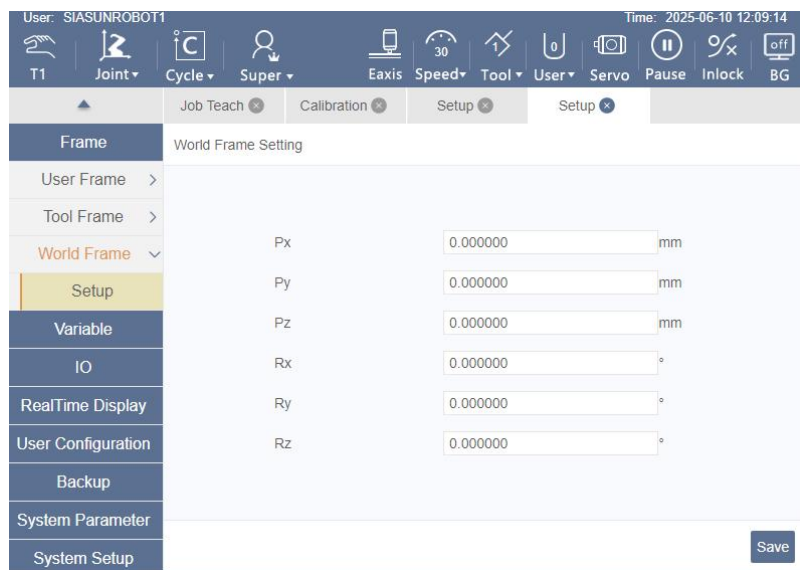


Figure 2-23 World frame setup page

2.2 Model configuration

Click [System Setup] - [Model Configuration] on the main menu, enter the password: sysiasun, and enter the robot model configuration interface. You can view and reselect the model of this robot. After selecting the model, click Save, and a restart will be prompted. After the restart, the system parameters will be automatically cleared. Before saving or restarting, confirm the previous system configuration or back up the system parameters in advance.

This parameter has been configured before delivery and does not require user modification.
Note: The model of this parameter must be consistent with the model of the actually connected robot.

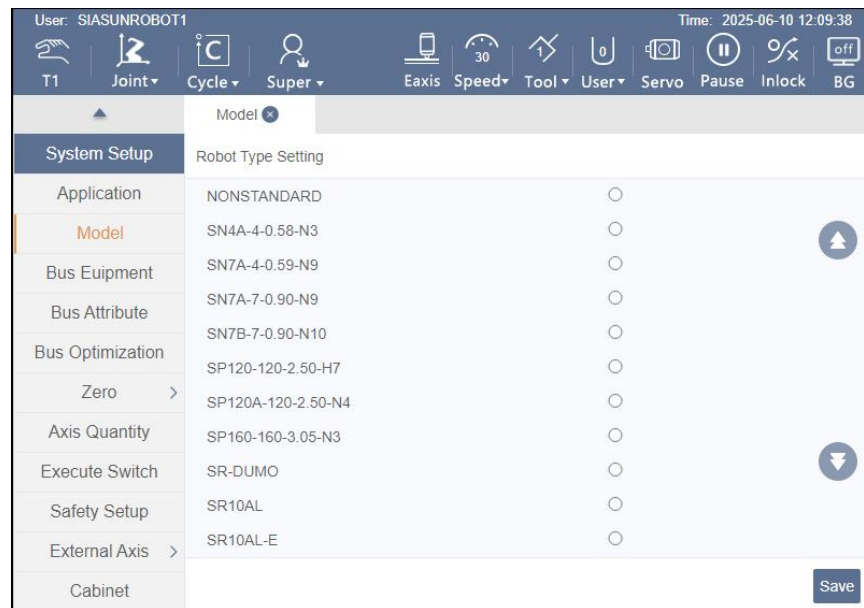


Figure 2- 24 Robot model configuration interface

2.3 Cabinet configuration

Click [System Setup] - [Cabinet Configuration] on the main menu, enter the password: sysiasun, and enter the robot cabinet model configuration interface. You can view and reselect the cabinet model of this robot. This parameter has been configured before delivery and does not require user modification. Note: The model of this parameter must be consistent with the actual cabinet model.

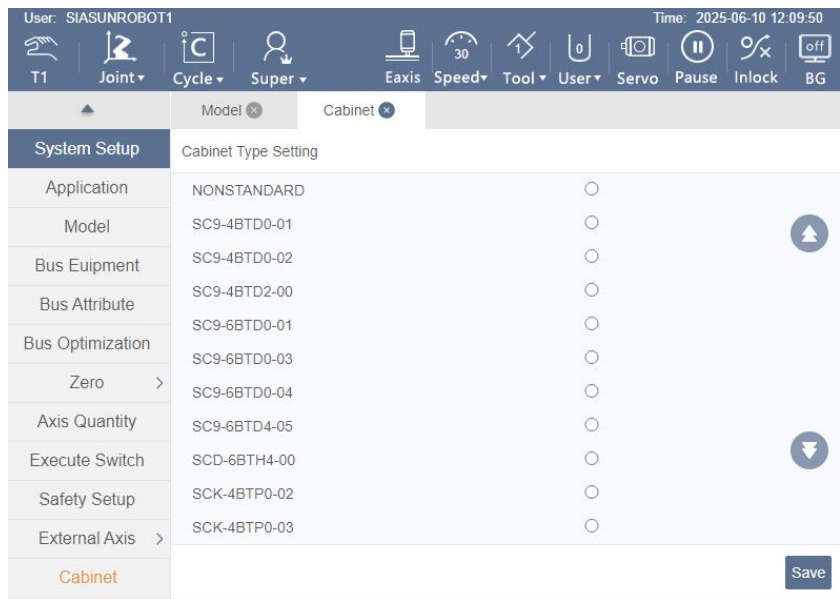


Figure 2- 25 Robot cabinet model configuration interface

3 Program Description

The program of the robot is composed of various commands edited by the staff, namely additional information, which can enable the robot to complete the desired jobs.

Note:

This chapter explains the program itself. How to create and teach each operation step of the program will be explained in the next chapter.

Job editing interface:

After opening the selected program, enter the job editing interface.

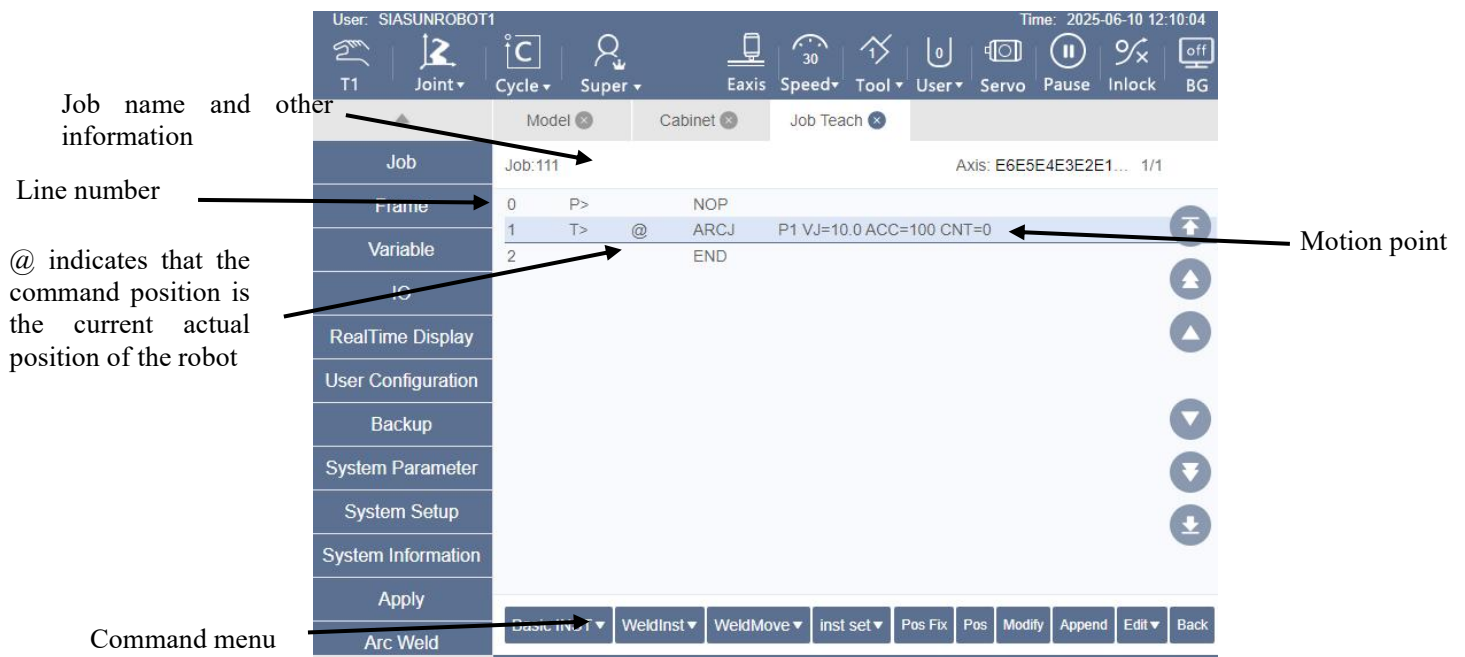


Figure 3-1 Job Editing Interface

The program consists of the following information:

- 1) Line number of command.
- 2) Motion command that instructs the robot to move from the current position to the target position in the expected way.
- 3) Some program commands for robot.
- 4) I/O class commands for sending and receiving I/O signals to external devices.
- 5) Commands for assigning, calculating and operating on internal variables of the robot (including position variables).
- 6) Commands for changing the program flow (IF, GOTO, CALL, RET) when some

conditions are established.

- 7) Wait command (WAIT, DELAY) that delays the program process.
- 8) Interrupt command enabling the robot to interrupt the current movement under certain conditions and executing the desired program.
- 9) Other commands.

3.1 Program Information

In the job list, you can see the name of each program and the operations performed on the program.

Click [Job] on the main menu to open the job list. The order of the job list is based on the first character of the program name. When the first character is a number, the job is placed at the front; when the first character is a letter, the job is placed after the one that the first character is a number.

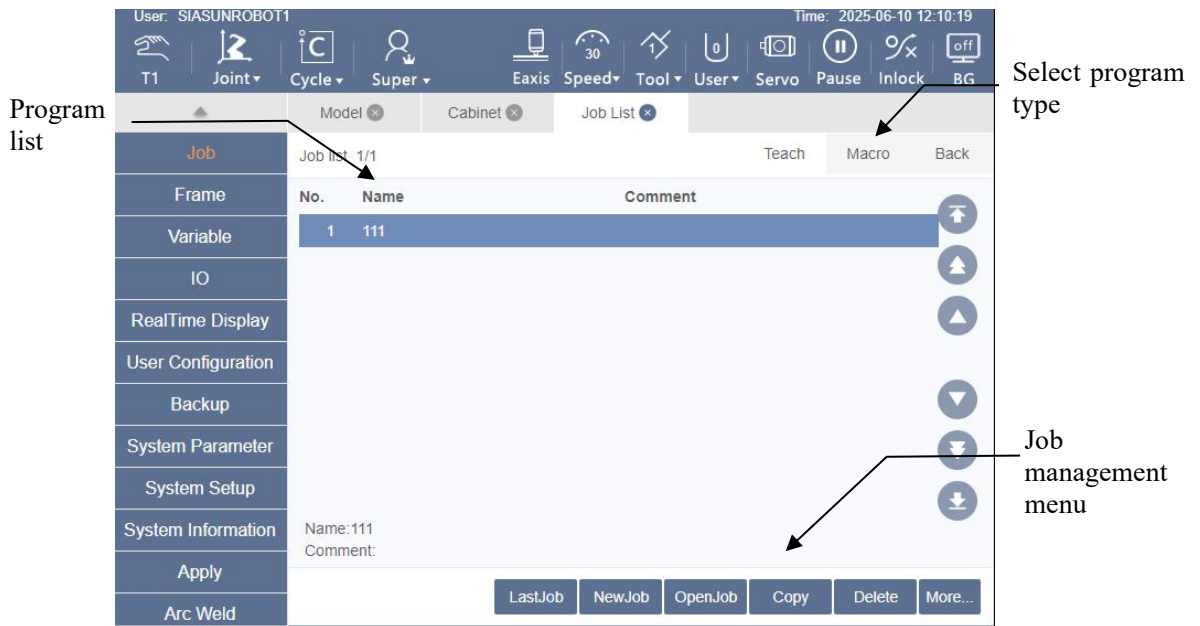


Figure 3-2 Job List Interface

In the job list, the programs that have been created are listed. By program types, the jobs are listed in different lists. In the list, you can view how many jobs have been created in total.

At the same time, you can also press LastJob, NewJob, OpenJob, etc. on this interface.

3.1.1 Program Name

Each program has a program name and cannot be named repeatedly. The program name

supports a maximum of 32 characters and can be named by either numbers or letters. Special characters cannot be included in the program name. Comments such as Chinese/English letters/letters can be added to the program, with a maximum support of 36 characters.

3.1.2 Program Type

The robot program can be viewed by job type.

- 1) Main program: The program specified by the user serves as the main program, which can be directly initiated from the teaching pendant or external devices.
- 2) Teaching program: When the robot executes, it follows the teaching program to carry out actions and logical control commands.
- 3) Macro program: It refers to a job initiated through macro commands.
- 4) Background program: A program that runs in the background. Once initiated, it can monitor the status at all times and complete logical control commands.

Note:

The main job can be set in [Job Management] by creating a created teaching program as the main job. Other jobs require the setting of the program type during the creation of a new job.

3.1.3 Job Management allows for routine operations on jobs

- 1) Job Property: You can perform operations on this job: view the creation and modification time, re-edit the job name, add and modify job comments, view and modify job axis groups, and the job with/without write-protection.
- 2) Transformation: The entire job can be offset/mirrored.
- 3) Main job registration/cancellation: You can set a selected job as the main job or cancel the main job.
- 4) Copy and delete: Perform copy and delete operations on the selected job.
- 5) OpenJob: Open the selected job.
- 6) New job: Create a new job.
- 7) LastJob: Open the job you entered last time, and the cursor position remains unchanged.
(Ineffective after the control cabinet is powered off)

3.2 Line Number and Parameters

Line number:

It represents the number of the program command line.

The line numbers will be automatically added before the commands we write in the program. When transforming program commands such as deletion or insertion, the line numbers will change automatically, ensuring that the line numbers of the entire program start with 1, 2, 3... from beginning to end.

Parameters:

Parameters are the data that users need to add and use in programming commands. When inserting commands via the command menu button, most commands require engineers to select/input parameter values by themselves.

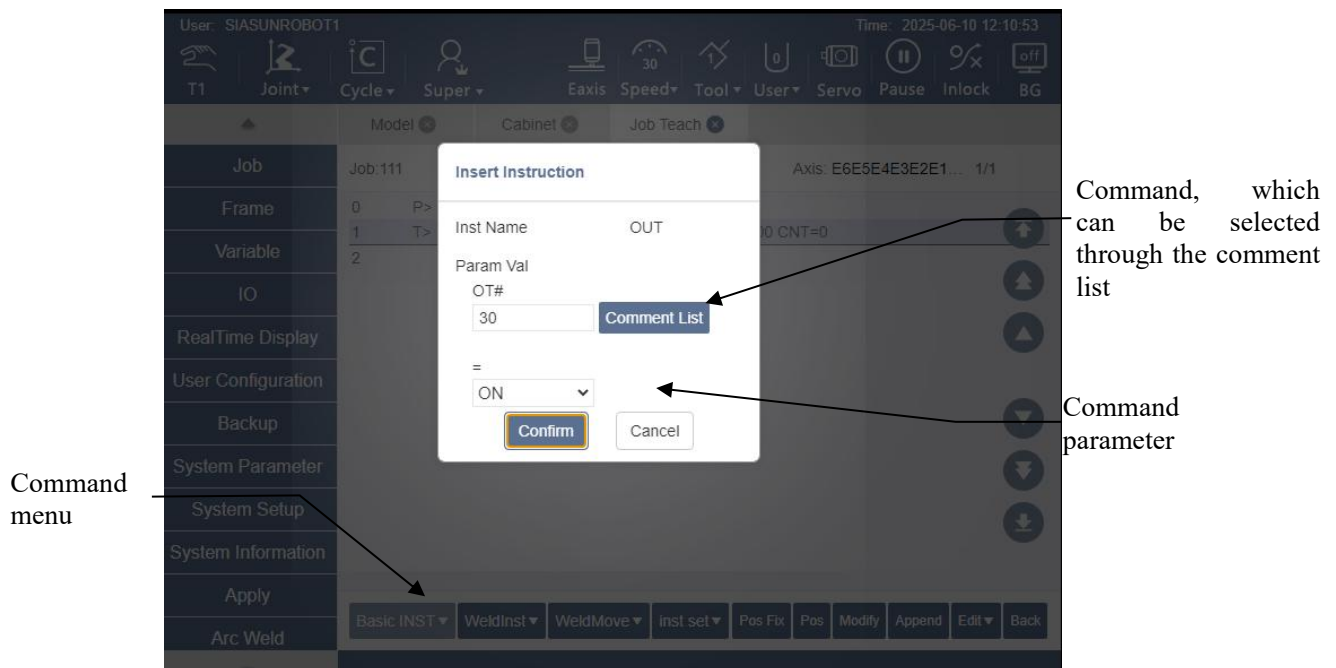


Figure 3-3 Command input with parameter

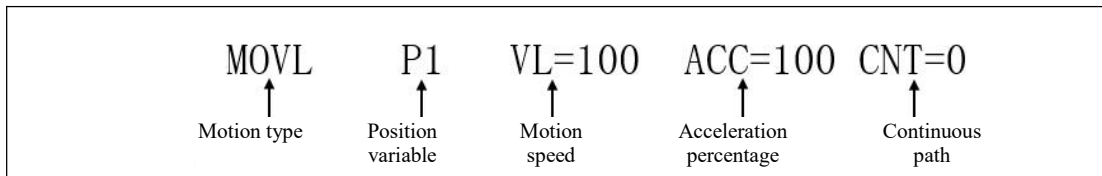
In the teaching program, commands need to be selected by clicking the menu rather than directly writing code for programming. Therefore, after the commands are selected, the relevant parameters will be filled in through the command interface. Users can add parameters by themselves according to the actual situation.

3.3 Motion Command

Motion command is a type of command that enables the robot to move from the current position to the target position at the speed and motion mode specified by the operator and record.

A motion command will record the following information.

Motion command:



- 1) Motion type: It refers to the moving path of the robot (or TCP) when it moves towards the specified position.
- 2) Position variables: Variable parameters that store position information, which are divided into local variables and global variables.
- 3) Motion speed: The speed at which the robot moves in a straight line or along an axis.
- 4) Acceleration percentage: The acceleration parameter for executing motion statements.
- 5) Continuous path: Whether it stops at the target point or slides rapidly near the target point.

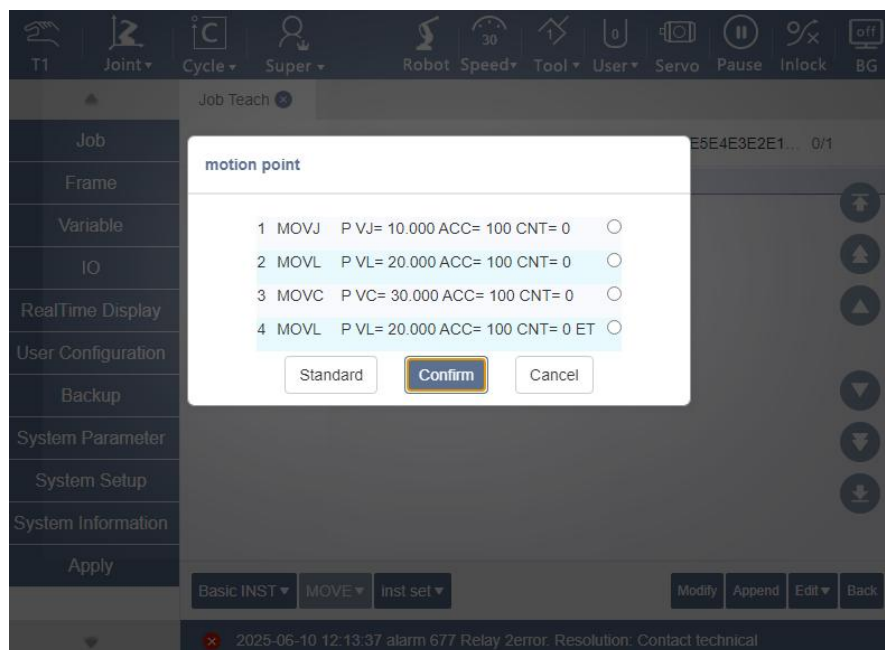


Figure 3-4 Motion point menu

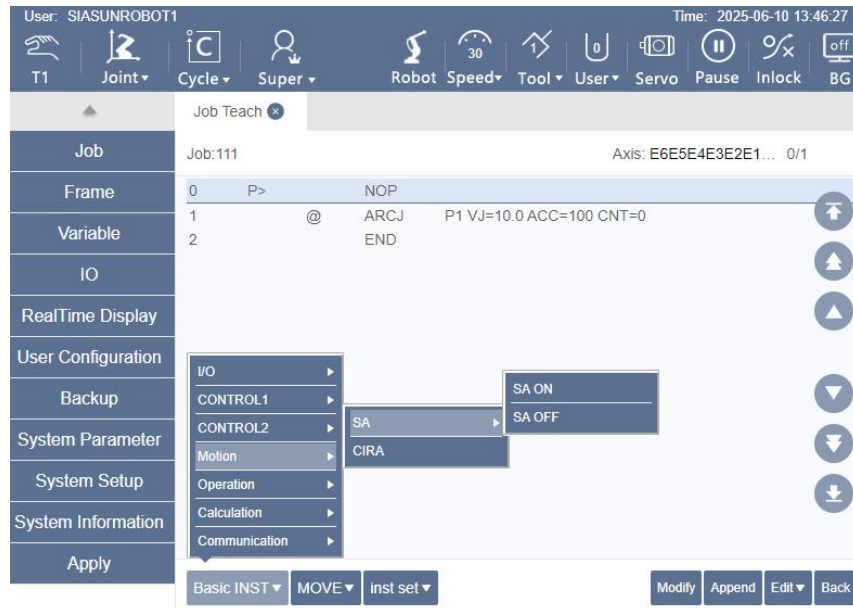


Figure3- 5 Motion Command Menu

3.3.1 Motion type

The motion type is the moving path of the robot towards the target point. The motion type includes joint motion without path or pose control, linear motion with path or pose control, and circular motion. In the command, it is represented as P/PR, where P is the position information storage in a single program and PR is the position information storage in all programs.

Joint motion	J
Linear motion	L
Circular motion	C

3.3.1.1 Joint motion

Joint motion is a method for a robot to move to the target position, and a method for all axes of the robot to accelerate simultaneously to the speed specified by the program and then decelerate simultaneously to reach the target point. The path of axis motion is generally nonlinear. The speed of axis motion is a percentage of the rated speed of the motor and is entered during teaching. Program using **MOVJ** command.

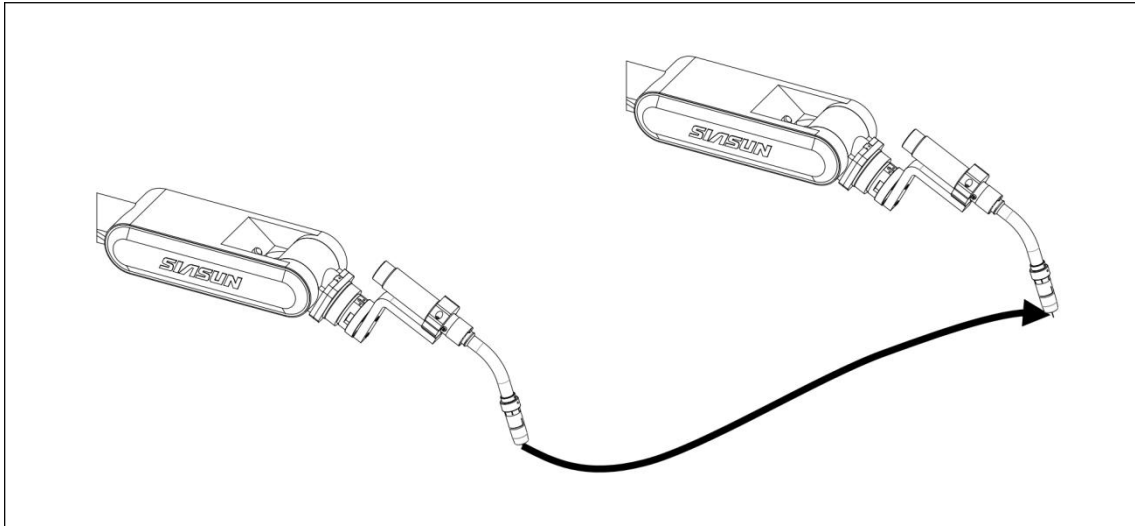


Figure 3-6 Axis motion path

MOVJ P command:

Function description: Movement between two points is executed in joint interpolation mode;

Format: MOVJ P<Parameter 1> VJ=<Parameter 2> ACC=<Parameter 3> CNT=<Parameter 4>

Parameter description:

<Parameter 1> Meaning: Position variable number, range: 1-7500.

<Parameter 2> Meaning: Speed of joint movement, unit: %, range: 1-100.

<Parameter 3> Meaning: Percentage of maximum acceleration, unit: %, range: 1-120, 120% is the maximum acceleration.

<Parameter 4> Meaning: The path is continuous. Input values: 0 and 1. 0 indicates the path is discontinuous, and 1 indicates the path is continuous.

Example: MOVJ P1 VJ=10 ACC=100 CNT=0.

3.3.1.2 Linear Motion

Linear motion is a motion method that moves the TCP point of a robot from the starting point to the target point linearly. The speed of linear motion is in mm/s, and the speed parameter is added after the teaching point. Program using **MOVL** command.

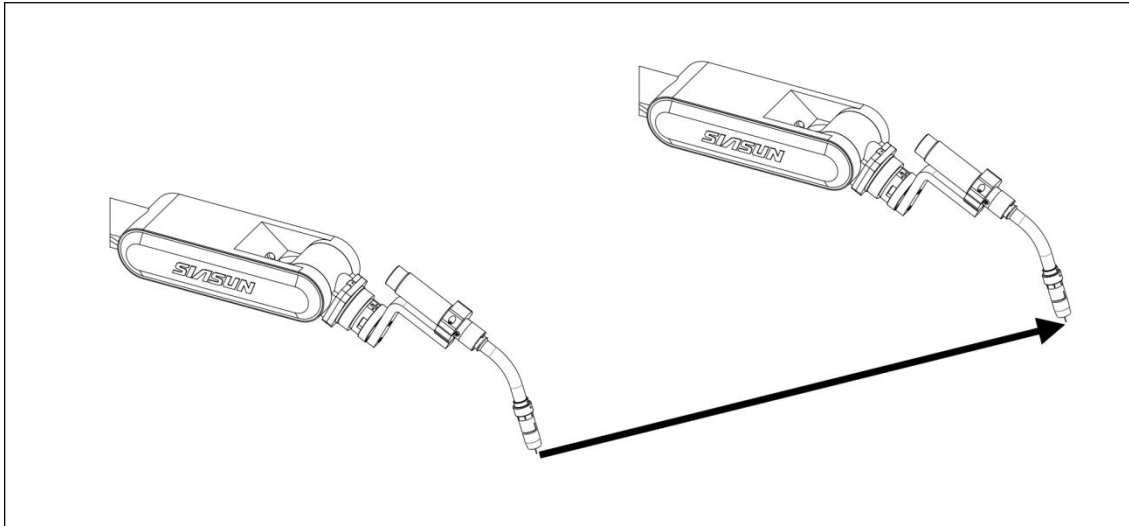


Figure 3-7 Linear motion path

The TCP points of the robot can rotate in the direction of the axis. Rotate the pose of the tool from the initial pose centered on the TCP point. The motion mode is controlled through linear motion.

MOVL P command:

Function description: Movement between two points is executed in linear interpolation mode.

Format: MOVL P<Parameter 1> VL=<Parameter 2> ACC=<Parameter 3> CNT=<Parameter 4>

Parameter description:

<Parameter 1> Meaning: Position variable number, range: 1-7500.

<Parameter 2> Meaning: Speed of linear motion, in mm/s, the range is limited by the robot model (for details, see 3.23.4 Motion Parameter Description).

<Parameter 3> Meaning: Percentage of maximum acceleration, unit: %, range: 1-120, 120% is the maximum acceleration.

<Parameter 4> Meaning: The path is continuous, with a range of 0 to 100. 0 indicates a discontinuous path, and 100 is the point farthest from the target.

Example: MOVL P1 VL=600 ACC=20 CNT=100

MOVL P ET command:

Function Description: External TCP command. Under normal circumstances, fixtures are installed on the flange of the robot, and then the robot performs operations such as picking and placing the workpiece. However, in some special cases, the tool needs to be installed on the outside of the robot, and the robot picks up the workpiece onto the external tool for processing.

For example, in grinding, belt sanders are typically installed on the ground, separate from the robot. The robot picks up the workpiece onto the sanding belt for grinding.

Setting commands: When using external TCP commands, it is necessary to configure the external tool TCP in advance. For specific configuration, please refer to Section 2.2.4.

Format: `MOVL P<Parameter 1> VL=<Parameter 2> ACC=<Parameter 3> CNT=<Parameter 4> ET`

Parameter description:

<Parameter 1> Meaning: Position variable number, range: 1-7500.

<Parameter 2> Meaning: Speed of linear motion, in mm/s, the range is limited by the robot model.

<Parameter 3> Meaning: Percentage of maximum acceleration, unit: %, range: 1-120, 120% is the maximum acceleration.

<Parameter 4> Meaning: The path is continuous, with a range of 0 to 100. 0 indicates a discontinuous path, and 100 is the point farthest from the target.

Example: `MOVL P1 VL=600 ACC=100 CNT=50 ET`

Note: External TCP enable is only effective for axis motion in the tool frame, ineffective for other axis operations, and also ineffective for automatically executed commands. Even if the external TCP is not enabled, if the command `MOVL P2 VL=1000 ACC=100 CNT=100 ET` is taught, the external TCP motion planning path is executed.

3.3.1.3 Circular Motion

Circular motion is a motion mode in which the TCP of the robot performs circular motion on three points. It is necessary to demonstrate the passing points and the target points. The speed of circular motion is in mm/s. Program using **MOVC commands**.

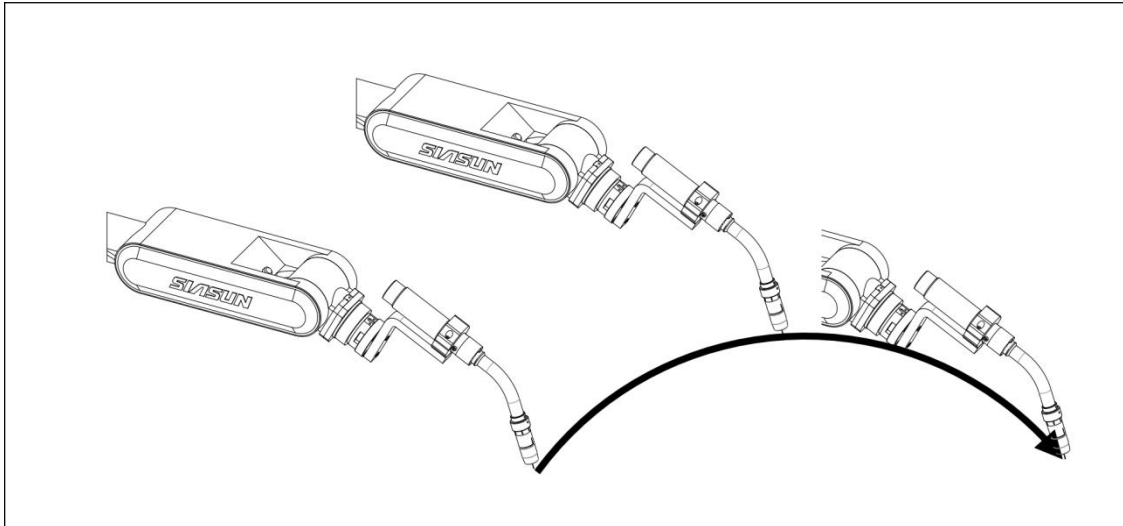


Figure 3- 8 Circular Motion

MOV C command:

Function description: Movement among three points is executed in circular interpolation mode.

Format: MOV C P<Parameter 1> VC=<Parameter 2> ACC=<Parameter 3> CNT=<Parameter 4>

Parameter description:

<Parameter 1> Meaning: Position variable number, range: 1-7500.

<Parameter 2> Meaning: Speed of circular motion, in mm/s, the range is limited by the robot model.

<Parameter 3> Meaning: Percentage of maximum acceleration, unit: %, range: 1-120, 120% is the maximum acceleration.

<Parameter 4> Meaning: The path is continuous, with a range of 0 to 100. 0 indicates a discontinuous path, and 100 is the point farthest from the target.

Examples:

MOV L P1 VL=600 ACC=100 CNT=50 Starting point of arc.

MOV C P2 VC=100 ACC=100 CNT=0 Process point of arc.

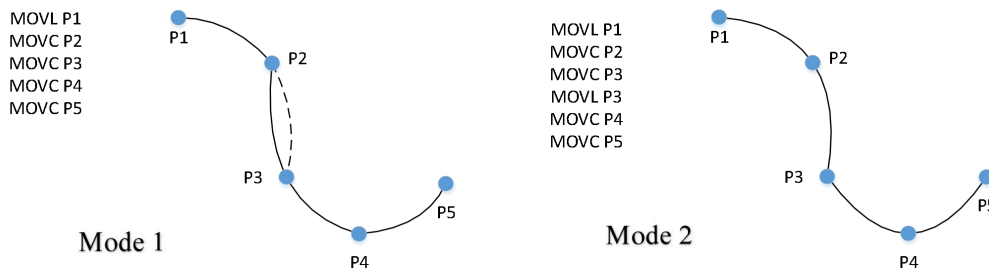
MOV C P3 VC=100 ACC=100 CNT=0 End point of arc.

For circular motion, three position points need teaching. The teaching arc angle between any two adjacent points must be less than 180°; otherwise, the arc path will deviate.

Arc specification:

Correct program:	Correct program:	Wrong program:	Wrong program:
NOP	NOP	NOP	NOP
MOVL P1	MOVL P1	MOVL P1	MOVC P1
MOVC P2	MOVC P2	MOVC P2	MOVC P2
MOVC P3	MOVC P3	MOVL P3	MOVC P3
END	MOVC P4	END	END
	MOVC P5		
	END		

When executing the program, the path follows the dashed line if not stopped. If paused at point P2, the path will follow the solid line. To ensure the trajectory follows the solid line even if stopped, the program must be edited in Mode 2.



3.3.1.4 Small Circle Command

It can be used to implement the application execution of path planning and parameterization of the entire circle type.

CIRA command:

Function Description: By using the small circle command to teach the center point, the path of the entire circle can be achieved.

Format: ARCCIRA P/PR<Parameter 1> VC=<Parameter 2> ACC=<Parameter 3>
CNT=<Parameter 4> R=<Parameter 5> OFFSET=<Parameter 6> OVER=<Parameter
7> UF=<Parameter 8> FIFL=<Parameter 9>

<Parameter 1> Meaning: Center point position, local/global position variable number;

<Parameter 2> Meaning: Speed of arc, with a maximum limit of 2000, in mm;

<Parameter 3> Meaning: Percentage of maximum acceleration, unit: %, range: 1-120;

<Parameter 4> Meaning: Approach level. In this command, the default value is 0, and no other values are accepted.

<Parameter 5> Meaning: Radius of target circle, in mm, range: 0.1-100;

<Parameter 6> Meaning: Starting point offset (unit: mm): OFFSET>0 indicates that the

<Parameter 7> Meaning: End point offset; over weld quantity, in mm; range: (-100,100); by using R, convert the unit of OVER into the angle; if the offset angle is greater than 60 degrees, an alarm should be triggered, that is, OVER (°) <60°.

<Parameter 9> Meaning: Welding file number, 1-32;

MOVL P2 VL = 10 ACC=100 CNT=0

ARCCIRA PVC=30 ACC=10 CNT=0 R=300 OFFSET=5 OVER=50 UF=1 FILE=1

MOVL P3 VL = 10 ACC=100 CNT=0

END



Figure 3-9 Schematic diagram of small circle path

Note: The small circle command limits the "over welding radian" to no more than 60 degrees; it can be a negative angle, indicating that the circle is not fully made. If the radian exceeds 60 degrees, an alarm will be triggered. When the speed is 20, the radius R of the small circle is $\geq 5\text{mm}$. In the teaching mode, under the forward motion and interlock (ON) conditions, MOVCIRA/ARCCIRA normally performs circle drawing; under the forward movement and non-interlock (OFF)/reverse conditions,

MOVCIRA/ARCCIRA actually executes the motion command, moving to MOVL P in the center of the circle.

3.3.1.5 Spiral Line Motion

MOV SPI command:

Function description: The function of this command is to achieve the moving path of the robot end as an equidistant spiral line. The path is shown below:

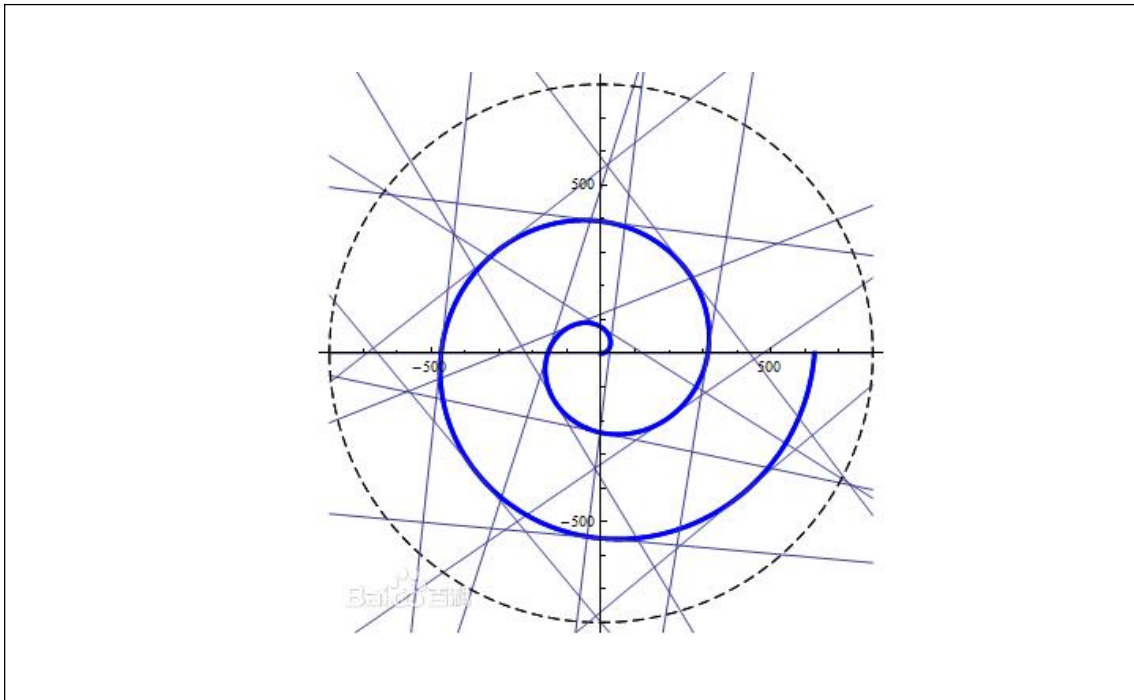


Figure 3-10 Spiral Motion

Format: MOV SPI<Parameter 1> VL=<Parameter 2> D=<Parameter 3> SD=<Parameter 4>
RTYPE=<Parameter 5> REFP<Parameter 6>

Parameter description:

<Parameter 1> Meaning: The teaching point is the coordinate position of the circle center of the spiral line.

<Parameter 2> Meaning: Speed of spiral line motion, in mm/s, the range is limited by the robot model.

<Parameter 3> Meaning: Spiral line spacing, unit: mm, range: 0.1-500.

<Parameter 4> Meaning: Stop distance of spiral line. This value is the radius of the distance from the stop position point to the center of the circle, with the unit being mm and the range being

0.1-1000.

<Parameter 5> Meaning: Range: 0, 1. Whether the pose changes synchronously with rotation.

0 - Change; 1 - Unchanged.

<Parameter 6> Meaning: The command teaching point is directly above the coordinate position of the circle center of the spiral line, that is, in the Z+direction of the circle center (ensuring that the plane of the spiral line motion is the XY plane).

Examples:

NOP

1 MOVJ P1 VL=600 ACC=100 CNT=0

2 MOV SPI VL=500.0 D=100.0 SD=120.0 RTYPE=0

3 REFP 1

END

Job details:

The current teaching point of the MOVJ command is the starting position point of the spiral line, and VJ=50 is the speed of the movement to this starting point. The MOVL command, etc. can also be used.

MOV SPI is the spiral line command. The current teaching point is the position point of the command center of the spiral line. VL=500.0 is the linear velocity of the spiral line motion, D=100.0 is the spiral line spacing, and SD=120.0 is the radius value of the spiral line stop position from the center of the circle. The pose will change accordingly.

The REFP command serves as the reference point, and the current teaching point is in the Z+direction at the center of the circle.

Note: This command supports pause, interrupt and offset. This command causes significant pose changes. Please use it with caution!

3.3.1.6 Singular Point Avoidance Command

For most 6-axis robots, the singular points of wrist posture ($J5=0$) can occur at almost all positions in the working area. When the robot passes through or near these singular points of the wrist, its motion performance is inappropriate for the following reasons.

J4 and J6 axes of the wrist axis will perform large rotational motions in a short period of time.

This limits the speed of the wrist axis. As a result, the speed of the TCP will slow down.

The singular point avoidance function enables to avoid singular points on the wrist during job execution. It can ensure that the wrist axis does not rotate excessively and can pass smoothly through the singular points of the wrist.

SA singular point avoidance command:

Function Description: After the function is enabled, the path singular point avoidance function after the job becomes effective.

Format: SA ON/OFF

Parameter description: Select ON/OFF

Examples:

SA ON

1 MOVJ P1 VL=600 ACC=100 CNT=0

MOVL P2 VL=600 ACC=100 CNT=50

SA OFF

Notes:

- 1) After the stack is cleared, SA OFF command is automatically executed by default to turn off the current singularity avoidance state.
- 2) The forward and reverse motions cannot pass through the singular point but remain in front of it, and an alarm is triggered: "3102 approaches the singular point and cannot move forward."
- 3) The offset motions such as SHITF/TRANS cannot pass through the singular point but remains in front of it, and an alarm is triggered: "3102 approaches the singular point and cannot move forward."
- 4) The external TCP motion cannot pass through the singular point but remain in front of it, and an alarm is triggered: "3102 approaches the singular point and cannot move forward."
- 5) The following application cannot pass through the singular point but remain in front of it, and an alarm is triggered: "3102 approaches the singular point and cannot move forward."

3.3.2 Position Information

Position information stores the position and pose of the robot. While the action commands are taught, the position information is written into the program.

Position information includes joint frame values based on the joint frame and rectangular frame values represented by the tool positions and poses within the job space. Under the standard setting, the rectangular frame value is used as the position.

Position variables are divided into two types: P and PR.

The position variable P is a local variable: in the same program, the serial number of the variable P is the same, while in different programs, the serial number of the variable P is the same, but the position information is different. P (1-7500) is a standard position storage variable. When action commands are taught, the position information is automatically recorded. The position number is used to refer to the position variable. The position number is automatically configured each time an action command is taught to the program. Configure P1 for the first time, P2 for the second time, and so on.

The position variable PR is a global variable: in a program, the serial number of the variable PR is unique. Using the same variable ID in any program will reference the same position information. PR (1-5630) is a variable used to store position information.

Notes:

On the motion command, click the position button to view and modify the information of the P/PR position variable. The position information of PR can also be viewed and modified through the position variable interface.

Cartesian frame value:

Cartesian frame value is defined by four elements: the position of the TCP based on the rectangular frame, the pose and direction of the tool, the form of the robot, and the Cartesian frame used.

Note:

The rectangular, tool and user frames of the robot body are all space Cartesian frames. The position variable records of SIASUN robot operating system are based on the rectangular frame, and the origin of the frame is located at the intersection of the axial line of the axis 1 and the center of the axis 2 of the robot by default.

User: SIASUNROBOT1 Time: 2025-06-10 13:48:14

T1 Joint Cycle Super Robot Speed Tool User Servo Pause Inlock BG

Job Teach System Float Position Shift POS Variable

Variable Global Position Variable

User Integer				
User Float	PR	1	Comment	
User Byte	Save/Lock	OFF	Current	Get
User String	Type	Joint		
POS Variable	Jt 1	-180.000	Jt 2	-123.141
	Jt 3	-77.931	Jt 4	0.000
System Integer	Jt 5	-68.928	Jt 6	-90.000
System Float	Jt 7	0.000	Jt 8	0.000
Position Shift	Jt 9	0.000	Jt 10	0.000
Local String	Jt 11	0.000	Jt 12	0.000

IO

RealTime Display

Previous Next Save

Figure 3- 11 Position variable in joint frame

User: SIASUNROBOT1 Time: 2025-06-10 13:47:27

T1 Joint Cycle Super Robot Speed Tool User Servo Pause Inlock BG

Job Teach System Float Position Shift POS Variable

Variable Global Position Variable

User Integer				
User Float	PR	1	Comment	
User Byte	Save/Lock	OFF	Current	Get
User String	Type	RPY		
POS Variable	X	0.000	Rx	0.000
	Y	0.000	Ry	0.000
System Integer	Z	0.000	Rz	0.000
System Float	E 1	0.000	E 2	0.000
Position Shift	E 3	0.000	E 4	0.000
Local String	E 5	0.000	E 6	0.000

IO

RealTime Display

Previous Next Save

Figure 3- 12 Position variable in user frame

Parameter setting:

Position variable number:

It refers to the position variable number used in the program, represented by PR in the program and ranging from 1 to 5630.

Note:

Comments such as Chinese/English letters/numbers can be added to this variable, with a maximum support of 36 characters. After adding a comment, the job will automatically bring the

comment into the command when using the variable. The display form is as follows:

```
MOVJ PR[2: Endpoint]VJ=10, ACC=100, CNT=0;
```

Lock:

When Lock is on, this position variable cannot be modified. When Lock is off, this position variable can be modified.

Current position:

Click [Get] button to save the current position of the robot to the position variable. If the form of this position variable is joint, the current joint value of the robot will be saved to the position variable; if the form of the position variable is orthogonal, the Cartesian frame of the robot is saved in the position variable.

Type:

The display forms of joint and orthogonal position variables can be switched.

Position and pose:

The position (x, y, z) is expressed by 3D frame values to indicate the position of the TCP point in the rectangular frame.

The pose (Rx, Ry, Rz) is expressed by the rotation angles of the X-axis, Y-axis and Z-axis in the rectangular frame, as shown in Figure 3- 16.

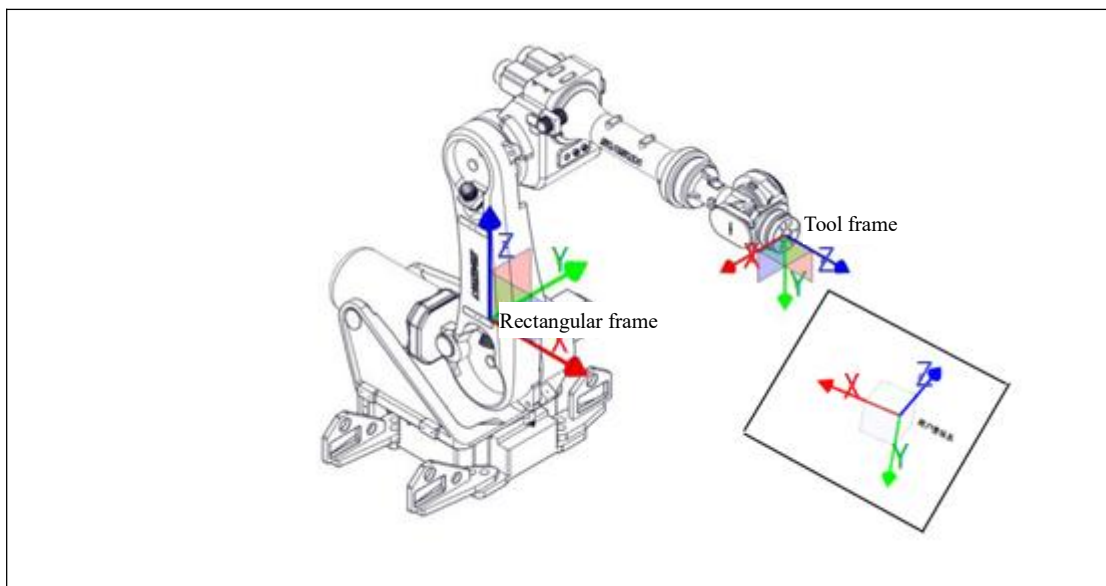


Figure 3- 13 Rectangular, Tool and User Frames

Determine frame number:

SIASUN robot has 1 to 8 tool frame numbers and 1 to 8 user frame numbers. Before teaching or execution, verify the frame numbers used to ensure that the frame numbers used during the

execution of the job are the same as those used during teaching.

Tool frame (TF) number:

SIASUN robot supports 8 tool frames, which are determined by 1 to 8 numbers. The default tool frame of the robot is the 6-axis flange frame of the mechanical interface.

User frame (UF) number:

SIASUN robot supports 8 user frames, which are determined by 1 to 8 numbers. The default user frame of the robot is the rectangular frame of the robot.

Frame position value:

Click [Frame] - [User Frame and Tool Frame] - [Setup] on the main menu, and select the frame number by pressing Previous Page or Next Page. You can then view the detailed values of the user and tool frames for the robot.

The screenshot displays the 'Tool Frame Setting' interface. The top status bar shows 'User: SIASUNROBOT1' and 'Time: 2025-06-10 13:59:06'. The main menu on the left includes options like 'T1', 'Joint', 'Cycle', 'Super', 'Robot', 'Speed', 'Tool', 'User', 'Servo', 'Pause', 'Inlock', 'BG', 'Frame', 'Variable', 'IO', 'RealTime Display', and 'User Configuration'. The 'Frame' menu is expanded, showing 'User Frame', 'Tool Frame', '5PtCalibration', '20PtCalibration', 'Pose Calibration', 'Setup', and 'World Frame'. The 'Tool Frame' menu is further expanded, showing 'Px', 'Py', 'Pz', 'Rx', 'Ry', 'Rz', and 'Comment'. The 'Setup' option is highlighted. The central area displays the following parameters and their values:

Parameter	Value	Unit
Px	0.000	mm
Py	0.000	mm
Pz	0.000	mm
Rx	0.000	°
Ry	0.000	°
Rz	0.000	°
Comment		

At the bottom of the interface, there are buttons for 'Import Comment', 'Export Comment', 'Prev Page', 'Next Page', 'Save', and 'Clear'.

Figure 3- 14 Detailed values of tool frame

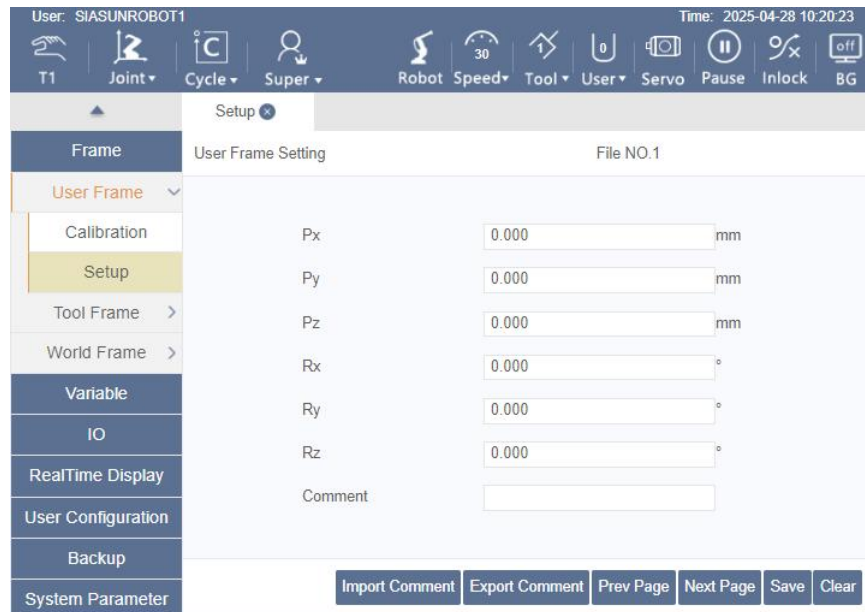


Figure 3- 15 Detailed values of user frame

Note:

The detailed values of the user coordinate system are recorded in the form of T6 matrix.

Joint frame values:

The position values of the joint frame are determined by the rotation angle values of each joint.

Robot body axis:

J1, J2, J3, J4, J5, J6

External axes:

E1, E2, E3, E4, E5, E6

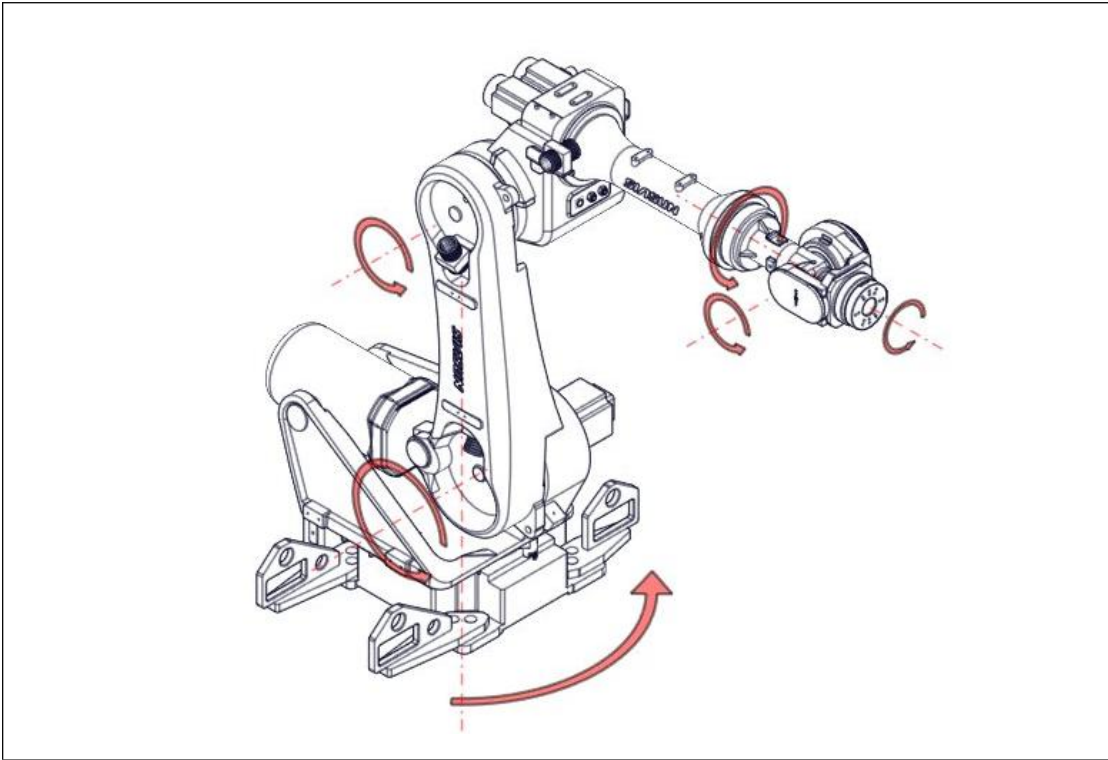


Figure 3- 16 Joint frame

Joint values:

On the main menu [RealTime Display] - [RealTime Display], you can view the current joint values.

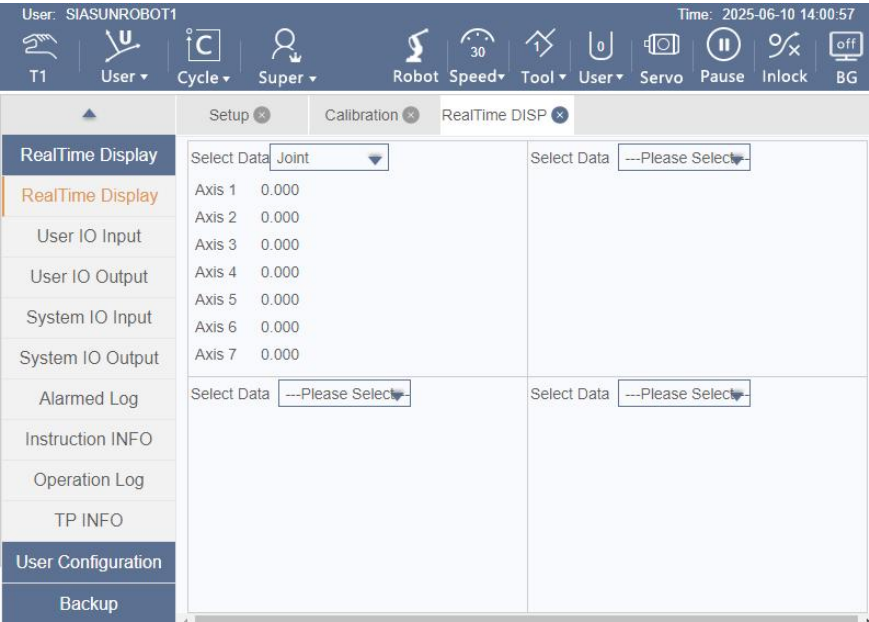


Figure 3- 17 Robot joint values

Position variable:

Viewing/editing mode of position variables:

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PR can be viewed/edited by clicking [Variable] - [Position Variable] on the main menu.

PR/P can be viewed and edited by selecting the position of the program in the job editing and clicking on the position.

The screenshot shows the 'POS Variable' configuration window in the SIASUN Robot software. The window is titled 'Global Position Variable'. On the left, there is a sidebar with various variable types: User Integer, User Float, User Byte, User String, POS Variable (highlighted in orange), System Integer, System Float, Position Shift, Local String, IO, and RealTime Display. The main area displays the 'PR' variable settings. The 'PR' field is set to '1'. The 'Save/Lock' button is 'OFF'. The 'Type' dropdown is set to 'Joint'. Below this, there are 12 joint position fields (Jt 1 to Jt 12) with their respective values: Jt 1: -180.000, Jt 2: -123.141, Jt 3: -77.931, Jt 4: 0.000, Jt 5: -68.928, Jt 6: -90.000, Jt 7: 0.000, Jt 8: 0.000, Jt 9: 0.000, Jt 10: 0.000, Jt 11: 0.000, Jt 12: 0.000. At the bottom right, there are 'Previous', 'Next', and 'Save' buttons.

Figure 3- 18 Position variable in joint frame

The screenshot shows the 'POS Variable' configuration window in the SIASUN Robot software, but with the 'Type' dropdown set to 'RPY'. The 'PR' field is still '1', and 'Save/Lock' is 'OFF'. The main area displays the 'RPy' (Rectangular Position Variable) settings. There are 6 fields for X, Y, Z, Rx, Ry, and Rz, all set to 0.000. Below these, there are 6 fields for E 1, E 2, E 3, E 4, E 5, and E 6, all set to 0.000. At the bottom right, there are 'Previous', 'Next', and 'Save' buttons.

Figure 3- 19 Position variable in rectangular frame

Parameter setting:

Position variable number:

It refers to the position variable number used in the program, which is represented by PR in the program and ranges from 1 to 5630.

Note:

Comments such as Chinese/English letters/letters can be added to this variable, with a

maximum support of 36 characters. Moreover, the comment information is directly displayed in the commands.

Lock:

When Lock is on, this position variable cannot be modified. When locked to off, this position variable can be modified.

Current position:

Click [Get] button to save the current position of the robot to the position variable. If the form of this position variable is joint, the current joint value of the robot will be saved to the position variable; if the form of the position variable is rectangular, the Cartesian frame of the robot is saved in the position variable.

Type:

The display forms of rectangular and joint position variables can be switched.

Position variables include six sub-variables: PX.1, PX.2, PX.3, PX.4, PX.5, and PX.6. They correspond to position and pose values (x, y, z, Rx, Ry, Rz) or the joint values of six axes (J1, J2, J3, J4, J5, J6) respectively. Commands can be used to directly assign the variable values to these six sub-variables.

Position variable assignment command:

GETTP PR1:

Through this command, the position and pose values of a motion point after this command can be obtained. They are recorded in PR1.

GETCP PR1:

Through this command, the position and pose values of the robot at the current stop position can be obtained. They are recorded in PR1.



Figure 3-20 GET CP\TP Commands

Table 3-1 Description of GET CP\TP Commands and Parameters

GETTP	Function	Obtain the teaching position in the next motion command of this command and store it in the position variable. If the next one is not a motion command, the system will trigger an alarm and stop operation.	
	Format	GETTP PR<Parameter 1>	
	Description	Parameter 1	<Position variable number>, range: 1-5630;
	Examples	GETTP PR1 MOVL PR1 VL=600 ACC=20 CNT=100	
GETCP	Function	When assigning the current position of the robot to the position variable, it is necessary to ensure that the robot has stopped moving; otherwise, the obtained position will be inaccurate.	
	Format	GETCP PR<Parameter 1>	
	Description	Parameter 1	<Position variable number>, range 1-5630;
	Examples	GETCP PR1	

Position variable calculation commands

Position variable components can be independently calculated or assigned via routine calculation commands.

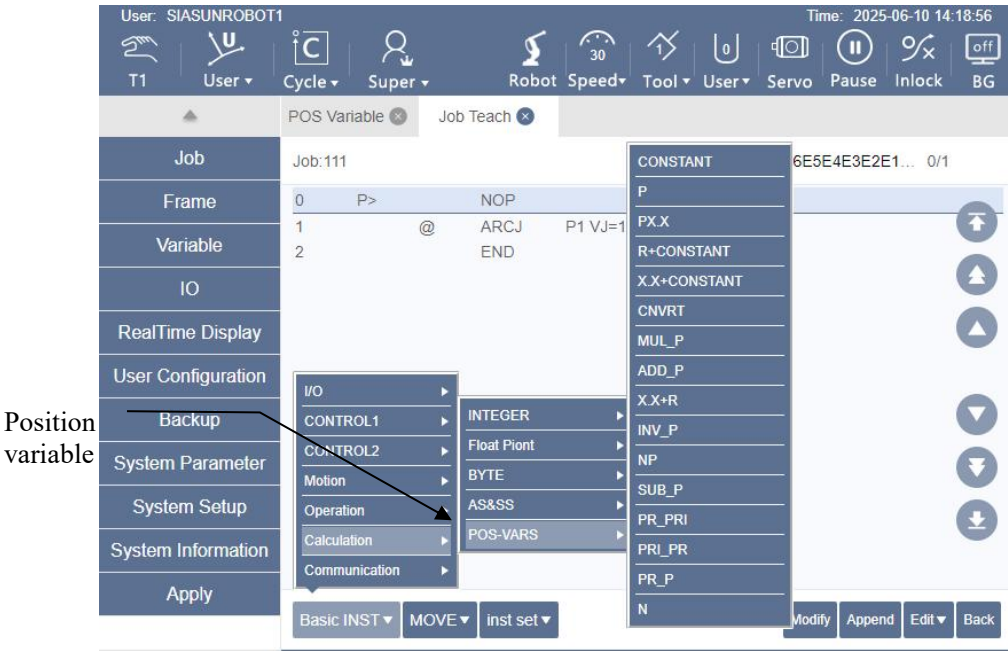


Figure 3-21 Position Variable Calculation Commands

Assignment of component values of position variables:

SET PX.X R:

Assign the float variable R to the component PX.X of the position variable.

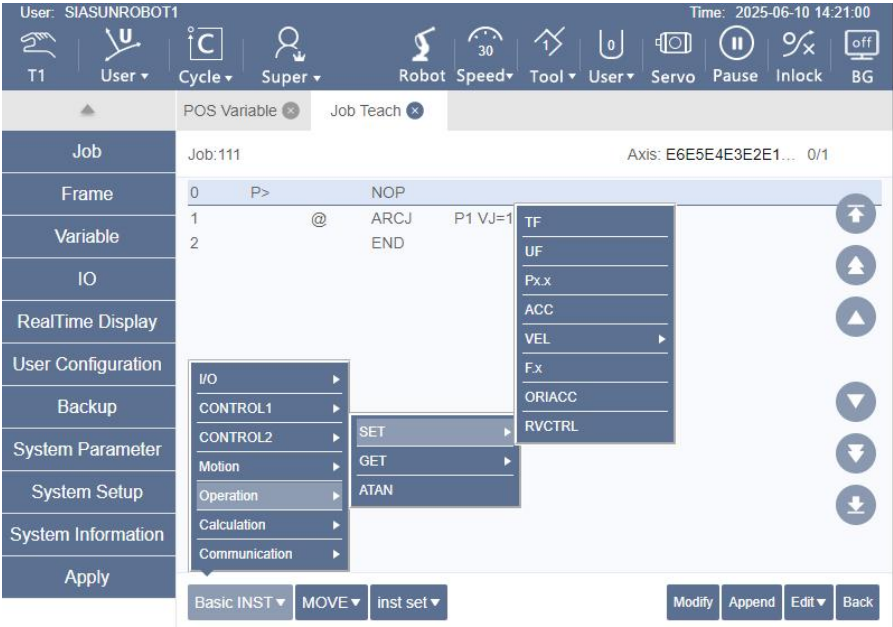


Figure 3-22 Position Component Assignment Commands

Table 3-2 Description of Position Component Assignment Commands and Parameters

SET PX.X	Function	Assignment operation, assigning a float variable to the component of a position variable.	
	Format	SET PR<Parameter 1>.<Parameter 2><Parameter 3>	
	Description	Parameter 1	<Position variable number>, range: 1-5630;
		Parameter 2	<Position variable component>, range: 1-12: 1:X 2:Y 3:Z 4:RX 5:RY 6:RZ 7:E1 8:E2 9:E3 10:E4 11:E5 12:E6
		Parameter 3	R<Variable number>, range: 1-520;
	Examples	SET PR1.1 R1	

3.3.3 Movement Speed

The movement speed of the robot can be determined in the program. During the automatic execution of the program, the movement speed varies with the specified percentage, the unit of the movement speed and the specified motion mode.

VJ MOVJ P1 VJ=10 ACC=100 CNT=0

When the action type is joint motion, the movement speed VJ represents percent of the maximum movement speed.

VL MOVL P1 VL=600 ACC=20 CNT=100

When the action type is linear motion, the unit is mm/s.

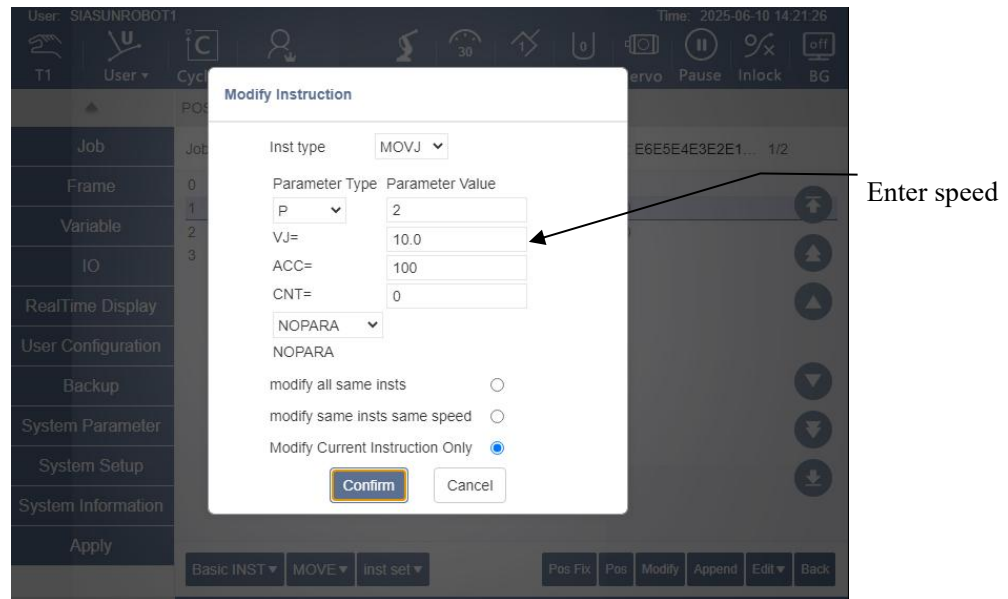


Figure 3-23 Motion Command Speed Input

VJ/VL/VC:

Modify the speed of the robot in the command. Only modify the speed of the current command, modify the speed of all commands of the same type (the current job cursor and those below the cursor), and modify the speed of commands of the same type and speed (the current job cursor and those below the cursor).

ACC:

Modify the percentage of acceleration in the command.

CNT:

Modify the continuous path level of the robot.

SET ACC:

Modify the acceleration percentage in the program.

SET VEL:

Modify the execution speed percentage in the program.

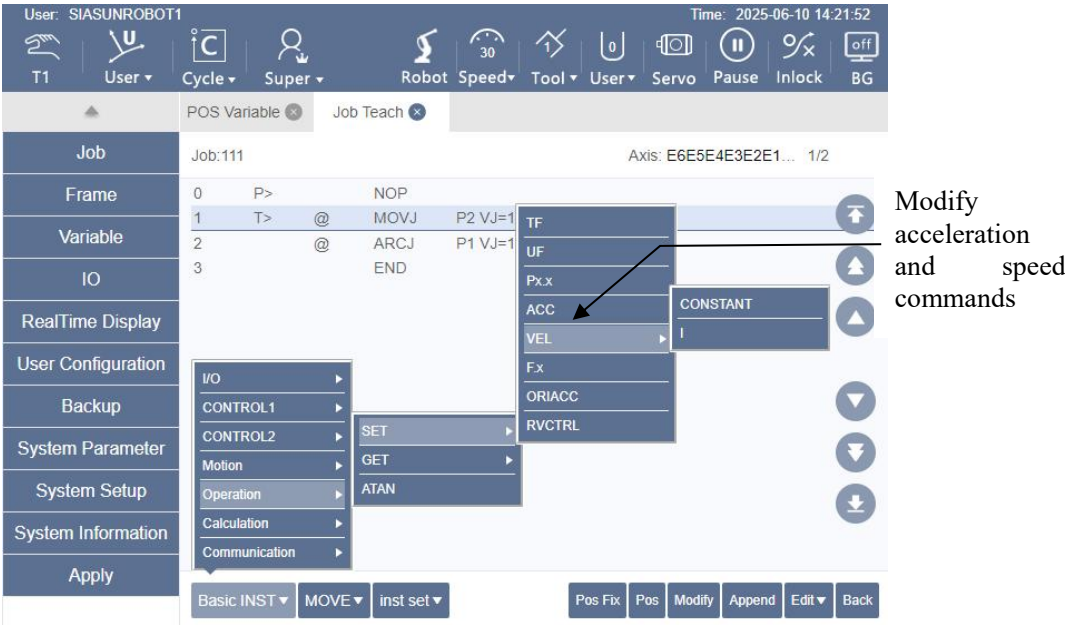


Figure 3-24 Modify Acceleration and Speed Percentage Commands

Table 3-3: Description of Acceleration and Speed Setting Commands and Parameters

SET ACC	Function	Set the acceleration percentage.	
	Format	SET ACC <Parameter 1>	
	Description	Parameter 1	Constant, range: 1-120; Final acceleration = Maximum acceleration * Parameter 1;
	Examples	SET ACC 80	
SET VEL	Function	Set the speed percentage.	
	Format	SET VEL <Parameter 1>	
	Description	Parameter 1	Constant, range: 1-100; <Integer user variable number>, range: 1-520; Final speed = Command speed * Parameter 1;
	Examples	SET VEL 80 SET VEL I10	

3.3.4 Continuous Path

There are two positioning methods for the robot to reach the designated position:

Discontinuous path;

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Continuous path;

Continuous path on/off need to be configured in Execute Switch. Click [System Setup] - [Execute Switch] on the main menu.

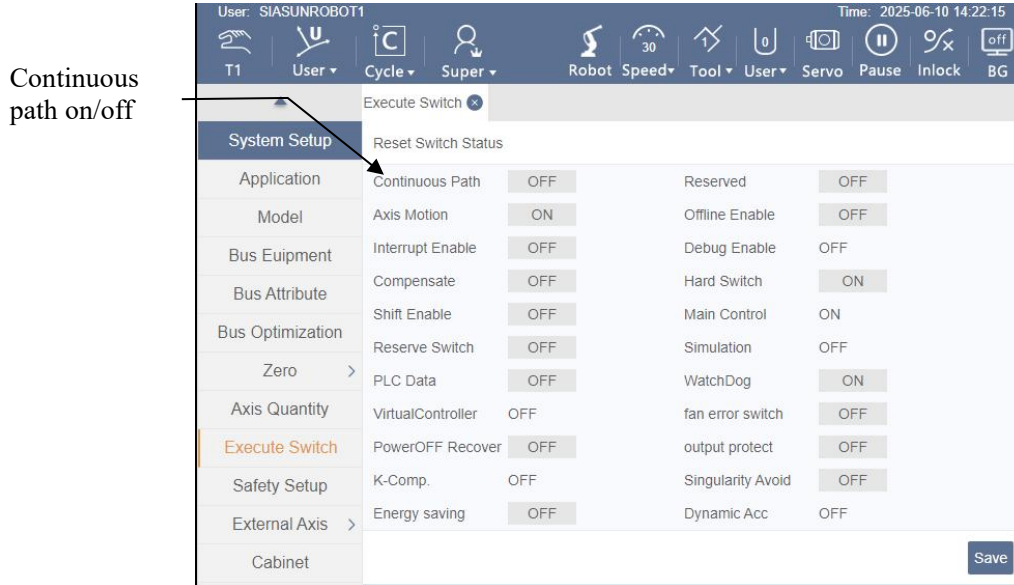


Figure 3-25 Execute Switch Interface

Discontinuous path:

When the continuous path is off, the robot completely stops at the target point and moves a little further down.

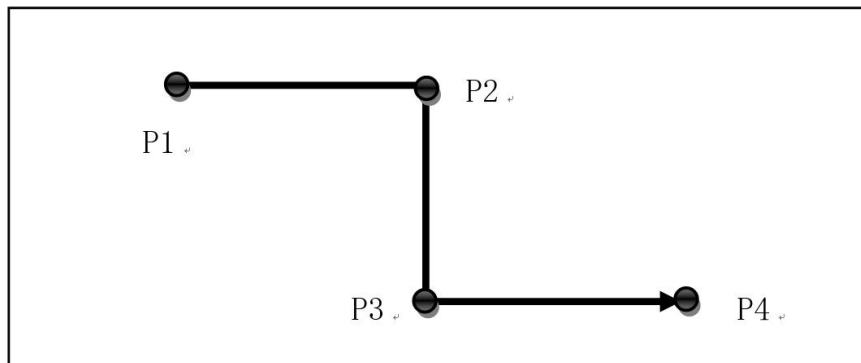


Figure 3-26 Continuous Path Off

Continuous path:

When the continuous path is enabled.

CNT positioning type:

MOVL P1 V=200 ACC=100 CNT= (0-100)

According to the CNT positioning type, the robot approaches the target position but does not

stop at that position and moves to the next position.

The degree to which a robot approaches the target position is defined by a value ranging from 0 to 100.

When 0 is specified, the robot stops moving when it reaches the target position. When 100 is specified, the robot does not slow down near the target position but immediately starts moving towards the next point and passes through the point farthest from the target position.

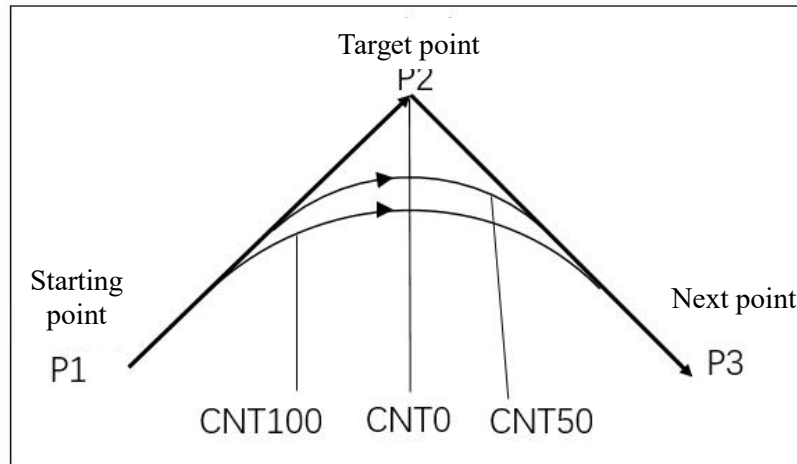


Figure 3-27 The path when the continuous path is enabled

Notes:

When the continuous path is enabled, some logical commands, such as "OUT" and other commands that can run continuously, will start to be executed in the deceleration section when the robot is about to reach the teaching point.

4 Program Teaching

This chapter explains the creation and modification of programs that can enable the robot to move.

Program commands refer to a series of control commands sent by the control system to the robot or peripheral devices. These commands can instruct the robot or peripheral devices to perform actions as instructed. For example,

- 1) Make the robot move in space along the expected path.
- 2) Grip and handle the workpiece.
- 3) Send signals to peripheral devices.
- 4) Process the signals input by peripheral devices.

Before creating the program, it is necessary to perform some summary designs of the program and consider the most effective way for the robot to complete the job.

Create the program through the robot menu and commands. When teaching the robot position, first move the robot to the desired position manually, and then record the position points through commands.

After the program is created, modify the program according to the menu commands when necessary.

4.1 Action Design

4.1.1 Action Commands

Workpiece gripping position:

The position of the workpiece should be gripped after positioning.

Move around the workpiece:

When the robot operates around the workpiece, a continuous path can be enabled. After initiation, the robot does not stay at the teaching point, but slides near it, which can accelerate the operation cycle of the robot.

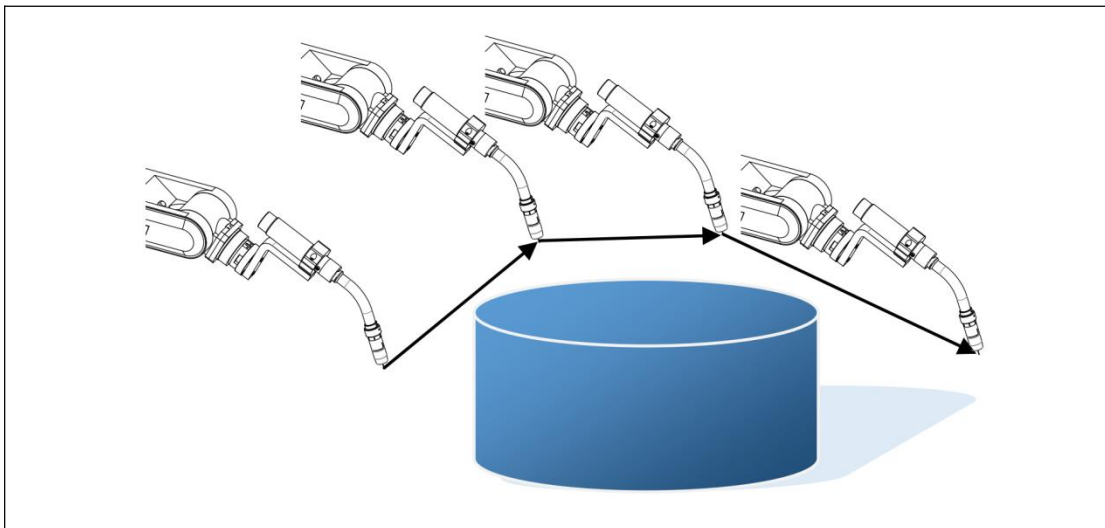


Figure 4-1 Moving around the workpiece

Note:

The above figure shows the motion path of the robot when the continuous path is not enabled. When the continuous path is enabled, the robot will quickly pass near the motion point, but the motion path will have a very small change near the motion point. It is

necessary to conduct a test to verify whether the path will interfere with the surrounding workpiece.

Robot pose:

When teaching the robot, a smooth pose should be adopted as much as possible to plan the path. If the correct pose is selected for the initial point, it can ensure that the subsequent points can reach the target point with minor changes. When teaching, you can manually move the robot using joint or linear frame to achieve the correct position and pose for teaching. After the program teaching is completed, it is necessary to manually move the program forward slowly to ensure that the path of the program is correct.

4.1.2 Origin Setup

The origin position is the fundamental position where all the programs of the robot are executed. This position is a safe one that does not interfere with all the surrounding peripheral devices. Output points can be set in remote I/O (for details, see 3.3 Instructions for Origin Setup).

4.2 Power-on and Manual Control

4.2.1 Robot Startup

Startup steps:

- 1) After connecting the power supply correctly, start the robot.
- 2) After startup, you can understand various statues of the robot according to the status bar above the teaching pendant.

4.2.2 Three Operation Modes

The key switch in the upper left corner of the teaching pendant can be used to select the appropriate robot operation mode.

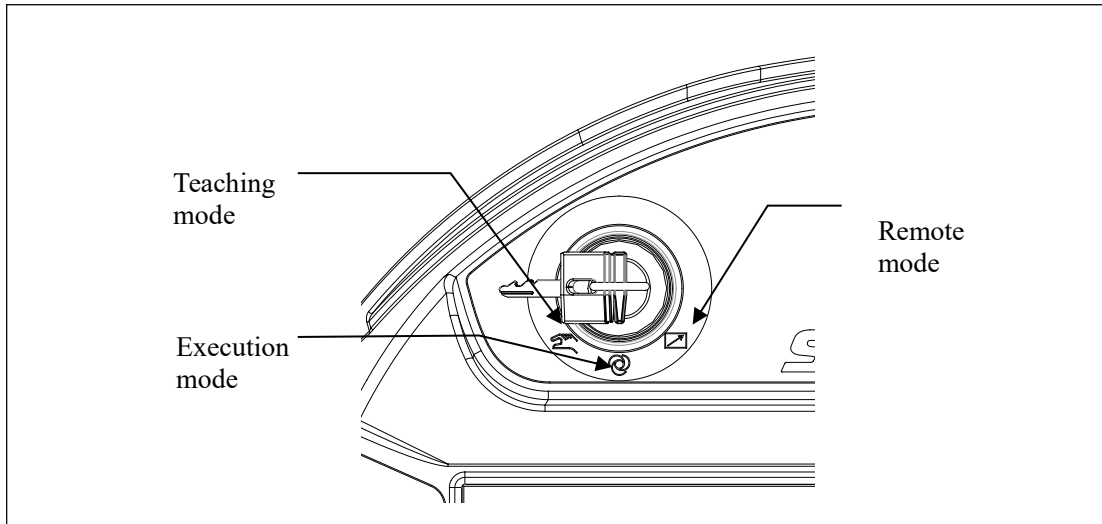


Figure 4-2 Mode Switch

1) T1 (low-speed manual) mode:

When teaching and manual testing are performed on the robot, the TCP of the robot is limited to a safe speed of less than 250mm/s in the selected mode.

a) Forward and reverse motions

The program can be manually run through forward or reverse motion.

b) Speed during forward and reverse motions

When moving forward and backward, even if the program speed is adjusted to 100, the TCP speed is still below the safe speed.

c) Safety door

In T1 or T2 mode, the robot can be manually operated with the safety door opened. The robot can only move when DEADMAN is pressed. When DEADMAN is not pressed, the robot cannot move.

d) Keep this mode

Remove the switch key to keep the teaching (T1 / T2) mode.

2) T2 (100% manual full speed) mode:

The T2 mode is used to confirm the motion of the robot at a normal speed. As the speed is limited in the T1 mode, the robot motion at full speed cannot be fully simulated. So sometimes it is necessary to confirm the full-speed motion and rhythm of the robot in the T2 mode.

a) Forward and reverse movements

The program can be manually run through forward or reverse motion.

Notes:

If unnecessary, please do not run the program continuously in reverse in T2 mode to avoid unexpected dangers due to excessive speed. This will cause damage to equipment or casualties.

b) Speed during forward and reverse motions

When moving forward, the robot speed is the same as its actual speed and is not subject to any special restrictions.

When moving backward, the maximum speed is 250mm/s.

When switching to T2 mode, for safety reasons, the speed percentage of the robot will change to **30%**, and the speed needs to be manually increased.

c) Safety door

In T1 or T2 mode, the robot can be manually operated with the safety door open. The robot can only move when the safety switch is pressed. When the safety button is not pressed, the robot cannot move.

d) Keep this mode

Remove the switch key to keep the teaching (T1 / T2) mode.

Warning:

- 1) T2 mode is a rather dangerous test mode. Please do not run the robot in T2 mode unless necessary.**
- 2) Before starting the robot, it is necessary to confirm that nobody is within the robot motion range, and to have a thorough understanding of any behavior of the robot after pressing Forward Motion.**

3) Execution mode

The execution mode is a mode used by the robot in automated production.

a) Program execution

The robot program can only be controlled through teaching power-on, start, pause and reset.

b) Speed

The speed of the robot can be full speed.

c) Manual motion

The robot cannot be moved manually.

4) Remote mode

The remote mode is a mode used by the robot in automated production.

a) Robot speed

The speed of the robot can be full speed.

4.2.3 Robot Axis Motion

The robot can be manually moved through the axis motion button on the right side of the teaching pendant. During program teaching, it is necessary to move the robot to the target position.

Warning:

- 1) To ensure the personal safety of engineers during the teaching process, please set the robot to manual mode throughout the teaching process, remove the mode key, and keep the safety door open and unable to be closed to avoid mistakenly pressing the start button or accidentally receiving an external start signal and starting the robot.
- 2) Try to use T1 mode for teaching as much as possible to avoid casualties or equipment damage caused by too high motion speed of the robot.

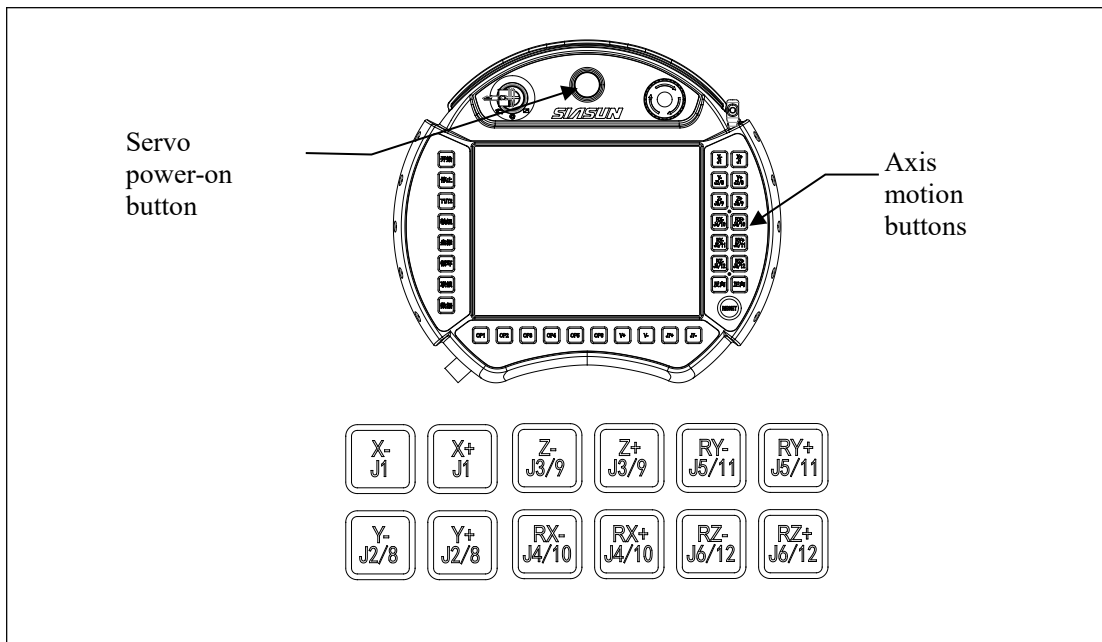


Figure 4-3 Manual axis motion buttons

The use of axis motion buttons to control the robot is related to the following two elements:

Robot speed;

Robot frame;

Note:

Before moving robot using the axis, it is necessary to use the servo power-on button to prepare the servo state.

Speed ratio:

The speed ratio is the maximum speed percentage for setting the robot motion. It is displayed on the status bar above the robot.

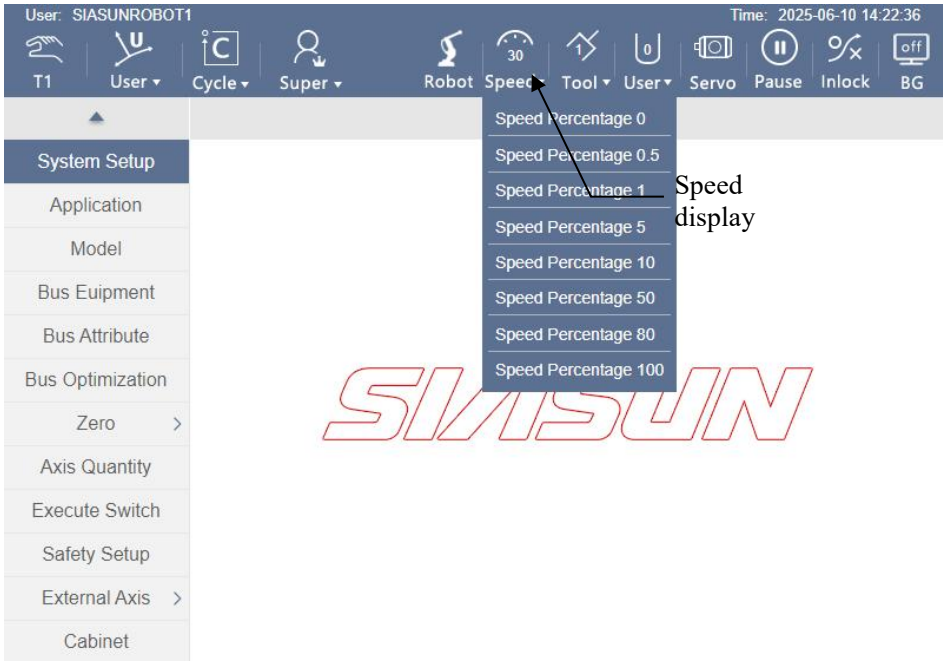


Figure 4-4 Speed Ratio Display

Manual ratio can be modified by clicking on the touchscreen of the teaching pendant or by pressing buttons. The minimum speed ratio is 0.2.

Table 4-1 Manual Ratio Switching

Manual ratio	Ratio: 0-0.2-0.5-1-5-10-30-50-80-100
--------------	--------------------------------------

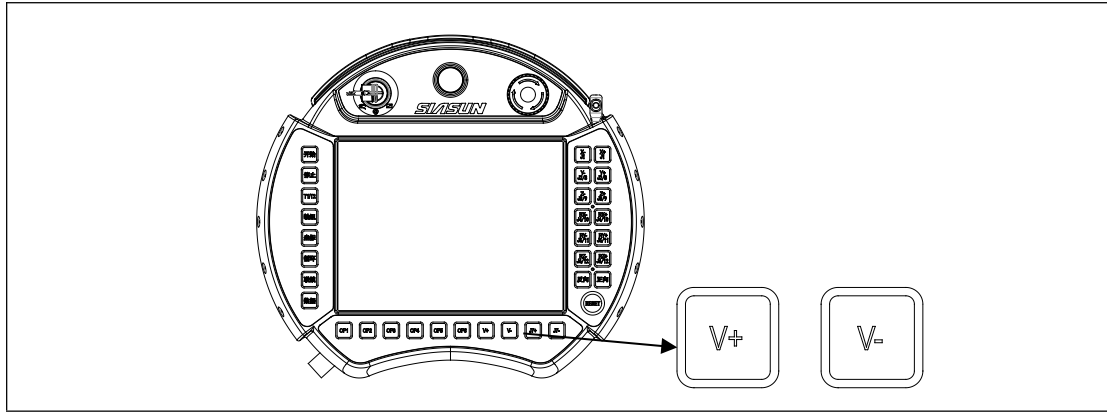


Figure 4-5 Modify speed ratio position

Frame for manual motion:

There are a total of four frames available for the manual motion of the robot. They are respectively joint frame, rectangular frame, tool frame and user frame.

Manual motion in the joint frame:

By using the axis motion button, the robot can run uniaxially in the direction of each axis. The direction of the arrow indicates the direction of the robot axis rotation after pressing the corresponding axis motion button.

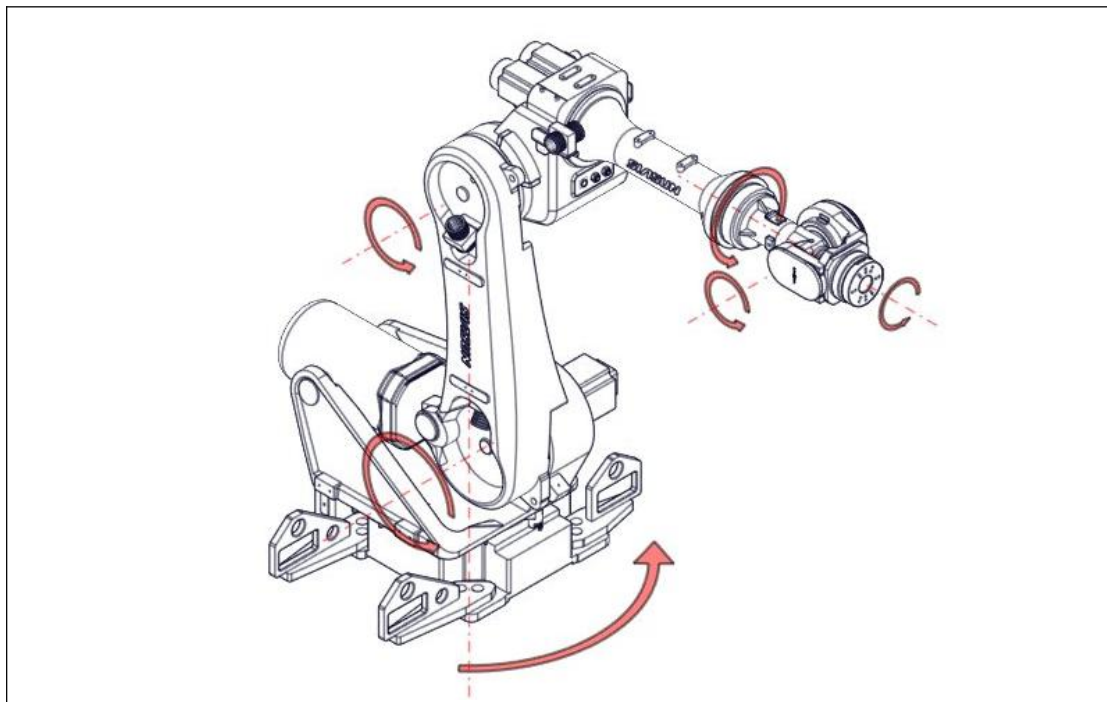


Figure 4-6 Joint Frame

Manual motion in the rectangular frame:

In the motion of the rectangular frame, the robot TCP will move along the X, Y and Z directions parallel to the axes. Meanwhile, when rotating, the robot TCP will also rotate around the

parallel lines parallel to the X-axis, Y-axis and Z-axis of the rectangular frame.

In the rectangular frame, when the corresponding direction button of the robot teaching pendant is pressed, the robot TCP will move in the direction parallel to the corresponding arrow, or rotate around the parallel lines of the rectangular frame axis with TCP as the center.

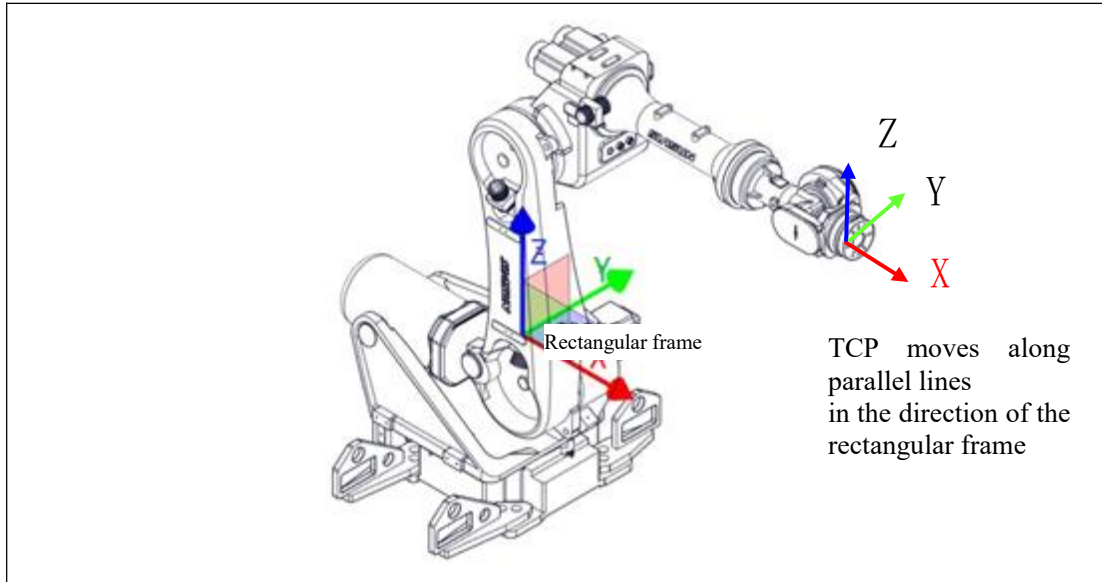


Figure 4-7 Rectangular Frame

Manual motion in the tool frame:

Manually move the robot in the tool frame, and the TCP move along X, Y and Z of the tool frame defined by the robot. Furthermore, the tool frame can also enable the robot to rotate around the X-axis, Y-axis and Z-axis of the tool frame.

In the tool frame, after pressing the axis motion button of the teaching pendant, the robot TCP will move in the direction pointed by the tool frame or rotate around TCP in the direction of the working frame.

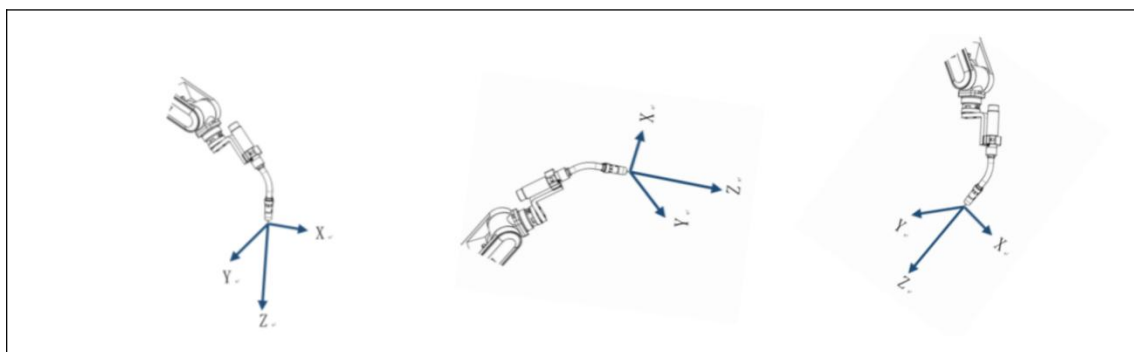


Figure 4-8 Tool Frame

Manual motion in the user frame:

Manually move the robot in the user frame, and the TCP move along the parallel lines of

General Operating Manual for SIASUN Robot

X-axis, Y-axis and Z-axis of the user frame defined by the robot. Furthermore, the user frame can also enable the robot TCP to rotate around the parallel lines of X-axis, Y-axis and Z-axis of the user frame.

In the user frame, after pressing the axis motion button of the teaching pendant, the robot TCP will move in the direction pointed by the user frame or rotate around TCP as the center along the parallel lines in the direction of the user frame axis.

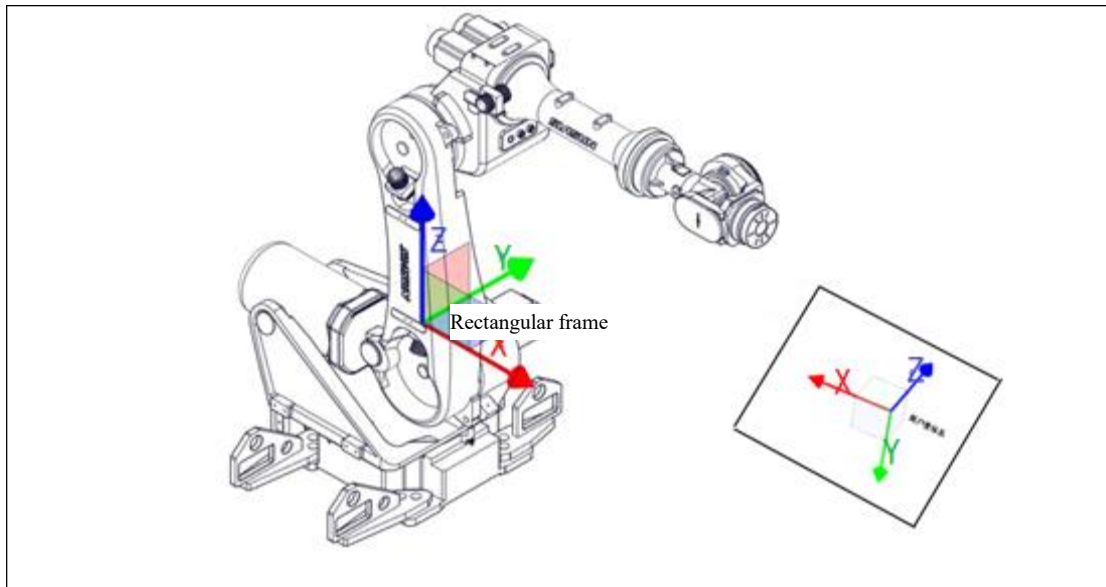


Figure 4-9 User Frame

Frame switching:

The robot frame can be selected in the status bar above the teaching pendant. It can be selected through the icon on the touchscreen.

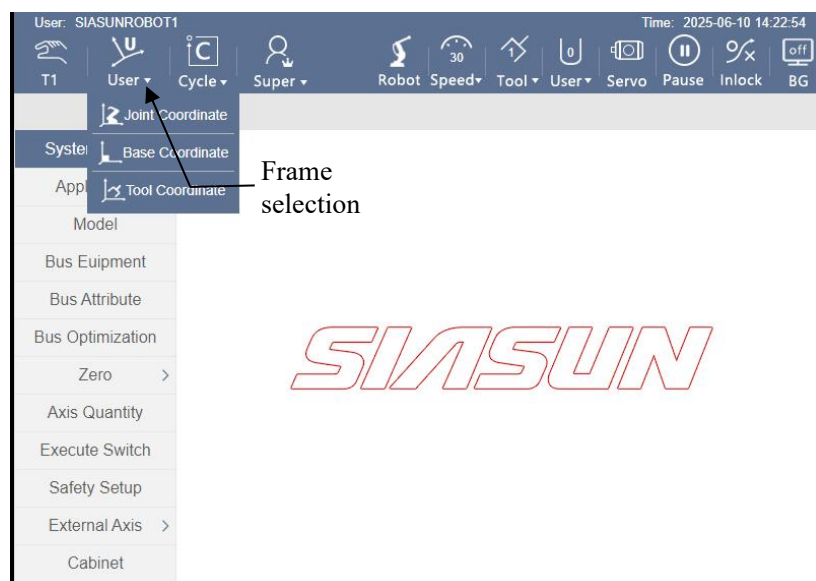


Figure 4-10 Frame Selection

It can also be selected through the frame button on the teaching pendant.

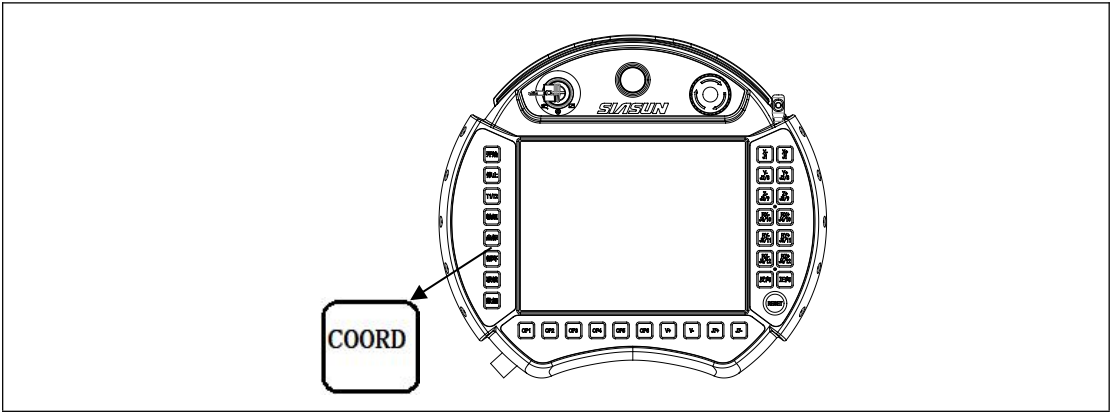


Figure 4-11 Frame Buttons

Table 4-2 Frame Switching

Frame switching	Joint frame - rectangular frame - tool frame - user frame
-----------------	---

Use of external axis:

Press the axis buttons of the teaching pendant to switch the motion of the external axis. Axis 7 can be moved directly on the teaching pendant. Axes 8 to 12 can be moved using the axis motion buttons after switching. The status of the axis group can be displayed on the teaching pendant screen.

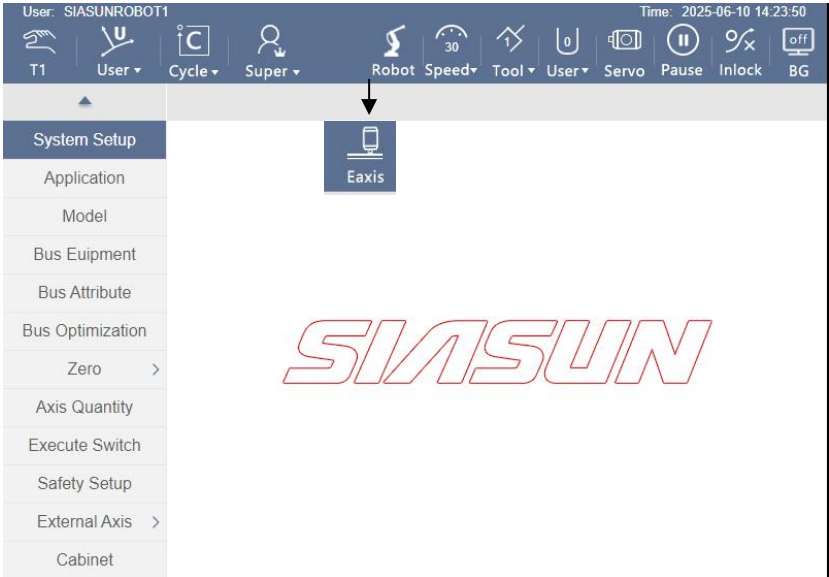


Figure 4-12 Robot body axis display icon of external axis

The external axis can be controlled by using the axis motion buttons, as shown in Figure 4-13.

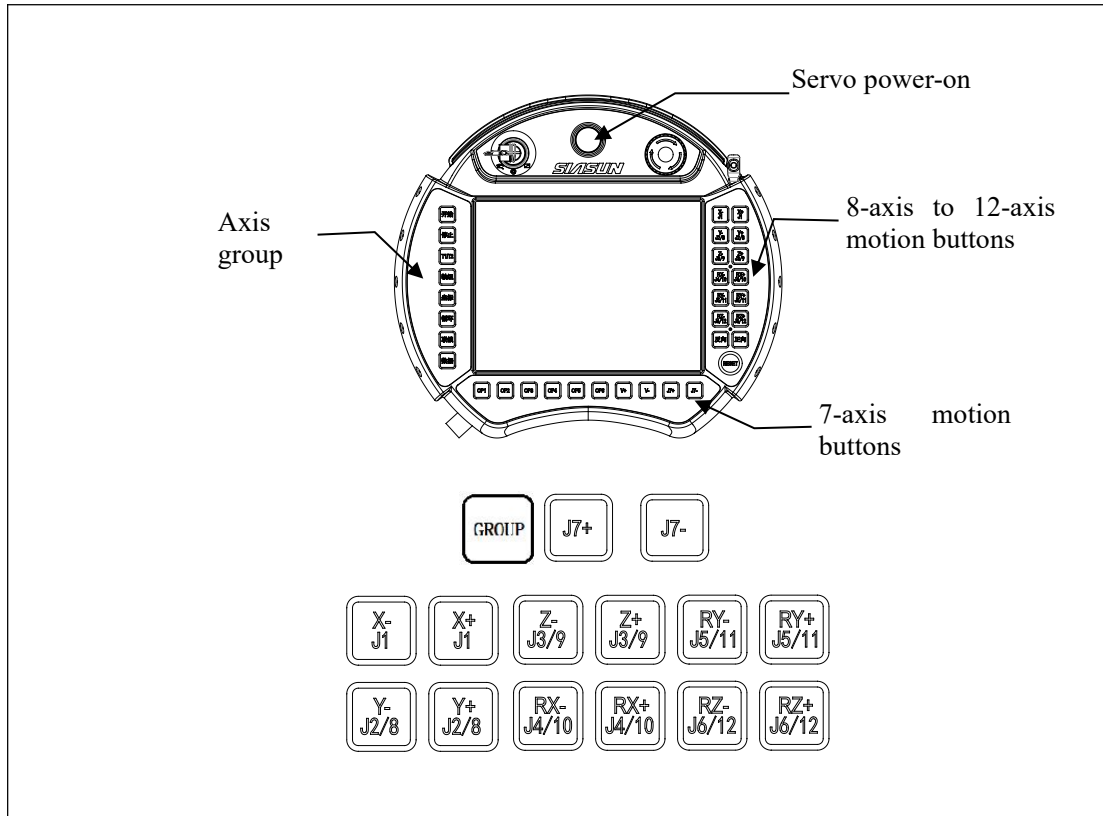


Figure 4-13 Axis group and axis motion buttons

Conditions for moving the robot using the axis motion buttons:

Before manually starting the robot, please ensure that there is no one or any object within the movement range of the robot, including the end effector.

Steps for moving the robot using the axis motion buttons:

- 1) Select the frame of the manual robot by clicking [Frame] or the icon.
- 2) Adjust the rate through [V+ \ V-] or the rate icon.
- 3) Hold the teaching pendant with your left hand, press DEADMAN to the middle position, and hold the down switch throughout the robot motion.
- 4) While holding down DEADMAN, hold down the axis motion button to control the robot to move in the expected direction.

Warning:

When operating manually, if you want to stop the robot urgently to avoid dangerous situations, you can release DEADMAN or press the emergency stop button.

Description of singular point:

In robot kinematics, there are three singular points that require special treatment. They are

respectively top singular point, extension singular point and wrist singular point. These singular points have no way to plan paths. So, the robot will stop moving and give an alarm near the singularity points. Joint frame can be used for planning during teaching.

Top singular point:

When the joint center point is the intersection of the axes 4, 5 and 6 and is located above the axial line of one axis, the robot is located at the top singular point.

Extension singular point:

When J2-J3 extension line passes through the center point of the wrist joint, the robot is located at the extension singular point.

Wrist singular point:

When the axis 4 is parallel to the axis 6, that is, when the joint value of the axis 5 is near 0, the robot is located at the wrist singular point.

4.3 Teaching Program

New program:

Create a new program and record the program name.

Set program information:

Set the relationship between the program type and the external axis group.

Teaching actions and logical commands:

Teach the robot program.

4.3.1 New job

Create a new teaching program.

Steps for creating a new program:

- 1) Click [New Job] to open the interface for creating a new job.

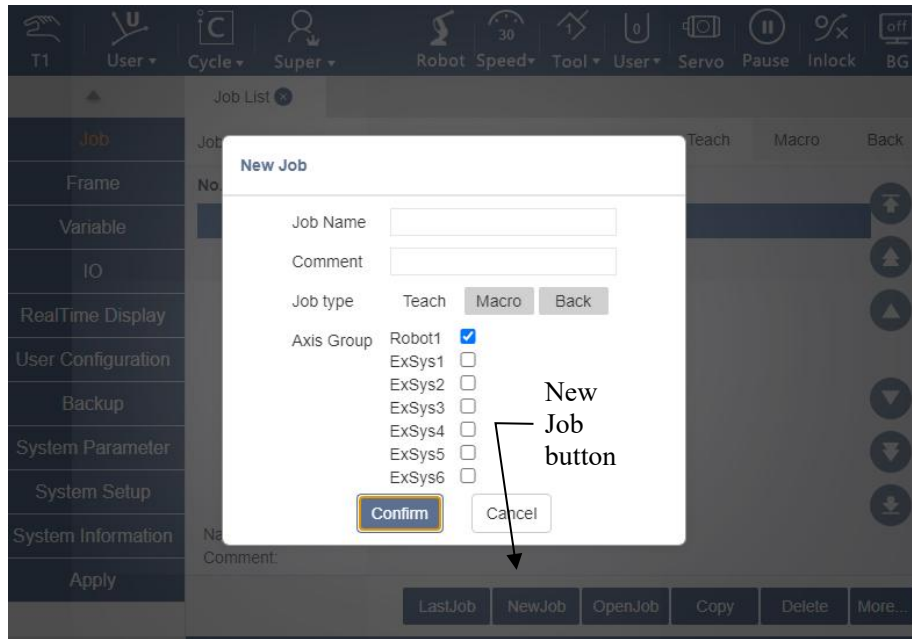


Figure 4-14 New Job Interface

- 2) Select the job type in the new interface. If the job type is selected as the teaching program, the axis group information needs to be set.

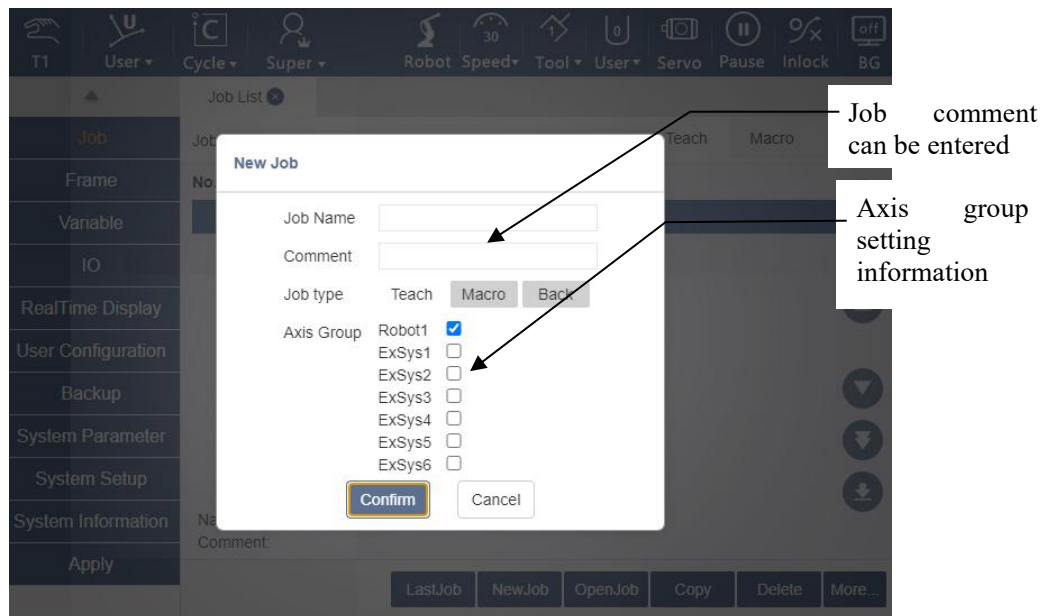


Figure 4-15 New Job Interface

- 3) Select the settings of the axis group. SIASUN robot supports up to 12 axes, including up to 6 body axes and up to 6 external axes. The external shaft system needs to be configured separately in the software according to the on-site application. When creating a new job, Robot1 in the axis group setting interface represents the robot body, and Robot2 in the axis group is currently reserved and not set. ExSys1-6 corresponds to the

configured external axis system numbers. ExSys1 corresponds to the external axis system number 1, and so on.

Only when the axis group information is configured correctly, the robot motion will be correct during the forward and reverse motion inspection and automatic execution of the teaching operation. Otherwise, an axis of the robot does not move, or the motion is abnormal.

After the job creation is completed, the combination of 8 groups of letters and numbers after "Axis Group" displayed in the data information area of the teaching pendant interface is the 8-bit axis group information. From right to left, they correspond to Robot1, Robot2 and ExSys1-6 in sequence. When the axis group switch is $\sqrt{}$, the corresponding data bits are displayed in orange. When the axis group switch is not $\sqrt{}$, only the robot body is orange.

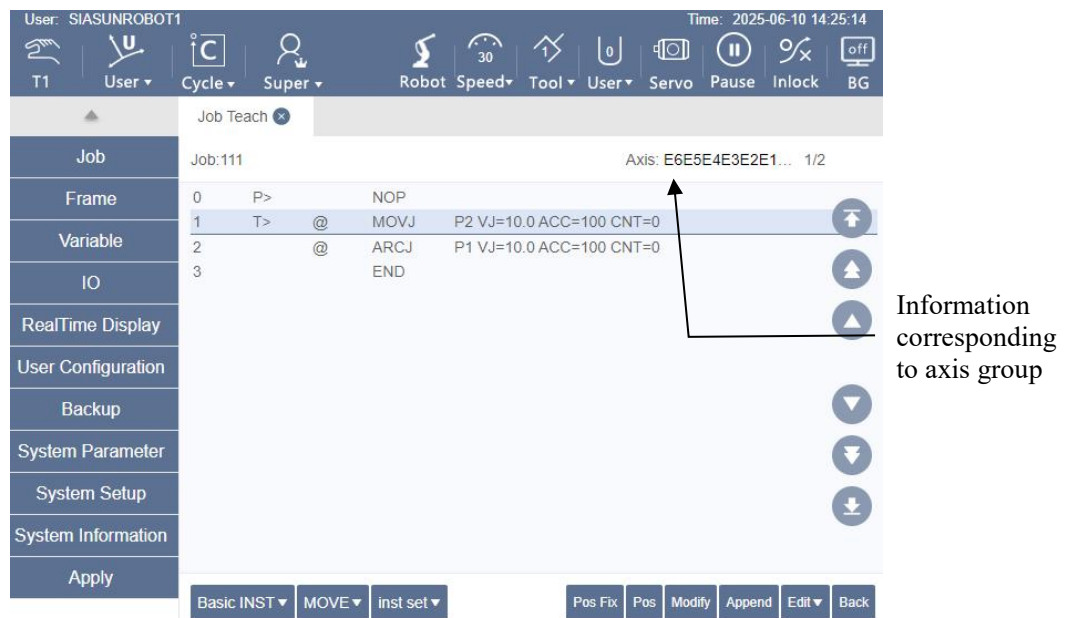


Figure 4-16 Data information corresponding to axis group

Job name: Supports capital letters, numbers and underscores. The job name cannot exceed 32 characters.

Note: Comment can be entered and displayed after the job name. Comment cannot exceed 36 characters.

Note:

- 1) After entering the job name, press [Go] button and then [OK] button.
- 2) The entered content is based on the content before the first trigger of [Go] button. After [Go] button is triggered for the first time, the content subsequently edited will be invalid.

- 3) Suggestion: Before triggering [Go] button, please carefully confirm the content you have edited. If [Go] button has been triggered and the edited content requires changing, click [Cancel] button on the soft keyboard.**

New macro job:

In the new job interface, select the macro job in the job type to create a new macro job.

Macro job is subject to the following restrictions:

- 1) The name of the macro job is in a fixed format of MACRO+1-99999999.... The number following it will correspond to the parameter after the MARCO command in the teaching program, and it will be called through the control class 1 - MARCO command.
- 2) Macro jobs cannot contain motion commands or motion-related commands (interrupt, offset, get position point, set tool, user frame), CALL, WAIT, IRQ, etc. Otherwise, an alarm will be triggered when calling macro commands.
- 3) Macro job is called through the MACRO command, and its menu position is [Basic Command] - [Control Command 1] - [MACRO]. The parameter range of the macro command is the integer ranging from 1 to 99999999.... For example, MACRO 100 command will invoke the job named "MACRO100".

New background job:

In the new job interface, select the background job in the job type to create a new background job.

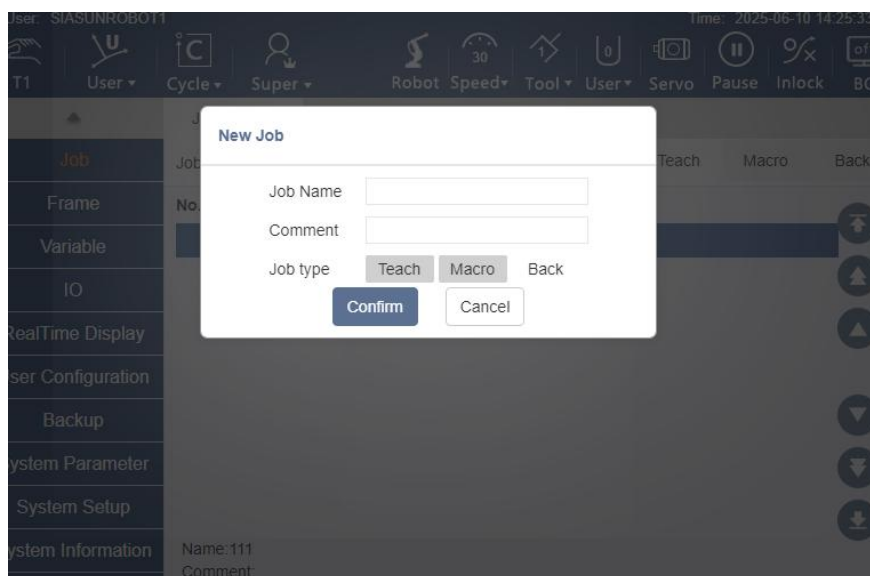


Figure 4-17 Establish background job

The background program cannot handle action commands and commands related to actions (such as interrupt, offset, obtaining position point, setting tool, user frame, etc.). When starting the background, the system will check the background programs. If there are commands that cannot be processed, it will give an alarm that the 2237 command is illegal, with line number of n. The background program cannot be started.

Multiple background programs can be created, with a maximum of 8 background programs allowed simultaneously. Background programs can invoke other background programs, with a maximum of 10 levels invoked. Background configuration is required before the background program is started. After the background job is successfully configured, the job name will change to orange. There are two ways to start background jobs: switch and foreground job command. If the background job is started through the menu before shutdown, the background job will be automatically started after startup. If the background job is stopped through the menu before shutdown, the background job will not be automatically started after startup. The background job icon is in ON state, indicating that the robot background job has been successfully started; it is in OFF state, indicating that the background program is closed.

Note:

- 1) The background job cannot invoke the teaching program. The invocation level is the same as that of the teaching program, with a maximum of 10 levels invoked.**
- 2) The maximum number of background job commands shall not exceed 200 (including the number of commands in the invoked sub-background jobs).**
- 3) The jump tag command must be added after the jump condition; otherwise, an alarm will be triggered when the background is started.**

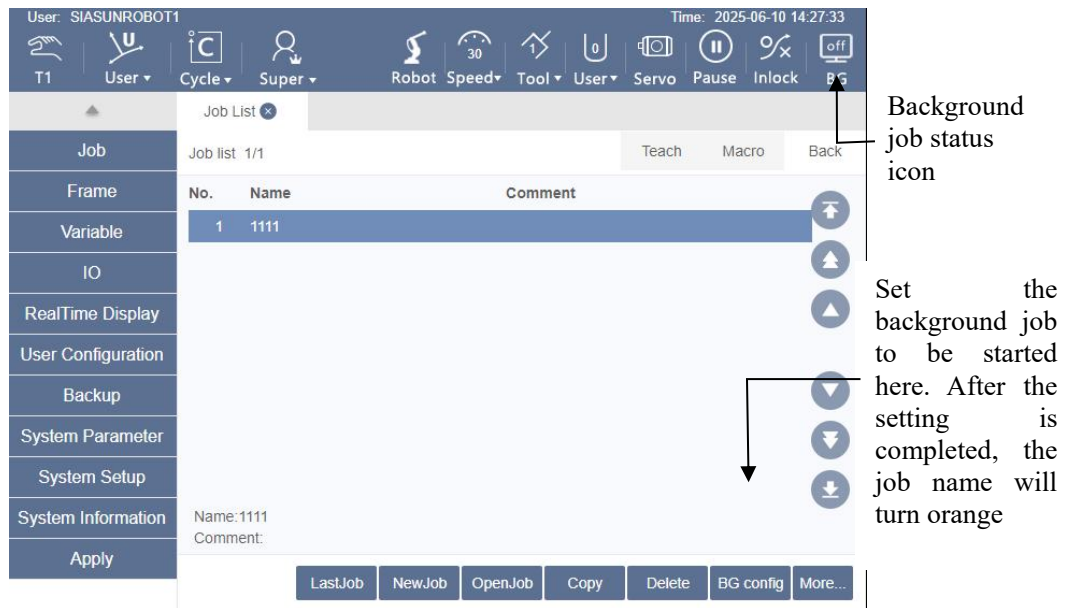


Figure 4-18 List of background programs

After viewing the background program, you can set, start and stop the background program through the background configuration. By setting it as a background command, you can select the background program to be started.

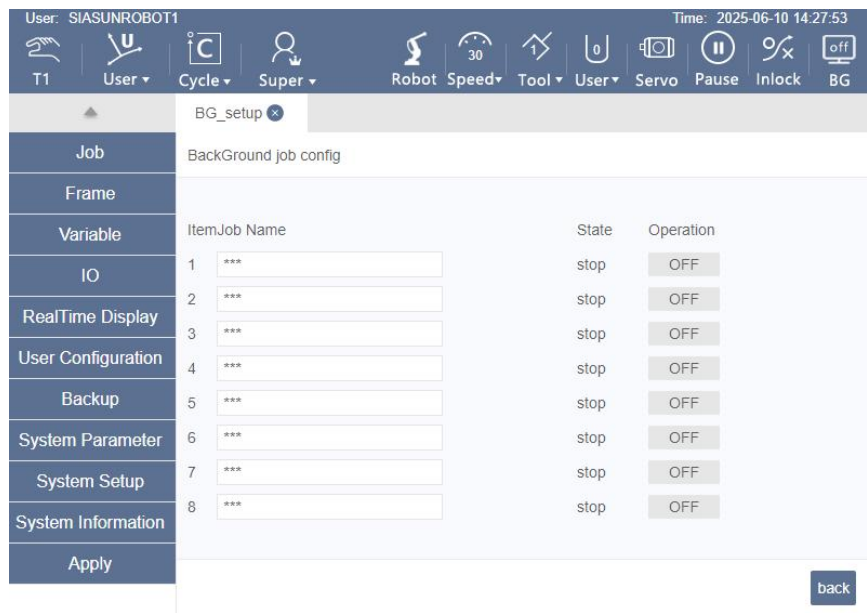


Figure 4-19 Background configuration interface

Job name: Enter the name of the background program. (As shown in the above figure, START BG#1 can be used to start/stop background jobs in the job)

Status: Display the current running status.

Switch: View and force background start/stop.

The background program that has been started will remain in the startup state if it is not closed. It has nothing to do with the robot mode, whether the robot teaching program is started, or whether the robot is servo-powered. When you do not want the background program to run, you can close it via the menu. After the background program is stopped, the icon in the upper right corner of the screen will display OFF. If the foreground and background programs are in the startup state after power-off, the background programs will automatically start and run after power-off and restart.

The background program can be stopped and restarted not only through the menu but also via commands in the foreground task (teaching program). The commands are as follows:

START BG: Start background;

STOP BG: Stop background;

They respectively correspond to restarting and stopping background jobs. When the foreground job executes this control command, the background job is stopped or restarted. To control the stop and restart of the background jobs using commands, the background program must first be started via the menu before executing the foreground jobs. If the background program is not started via the menu, an error will occur when the foreground jobs execute the commands controlling the background program.

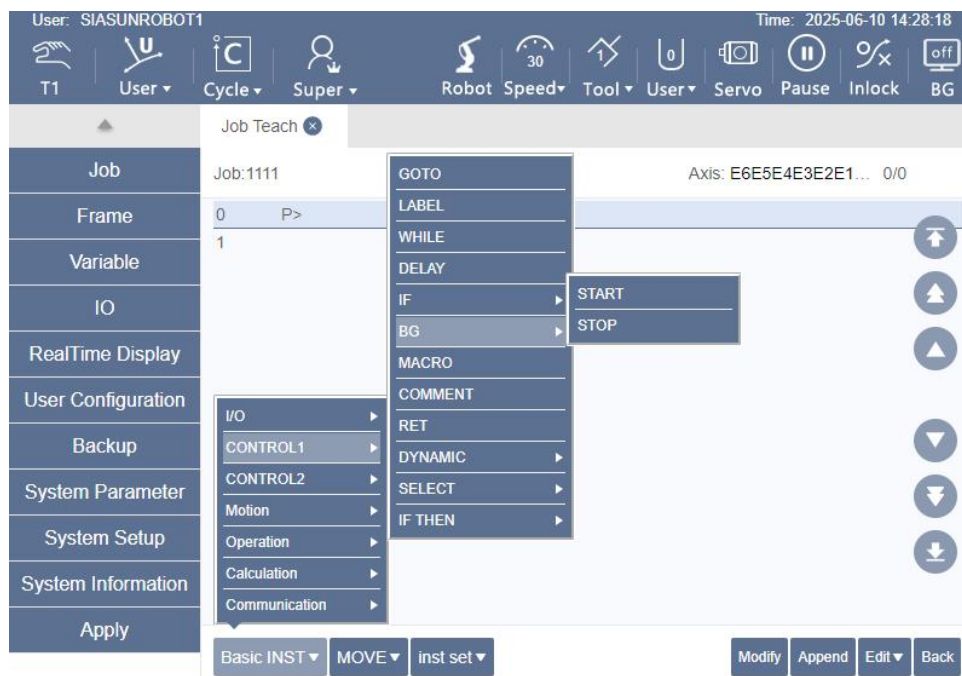


Figure 4-20 Background Command Menu

Table 4-3 Description of Background Commands

START BG	Function	Start background program.
	Format	START BG# (1-8)
	Description	Start background programs (1-8)
	Examples	NOP START BG#1: Start background job MOVJ P1 VJ=10 DELAY T=1 MOVJ P2 VJ=10 DELAY T=1 STOP BG#1: Stop background job END
STOP BG	Function	Stop background program.
	Format	STOP BG# (1-8)
	Description	Stop background programs 1-8
	Examples	NOP START BG#2: Start background job MOVJ P1 VJ=10 DELAY T=1 MOVJ P2 VJ=10 DELAY T=1 STOP BG#2: Stop background job END

4.3.2 Settings of Job Interface

Before teaching, it is necessary to set all the functions required for the teaching job.

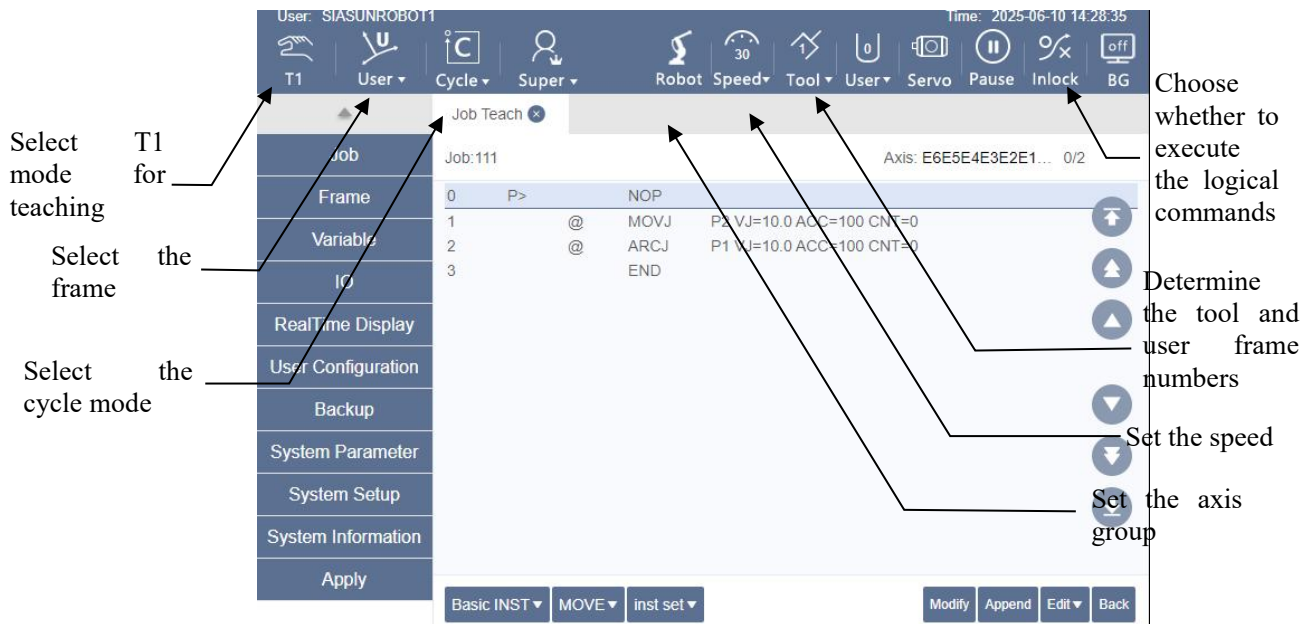


Figure 4-21 Function items to be set

Mode selection:

When the engineer teaches, please always keep the robot in T1 mode.

Warning

To ensure the personal safety of engineers during the teaching process, please set the robot to teaching mode throughout the teaching process, remove the mode key, and keep the safety door open and unable to be closed to avoid mistakenly pressing the start button or accidentally receiving an external start signal and starting the robot.

Frame selection:

The frame selection is related to the path and manner in which the robot performs single-axis or multi-axis linked actions.

Select the cycle mode:

Select different cycle modes. The cycle mode will be different when the robot moves forward.

Step: Each time Forward Motion is pressed and held down, the robot only executes one motion command or logical command.

Cycle: After holding down Forward Motion, the robot continuously executes motion or logical commands until the job is completed.

Auto: After holding down Forward Motion, the robot continuously executes motion or logical commands and keeps cycling the job commands.

Notes:

When moving backward, the robot can execute the program continuously or by step as set, but it cannot execute logical commands.

Select the axis group:

After the axis motion button of the robot is pressed, the teaching pendant controls the body axis or the external axis.

Determine the speed:

Determine the speed when the robot axis moves.

Select the frame number:

Select the tool and user frame number when the robot is in motion.

Interlock:

Determine whether the robot executes logical commands or merely executes action commands and skips the logical commands.

The O/o icon indicates that the robot executes motion commands and logical commands when moving forward.

The O/x icon indicates that the robot only executes motion commands and does not execute logical commands when moving forward.

4.3.3 Teach Action Commands

An action command is a command that enables a robot to move from its current position to the target position at specified motion type and speed.

Click Motion Point to enter.

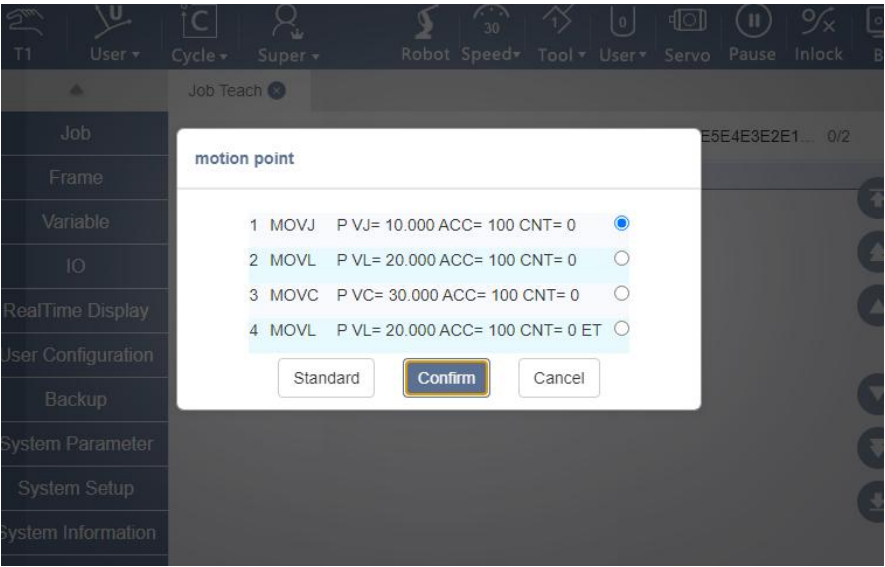


Figure 4-22 Motion Command Menu 1

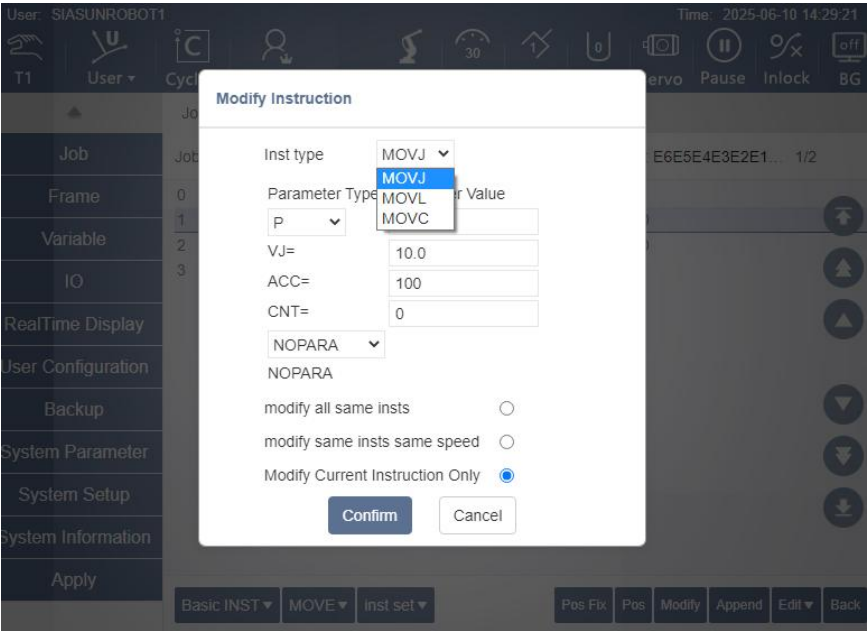


Figure 4- 23 Motion Command Menu 2

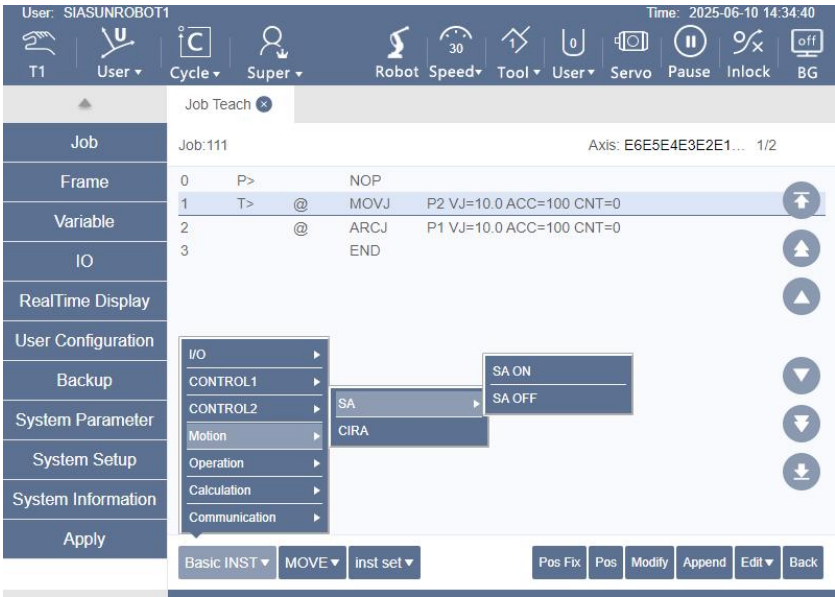


Figure 4-24 Motion Command Menu 3

In the menu, select the desired motion type. If you need to change the motion parameters, choose the command to be modified and click [Modify] at the bottom right.

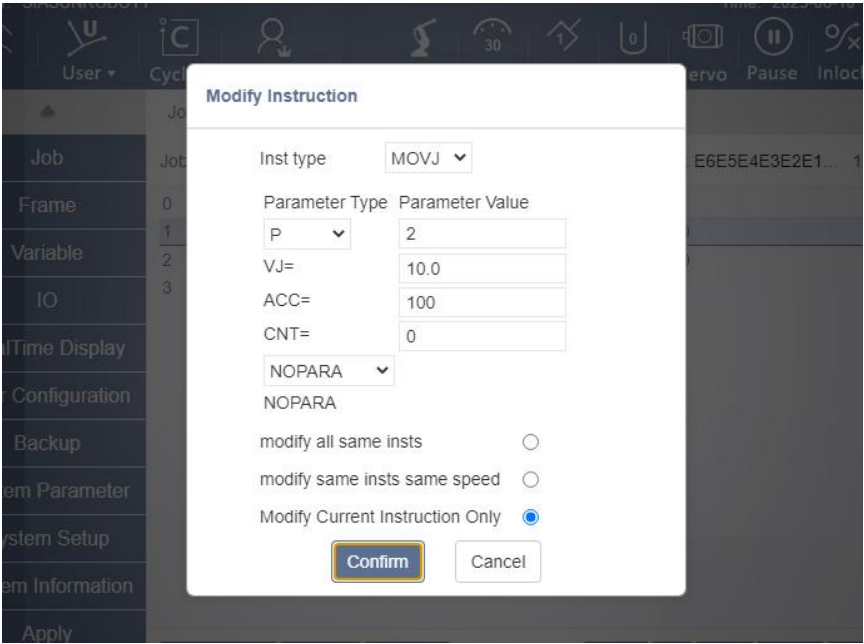


Figure4- 25 Enter Motion Command Parameters

Enter the parameters and click Confirm.

4.3.4 Logic Control Commands

Logic control commands refer to all robot commands other than motion commands,

including:

I/O commands;

Control class 1 commands;

Control class 2 commands;

Operation commands;

Calculation commands;

These commands can be directly invoked by clicking the corresponding menu of the teaching pendant and adding them by writing parameters.

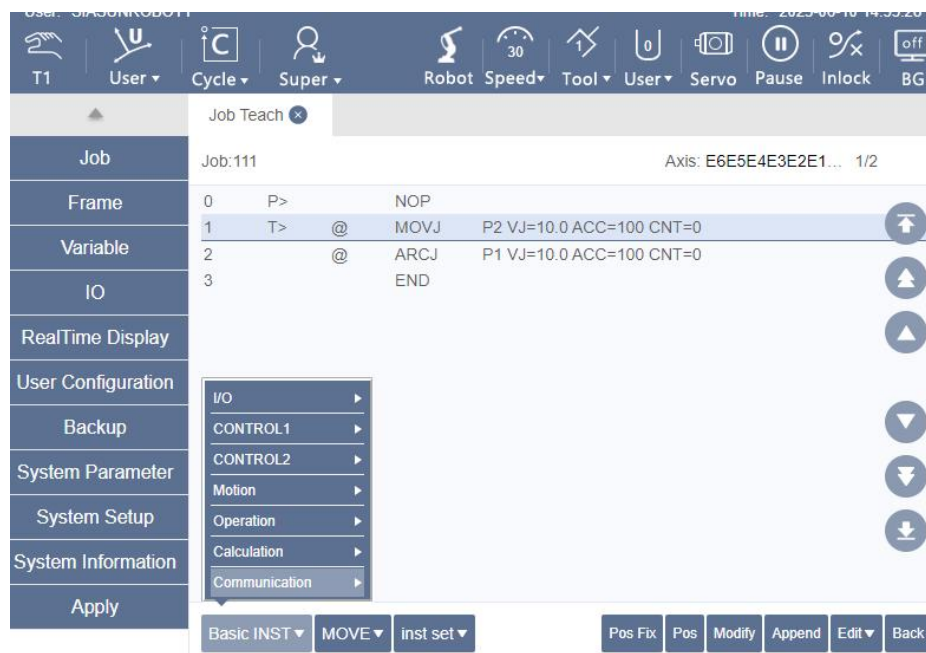


Figure 4-26 Logic Control Commands of Command Menu

4.3.5 Commands of Command Set

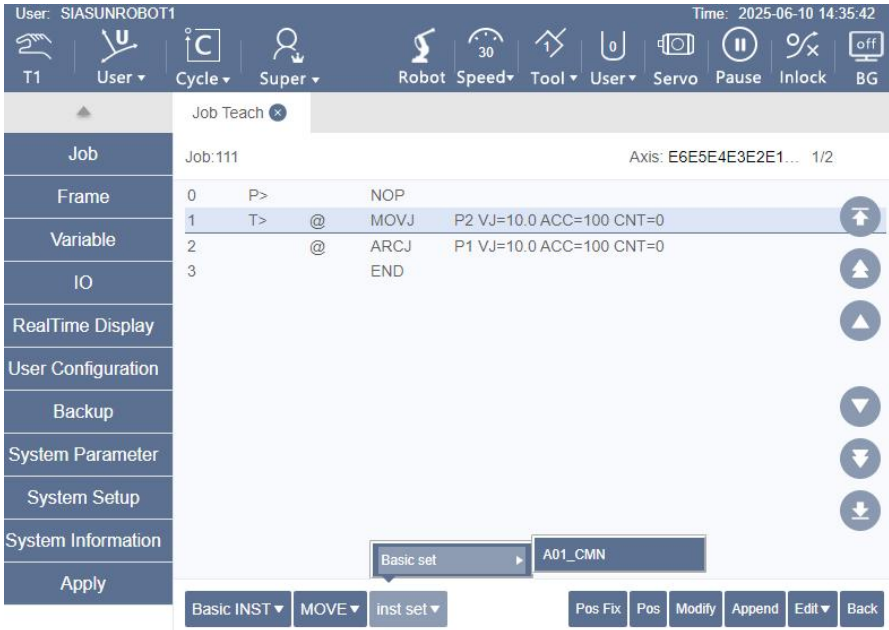


Figure 4- 27 Basic Command Set of Command Menu

4.4 Command Editing

The editing menu can perform operations such as delete, copy and cut on the edited commands.

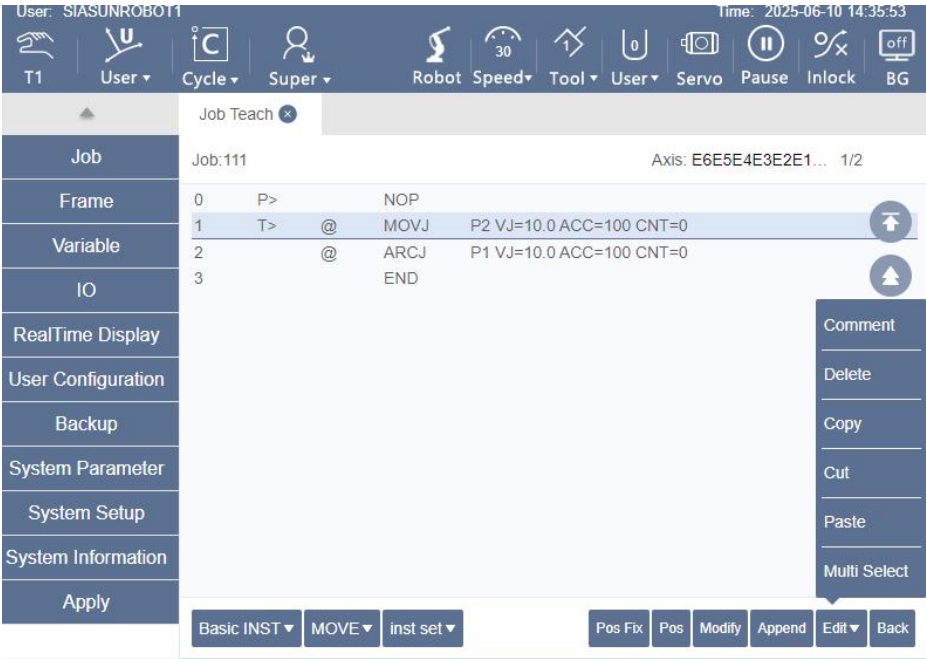


Figure 4- 28 Program Command Editing Menu

- 1) Pos Fix: The current command position can be modified to the current actual position of the robot.
- 2) Pos: You can view and modify the current position information of the robot.
- 3) Modify: It will be introduced below.
- 4) Append: When editing the same command continuously, you can click "Append", and the last used command will be directly added below.
- 5) Comment: Mask/demask this command (this command is not executed.
- 6) Delete: Delete the selected command.
- 7) Copy: Copy the selected command.
- 8) Cut: Cut the selected command.
- 9) Paste: Paste the copied/cut command.
- 10) Multi Select: Multiple commands can be selected simultaneously and edited as a whole.

4.4.1 Modify Motion Commands

To modify motion commands, you can modify the type of motion commands, local/global variables, variable numbers and related parameters, obtain the current position, and view and modify the position value. Click [Modify] on the motion command to be modified with the cursor, and a modification interface will pop up, that is, to modify the motion command.

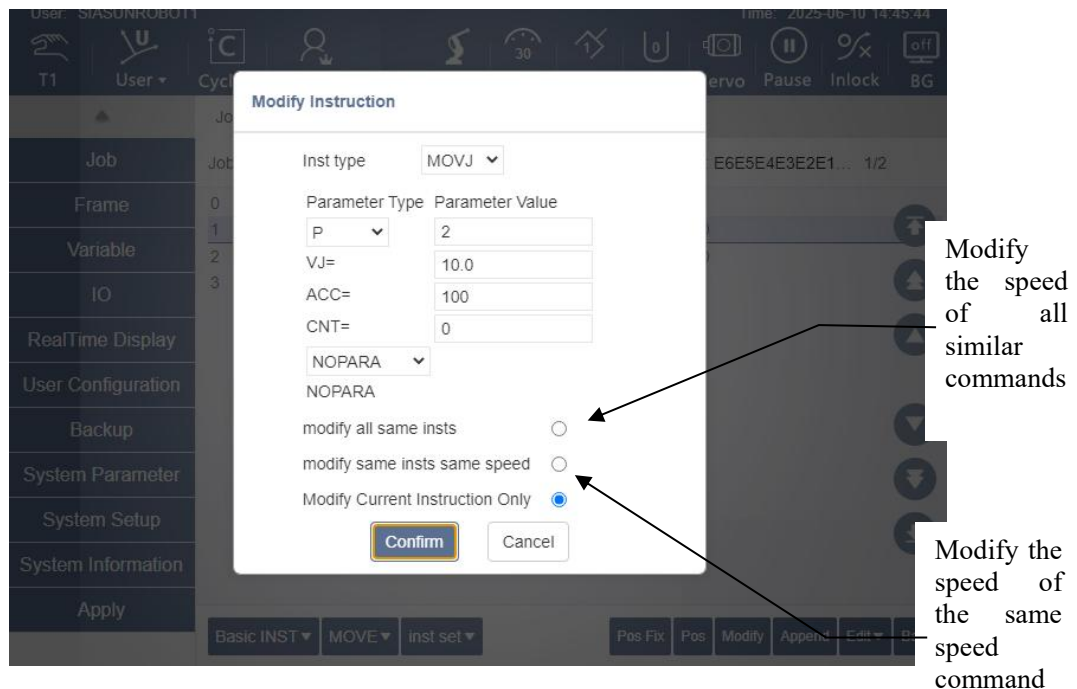


Figure 4- 29 Command Modification Interface

- 1) The command type can be modified as MOVJ/MOVL....
- 2) P/PR can be modified.
- 3) The speed can be modified.
- 4) The ACC acceleration percentage can be modified.
- 5) The CNT continuous path level can be modified.
- 6) NOPARA/INTERLOCK can be modified to determine whether the interference zone is distinguished or not.
- 7) External TCP can be modified. (None for joint motion)
- 8) If you need to modify the speed of all commands of the same motion type, please make a selection.
- 9) If you need to modify commands of the same type and the same speed, please make a selection.
- 10) If you need to modify only the current command speed, please make a selection.

4.4.2 Modify Logic and Control Commands

- 1) By modifying the logic control commands, the parameters of the commands can be modified. Place the cursor on the command to be modified, then click [Modify], and a command modification interface will pop up.

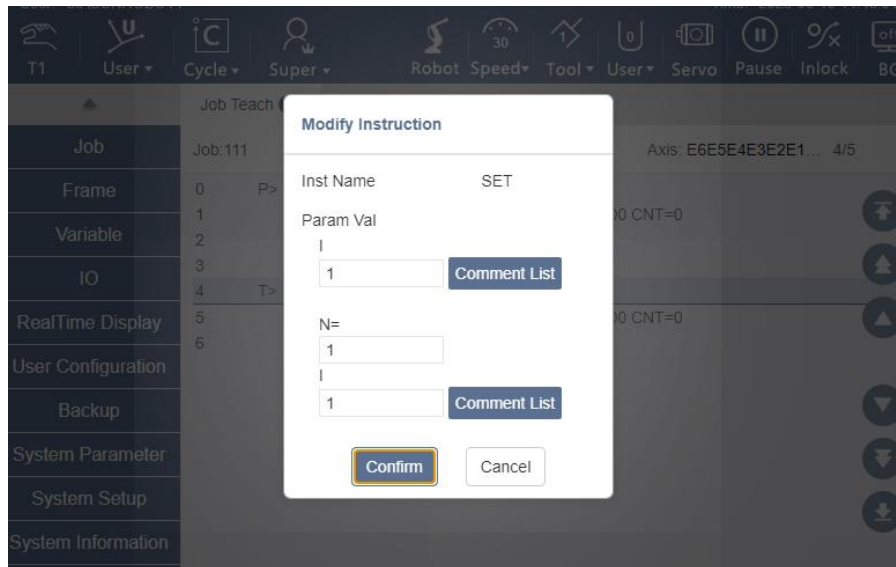


Figure 4-30 Modify Commands

After the menu pops up, the parameters of the logical commands can be modified.

- 2) The 2DELAY DELAY command supports batch modification of delay time. When

modifying the DELAY command, the following dialog box will pop up.

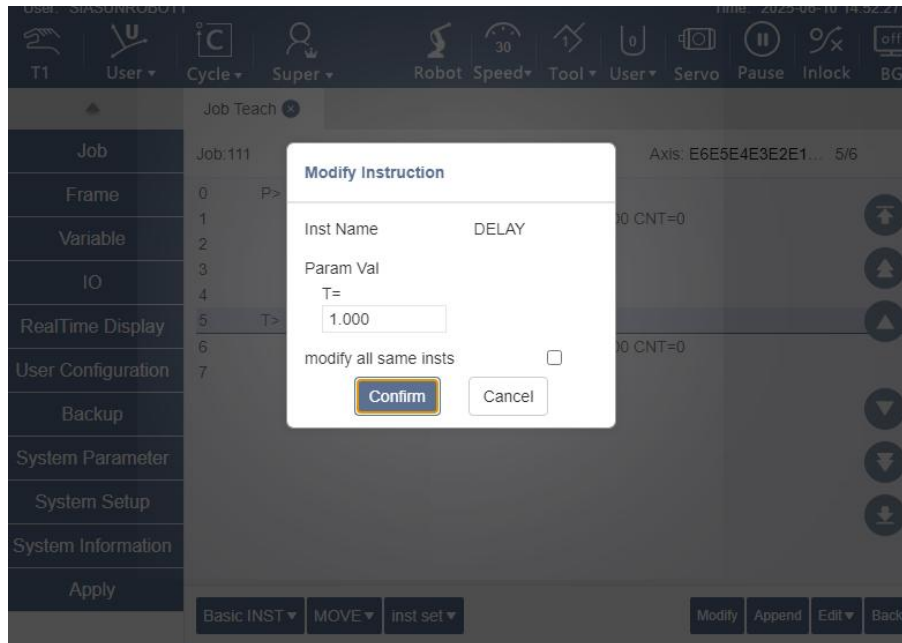


Figure 4- 31 Modify DELAY Command

If "Whether to modify all logic parameters of similar commands" is not checked, only the delay time of the current delay command will be modified, and the modification result will be the entered delay time. If "Whether to modify all logic parameters of similar commands" is checked, the delay time of all delay commands below the current line will be modified. After the modification is completed, the cursor position will stop at the position of the last delay statement, and the modification result will be the entered delay time.

4.5 Job Management

Job management is the function of setting the job information. On the job list interface, open the job management menu.

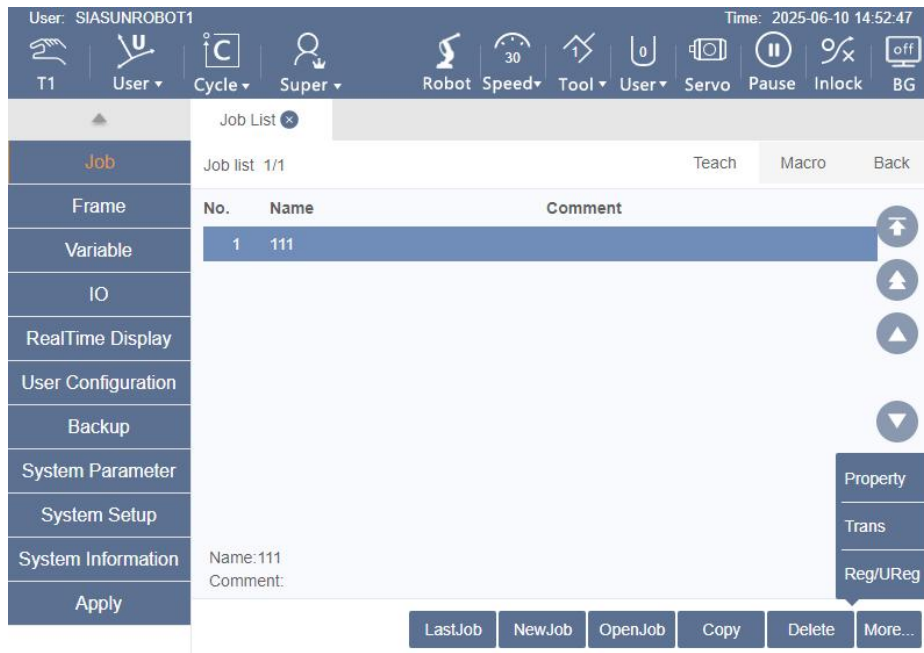


Figure 4-32 Job Management Interface

1) LastJob:

Open the job opened last time.

If the robot is turned on for the first time or no job is opened after a restart, the user clicks [LastJob], a pop-up box "No open program record" will appear on the teaching pendant.

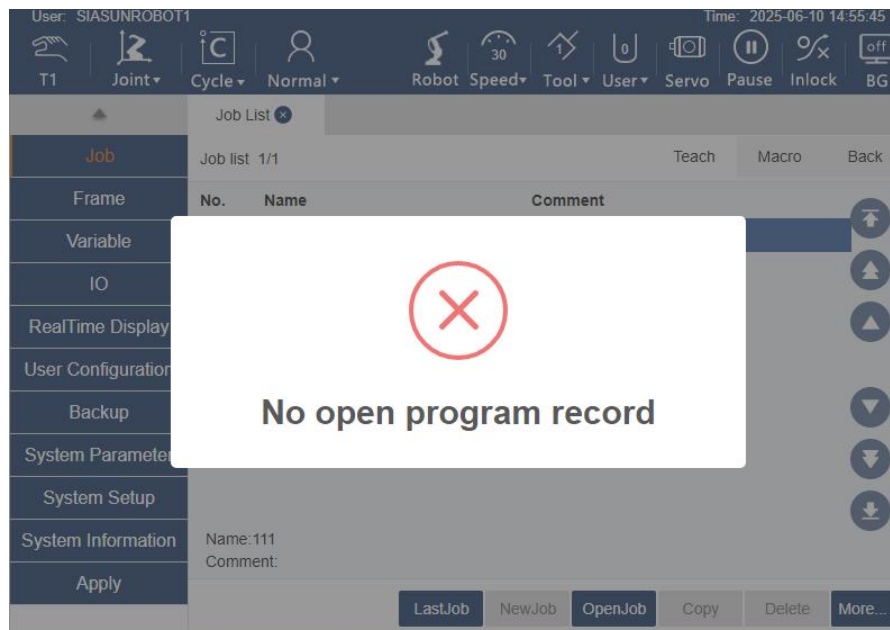


Figure 4-33 Prompt of No Open Program Record

2) OpenJob:

Open the currently selected job.

3) Copy:

Copy the selected job.

4) Delete:

Delete the selected job.

5) Job Property:

User: SIASUNROBOT1 Time: 2025-06-10 14:56:22

T1 Joint Cycle Super Robot Speed 30 Tool User Servo Pause Inlock BG

Job Property

JobProperty

Job	JobProperty
Frame	
Variable	CreateDate 2025-6-10 12:7
IO	ModifyDate 2025-6-10 14:52
RealTime Display	CopySource
User Configuration	JobName 111
Backup	Comment
System Parameter	AxisInformation 1,0,0,0,0,0,0
System Setup	WriteProtect OFF
System Information	
Apply	

Save Back

Figure 4- 34 Job Property Interface

6) CreateDate:

You can check when the program was created.

7) ModifyDate:

You can check the time of the last modification of this program.

8) CopySource:

You can check whether the program was copied from other programs. If yes, it will display which program was used to copy this program. If not, it will not display.

9) JobName:

Display the program name.

10) Note:

You can view and edit the comment.

11) AxisInformation:

You can check which axis groups are included in the program setup.

12) Writeprotect:

General Operating Manual for SIASUN Robot

This status includes ON/OFF.

ON: Indicates protected job. The protected job cannot be modified, which can ensure the safety of the job. A lock identifier will be displayed before the job name.

OFF: Indicates unprotected job.

13) Transform:

Perform an overall transformation of the positions within the job (for details, see 5.6 Job Transformation Instructions).

14) Register/Cancel:

Register: Register a job as the main job. After a job is registered as the main job, its name changes to orange, and the registered main job is moved to the first position in the job list.

Cancel: Cancel the main job.

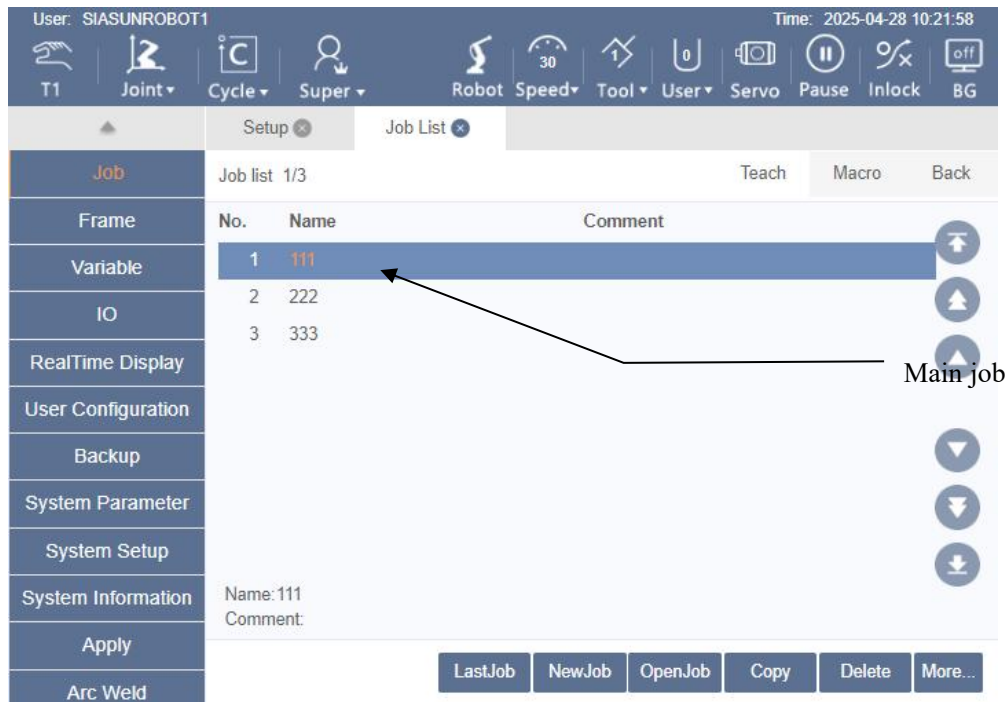


Figure 4- 35 Representation of Main Job in the Interface

Notes:

- 1) Protected jobs cannot be renamed, copied or deleted.
- 2) Jobs that have been registered as the main jobs cannot be renamed or deleted.
- 3) The running background job cannot be renamed, copied or deleted.

5 Program Run

This chapter explains the manual testing and automatic running of the program.

5.1 Program Stop and Recovery

Program stop means that the program in execution is stopped. The robot can be manually stopped by pause and emergency stop. There is also stop caused by alarms.

There are two deceleration methods when a robot stops during its operation:

- 1) Instantaneous stop: The robot stops moving rapidly.
- 2) Decelerated stop: After the robot decelerates, it stops moving.

The methods for artificially stopping a program are as follows:

- 1) Press the stop button on the teaching pendant.
- 2) Press the emergency stop button on the teaching pendant and the controller, and the controller will cut off the power supply.
- 3) Stop by external control, external pause input in remote IO mapping, external stop, and external power-off.
- 4) Press the external emergency stop button and open the safety door.

5.1.1 Stop and resume by emergency stop

The robot can be stopped by pressing the controller, the teaching pendant or the external emergency stop button.

Emergency stop method:

The robot will stop immediately by pressing the emergency stop button.

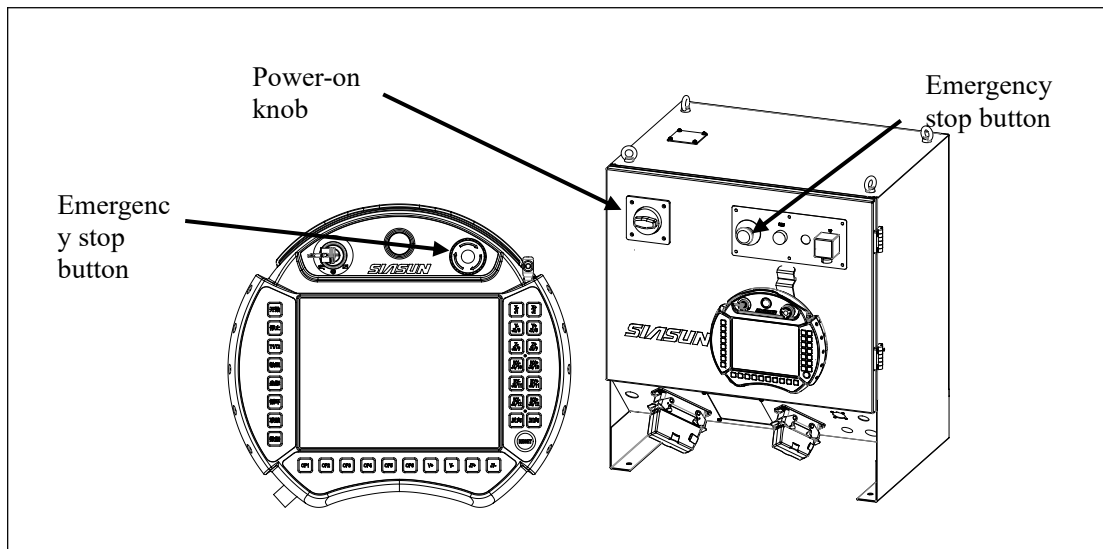


Figure 5-1 Robot Emergency Stop Button

The EXTI ports 5, 6, 7 and 8 on the rear panel of the SRC C5-A controller are external emergency stop dual-loop interfaces.

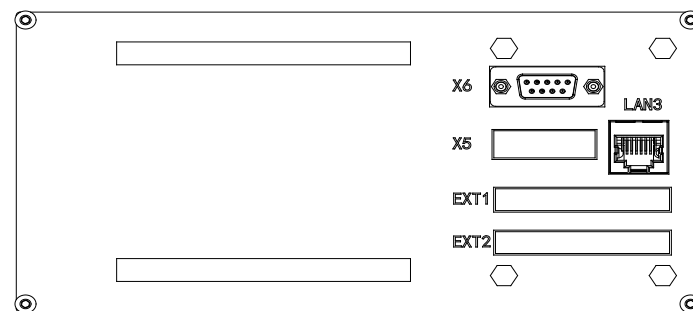


Figure 5-2 External Emergency Stop Dual-loop Interface

The ports 1, 2, 3 and 4 of the P12 terminal block on the inner panel of the M5-B/G5-A controller cabinet door are external emergency stop dual-loop interfaces.

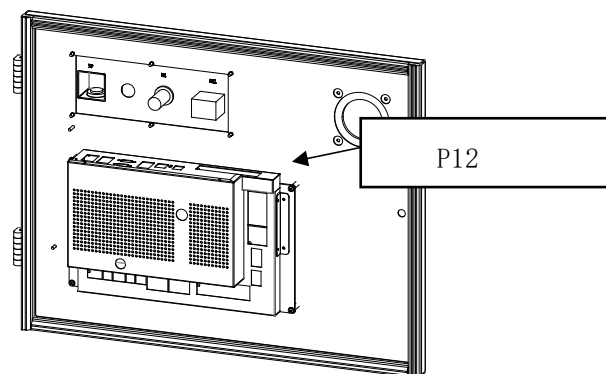


Figure 5-3 External Emergency Stop Dual-loop Interface

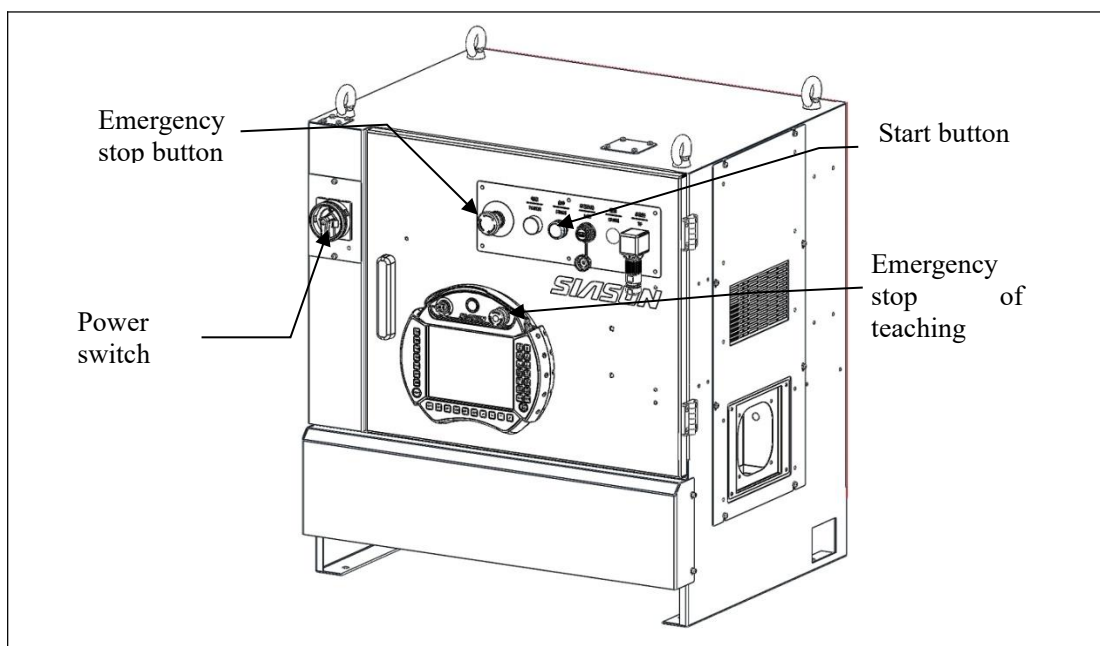


Figure 5-4 SRC M6 Robot Controller

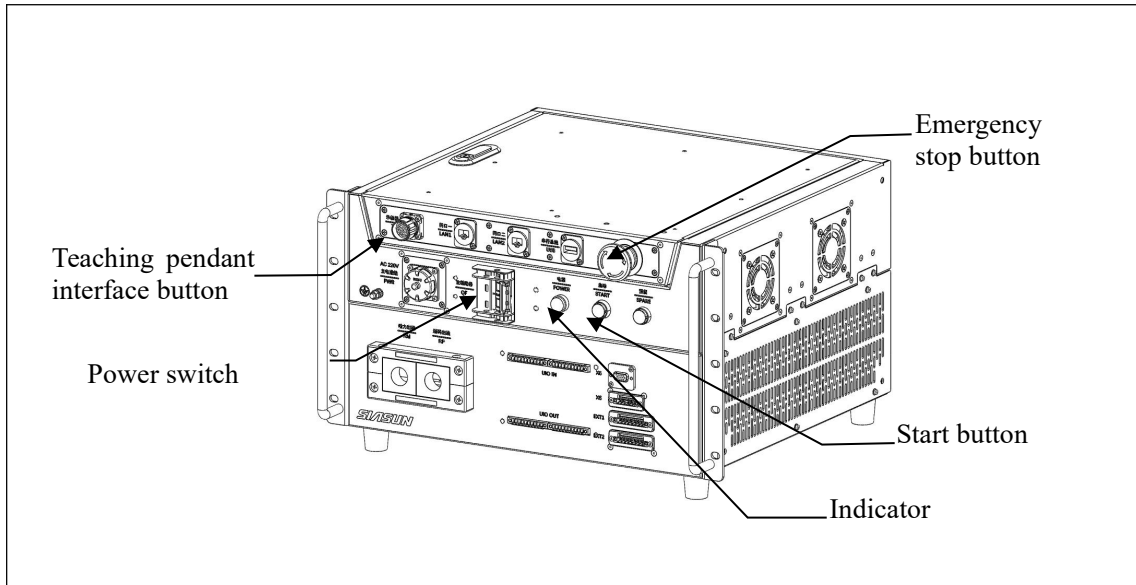


Figure 5-5 SRC C6 Robot Controller

Recovery methods:

Eliminate the causes that lead to the pressing of the emergency stop button:

- 1) Turn the emergency stop button to the right.

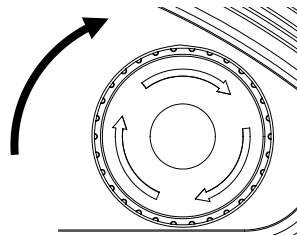


Figure 5-6 Emergency Stop Button

- 2) Press RESET (alarm release) of the teaching pendant and the alarm reset signal of the remote control IO. The alarm icon on the teaching pendant screen disappears. The alarm indicator on the teaching pendant disappears.

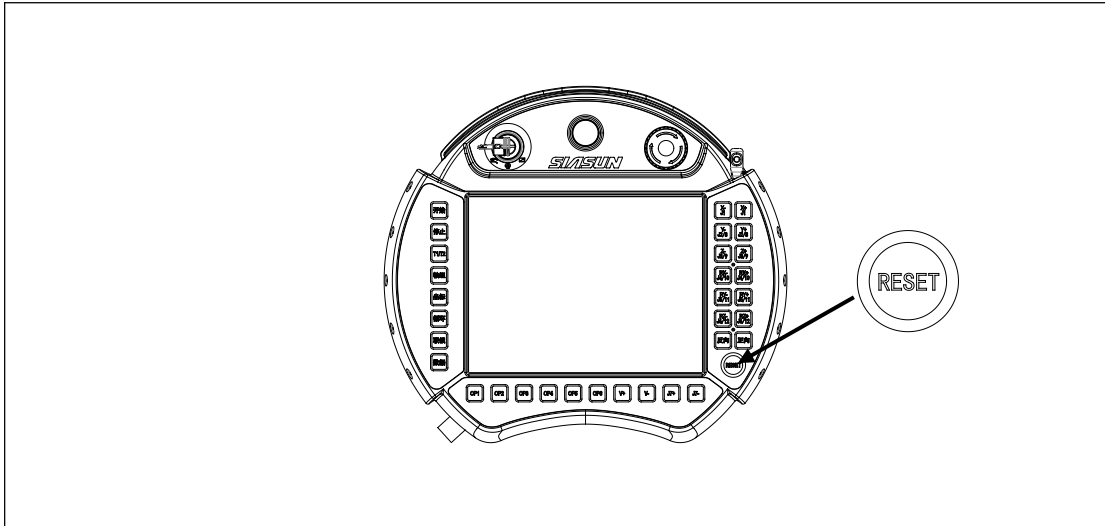


Figure 5-7 RESET Button

5.1.2 Stop and Recovery Method By Pressing Pause Button

Press the pause button, and the robot slows down and stops.

Pause method:

- 1) Press the stop button on the teaching pendant.

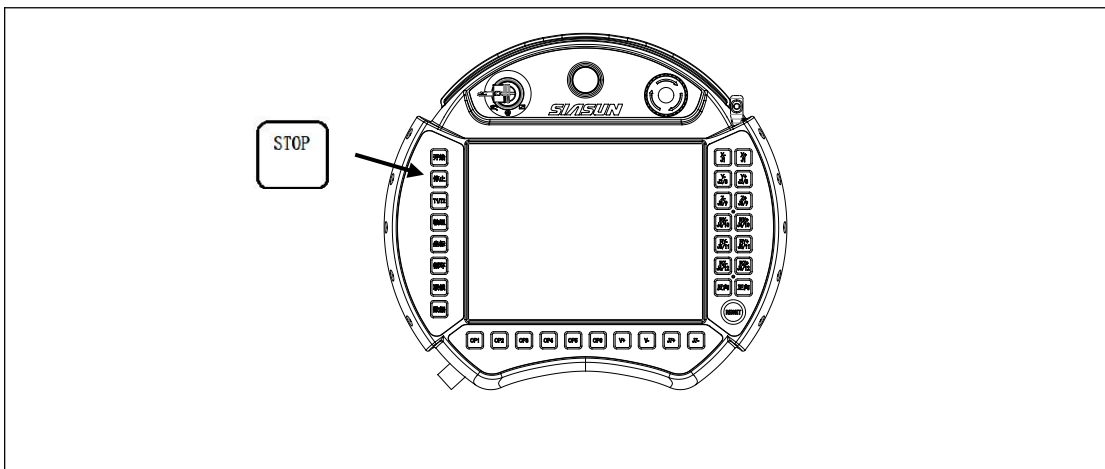


Figure 5-8 Stop Button

- 2) External control stop, external pause input in remote IO mapping.

Recovery method:

Start the program again, and the pause will be released.

5.1.3 Stop Program By Alarm

If the robot detects an exception or receives an emergency stop signal from the outside during program execution or teaching, an alarm will be triggered, and the robot will stop, waiting for the

alarm to be released.

Display alarm:

The alarm information will be displayed in the alarm information column below the robot. Click on the alarm information to open the robot alarm interface. Meanwhile, the LED light of the teaching pendant turns red.

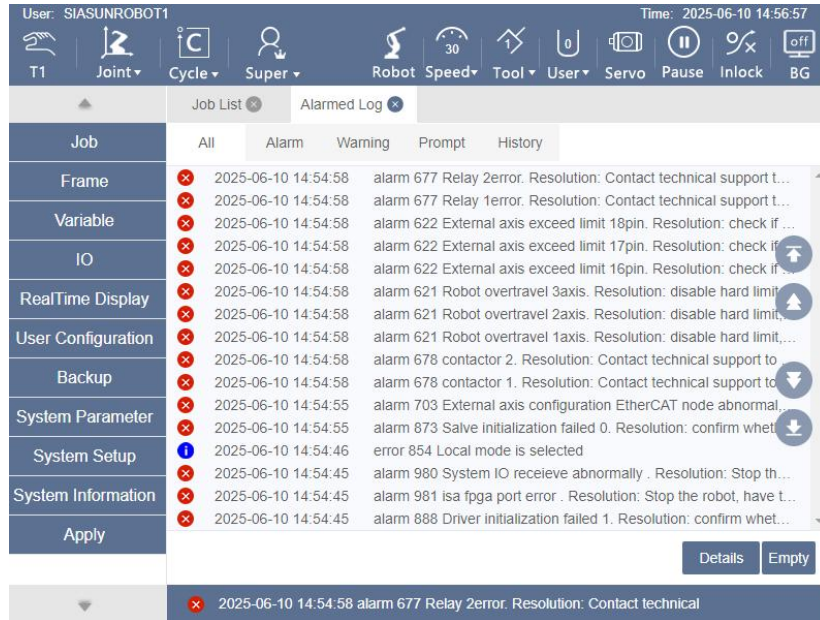


Figure 5-9 Alarm Interface

The alarm log can display all information of alarms and distinguish their importance in specified colors.

Table 5-1 Table of LED Light Colors Corresponding to Importance of Alarms

Color	Importance
Blue	Prompt message
Yellow	Warning
Red	Alarm

Release alarm:

Press RESET (alarm release) of the teaching pendant and the alarm reset signal of the remote control IO. The alarm icon on the teaching pendant screen disappears. The alarm indicator on the teaching pendant screen disappeared.

5.2 Program Execution

Execute Program means that the robot repeatedly runs the edited program and will execute the commands specified by the program. The program execution is divided into three cycle modes: Step, Cycle and Auto (infinite loop).

5.2.1 How to start the program

There are three ways to start the program:

- 1) Manual mode: The program can be manually executed through forward or reverse motion on the teaching pendant.
- 2) Execution mode: The program can be automatically started by pressing the start button on the teaching pendant. **(For M6/C6 control cabinet /ARM board, first press the start button on the teaching pendant, then press the start button on the control cabinet, and the program can be automatically started.)**
- 3) Remote mode: The robot is remotely controlled to start the program through remote I/O using peripheral devices.

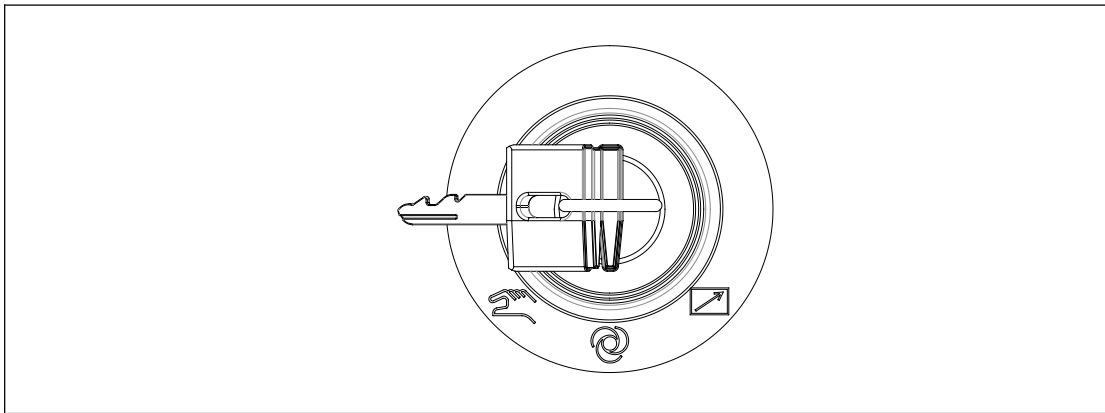


Figure 5-10 Mode Selection Switch

There is a mode selection key on the robot teaching pendant. In different modes, use the corresponding start methods.

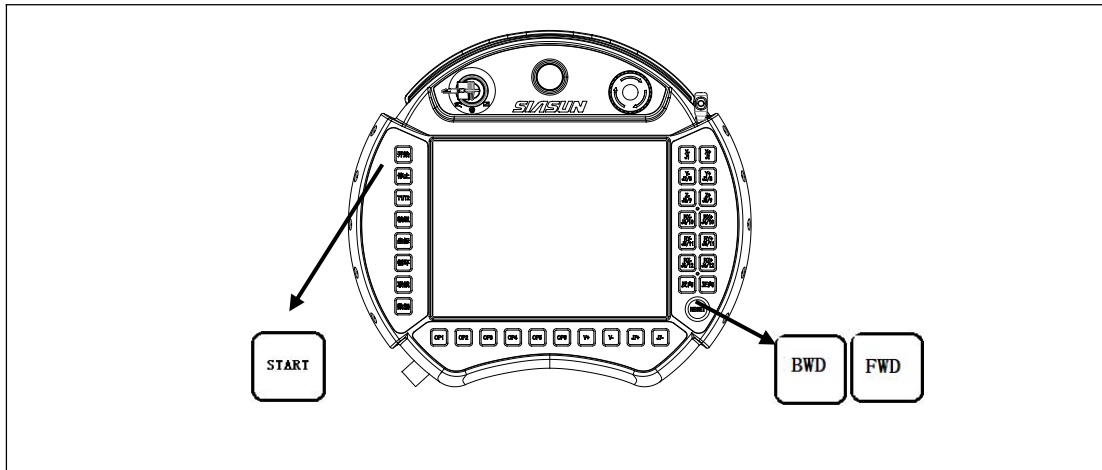


Figure 5-11 Start Button, Forward and Reverse Motion Buttons

5.2.2 The robot operates by the program

When the robot performs jobs in forward motion, the program will execute commands in the order from top to bottom. The motion commands will indicate in what way the robot moves to reach the expected position.

Before starting the robot to perform actions, be sure to confirm some elements of the robot to ensure that it moves along the expected path and at the expected speed, and confirm whether the robot operates in the desired logic and cycle mode.

Robot speed: Speed of robot motion = command speed $\times$ speed ratio/100. The maximum speed in manual TI mode is 250mm/s.

Robot frame: Confirm that the currently selected frame, and check that the commands for switching frames in the program are correct. The motion of the robot will change when the program is executed in different frames.

Whether the logical command is executed: When manually moving the robot forward, check whether the interlock function is enabled through Interlock icon. If Interlock is enabled, the logical commands and motion commands will be executed. If Interlock is not enabled, no logical commands will be executed, and only motion commands will be executed.

Determine the cycle state of the robot: In Step mode, the robot only executes one command after starting. In Cycle mode, the robot continuously executes the job after starting until the job is completed. In Auto mode, the robot continuously executes the job in a loop after starting.

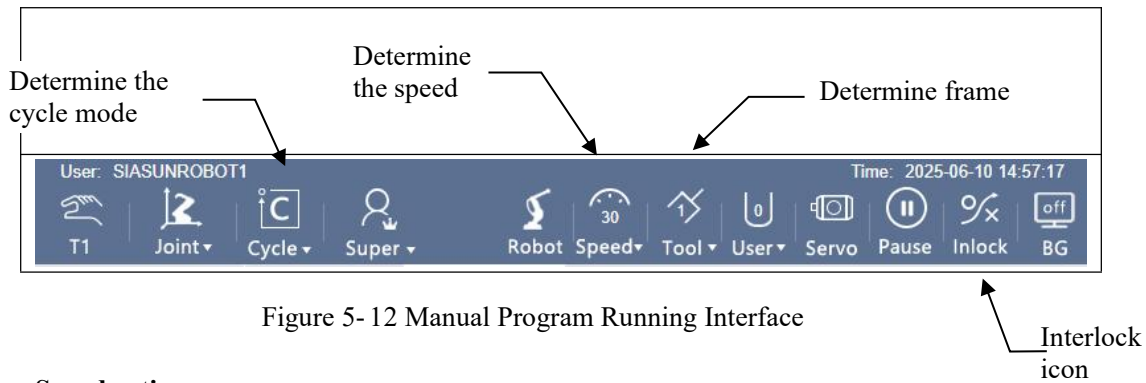


Figure 5-12 Manual Program Running Interface

Speed ratio:

The speed of the robot depends on the speed ratio specified by the teaching pendant.

When the ratio is 100%, the robot will run at the speed instructed by the program.

The running speed can be changed through the acceleration and deceleration buttons on the teaching pendant. The speed can also be changed by clicking the icons on the touchscreen. The speed can be changed by the SET VEL command.

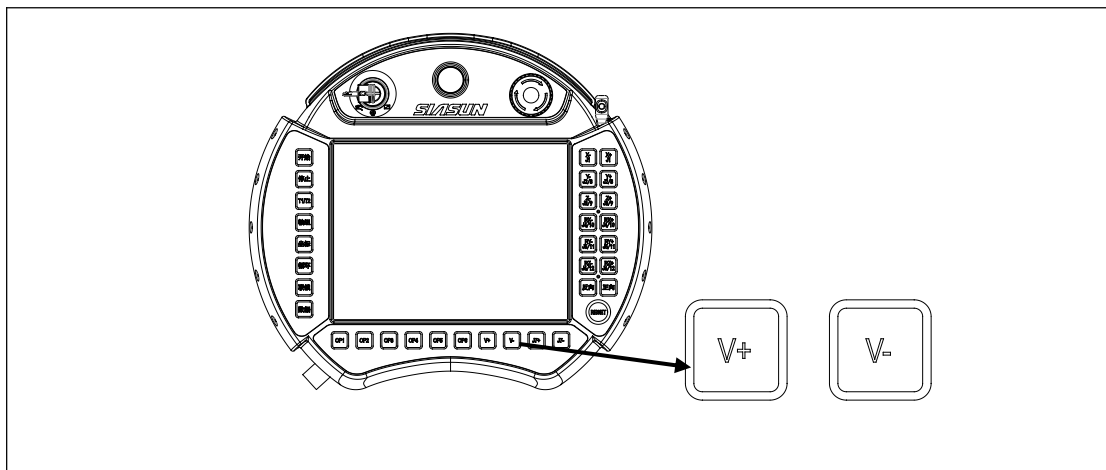


Figure 5-13 Robot Acceleration and Deceleration

Warning:

In T1 and T2 modes, the actual operating speed is different when the robot operates at the same speed ratio. In T1 mode, the maximum speed TCP does not exceed the safe speed. In T2 mode, the robot operates at a percentage of the actual maximum operating speed. Please make sure that there is no one and no irrelevant items around the robot before moving the robot to avoid serious personal injury or equipment damage.

Frame determination:

Before executing the robot program, please confirm the frame of the robot. Make sure the frame used by the robot during execution is consistent with the one used for teaching. Otherwise,

the position of the robot will change due to the difference in the frame.

1) Tool frame (TF) number:

Eight tool frames can be defined, which are distinguished by numbers 1 to 8. They can be switched according to the situation by directly selecting on the teaching pendant and adding SET TF 1-8 in the program.

2) User frame (UF) number:

Eight user frames can be defined, which are distinguished by numbers 1 to 8. They can be switched according to the situation by directly selecting on the teaching pendant and adding SET UF 1-8 in the program.

Warning:

- 1) When the robot program is started, the robot will move according to the frame selected from the frame icon above the teaching pendant. If the selected frame is not the one corresponding to the teaching, the position and pose of the robot during the program operation will be different from those during the teaching. Please make sure to determine the frame number before moving.**
- 2) If different frames are used for program teaching in a program, SET TF or SET UF commands can be added before the program teaching position. When manually running the program, Interlock needs to be enabled, and the logical command execution mode should be enabled to switch to the required frame file number for operation when the program is running.**

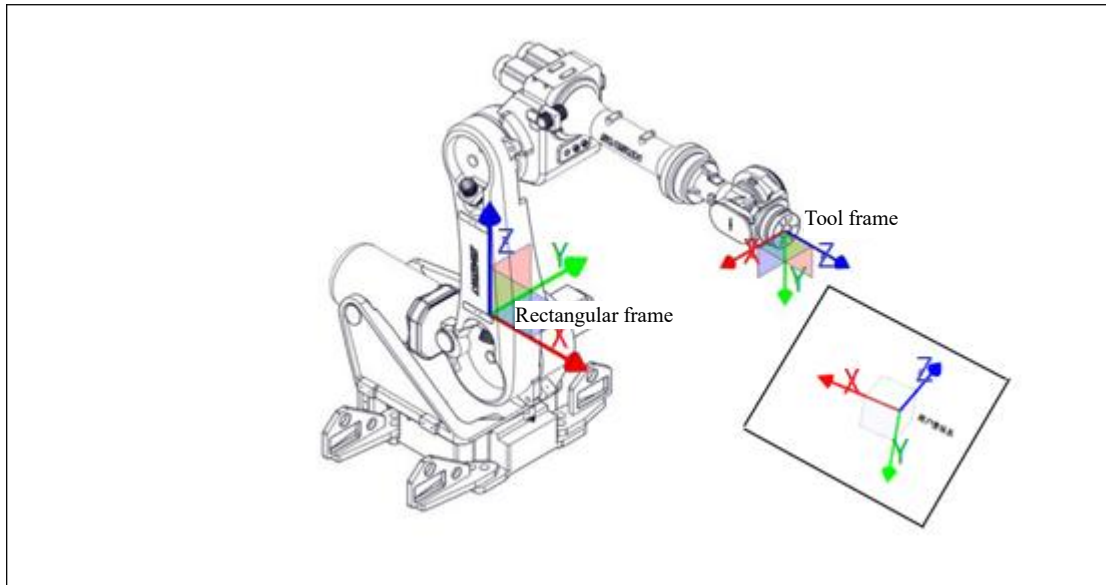


Figure 5- 14 Selection of Tool and User Frames

5.2.3 Recover from Pause

When the forward motion of the robot is interrupted, press Forward Motion again, and the robot can continue to move according to the program.

When the robot automatically stops running, press the start button again or start externally, and the robot can continue to move according to the program.

5.3 Test Program

Test Program refers to determining the robot motion and logic before the robot is automatically executed on site. Program testing is of great significance for the safe operation of the robot.

Program testing is divided into two steps:

- 1) Use the teaching pendant, select Step as the cycle mode, and conduct the test manually (moving forward). It is to verify the position points.
- 2) Use the teaching pendant, select Cycle as the cycle mode, and run manually (select manual mode and turn on the interlock switch) or automatically (select execution mode and select low speed) to conduct the test. It is to verify the position continuity and overall logic.

The cycle mode can be selected through the drop-down menu of the loop status icon on the teaching pendant, or the cycle mode can be switched through the loop button on the teaching

pendant. The cycle mode of the robot is divided into three types:

- 1) Step: The robot executes each command by step.
- 2) Cycle: The robot moves continuously for one cycle.
- 3) Auto: The robot operates continuously in a loop.

Whether the robot executes logical commands can be changed through Interlock icon.

- 1) o/o is the on state of the interlock switch, in which both logical commands and motion commands are executed.
- 2) o/x is the on state of the interlock switch, in which logical commands are not executed while motion commands are executed.

To test the robot through external startup, the remote mode needs to be selected.

Furthermore, if the robot needs to communicate with external devices through I/O and control peripheral devices, it is necessary to ensure that the peripheral devices are in good working condition, that I/O is in an effective state, and that there is no alarm display.

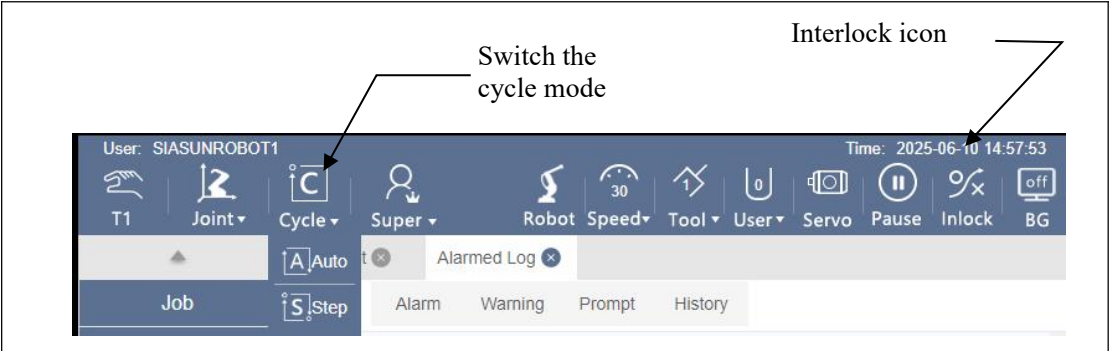


Figure 5- 17 Cycle Mode and Interlock Icon

Warning:
Before testing the robot, please ensure that the selected frame number is consistent with the one used for teaching.

5.3.1 Settings To Be Determined Before Test

Table 5-2 Test the items to be confirmed

Setup Item	Description
Cycle status 	Please confirm the cycle status executed by the robot before running: Step: The robot executes each command by

	step. In Cycle mode, hold down [Forward Motion], and the robot will move continuously for one cycle. In Auto mode, hold down [Forward Movement], and the robot will keep moving in a continuous loop. Warning: When the interlock is enabled and executed continuously, the logical commands are also executed continuously. Please pay attention to confirmation during manual operation to avoid the surrounding equipment being triggered to move due to signals, which may cause equipment damage or personal injury.
Interlock 	Interlock status indicates whether the robot skips logical commands when operating manually: o/o is the on state of the interlock command, in which logical commands are executed. "o/x" represents that the logical commands are not executed.

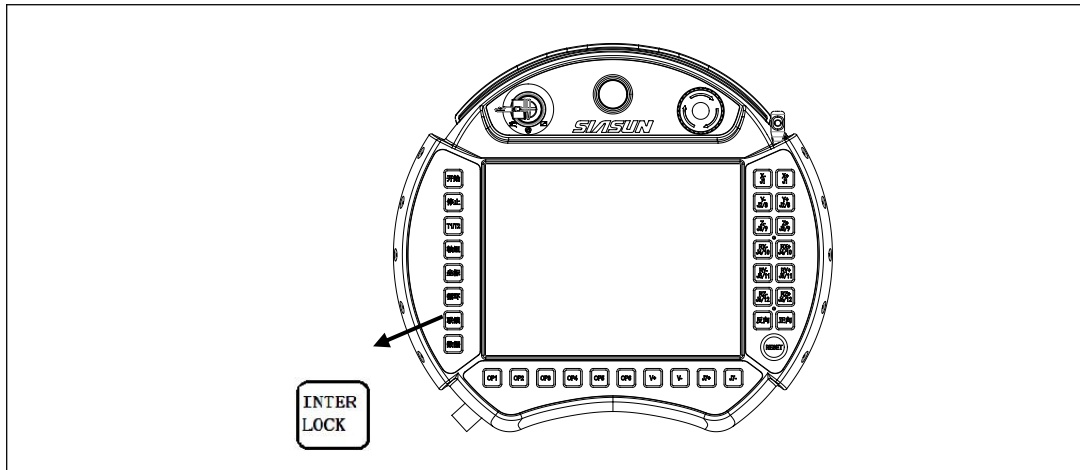


Figure 5- 18 Continuous Buttons of Robot

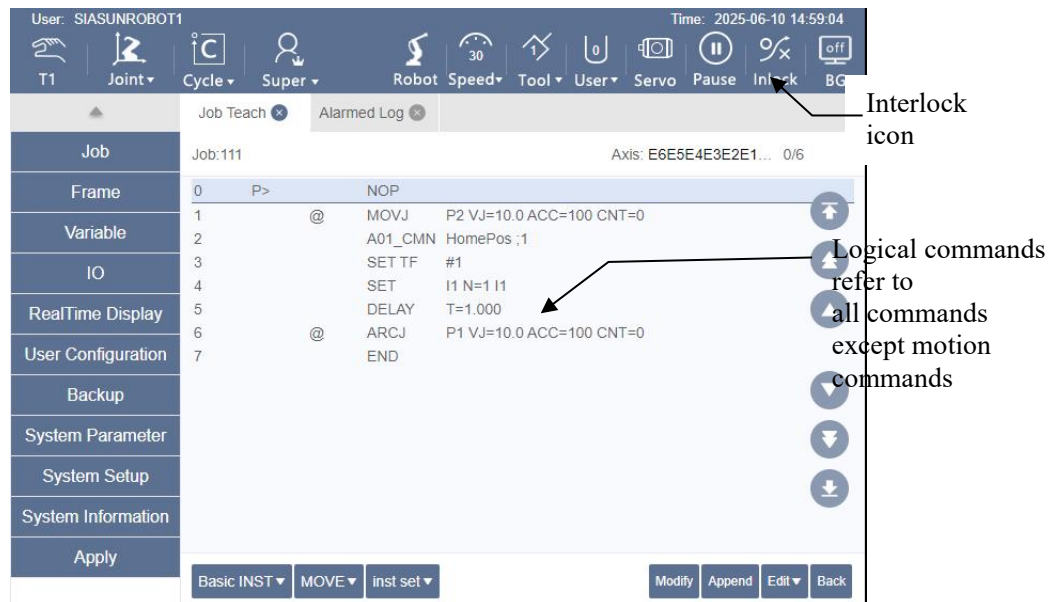


Figure 5- 19 Logical Commands

After confirming the project before the test, the robot can:

- 1) Execute each command in the order of the program from top to bottom according to the logic of the program, and change the state of the robot and the execution sequence of the program according to the logical commands.
- 2) Only execute motion commands, automatically skip logical control commands, execute the robot actions one by one, and also execute unconditioned CALL subprograms if any.

Whether the robot's logical commands are executed needs to be confirmed by whether the interlock switch is turned on.

Notes:

Calling the unconditioned CALL command of a sub-job is valid when executing motion commands. When this command is executed, the sub-job can be called for execution.

5.3.2 Single-step Test Program

In the single-step test program, the robot executes the programs item by item. After completing one program, the robot pauses, and the cursor will rest on this command. When the robot executes again, the cursor will automatically jump to the next statement for execution.

Set the operation mode of the robot:

The operation mode of the robot can be changed through the buttons on the interface of the robot teaching pendant.

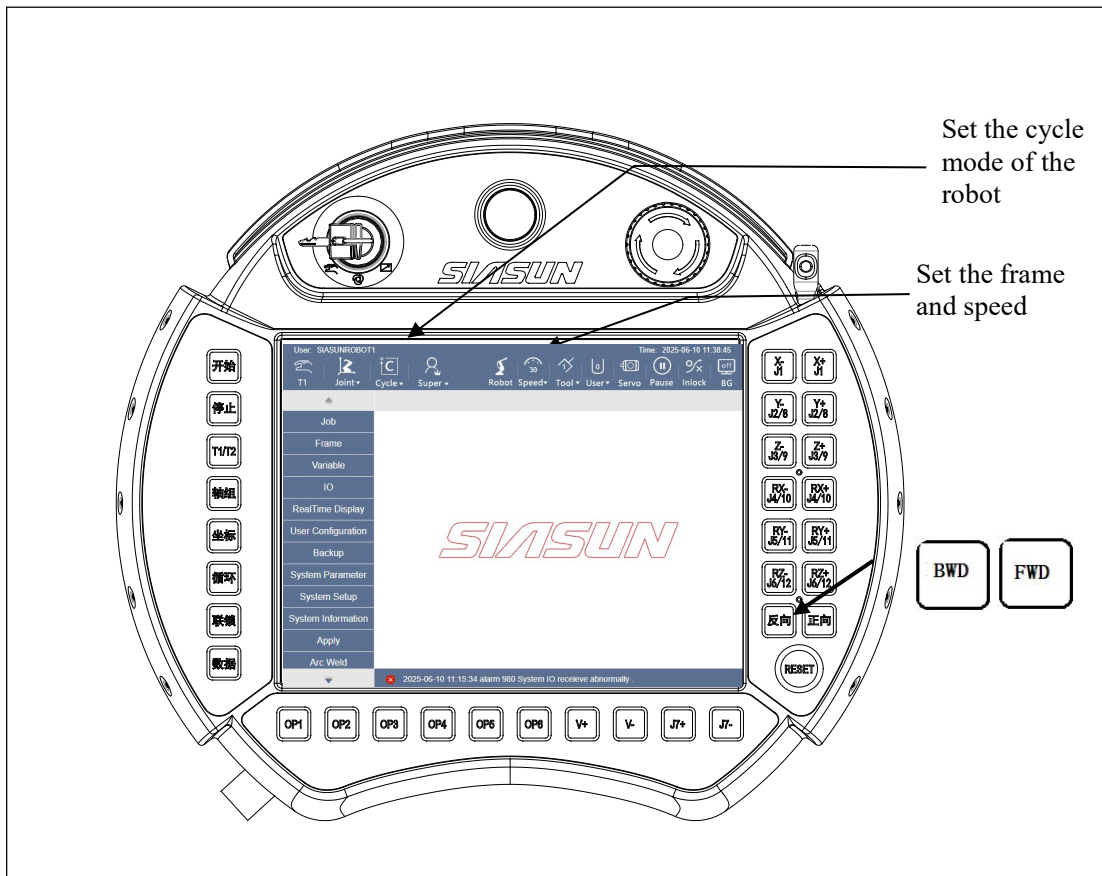


Figure 5-20 Single-step Operation Settings and Execution

The program can be manually run through forward or reverse motion.

Forward motion:

Forward motion refers to the forward execution of robot program commands in the order

from top to bottom. You need to hold down the safety switch [DEADMAN] and [Forward] of the robot simultaneously. Release either button, and the robot will stop.

The robot executes from the line where the program cursor is located and pauses after completion. However, when the [Forward] button is pressed at the end of the subprogram, the robot will return to the previous layer of main program and move to the CALL command, moving forward and downward again.

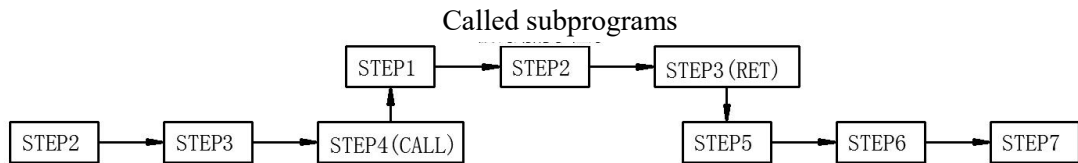


Figure 5-21 Single-step Operation Settings and Execution

Warning:

Before operating the robot, it is necessary to confirm that there are no personnel or unnecessary equipment within the robot's movement range to avoid unexpected problems that may cause casualties or equipment damage.

Reverse motion:

Reverse motion refers to reverse execution of a program by the robot. You need to hold down the safety switch [DEADMAN] and [Reverse] of the robot simultaneously. Release either button, and the robot will stop.

In reverse motion, the robot only executes motion commands and does not execute logical commands (regardless of the state of the interlock switch). A subprogram is reopened. Reverse motion cannot return to the previous level of the program.

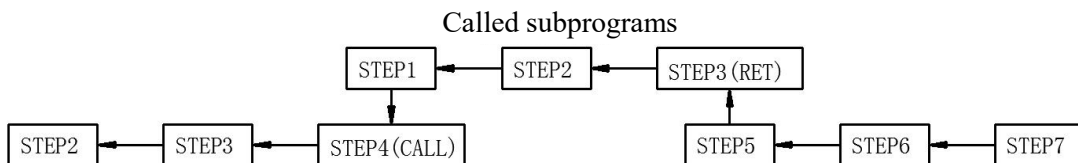


Figure 5-22 Reverse Motion Diagram of Program

Notes:

When moving backward, the robot can execute the program continuously or by step as set, but it cannot execute logical commands.

Warning:

Before operating the robot, it is necessary to confirm that there are no personnel or unnecessary equipment within the robot's movement range to avoid unexpected problems that may cause casualties or equipment damage.

When the robot performs circular motion and stops in the middle, continue to press [Forward] and [Reverse] if the position point of the robot remains unchanged, and the robot will still perform the original circular motion. However, when the robot's position point moves, continue to press [Forward], and the robot's motion path will be re-planned.

Steps of single-step execution:

- 1) Make sure there are no personnel within the movement range of the robot.
- 2) Select the single-step operation mode of the robot.
- 3) Determine various states of the robot during its execution.
- 4) Move the cursor to the beginning of the program.
- 5) Press the power-on button to power on the robot, and press and hold [DEADMAN].
- 6) Run the program: Press and hold the safety switch [DEADMAN] and [Forward] or [Reverse], and execute the program by step.
- 7) After a line of program is executed, the program pauses, and then press [Forward] or [Reverse] to continue execution.
- 8) After completion, release DEANMAN.

5.3.3 Run the test program continuously

Continuous run testing means that the robot runs the program continuously in the order of the program.

Select Cycle and Auto to conduct continuous testing.

When manually running a program continuously using the teaching pendant, press [DEADMAN] and [Forward], and the program will start executing from the current line.

When the robot automatically runs the continuous test program, press [Start] on the teaching pendant or remotely start to make the robot program executed.

Warning:

Before operating the robot, it is necessary to confirm that there are no personnel or unnecessary equipment within the robot's movement range to avoid unexpected problems that may cause casualties or equipment damage.

Manual continuous test steps using the teaching pendant:

- 1) Check that there are no personnel within the movement range of the robot.
- 2) Select Cycle or Auto mode.
- 3) Determine the speed, frame number, interlock and other states of the robot during execution.
- 4) Press the power-on button to power on the robot servo.
- 5) While pressing [DEADMAN], press [Forward] to continuously execute the program.
Before the program ends, hold down [DEADMAN] and [Forward]. If any button is released, the program will pause halfway.

Use the start button or remotely start to perform continuous test steps:

- 1) Make sure there are no personnel within the movement range of the robot.
- 2) Select Cycle or Auto mode.
- 3) Determine whether all settings (cycle mode, speed, frame number) are correct.
- 4) Change the execution mode to execution or remote.

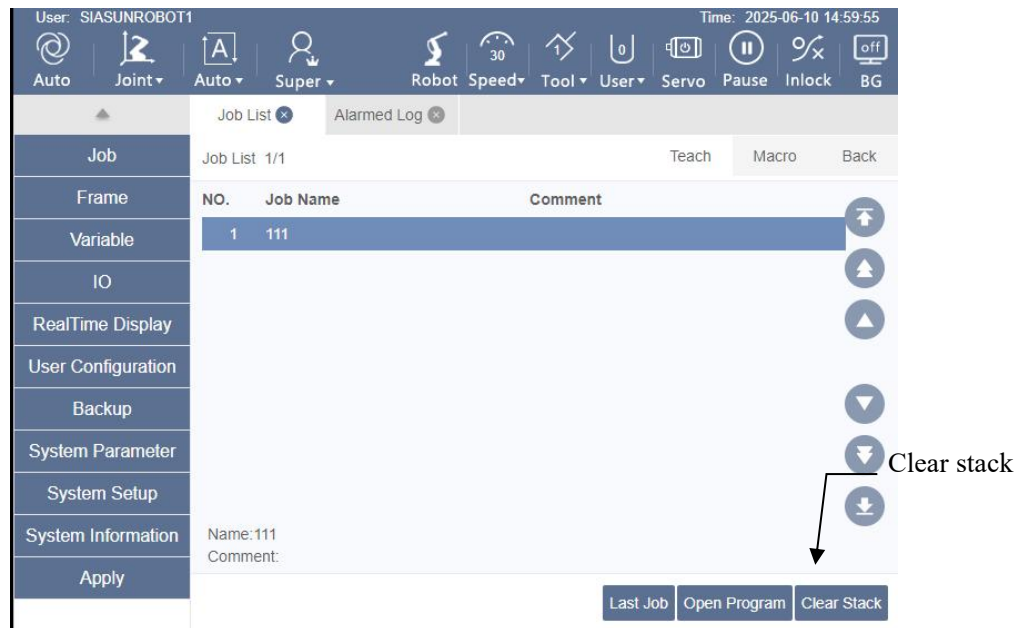


Figure 5-23 Job List in Execution Mode

Notes:

The stack stores information such as job name, line number, cursor, path marker, compensation flag bit, offset, and interrupt. After clearing, the restrictions or records related to this information will be cancelled. Please avoid the danger caused by the robot not moving as expected after these restrictions are cancelled. Avoid personal injury or equipment damage.

- 5) Move the cursor to the first line.
- 6) Press [Start] or start the program remotely.

5.3.4 Program Confirmation and Monitoring

When the program is running, the job interface on the teaching pendant is the main medium for monitoring the program running process. On the job execution screen, the cursor moves to the line being executed along with the job's execution. In the execution mode, the job cannot be edited.

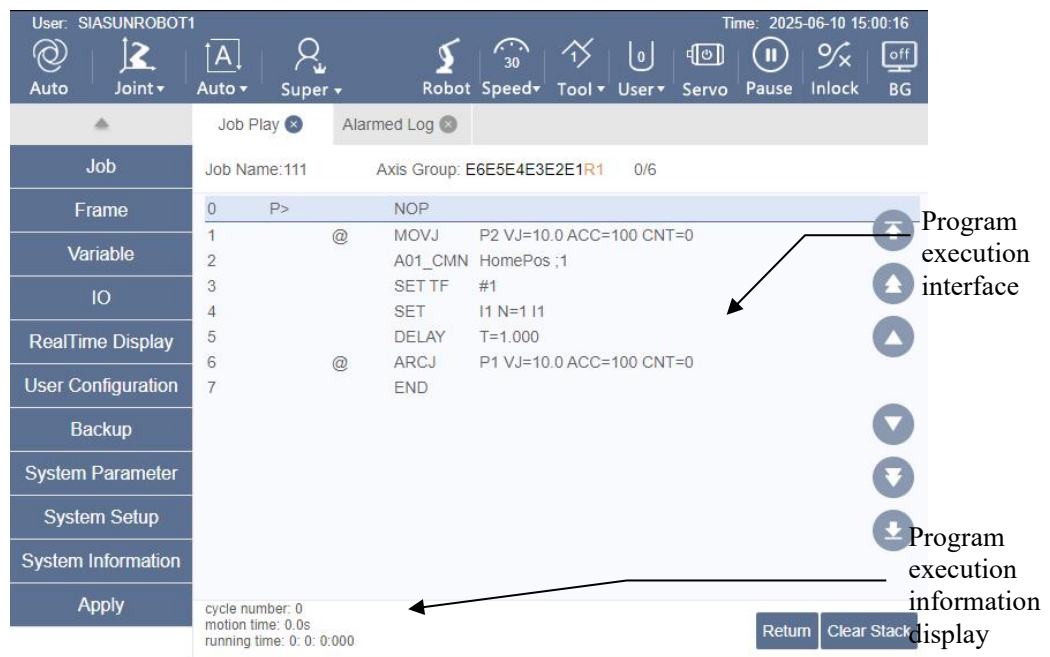


Figure 5- 24 Program Execution

5.4 Automatic Operation

Through the automatic operation of the program, the program is automatically started by the local start button or external remote I/O control, thereby achieving the automation of the production line.

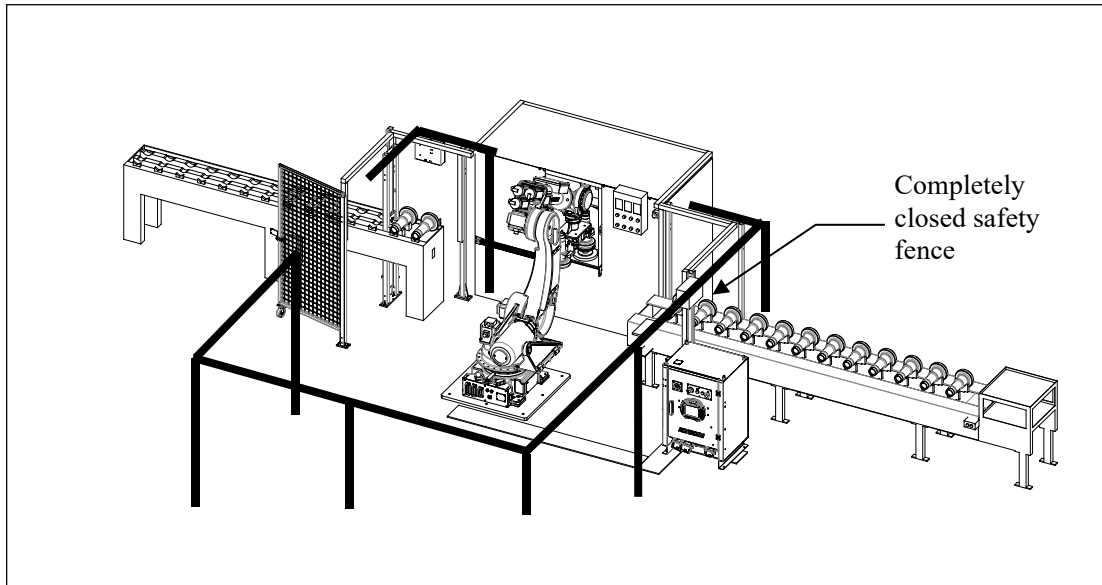


Figure 5-30 Automatic Operation of Robot System

Precautions before automatic execution:

- 1) 1. All the settings of the robot have been completed.
- 2) 2. The robot is in the execution or remote mode.
- 3) 3. There is nobody in the working space. No other objects.

Operation steps for automatic execution:

- 1) Select the execution mode or remote mode through the key.
- 2) Press the Start button in the execution mode.

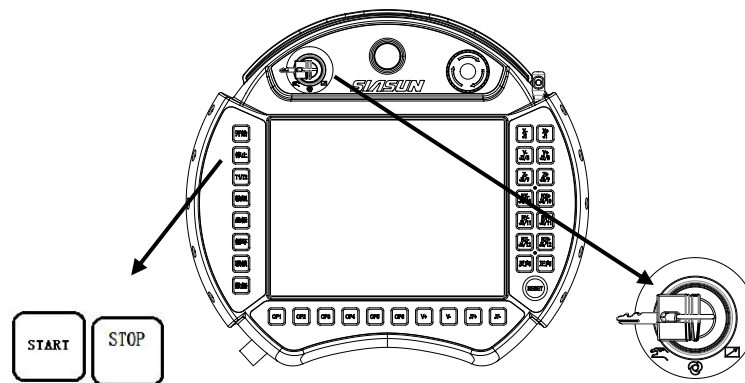


Figure 5-31 Start/Stop Button and Mode Selection Switch

- 3) In remote mode, the robot is started using an external start signal based on the configuration of remote I/O.
- 4) In execution mode, the robot can be stopped by the pause button.
- 5) In remote mode, the robot can be stopped by external pause.

Notes

In remote mode, when starting through external signals, certain preconditions are required. These preconditions need to take into account the timing sequence, aiming to ensure that no accidents occur after the robot is started.

Stack:

The stack stores program information such as opened job name, line number, cursor, path marker, compensation flag bit, offset, and interrupt. Before the robot is started normally, it will determine whether the current position of the robot is at the same position as the line where the cursor is located in the program. If they are not at the same position, it cannot be started, and relevant prompt information will be given. So please use the stack clearing function with caution. [Clear Stack] button is in the lower right corner of the job list in the execution mode, as shown in Figure 5-32.

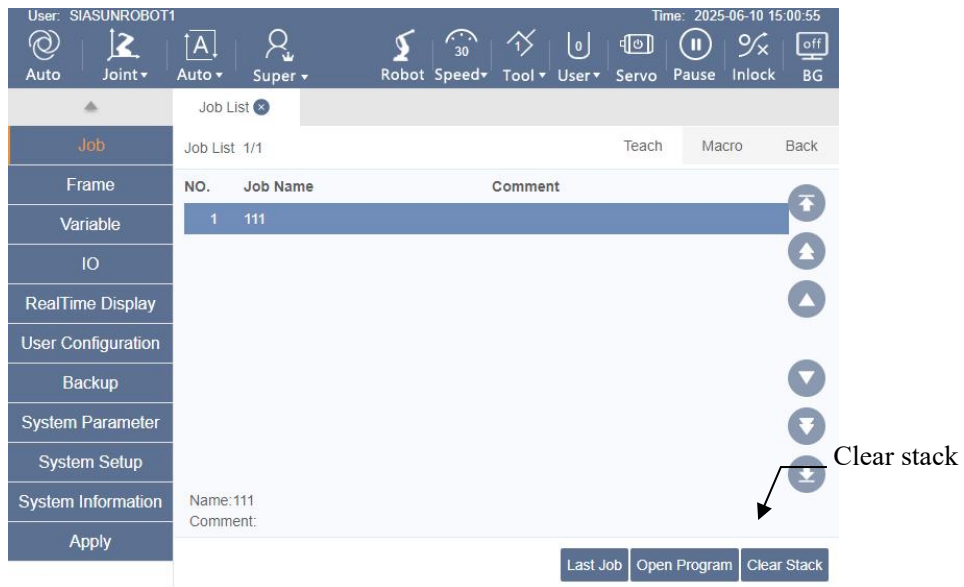


Figure 5- 32 Clear Stack Button

Notes:

The stack stores information such as job name, line number, cursor, path marker, compensation flag bit, offset, and interrupt. After clearing, the restrictions or records related to this information will be cancelled. Please avoid the danger caused by the robot not moving as expected after these restrictions are cancelled. Avoid personal injury or equipment damage.

6 Status Display

The robot teaching pendant can display various states and data of the robot itself.

6.1 Teaching Pendant LED

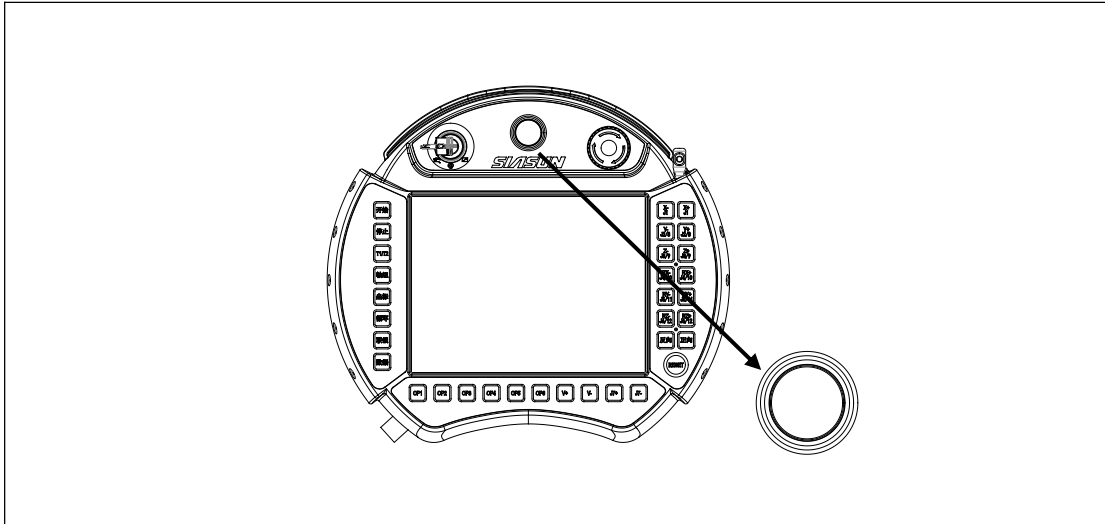


Figure 6-1 Teaching Pendant LED

Table 6-1 LED Status

LED color	Description
Yellow	The robot is in paused state.
Green	The program of the robot is in operation.
Red	The robot is in alarm state.

6.2 Job Interface

The information of the program commands in execution will be displayed on the job interface.

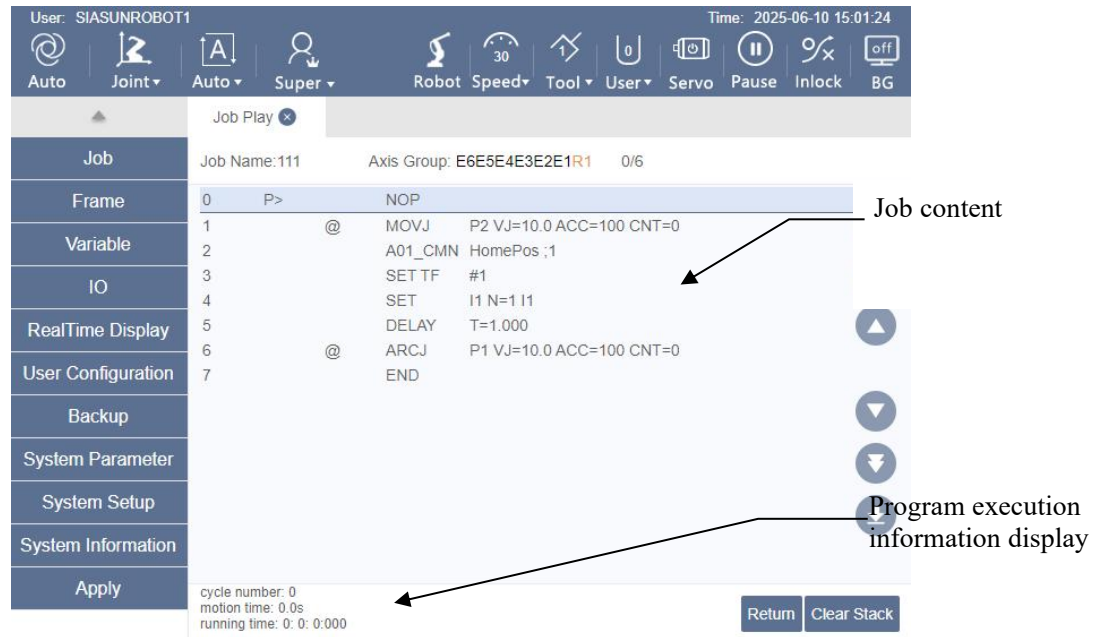


Figure 6-2 Job Display Interface

6.3 Variable

The required variable values can be viewed and written in the variable interface of the teaching pendant. These variables are global variables.

Click [Variable] on the main menu and select the variable value you wish to view.

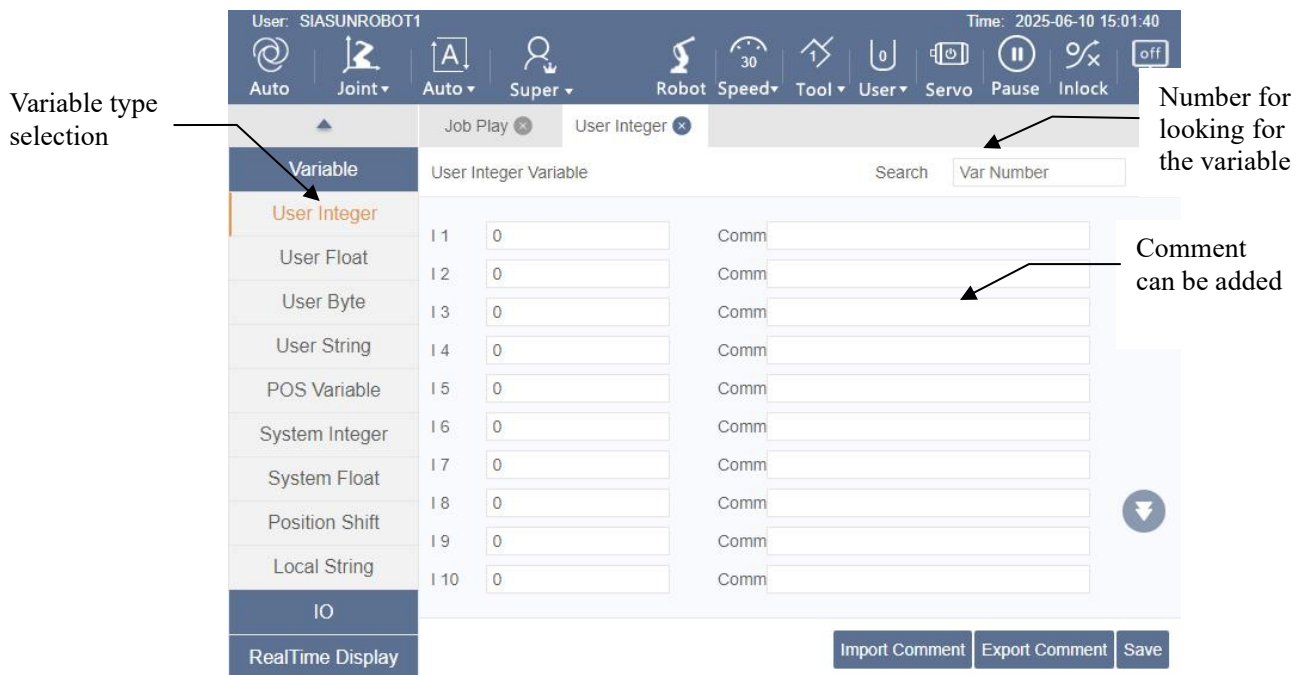


Figure 6-3 User Integer Variable Interface

There are a total of 520 user integer variables, ranging from I1 to I520, and the range of

stored values is -2147483648 to 2147483648.

Comment: Comment can be added.

Search: You can use the search variable number in the upper right corner of the interface.

Import comment: A USB flash drive needs to be inserted into the robot controller, and there is corresponding comment text in the USB flash drive. Click "Import Comment".

Export comment: A USB flash drive needs to be inserted into the robot controller. Click "Export Comment". The exported comments are in text format and can be edited. The name of the exported comment text is intVarComment.txt.

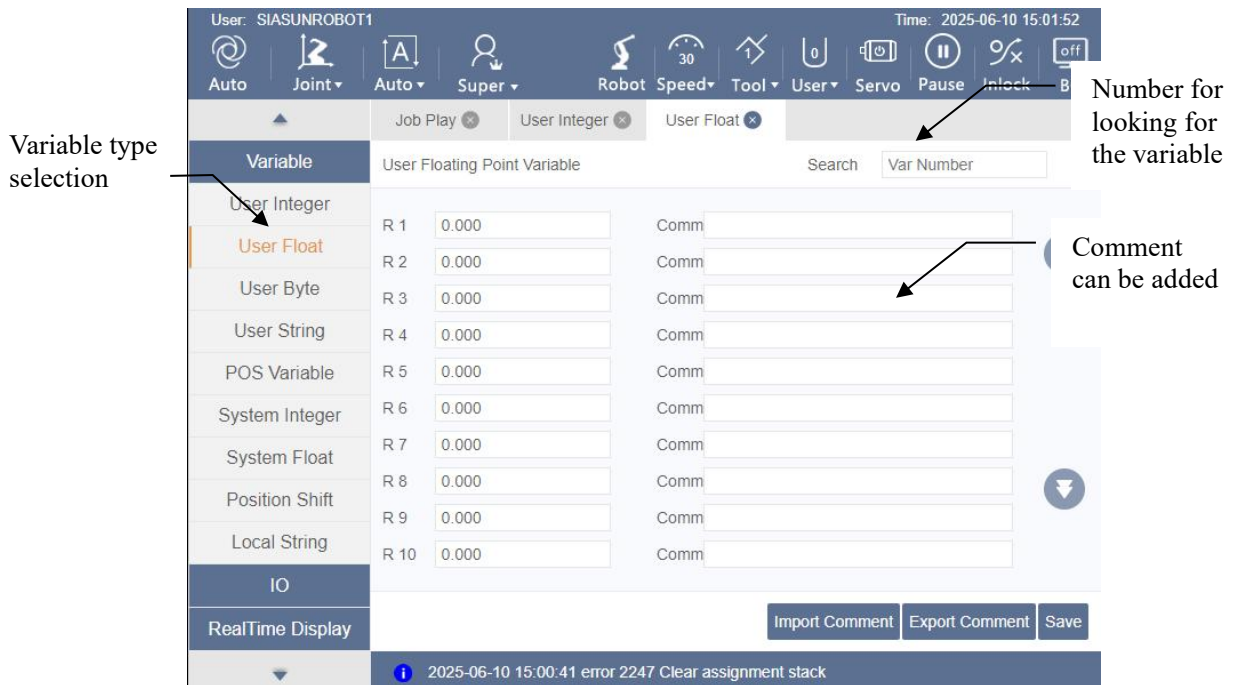


Figure 6-4 User Float Variable Interface

There are a total of 520 user float variables, R1-R520. The range of stored values is -9999.999-9999.999, accurate to three decimal places.

Comment: Comment can be added.

Search: You can use the search variable number in the upper right corner of the interface.

Import comment: A USB flash drive needs to be inserted into the robot controller, and there is corresponding comment text in the USB flash drive. Click "Import Comment".

Export comment: A USB flash drive needs to be inserted into the robot controller. Click

"Export Comment". The exported comments are in text format and can be edited. The name of the exported comment text is floatVarComment.txt.

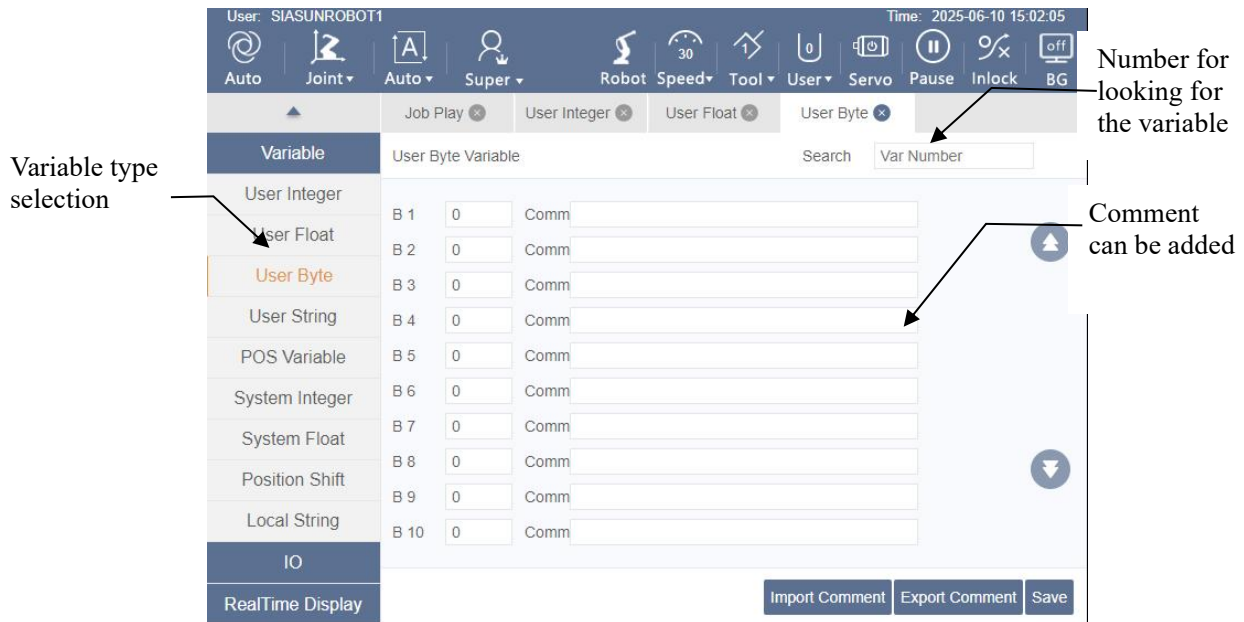


Figure 6- 5 User Byte Type Variable Interface

There are a total of 100 user byte type variables, B1-B100, and the range of stored values is 0-255.

Comment: Comment can be added.

Search: You can use the search variable number in the upper right corner of the interface.

Import comment: A USB flash drive needs to be inserted into the robot controller, and there is corresponding comment text in the USB flash drive. Click "Import Comment".

Export comment: A USB flash drive needs to be inserted into the robot controller. Click "Export Comment". The exported comments are in text format and can be edited. The name of the exported comment text is byteVarComment.txt.

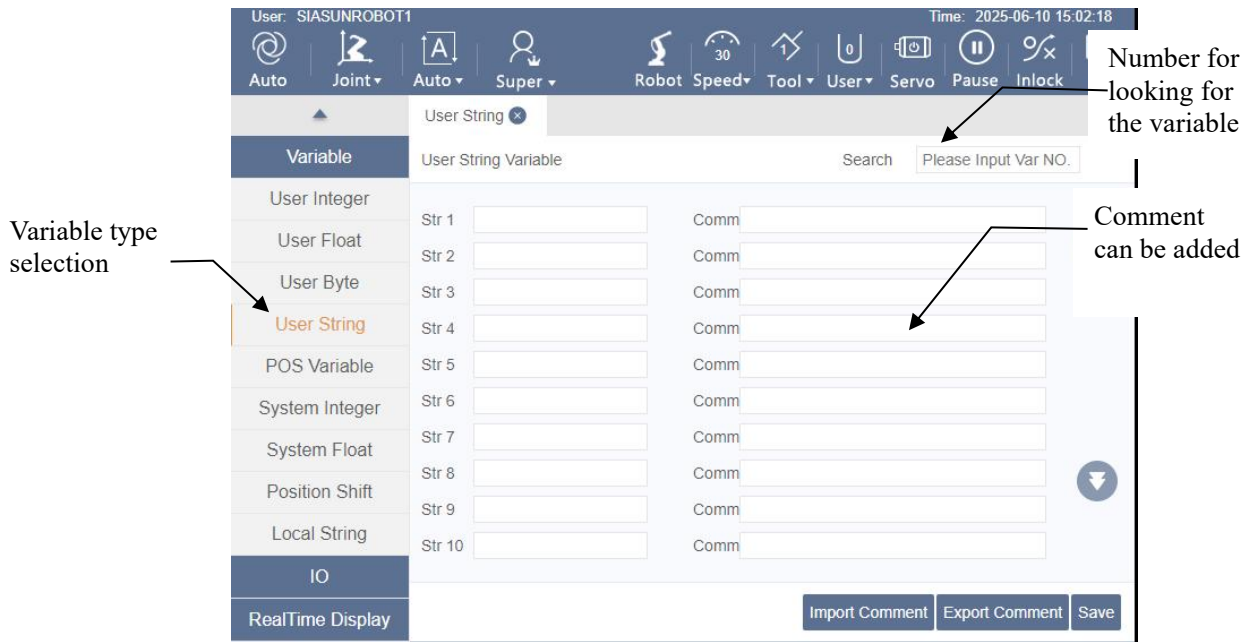


Figure 6-6 String Variable Interface

There are a total of 60 string variables, Str1-Str60, and the input length range is 0-36 characters.

Comment: Comment can be added.

Search: You can use the search variable number in the upper right corner of the interface.

Import comment: A USB flash drive needs to be inserted into the robot controller, and there is corresponding comment text in the USB flash drive. Click "Import Comment".

Export comment: A USB flash drive needs to be inserted into the robot controller. Click "Export Comment". The exported comments are in text format and can be edited. The name of the exported comment text is stringVarComment.txt.

6.4 Value of Position Variable

Position variable is a variable used to store position information. There are a total of 5,630 position variables.

The position variable interface allows you to view the values in the position variables and store the current positions of the robot in the position variables.

Click [Variable] - [Position Variable] on the menu to open the position variable interface.

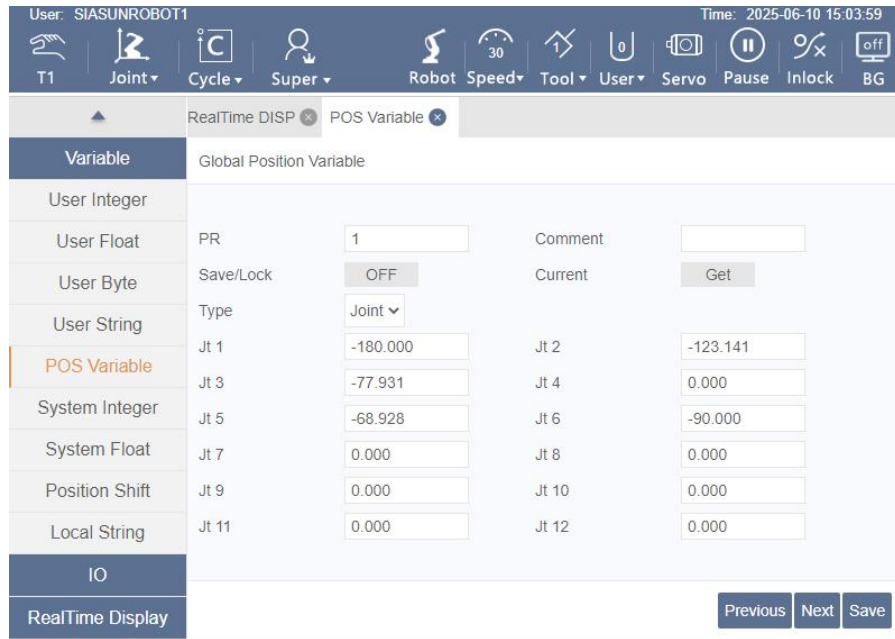


Figure 6- 7 Position Variable Interface (Joint Value)

Parameter description:

1) PR:

Enter the number in the dialog box to search for the location variable number.

2) Note:

This variable can be noted and displayed in the variables used in the job.

3) Lock:

This is to protect this position variable.

4) Current position:

The current position of the robot can be stored in this position variable by Get.

5) Type:

Joint or orthogonal: Joint indicates that the stored value is a joint value. Orthogonal indicates that the stored value is a rectangular coordinate value.

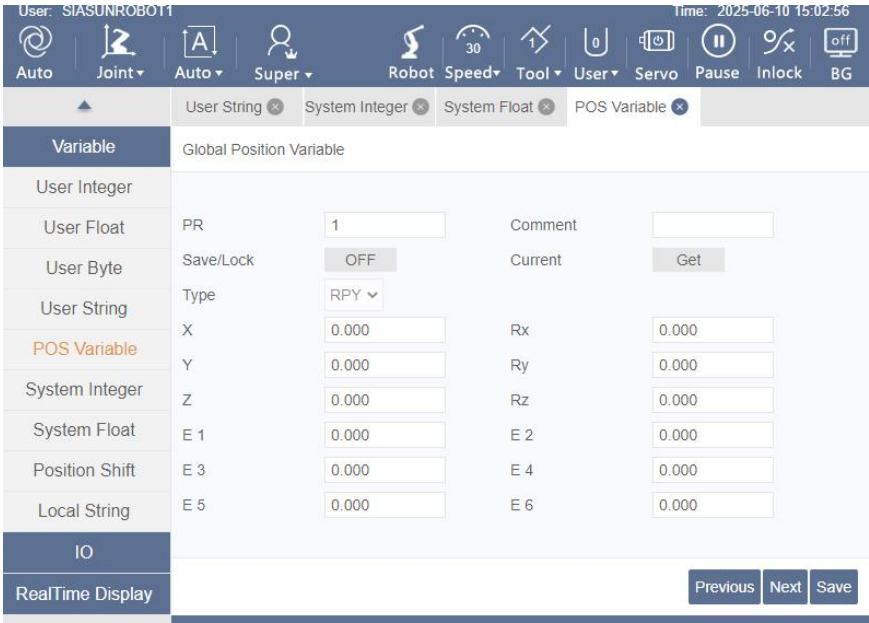


Figure 6- 8 Position Variable Interface (Coordinate Value)

6.4.1 Position Variable Display Form

If the instruction for recording this position in the job is:

MOVL PR/P3...

the value of PR/P3 is the rectangular coordinate value because it is recorded using the linear motion instruction.

If the motion instruction used is:

MOVJ PR/P1...

the value of PR/P1 is the joint value.

6.4.2 Get Position Variables

If you directly get from the position variable interface at [Current Position] - [Get], the value displayed by the position variable will be the value in the original frame. If the position variable was previously empty, the value got in this way would be the value in the rectangular frame. Joint and orthogonal storage form can be converted through Form.

6.4.3 Purpose of Position Variables

The position variables can be used in the program through MOVJ PR, MOVL PR, SHIFT ON PRn F#N, TRANSON PRn F#4, etc. Refer to the instruction list for the specific purpose of the

instructions.

Motion instructions with position variables are not restricted by the form in which the position variables are stored and can be executed.

For example, the storage position of MOVJ PR/P1 can also be moved in the form of MOVL PR/P1.

For example, the storage position of MOVL PR/P2 can also be moved in the form of MOVJ PR/P2.

6.5 Real-time display

By pressing **[Data] button on the teaching** pendant or entering the real-time display interface from the main menu, you can view the current position value, IO status, etc. of the robot. It can display 4 parameters simultaneously. Users need to select the desired parameters, and close the real-time display, and the previously selected interface will reappear upon the next opening. After a certain parameter is selected, it can be switched, but the option cannot be turned off. After restarting, re-select.

Click [RealTime Display] - [RealTime DIS] or [Data] in the main menu to open the real-time display interface.

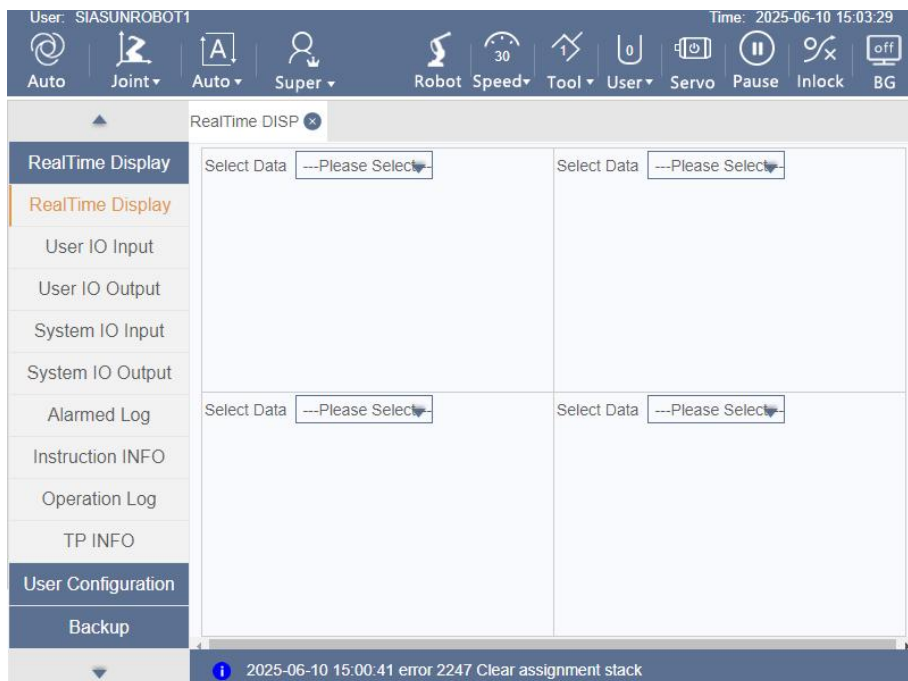


Figure 6-9 Real-time Display Interface

Parameter description:

1) Joint value:

Display the current angle value of each joint of the robot, with the unit being degrees.

2) Code disk input value:

The code disk input is the actual code disk value of the drive read from the drive.

3) Code disk output value:

The code disk output refers to the code disk value sent by the robot to the drive.

4) Code disk tracking error:

Display the difference between the code disk input value and output value of the robot.

5) T6 array of world frame position:

T6 matrix of the robots position in the world frame, where N, O and A are matrices, and P represents the position, with the unit being mm.

6) User Rpy:

TCP position and attitude values in the user frame, with the position unit being mm and the attitude unit being degrees.

7) T6 array of current position:

T6 array of the current position of the robot in the rectangular frame.

8) Rpy of current position:

TCP position and attitude values at the current position of the robot in the rectangular frame.

9) Speed value:

Display the real-time speed of each axis when the robot moves.

10) Torque value:

Display the real-time torque of each axis when the robot is moving.

11) Input voltage:

Used to check the external input voltage of the motor. (Specially customized for specific projects. Inquiries must be made in advance to use this feature.)

12) Output voltage:

Used to check the output voltage of the motor. (Specially customized for specific projects. Inquiries must be made in advance to use this feature.)

13) Servo temperature:

Applicable to the 500kg model only; not valid for other models.

14) True value of external axis code disk:

Display the encoder value of the external axis. (Valid after configuring the external axis)

15) Tool angle:

Display the angle between the TCP position in the rectangular frame and the Z direction of the current tool frame. Angle 1: Angle between the projection of the tool on the O-YZ plane and the Y-axis; Angle 2: Angle between the projection of the tool on the O-XZ plane and the X-axis; Angle 3: Angle between the projection of the tool on the O-XY plane and the X-axis; Angle 4: Angle between the tool and the O-XY surface;

16) Theoretical torque value:

Planned torque value used during the factory testing of robot.

7 Alarm information

7.1 Alarm

When the alarm listed below is raised, the servo driver will be powered off, and it is not allowed to start directly. If the user intends to power on the servo, make sure that there is no alarm in the system, and press the Cancel button to cancel the alarm. The alarm is included in the error message.

Alarm number	Alarm information	Troubleshooting
101	Serve on failed on Energy Saving Mode	Power-on check failed. Contact technical support.
109	LAN disconnect, check wire	Resolution: Check communication lines
111	Modbus TCP disconnect	Resolution: check master station and network
112	Modbus TCP Master connection error	Resolution: Check master station configuration, slave station configuration, and wiring
230	Hash table error	Resolution: Contact technical support
232	Parameter not be 0:type	Resolution: enter straight line configuration interface, modify maximum speed to reasonable value
234	Parameter not be 0: type	Resolution: check acceleration or speed page configuration

Alarm number	Alarm information	Troubleshooting
237	Vibration suppression planning pause speed processing timeout	Resolution: Contact technical support
238	Bake Stop Speed Value Not Normal	Resolution: Contact technical support
242	Interpolation Function Process Not Normal:	Resolution: Contact technical support
244	VS Pause Process Not Normal	Resolution: Contact technical support
314	Output data is out of range	Resolution: serial command length out of range, check command format sent
614	Motion type isn't supported	Resolution: Contact technical support
615	Safe gate open	Resolution: check safety door
621	Robot overtravel	axis. Resolution: disable hard limit, cancel alarm and move robot
622	External axis exceed limit	pin. Resolution: check if the external axis corresponding to the pin has reached the hard limit
623	Position limit cable damage	Resolution: check cables
624	SD data have changed	Resolution: Contact technical support
642	Poweron response reset failure	Resolution: Inspect the control cabinet, retry power-on, Contact technical support if still fails
645	Brake response reset failure	Resolution: Inspect the control cabinet, retry power-on, Contact technical support if still fails
646	Driver Enable ON Failure	Resolution: Inspect the control cabinet, retry power-on, Contact technical support if still fails
651	Driver Enable OFF Error	Resolution: Contact technical support
664	Driver Enable OFF Failure	Resolution: Inspect the control cabinet, retry power-on, Contact technical support if still

Alarm number	Alarm information	Troubleshooting
		fails
677	Relay	error. Resolution: Contact technical support to replace hardware
678	contactor	Resolution: Contact technical support to replace hardware
679	Mode relay error	Resolution: contact technical personnel
680	FPGA error lock	Resolution: contact technical personnel
700	Singularity Avoid Failed,ErrCode	Resolution: contact technical personnel
701	Singularity Avoid Type Set Err!	Resolution: contact technical personnel
702	Virtual Controller is turned on	Resolution: no need to handle, just be aware
703	External axis configuration EtherCAT node abnormal, axis number:	Resolution: check configuration
704	External axis configuration multi EtherCAT node abnormal, axis number:	Resolution: check configuration
710	Invalid value	Resolution: contact technical personnel
720	Second CollDetecation Aixs:	Resolution: disable collision protection switch, cancel alarm, move robot away
721	Mode Error	Resolution: system error, need to restart
754	Robot collision: axis	Resolution: cancel alarm, move robot away
755	Dynamic Acc Calc Err:	Resolution: contact technical personnel
813	813 Alam: axis	Resolution: check servo drive alarm cause, eliminate
814	speed unreachable J	Resolution: 1check (RC parameter) speed limit setting value2check reduction ratio
815	F_error exceed limit J	Resolution: 1check (RC parameter) speed limit setting value2check reduction ratio

Alarm number	Alarm information	Troubleshooting
818	Overlarge pulse deviation:	Resolution: try increasing the deviation setting value, if still alarming, contact SIASUN personnel
850	Remote/local switch fault	Resolution: check hardware line of remote/local mode switch
853	Choose local/remote mode	Resolution: check teaching pendant mode switch
858	RR, joint limit Error	Resolution: contact technical personnel
859	Joint value exceed limit	Resolution: adjust soft limits; turn off limit switches, adjust robot; adjust conversion parameters
860	RDY signal error	Resolution: contact professional technical personnel
861	Cannot move the axis further in the over-limit direction	Resolution: can only move in the direction away from the limit
868	Off-line external stop	Resolution: check if offline external emergency stop signal is input
870	Encoder value limit error	Resolution: check disk limit setting in machine parameters
871	Position command invalidity	Resolution: Method of handling: contact technical personnel
872	Reduction ratio modified	Speed exceed limit. Resolution: check joint speed setting value
873	Salve initialization failed	Resolution: confirm whether the system axis number and bus device configuration are correct
874	Abnormal equipment number	Resolution: confirm if there is a disconnected network cable
875	Master network disconnected	Resolution: Contact technical support
876	Master network error	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
877	Salve node abnormal:	Resolution: Contact technical support
878	Master activate failed	Resolution: Contact technical support
879	Master is not available:	Resolution: Contact technical support
881	IO initialization failed	Resolution: confirm whether the configuration of IO board in 'bus device' is correct
885	Ethercat IO slave config error	Resolution: confirm whether the configuration of IO board in 'bus device' is correct
888	Driver initialization failed	Resolution: confirm whether the driver configuration is correct
890	Robot abnormal: axis	Resolution: check the physical connection of the driver's network cable for loose connections
891	Acceleration or part bug	Resolution: check torque limit settings, if no problem, contact technical personnel
892	Point component exceed limit	Resolution: check position component X assignment instruction and related communication protocol
895	Loose breake response lost	Resolution: power cycle, if still fails, contact technical personnel
897	Driver configuration ABN:	Resolution: check 'bus device configuration' page configuration
899	Multi-Motor sync error:axis	Resolution: modify the 'multi-drive deviation' value of the corresponding external axis, increase by about 500 before powering on
903	External emergency stops	Resolution: check if external emergency stop signal is input
909	Profinet break,please check!	Resolution: check network cable, network configuration, etc., related to communication
910	Explosion-proof pneumatic system ABN	Resolution: contact designer

Alarm number	Alarm information	Troubleshooting
916	Master disconnect, check wire	Resolution: check hardware communication lines
917	Explosion-proof mode is OFF, Please do not use robot in explosive environment	Resolution: Inform the user only
919	EtherCAT commu task request poweroff	Resolution: Check the driver
920	Emergency stops	Resolution: reset after lifting the button
921	lookahead cache only support MOVL/MOVJ	Resolution: Contact technical support
922	Motor block:SV	Resolution: Investigate mechanical jamming conditions; Check whether the brake circuit is normal
923	Motor overheating:SV	Resolution: Lower the ambient temperature, such as improving the motor cooling conditions; Replace the motor or driver
924	Incremental encoder Z signal abnormality:SV	Resolution: Check the motor encoder wiring; Add magnetic rings to the encoder cable and motor power cable
925	lookahead cache not support exp plan type!	Resolution: Contact technical support
926	External Emergency stops	Resolution: Check the external emergency stop signal in remote IO mapping input
930	The index of array is out of bound	Resolution: contact technical personnel
980	System IO receive abnormally	Resolution: Stop the robot, have the main board inspected by after-sales personnel, replace the main board if necessary
981	isa fpga port error	Resolution: Stop the robot, have the main board inspected by after-sales personnel, replace the main board if necessary

Alarm number	Alarm information	Troubleshooting
982	System IO heartbeat output error	Resolution: Internal robot error, please stop the robot and contact after-sales personnel
986	Confirm ISA or SIM_SCB communication configuration	Resolution: Check for faults in emergency stop, safety door, and limit switches
987	hardware device failed to open!	Resolution: Restart the robot, Contact technical support if still fails
996	Teachbox connect error	Resolution: Check the teaching pendant connection
997	Short-circuit plug error	Resolution: Have electrical personnel check the shorting plug
1101	TCP connection failure	Contact technical support
1102	TCP closing failure	Contact technical support
1103	Local string variable exception	Contact technical support
1104	The TCP server connection is not created	Contact technical support
1105	The TCP connection was closed	Contact technical support
1106	TCP communication timeout	Contact technical support
1107	Out of memory	Contact technical support
1108	String operation failed	Contact technical support
1109	TCP data sending length error	Contact technical support
1201	Stall torque overlord:	Resolution: Driver alarm code 42If it recurs, please contact the SIASUN customer service
1202	Main supply Fail Phase:	Resolution: Driver alarm code 63
1203	Over-voltage:	Resolution: Driver alarm code 61
1204	Regenerative Overload:	Resolution: Driver alarm code 43
1205	DC Main circuit low voltage	Resolution: Driver alarm code 62
1206	High Ambient temperature:	Resolution: Driver alarm code 51

Alarm number	Alarm information	Troubleshooting
1207	Resistance Overheat 0:	Resolution: Driver alarm code 52
1208	Resistance Overheat 3:	Resolution: Driver alarm code 53
1209	drive module Overheating:	Resolution: Driver alarm code 56
1210	Resistance Overheat 1:	Resolution: Driver alarm code 54
1211	Resistance Overheat 2:	Resolution: Driver alarm code 55
1212	Under voltage of 5V:	Resolution: Driver alarm code 73
1213	Control power Under-voltage	Resolution: Driver alarm code 71
1214	Under voltage of 12V:	Resolution: Driver alarm code 72
1215	Current Detection Error 0:	Resolution: Driver alarm code 22
1216	Current Detection Error 1:	Resolution: Driver alarm code 23
1217	Current Detection Error 2:	Resolution: Driver alarm code 24
1218	CPU CIRC parameter Error	Resolution: Driver alarm code E8
1219	System Code Error:	Resolution: Driver alarm code E9
1220	Drive current Abnormal:	Resolution: Driver alarm code 21
1221	Error in CPU built in ROM:	Resolution: Driver alarm code E3
1222	EEPROM Error	Resolution: Driver alarm code E1
1223	Initial Process Time-Out	Resolution: Driver alarm code F2
1224	EEPROM Check Sum Error	Resolution: Driver alarm code E2

Alarm number	Alarm information	Troubleshooting
1225	System Parameter Error1:	Resolution: Driver alarm code E5
1226	System Parameter Error2:	Resolution: Driver alarm code E6
1227	Motor Parameter Error:	Resolution: Driver alarm code E7
1228	Motor code setting Error:	Resolution: Driver alarm code EA
1229	Sensor code setting Error:	Resolution: Driver alarm code EB
1230	Motor parameter 1	Resolution: Driver alarm code EE
1231	Motor parameter 2	Resolution: Driver alarm code EF
1232	motor power disconnection:	Resolution: Driver alarm code C3
1233	CS detection error:	Resolution: Driver alarm code 44
1234	Encoder Communication Error:	Resolution: Driver alarm code 84
1235	Encoder Initial Process Error:	Resolution: Driver alarm code 85
1236	Position skip of CS data:	Resolution: Driver alarm code 86
1237	CS signal line break:	Resolution: Driver alarm code 87
1238	encoder rotation overflow:	Resolution: Driver alarm code A0
1239	Multi-turn error:	Resolution: Driver alarm code A1
1240	Encoder Internal Error 4	Resolution: Driver alarm code A4
1241	Encoder Internal Error 5	Resolution: Driver alarm code A5
1242	Encoder Internal Error 6	Resolution: Driver alarm code A6

Alarm number	Alarm information	Troubleshooting
1243	Encoder Internal Error 9	Resolution: Driver alarm code A9
1244	Encoder Internal Error 12	Resolution: Driver alarm code AC
1245	Encoder Internal Error 13	Resolution: Driver alarm code AD
1246	Encoder Internal Error 14	Resolution: Driver alarm code AE
1247	Resolver disconnection:	Resolution: Driver alarm code AF
1248	Encoder 1 Disconnection:	Resolution: Driver alarm code 81
1249	Encoder 2 Disconnection:	Resolution: Driver alarm code 83
1250	Accelerate error:	Resolution: Driver alarm code A2
1251	Over-speed error:	Resolution: Driver alarm code A3
1252	Position data error:	Resolution: Driver alarm code AA
1253	Port 0 Invalid Frame Error:	Resolution: Driver alarm code 10
1254	Port 1 Invalid Frame Error:	Resolution: Driver alarm code 11
1255	Port 0 Successive Rx error:	Resolution: Driver alarm code 12
1256	Port 1 Rx occurrence error:	Resolution: Driver alarm code 13
1257	Port 0 Tx Error:	Resolution: Driver alarm code 14
1258	Port 1 Tx Error:	Resolution: Driver alarm code 15
1259	Communication time out:	Resolution: Drive alarm code 1A
1260	Port 0 Lost link:	Resolution: Drive alarm code 18

Alarm number	Alarm information	Troubleshooting
1261	Port 1 Lost link:	Resolution: Drive alarm code 19
1262	Overload 1:	Resolution: Drive alarm code 41
1263	Safe Torque Off Error 1:	Resolution: Drive alarm code 25
1264	Safe Torque Off Error 2:	Resolution: Drive alarm code 26
1265	over rotational speed:	Resolution: Drive alarm code 45
1266	Over-speed:	Resolution: Drive alarm code C1
1267	Velocity Control Error:	Resolution: Drive alarm code C2
1268	Model control error:	Resolution: Drive alarm code C5
1269	Position Command Error 1:	Resolution: Drive alarm code D2
1270	Position Command Error 2:	Resolution: Drive alarm code D3
1271	Excessive Position:	Resolution: Drive alarm code D1
1272	CPU-defined exception	Resolution: Drive alarm code F1
1273	Test Run Close:	Resolution: Drive alarm code DF
1274	Parameter change:	Resolution: Drive alarm code DE
1286	Clock Synchronization error	Resolution: Drive alarm code 80.0
1287	Parameter setting error:	Resolution: Drive alarm code 81.0
1288	PDO assignment error:	Resolution: Drive alarm code 85.0
1289	Improper operation error:	Resolution: Drive alarm code 88.0

Alarm number	Alarm information	Troubleshooting
1290	under-voltage protection 0:	Resolution: Drive alarm code 11.0
1291	Over-voltage protection:	Resolution: Drive alarm code 12.0
1292	under-voltage protection 1:	Resolution: Drive alarm code 13.0
1293	Over-current protection:	Resolution: Drive alarm code 14.0
1294	over-heat protection:	Resolution: Drive alarm code 15.0
1295	Torque saturation error:	Resolution: Drive alarm code 16.0
1296	over-regeneration load:	Resolution: Drive alarm code 18.0
1297	Encoder communication error	Resolution: Drive alarm code 21.0"
1298	ENC: Encoder	Resolution: Drive alarm code 23.0
1299	Position deviation excess:	Resolution: Drive alarm code 24.0
1300	Hybrid deviation excess:	Resolution: Drive alarm code 25.0
1301	Over-speed protection	Resolution: Drive alarm code 26.0
1302	Directive error protection:	Resolution: Drive alarm code 27.0
1303	Limit of pulse relay error:	Resolution: Drive alarm code 28.0
1304	Deviation counter overflow:	Resolution: Drive alarm code 29.0
1305	Safety detection:	Resolution: Drive alarm code 30.0
1306	Input function error:	Resolution: Drive alarm code 33.0
1307	software limit protection:	Resolution: Drive alarm code 34.0

Alarm number	Alarm information	Troubleshooting
1308	EEPROM parameter error:	Resolution: Drive alarm code 36.0
1309	EEPROM check code error:	Resolution: Drive alarm code 37.0
1310	Over-travel inhibit input:	Resolution: Drive alarm code 38.0
1311	Absolute system close:	Resolution: Drive alarm code 40.0
1312	Absolute counter over error	Resolution: Drive alarm code 41.0
1313	Absolute over-speed error:	Resolution: Drive alarm code 42.0
1314	Incremental encoder error:	Resolution: Drive alarm code 43.0
1315	counter error 1:	Resolution: Drive alarm code 44.0
1316	counter error 2:	Resolution: Drive alarm code 45.0
1317	Absolute status error:	Resolution: Drive alarm code 47.0
1318	Incremental encoder Z error	Resolution: Drive alarm code 48.0
1319	Incremental encoder CS error	Resolution: Drive alarm code 49.0
1320	External scale error:	Resolution: Drive alarm code 50.0
1321	External scale ST error:	Resolution: Driver alarm code 51.0
1322	Phase connection error:	Resolution: Driver alarm code 55.0
1323	Sync establishment error:	Resolution: Driver alarm code 84.0
1324	Compulsory alarm input:	Resolution: Driver alarm code 87.0
1325	Command error protection:	Resolution: Driver alarm code 91.0

Alarm number	Alarm information	Troubleshooting
1326	Encoder data recovery error	Resolution: Driver alarm code 92.0
1327	Parameter setting error:	Resolution: Driver alarm code 93.0
1328	Home position Homing error:	Resolution: Driver alarm code 94.0
1329	Motor recognition error:	Resolution: Driver alarm code 95.0
1330	communication IC error:	Resolution: Driver alarm code 98.0
1301	Over-speed protection:SV	Resolution: Driver alarm code 26.0
1302	Command anomaly protection:SV	Resolution: Driver alarm code 27.0
1333	Position control overflow:	Resolution: Execute 'encoder multi-turn zero' command; enable infinite position control function
1334	ECAT synchronization error:	Resolution: Correctly set servo parameter 0x20D3; correctly configure synchronization mode
1335	Running error	Resolution: Correctly set position planning parameters
1336	Brake circuit error	Resolution: Check the drive brake output wiring and ensure it is correct and reliable
1337	Multi-axis sync error	Resolution: Replace the driver
1338	PDO Configuration error	Resolution: Correct EtherCAT PDO configuration
1339	Rotation circuit error	Resolution: Correctly set drive resolver parameters, invalid replace the driver
1340	Control mode error	Resolution: Before servo enable, the controller first sets the correct control mode

Alarm number	Alarm information	Troubleshooting
1341	Amplifier short circuit	Resolution: Cable to ground short circuit, handle cabling; motor to ground short circuit, replace the motor
1342	Amplifier earth leakage	Resolution: Cable to ground short circuit, handle cabling; motor to ground short circuit, replace the motor
1343	Amplifier UV short circuit	Resolution: Repair or replace the driver
1344	Amplifier UW short circuit	Resolution: Repair or replace the driver
1345	Amplifier WU short circuit	Resolution: Repair or replace the driver
1346	AD sampling circuit error	Resolution: Repair or replace the driver
1347	EEPROM data error	Resolution: Check the driver hardware
1348	Stack overflow	Resolution: Repair or replace the driver
1349	Parameter not initialized	Resolution: Repair or replace the driver
1350	Encoder data error	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1351	Rotor(Magnetic pole) positioning failed	Resolution: Repair or replace the driver
1352	Amplifier internal error	Resolution: Check parameter settings; replace or repair the driver
1353	Motor brake output error	Resolution: Repair or replace the driver
1354	Charge relay error	Resolution: Replace or repair the driver"
1355	Brake connection error	Resolution: Check braking resistor wiring; replace or repair the driver
1356	AD correction factor error	Resolution: Reset the AD calibration coefficient
1357	Mismatch 1	Resolution: Repair or replace the driver
1358	Mismatch 2	Resolution: Repair or replace the driver

Alarm number	Alarm information	Troubleshooting
1359	Phase sequence is wrong	Resolution: Repair or replace the driver
1360	Fail to initialize amplifier	Resolution: Repair or replace the driver
1361	internal communication error	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1362	Encoder internal fault	Resolution: Check encoder wiring, cabling; check encoder parameters
1363	Encoder type change:	Resolution: Restart the driver or soft reset
1364	Amplifier over-current U	Resolution: Check U-phase wiring; reduce load; check U-phase cable
1365	Amplifier over-current V	Resolution: Check U-phase wiring; reduce load; check U-phase cable
1366	Amplifier over-current W	Resolution: Check U-phase wiring; reduce load; check U-phase cable
1367	DC link over-voltage:	Resolution: Reduce start-stop frequency; increase braking resistor power; repair or replace the basic power module
1368	Under voltage of 24V:	Resolution: Check control power voltage
1369	Watchdog overflow	Resolution: Re-power; repair or replace the driver
1370	Amplifier is Overloaded	Resolution: Check mechanical transmission; check motor load, or increase motor drive module capacity
1371	Encoder wiring is incorrect	Resolution: Check encoder wiring, cabling; replace the encoder
1372	CPU overload	Resolution: Reduce control instruction

Alarm number	Alarm information	Troubleshooting
		operation frequency; replace or repair the driver
1373	Amplifier output open phase	Resolution: Check motor wiring; replace or repair the motor; replace or repair the driver
1374	Access error of encoder	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1375	instantaneous overload 1	Resolution: Check output side cable wiring; reliable grounding; readjust control parameters
1376	Extenal sending fault	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1377	External receiving fault	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1378	hardware over-current	Resolution: Check encoder wiring, cabling; check encoder parameters; replace the encoder
1379	Input fail phase:	Resolution: Check power circuit, main circuit in connected state one phase voltage too low
1380	DC link under-voltage:	Resolution: Check power circuit, DC bus insulation, driver cabling; reduce load
1381	Module overheating 2	Resolution: Check drive cooling system, output cabling; replace larger capacity inverter
1382	module too cold	Resolution: Repair or replace the driver
1383	Dynamic Brake Overheating"	Resolution: Reduce start-stop frequency; increase acceleration/deceleration time constant; reduce load inertia

Alarm number	Alarm information	Troubleshooting
1384	Motor is overloaded	Resolution: Check mechanical transmission; check load, reduce acceleration/deceleration, or replace the motor
1385	Brake Resistor Overheat	Resolution: Increase external cooling; extend acceleration/deceleration time; replace the braking resistor
1386	Module overheating 2	Resolution: Check drive cooling system; repair or replace the driver
1387	instantaneous overload 2	Resolution: Set rotor compensation angle; reduce motor acceleration/deceleration; optimize motor control parameters
1388	instantaneous overload 3	Resolution: Set rotor compensation angle; reduce motor acceleration/deceleration; optimize motor control parameters
1389	instantaneous overload 4	Resolution: Set rotor compensation angle; reduce motor acceleration and deceleration; optimize motor control parameters
1390	Hardware STO1 trigger	Resolution: Troubleshoot external faults
1391	Hardware STO2 trigger	Resolution: Troubleshoot external faults
1392	STO wiring abnormality	Resolution: Check STO wiring
1393	Amplifier external fault	Resolution: Check other axes, identify and eliminate faults
1394	Parameter data error	Resolution: Check if parameter settings exceed the set parameter range; check position unit settings
1395	Position following error	Resolution: Check encoder wiring; re-adjust control parameters; increase anti-interference measures
1396	Position control overflow	Resolution: Zero the encoder and perform a soft reset or restart the drive

Alarm number	Alarm information	Troubleshooting
1397	Velocity following error	Resolution: Check encoder wiring; re-adjust control parameters; increase anti-interference measures
1398	Control Cycle PARM error	Resolution: Adjust EtherCAT communication cycle or servo control cycle
1399	ECAT Process data error	Resolution: Repair or replace the drive
1400	Fail to write into EEPROM	Resolution: Repair or replace the drive
1401	Fail to find home position	Resolution: Repair or replace the drive
1402	Invalid ECAT bus command	Resolution: Master station misoperation, fault reset
1403	DS communication error	Resolution: Check RS485 cable, RS485 converter for normal operation
1404	ECAT communication error	Resolution: Check bus wiring; check if cable connectors are properly crimped
1405	Hardware limit reached	Resolution: Check position command planning range
1406	Forward software limit	Resolution: Remove positive soft limit; check position command planning range
1407	Backward software limit	Resolution: Remove negative soft limit; check position command planning range
1408	Excessive Position Deviation	Resolution: Check if mechanical position has changed, confirm no abnormal mechanical zero point, then clear"
1409	Change parameter illegally	Resolution: Modify parameter values within the range of modifiable servo parameters
1410	Encoder under-voltage	Resolution: Replace encoder battery
1411	Motor over-speed	Resolution: Optimize motor parameters and control parameters; check encoder settings and encoder wiring

Alarm number	Alarm information	Troubleshooting
1412	Voltage following error	Resolution: Reduce motor load change rate; repair or replace the drive
1413	Encoder over-speed	Resolution: Appropriately reduce motor operating speed; check encoder parameters and motor control parameters
1414	EEPROM version change	Resolution: Restart the drive or perform a soft reset
1415	Motor overlord warning	Resolution: Check mechanical parts for lubrication issues or blockages; replace motor
1416	Dynamic Brake overlord	Resolution: Replace with a larger power consumption brake resistor
1417	rotation speed warning	Resolution: Check input power voltage
1418	DC link under-voltage:	Resolution: Check DC bus voltage
1419	History fault record error	Resolution: Restart the drive or perform a soft reset; repair or replace the drive
1420	Can't set Control mode	Resolution: Re-set parameter 0x6060
1421	Re-power parameter changed	Resolution: Restart the drive or perform a soft reset
1422	CPU overload warning	Resolution: Repair or replace the drive
1423	Encoder under-voltage	Resolution: Replace encoder battery
1424	Amplifier internal warning	Resolution: Replace the drive
1425	Uncalibrated home position	Resolution: Re-calibrate mechanical origin
1426	Amplifier not ready	Resolution: Repair or replace the drive
1427	communication error	Resolution: Execute 'encoder multi-turn zeroing' command; switch to enable infinite position control mode

Alarm number	Alarm information	Troubleshooting
1428	External receiving warning	Resolution: Check motor encoder wiring; add magnetic rings to encoder cable, motor power cable
1429	External sending warning	Resolution: Check motor encoder wiring; add magnetic rings to encoder cable, motor power cable
1430	communication warning:	Resolution: Check motor encoder wiring; add magnetic rings to encoder cable, motor power cable
1431	Software limit alarm:	Resolution: Appropriately increase servo parameter 0x2004 and 0x2005 settings
1432	Invalid AD correction factor	Resolution: Reset drive AD correction coefficient
1433	Abnormal path planning	Resolution: Correctly set position planning parameters
1434	Excessive Position Deviation	Resolution: Drive performs fault reset command; drive restarts or performs a soft reset
1435	Short circuit:	Resolution: Check motor cable length; use shorter or lower capacitance motor cable
1436	Short circuit on motor side	Resolution: Check if motor cable is short-circuited

Alarm number	Alarm information	Troubleshooting
1437	Motor utilisation too high:	Resolution: Check drive size; check warning threshold or error threshold settings
1438	Fault-motor Ixt too high:	Resolution: Check drive size
1439	Warning-motor Ixt too high:	Resolution: Check drive size
1440	Fault-device Ixt too high:	Resolution: Acceleration process too long; check drive size; reduce inverter maximum current
1441	Warning-device Ixt too high:	Resolution: Check drive size
1442	At ultimate motor current:	Resolution: Reduce maximum inverter current 0x6073; optimize current controller
1443	Output current > maximum:	Resolution: Reduce maximum motor current 0x6073
1444	Clamp is active:	Resolution: Choose flatter speed ramp; reduce load; set Imax controller
1445	Clamp responded too often:	Resolution: Choose flatter speed ramp; reduce load; set Imax controller
1446	DC link over-voltage:	Resolution: Reduce dynamic performance of load curve; check power voltage
1447	DC link under-voltage:	Resolution: Check power voltage, DC bus voltage, power settings, fuses
1448	Module temperature too high	Resolution: Check power voltage; provide adequate cooling for equipment; clean fans and ventilation slots
1449	Module temperature fault:	Resolution: Hardware error: Must contact manufacturer due to necessary replacement of

Alarm number	Alarm information	Troubleshooting
		equipment
1450	Motor Over-temperature:	Resolution: Check drive size; check motor temperature sensor and wiring
1451	Motor temperature sensor:	Resolution: Connect PTC or disable monitoring
1452	24V supply voltage error:	Resolution: Check optional external 24V power supply
1453	Watchdog restart	Resolution: Restart device. If error persists, contact manufacturer
1454	fault while download:	Resolution: Recreate parameter set. Repeat parameter set download
1455	Parameter error:	Resolution: Recreate the parameter set
1456	PLI-motor blocked:	Resolution: Check motor data; ensure the motor is stationary during pole position recognition
1457	Error in feedback system:	Resolution: Check encoder cable and encoder; if not using an encoder, disable monitoring
1458	Encoder communication error	Resolution: Check power voltage and encoder; check encoder cable
1459	motor speed too high:	Resolution: Allow the motor to coast
1460	ECAT DC-sync required:	Resolution: Activate DC mode
1461	ECAT communication error:	Resolution: Check bus cable; select a longer PDI watchdog time
1462	Sync Manager-address	Resolution: Check bus devices and wiring order; remove i700 servo inverter

Alarm number	Alarm information	Troubleshooting
1463	Sync Manager-actual size	Resolution: Check bus devices and wiring order; remove i700 servo inverter
1464	Sync Manager - settings	Resolution: Check bus devices and wiring order; remove i700 servo inverter
1465	PDO Mapping: Object unknown	Resolution: Check PDO mapping
1466	No PDO mapping object	Resolution: Check PDO mapping
1467	mapped objects>64	Resolution: Reduce the number of mapped objects in 'free mapping
1468	PDO mapping error	Resolution: Check PDO mapping"
1469	Sync controller error	Resolution: Check if the master station configuration has set DC mode; restart configuration
1470	EtherCAT Sync time out:	Resolution: Check if there is a DC bias in the master station configuration; restart configuration
1471	Fatal internal error	Resolution: Restart the device; if the error persists, contact the manufacturer
1472	internal communication error	Resolution: Restart the device; if the error persists, contact the manufacturer
1473	Brake error:	Resolution: Check brake. Check wiring
1474	overflow task	Resolution: Restart the device; if the error persists, contact the manufacturer
1475	motor movement too large	Resolution: Check controller settings at 0x2C62
1476	STO inhibited	Resolution: Re-enable 'Safe Torque Off (STO)' or disable the controller
1477	Maxspeed reached	Resolution: Adjust maximum motor speed and warning threshold or error threshold

Alarm number	Alarm information	Troubleshooting
1478	Forbid to change Para:	Resolution: Do not change parameters until identification is complete or test mode is deactivated
1479	Forbid to change Para 2:	Resolution: Do not change parameters until identification is complete
1480	Motor fail phase:	Resolution: Check wiring between inverter and motor
1481	Failure of motor phase U:	Resolution: Check wiring between inverter and motor
1482	Failure of motor phase V:	Resolution: Check wiring between inverter and motor
1483	Failure of motor phase W:	Resolution: Check wiring between inverter and motor
1484	No resolver connected:	Resolution: Resolver errors can only be identified when a resolver is used as the feedback system
1485	Speed too low	Resolution: Repeat identification at a higher speed
1486	No resolver connected:	Resolution: Resolver errors can only be identified when a resolver is used as the feedback system
1487	Time-out:	Resolution: Repeat identification at a higher speed
1488	Numerical problem:	Resolution: Repeat identification at a higher speed
1489	Inverter error too big	Resolution: Check shielding of motor cables. Connect the shield to as large a surface as possible

Alarm number	Alarm information	Troubleshooting
1490	Identification aborted:	Resolution: Repeat pole position identification
1491	Not allowed in state	Resolution: Disable the controller before reading
1492	Encoder IS not connected	Resolution: Connect HIPERFACE encoder
1493	Encoder is not supported 1	Resolution: Enter the correct type code at 0x2C41:002
1494	Encoder is not supported 2	Resolution: Connect encoder with supported type code
1495	Communication timeout	Resolution: Restart the manual jogging dialog box (0x2836:005)
1496	Internal error 3	Resolution: Set motor data according to the data on the motor nameplate; check motor wiring
1497	Protection failed	Resolution: Enter the correct PIN code; replace the device and reprogram the pins
1501	EEPROM info:SV	Resolution: Contact technical support
1502	Driver info:SV	Resolution: Contact technical support
1503	PDO data table:SV	Resolution: Contact technical support
1504	Axis Status ESM:SV	Resolution: Contact technical support
1505	Driver sv para:SV	Resolution: Contact technical support
1506	Driver svon para:SV	Resolution: Contact technical support
1507	Axis internal data:SV	Resolution: Contact technical support
1508	Axis trade mark:SV	Resolution: Contact technical support
1509	Sys encoder:SV	Resolution: Contact technical support
1510	Axis motor para:SV	Resolution: Contact technical support
1511	Axis basic para:SV	Resolution: Contact technical support
1512	Axis original para:SV	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1513	Axis driver para:SV	Resolution: Contact technical support
1514	Axis switch ratio:SV	Resolution: Contact technical support
1515	Axis control cycle:SV	Resolution: Contact technical support
1516	Axis high rotation:SV	Resolution: Contact technical support
1517	Axis error protect limit:SV	Resolution: Contact technical support
1518	Axis internal switch para:SV	Resolution: Contact technical support
1519	Axis PDO data:SV	Resolution: Contact technical support
1520	Axis self adapt para:SV	Resolution: Contact technical support
1521	Axis adjuster limit:SV	Resolution: Contact technical support
1522	Axis filter type:SV	Resolution: Contact technical support
1523	Axis filter para:SV	Resolution: Contact technical support
1524	Axis PP control mode:SV	Resolution: Contact technical support
1525	Axis error log:SV	Resolution: Contact technical support
1526	ADC apply:SV	Resolution: Contact technical support
1527	PWM apply:SV	Resolution: Contact technical support
1528	EtherCAT init:SV	Resolution: Contact technical support
1529	Version info:SV	Resolution: Contact technical support
1530	EEPROM info:SV	Resolution: Contact technical support
1531	DriveStarter stack:SV	Resolution: Contact technical support
1532	Driver info init:SV	Resolution: Contact technical support
1533	Error log init:SV	Resolution: Contact technical support
1534	Driver svon para:SV	Resolution: Contact technical support
1535	Driver para version:SV	Resolution: Contact technical support
1536	Driver para read:SV	Resolution: Contact technical support
1537	Re-svon para:SV	Resolution: Contact technical support
1538	Main circuit type:SV	Resolution: Contact technical support
1539	PDO data table:SV	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1540	EtherCAT main func:SV	Resolution: Contact technical support
1541	EtherCAT interrupt:SV	Resolution: Contact technical support
1542	EtherCAT master select:SV	Resolution: Contact technical support
1543	IO operation:SV	Resolution: Contact technical support
1544	Sys encoder:SV	Resolution: Contact technical support
1545	Svon position deviation:SV	Resolution: Contact technical support
1546	Axis trade mark:SV	Resolution: Contact technical support
1547	Adaptor para:SV	Resolution: Contact technical support
1548	Motor para:SV	Resolution: Contact technical support
1549	Motor switch para:SV	Resolution: Contact technical support
1550	AD adjust ratio:SV	Resolution: Contact technical support
1551	AD switch ratio:SV	Resolution: Contact technical support
1552	Hardware version:SV	Resolution: Contact technical support
1553	AD adjust ratio check:SV	Resolution: Contact technical support
1554	Driver rated electricity:SV	Resolution: Contact technical support
1555	Driver max electricity:SV	Resolution: Contact technical support
1556	Driver main power:SV	Resolution: Contact technical support
1557	AD voltage switch:SV	Resolution: Contact technical support
1558	Driver error protect limit:SV	Resolution: Contact technical support
1559	Position adjuster limit:SV	Resolution: Contact technical support
1560	Speed adjuster input limit:SV	Resolution: Contact technical support
1561	Speed adjuster output limit:SV	Resolution: Contact technical support
1562	Electricity adjuster input limit:SV	Resolution: Contact technical support
1563	Cycle time:SV	Resolution: Contact technical support
1564	Motor overload para:SV	Resolution: Contact technical support
1565	Resistance overload para:SV	Resolution: Contact technical support
1566	Driver overload once para:SV	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1567	Driver overload continually para:SV	Resolution: Contact technical support
1568	Filter init:SV	Resolution: Contact technical support
1569	Sv internal switch para:SV	Resolution: Contact technical support
1570	Sys interrupt config:SV	Resolution: Contact technical support
1571	Sys config finish:SV	Resolution: Contact technical support
1572	Abnormal main power input:SV	Resolution: Check the three-phase AC power supply to ensure the input power is normal
1573	CPU1 watchdog overflow:SV	Resolution: 1. Restart the driver 2. Replace the driver
1574	CPU2 watchdog overflow:SV	Resolution: 1. Restart the driver 2. Replace the driver
1575	First encoder data overflow:SV	Resolution: Ensure that the actual value change of the encoder does not exceed the allowable range
1576	Second encoder data overflow:SV	Resolution: Ensure that the actual value change of the encoder does not exceed the allowable range
1577	Abnormal CPU1 operation:SV	Resolution: 1. Restart the driver 2. Replace the driver
1578	Abnormal CPU2 operation:SV	Resolution: 1. Restart the driver 2. Replace the driver
1579	CPU1 memory exception:SV	Resolution: 1. Restart the driver 2. Replace the driver
1580	CPU2 memory exception:SV	Resolution: 1. Restart the driver 2. Replace the driver
1581	CPU memory conflict:SV	Resolution: 1. Restart the driver 2. Replace the driver
1582	First encoder data anomaly:SV	Resolution: Replace the encoder; Check the encoder wiring; Restart the driver; Replace the

Alarm number	Alarm information	Troubleshooting
		driver
1583	First encoder communication anomaly:SV	Resolution: Check the motor encoder wiring; Add magnetic rings to the encoder cable, motor power cable
1584	First encoder communication timeout:SV	Resolution: Replace the encoder; Check the motor encoder wiring and ensure correct wiring
1585	First encoder internal anomaly1 :SV	Resolution: Perform encoder zeroing operation; Restart the driver
1586	Second encoder data anomaly:SV	Resolution: Replace the encoder; Check the encoder wiring and ensure correct wiring
1587	Second encoder communication anomaly:SV	Resolution: Check the motor encoder wiring; Add magnetic rings to the encoder cable, motor power cable
1588	Second encoder communication timeout:SV	Resolution: Replace the encoder; Check the motor encoder wiring and ensure correct wiring"
1589	Second encoder internal anomaly1:SV	Resolution: Perform encoder zeroing operation; Restart the driver
1590	Control encoder overspeed:SV	Resolution: Optimize regulator parameters; Strengthen the anti-interference of the encoder cable; Replace the encoder
1591	Driver output absence:SV	Resolution: Check the motor U, V, W wiring; Replace the motor; Replace the driver

Alarm number	Alarm information	Troubleshooting
1592	Motor stall:SV	Resolution: Optimize regulator parameters; Appropriately increase the setting value of parameter 0x20A0
1593	Coprocessor communication anomaly:SV	Resolution: Add magnetic rings to the encoder cable and motor power cable; Ensure reliable connection of the driver ground wire
1594	Abnormal change of encoder AB signal:SV	Resolution: Check the motor encoder wiring; Add magnetic rings to the encoder cable and motor power cable
1595	Excessive current following error:SV	Resolution: Check the mechanical load transmission method; Optimize regulator parameters; Replace the driver
1596	Abnormal position target:SV	Resolution: Plan position target values and acceleration parameters reasonably
1597	First encoder power-on data overflow:SV	Resolution: Ensure that the actual value change of the motor at power-on does not exceed the allowable range
1598	Position target overflow:SV	Resolution: Execute the multi-turn zeroing command of the encoder to ensure that the motor operation does not exceed the maximum range
1599	Abnormal motor braking:SV	Resolution: Replace the motor brake; Check the motor brake circuit
1600	Control power undervoltage:SV	Resolution: Check the control power wiring and load; Replace the control power and driver
1601	Forward hardwire switch trigger:SV	Resolution: Check the limit switch status and ensure that the limit switch is not triggered
1602	Negative hardwire switch	Resolution: Check the limit switch status and

Alarm number	Alarm information	Troubleshooting
	trigger:SV	ensure that the limit switch is not triggered
1603	Emergency stop input switch trigger:SV	Resolution: Confirm that the DI configured as the 'emergency stop switch' is not in the triggered state
1604	Torque monitoring saturation fault:SV	Resolution: Appropriately increase the setting values of parameters 0x20A7 and 0x20A8
1605	Speed following error is too large:SV	Resolution: Check the mechanical load transmission method; Reduce the upper machine speed target value planning acceleration
1606	Full closed loop position deviation large:SV	Resolution: Check the mechanical load transmission method; Reduce the upper machine speed target value planning acceleration
1607	Origin search failed:SV	Resolution: Set the homing parameters 0x6098, 0x6099, 0x609A
1608	EtherCAT communication cycle error:SV	Resolution: Ensure that the communication cycle is greater than the position control cycle, speed control cycle, torque control cycle
1609	Location planning error:SV	Resolution: Correctly set the position planning parameters
1610	EtherCAT illegal sync mode:SV	Resolution: Ensure that the synchronization mode settings of the driver and the upper machine are consistent
1611	Position target value exceeds set	Resolution: Avoid position instruction values

Alarm number	Alarm information	Troubleshooting
	range:SV	exceeding the range of parameter 0x607B
1612	Radiator overheating:SV	Resolution: Reduce the actual mechanical load of the motor; Lower the ambient temperature; Replace the driver
1613	Abnormal limit switch:SV	Resolution: Detect wiring and parameter settings to ensure correct polarity definition and wiring of the limit switch
1614	Interface encoder resolution change:SV	Resolution: Restart the driver
1615	First encoder overheating:SV	Resolution: Lower the ambient temperature, such as improving the cabinet cooling conditions; Replace the driver
1616	First encoder battery undervoltage fault:SV	Resolution: Check the encoder battery wiring; Replace the battery; Perform the multi-turn zeroing command of the encoder
1617	Second encoder overheating:SV	Resolution: Lower the ambient temperature, such as improving the cabinet cooling conditions; Replace the driver
1618	Second encoder battery undervoltage fault:SV	Resolution: Check the encoder battery wiring; Replace the battery; Perform the multi-turn zeroing command of the encoder
1619	Control mode setting error:SV	Resolution: Before servo enabling, the controller sets the correct control mode first
1620	Excessive power-on position deviation:SV	Resolution: For motors with batteries, ensure that the battery is connected and the battery voltage is normal
1621	Abnormal acceleration failure first encoder:SV	Resolution: Plan the acceleration reasonably; Set reasonable servo parameters 0x2037
1622	Motor block:SV	Resolution: Investigate mechanical jamming

Alarm number	Alarm information	Troubleshooting
		conditions; Check whether the brake circuit is normal
1623	Motor overheating:SV	Resolution: Lower the ambient temperature, such as improving the motor cooling conditions; Replace the motor or driver
1624	Incremental encoder Z signal abnormality:SV	Resolution: Check the motor encoder wiring; Add magnetic rings to the encoder cable and motor power cable
1625	Write EEPROM data exception:SV	Resolution: Replace the driver; Power on the driver again, strengthen the anti-interference of the driver
1626	Read EEPROM data exception:SV	Resolution: Replace the driver; Power on the driver again, strengthen the anti-interference of the driver
1627	Abnormal energy braking circuit:SV	Resolution: Replace the driver; Strengthen the anti-interference of the driver
1628	Abnormal brake control circuit:SV	Resolution: Check the brake circuit wiring, whether the 24V supply voltage is normal; Replace the driver
1629	CPU overheated:SV	Resolution: Lower the ambient temperature; Restart or replace the driver
1630	CPU1 overload:SV	Resolution: Standardize the peripheral wiring of the driver, add anti-interference measures; Replace the driver
1631	CPU2 overload:SV	Resolution: Standardize the peripheral wiring of the driver, add anti-interference measures; Replace the driver
1632	CPU1 handshake failed:SV	Resolution: Restart or replace the driver

Alarm number	Alarm information	Troubleshooting
1633	DriverMaster communication timeout:SV	Resolution: Check the driver debug cable connection; Replace with an isolated debug cable
1634	ESC internal access error:SV	Resolution: Restart or replace the driver
1635	Servo enabled not ready:SV	Resolution: Ensure the driver selects the actual encoder and motor control; Ensure the motor is stationary
1636	CPU2 handshake failed:SV	Resolution: Restart or replace the driver
1637	CPU1 main task timeout:SV	Resolution: Restart or replace the driver
1638	Main power down:SV	Resolution: Check the main power supply and connections; Ensure stable power when servo is ON
1639	Internal CPU error:SV	Resolution: Restart or replace the driver
1640	Position actual value overflow:SV	Resolution: Execute the 'encoder multi-turn reset' command; Enable infinite position control
1641	Second encoder power-on data overflow:SV	Resolution: Ensure the motor's actual value does not exceed the allowed range at power-on; Enable infinite position control
1642	First encoder internal anomaly2:SV	Resolution: Perform encoder reset operation; Restart the driver
1643	Second encoder internal anomaly2:SV	Resolution: Perform encoder reset operation; Restart the driver
1644	First encoder internal anomaly3:SV	Resolution: Perform encoder reset operation; Restart the driver
1645	Second encoder internal anomaly3:SV	Resolution: Perform encoder reset operation; Restart the driver
1646	Abnormal acceleration fault of the second encoder:SV	Resolution: Plan acceleration reasonably; Set reasonable servo parameter 0x2037

Alarm number	Alarm information	Troubleshooting
1647	Abnormal diagnosis of STO1 circuit:SV	Resolution: Check STO1/STO2 connections and ensure reliable wiring
1648	Abnormal diagnosis of STO2 circuit:SV	Resolution: Check STO1/STO2 connections and ensure reliable wiring
1649	Hall signal is abnormal:SV	Resolution: Add a ferrite core to Hall signal wiring; Separate motor power lines from signal lines
1650	Abnormal phase of encoder AB signal:SV	Resolution: Check encoder wiring; Add a ferrite core to wiring; Separate motor power lines from signal lines
1651	Second position follow error large:SV	Resolution: Check mechanical load transmission method; Reduce the acceleration of the position target value from the host computer
1652	Abnormal system parameters:SV	Resolution: Ensure the power voltage is within specifications; Restore factory parameters
1653	Product model selection failure:SV	Resolution: Check encoder wiring; Replace with a valid motor or driver
1654	Parameter storage failure:SV	Resolution: Check if the program frequently modifies parameters and writes them to EEPROM
1655	FPGA failure:SV	Resolution: Verify the software version number
1656	Product matching failure:SV	Resolution: Check encoder wiring; Replace incompatible products
1657	Program exception:SV	Resolution: Restore factory parameters (set P20.06 to 1), then power cycle

Alarm number	Alarm information	Troubleshooting
1658	Encoder initialization failed:SV	Resolution: Check encoder wiring, or replace the encoder cable
1659	Ground short circuit detection fault:SV	Resolution: Check if UVW cables are shorted to ground; Replace cables if necessary; Replace the motor
1660	Overcurrent fault A:SV	Resolution: Check instruction input timing, ensure instructions are given after the servo is ready ('rdy')
1661	Overcurrent fault B:SV	Resolution: Check instruction input timing, ensure instructions are given after the servo is ready ('rdy')
1662	Abnormal number of absolute encoder turns:SV	Resolution: Reconnect or replace the cable; Replace the encoder; Power cycle"

Alarm number	Alarm information	Troubleshooting
1663	Abnormal encoder communication:SV	Resolution: Check encoder wiring, or replace the cable; Ensure the encoder is properly grounded
1664	Encoder data exception:SV	Resolution: Check wiring, or replace the encoder cable
1665	Abnormal low voltage of encoder battery:SV	Resolution: Replace the encoder battery
1666	Excessive speed deviation:SV	Resolution: Increase the setting value of P06.45; Extend the acceleration/deceleration time for position commands
1667	Torque saturation timeout:SV	Resolution: Increase the setting duration of P06.46; Check if UVW lines are disconnected
1668	Control undervoltage:SV	Resolution: Check the input power and connections; Replace the driver
1669	Speed failure:SV	Resolution: Check UVW and encoder wiring; Inspect the driver and motor
1670	Overvoltage:SV	Resolution: Input correct voltage range; Check if external resistor is connected
1671	Undervoltage:SV	Resolution: Increase power supply capacity to ensure stable voltage; Check undervoltage protection threshold
1672	Current sampling fault:SV	Resolution: Replace the servo driver
1673	AI sampling voltage is too high:SV	Resolution: Correctly connect AI input, set input voltage within $\pm 10V$

Alarm number	Alarm information	Troubleshooting
1674	Speed:SV	Resolution: If enabled, reduce speed command; Check UVW phase sequence
1675	Electrical angle recognition failed:SV	Resolution: Reduce load or increase current loop gain; Replace encoder cable
1676	Inertia identification failure:SV	Resolution: Reduce load or increase current loop gain; Ensure proper identification process
1677	DI terminal parameter setting failure:SV	Resolution: Do not configure the same function to multiple physical DI terminals
1678	DO terminal parameter setting failure:SV	Resolution: Do not configure the same function to multiple DOs; Reallocate DO
1679	Servo ON command invalid fault:SV	Resolution: Change improper operation methods
1680	Dividing pulse output overspeed:SV	Resolution: Modify frequency divider output settings to prevent exceeding limits

Alarm number	Alarm information	Troubleshooting
1681	Failure with excessive positional deviation:SV	Resolution: Reconnect motor main circuit cables; Increase servo drive gain
1682	Main loop input missing phase:SV	Resolution: Check three-phase power input cables; Ensure balanced three-phase input
1683	Driver output phase loss:SV	Resolution: Check UVW wiring; Replace the servo motor
1684	Drive overload:SV	Resolution: Confirm UVW and encoder wiring; Ensure the motor is not stalled

Alarm number	Alarm information	Troubleshooting
1685	Motor overload:SV	Resolution: Confirm UVW and encoder wiring; Ensure the motor is not stalled
1686	Electronic gear setting error:SV	Resolution: Set correct gear ratio range
1687	Radiator overheating:SV	Resolution: Replace fan or driver; Improve cooling conditions for the servo driver
1688	Abnormal pulse input:SV	Resolution: Adjust maximum allowable frequency; Ensure reliable grounding and use twisted shielded cables
1689	Position deviation full closed loop too large:SV	Resolution: Replace external encoder; Reset related parameters
1690	User forced failure:SV	Resolution: Normal DI function input, configured DI function 32 and effective input
1691	Absolute position reset fault:SV	Resolution: Contact manufacturer for technical support

Alarm number	Alarm information	Troubleshooting
1692	Main power disconnected:SV	Resolution: Check for momentary power loss in the main power input; Increase power supply capacity
1693	Start first time after writing a custom version :SV	Resolution: Restore factory values to load custom parameters
1694	CAN terminal closed:SV	Resolution: Check connections, reconnect
1695	Abnormal NMT command:SV	Resolution: Reset NMT node; Do not stop or reset CAN nodes while servo is ON
1696	CAN terminal failure:SV	Resolution: Check connections, reconnect
1697	Node protection or heartbeat timeout:SV	Resolution: Check if the node is online; Reset NMT node
1698	Synchronization failure:SV	Resolution: Reset NMT node, or send fault reset command to 6040
1699	CANOpen trajectory buffer underflow :SV	Resolution: Confirm that the host computer is running normally. Reset the NMT node
1700	CANOpen trajectory buffer overflow:SV	Resolution: Confirm that the host computer is running normally and the clock frequency matches the configured value
1701	Slave initialization failed:SV	Resolution: Try re-flashing the XML configuration file and then power cycle
1702	Sync failed:SV	Resolution: Reset the NMT node or send a fault reset command to 6040
1703	EtherCAT communication interrupted:SV	Resolution: Solution: Check if the network cable is securely connected or replace it; increase the P09.16 value
1704	Instruction given exception:SV	Resolution: Reset the NMT node or send a fault reset command to 6040
1705	No control mode when enabled:SV	Resolution: Reset the NMT node or send a

Alarm number	Alarm information	Troubleshooting
		fault reset command to 6040
1706	Undervoltage warning:SV	Resolution: Check the main input power; lower the undervoltage detection point parameter P06.36
1707	Drive overload warning:SV	Resolution: Verify that the motor UVW wires and encoder connections are correct
1708	Motor overload warning:SV	Resolution: Verify that the motor UVW wires and encoder connections are correct
1709	Parameter changes require power reconnection:SV	Resolution: Solution: Power cycle
1710	Servo not ready:SV	Resolution: Enable only when the servo is READY
1711	Write E2PROM frequent operation warning:SV	Resolution: Use communication write commands that do not store in EEPROM

Alarm number	Alarm information	Troubleshooting
1712	Forward overrun warning:SV	Resolution: Check the operating mode, give a negative command or manually rotate the motor out of the positive limit
1713	Negative overrun warning:SV	Resolution: Check the operating mode, give a positive command or manually rotate the motor out of the negative limit
1714	Position command overdrive:SV	Resolution: Reduce the set electronic gear ratio; reduce the input pulse frequency
1715	Absolute encoder angle initialization warning:SV	Resolution: Replace the motor
1716	Energy consumption brake overload:SV	Resolution: Check the brake resistor connection; check the internal resistor connection
1717	External regeneration discharge resistance small:SV	Resolution: Configure the resistor power according to specifications; check parameters
1718	Emergency stop:SV	Resolution: Normal DI function input, disconnect the input to clear the warning
1719	Origin regression error:SV	Resolution: Increase the P08.95 setting; reduce the homing search speed
1720	Encoder battery undervoltage:SV	Resolution: Check and replace the encoder battery
1721	Limit alignment warning:SV	Resolution: Issue a reverse command to exit the limit area; prohibit manual rotation of the motor

Alarm number	Alarm information	Troubleshooting
1722	EEPROM message initialization failed:SV	Resolution: Contact technical support
1723	EEPROM version update initialization failed:SV	Resolution: Contact technical support
1724	Slave module address initialization failed:SV	Resolution: Contact technical support
1725	Device model reading initialization failed:SV	Resolution: Contact technical support
1726	Device model configuration initialization failed:SV	Resolution: Contact technical support
1727	Maintenance date initialization failed:SV	Resolution: Contact technical support
1728	Device serial number initialization failed:SV	Resolution: Contact technical support
1729	Device name initialization failed:SV	Resolution: Contact technical support
1730	Power-on position deviation flag initialization failed:SV	Resolution: Contact technical support
1731	Encoder save value initialization failed:SV	Resolution: Contact technical support
1732	Failure configuration version initialization failed:SV	Resolution: Contact technical support
1733	Mechanical origin calibration mark initialization failed:SV	Resolution: Contact technical support
1734	Interface encoder initialization failed:SV	Resolution: Contact technical support
1735	Board-level test flag initialization failed:SV	Resolution: Contact technical support
1736	Machine test flag initialization failed:SV	Resolution: Contact technical support
1737	Servo parameter version	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
	initialization failed:SV	
1738	User comment message initialization failed:SV	Resolution: Contact technical support
1739	Servo parameter number initialization failed:SV	Resolution: Contact technical support
1740	Motor model initialization failed:SV	Resolution: Contact technical support
1741	Encoder origin offset value initialization failed :SV	Resolution: Contact technical support
1742	AD correction factor initialization failed:SV	Resolution: Contact technical support
1743	Fault history initialization failed:SV	Resolution: Contact technical support
1744	Running statistics initialization failed:SV	Resolution: Contact technical support
1745	COE communication parameter1 initialization failed:SV	Resolution: Contact technical support
1746	COE communication parameter2 initialization failed:SV	Resolution: Contact technical support
1747	COE communication parameter3 initialization failed:SV	Resolution: Contact technical support
1748	COE communication parameter4 initialization failed:SV	Resolution: Contact technical support
1749	COE communication parameter5 initialization failed:SV	Resolution: Contact technical support
1750	COE communication parameter6 initialization failed:SV	Resolution: Contact technical support
1751	COE communication parameter7 initialization failed:SV	Resolution: Contact technical support
1752	COE communication parameter8 initialization failed:SV	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1753	COE communication parameter9 initialization failed:SV	Resolution: Contact technical support
1754	Failure configuration value initialization failed:SV	Resolution: Contact technical support
1755	Servo parameter content re-information initialization failed:SV	Resolution: Contact technical support
1756	Servo parameter version re-informationinitialization failed:SV	Resolution: Contact technical support
1757	Servo parameter quantity re-information initialization failed:SV	Resolution: Contact technical support
1758	COE communication object initialization failed:SV	Resolution: Contact technical support
1759	CiA402 object initialization failed:SV	Resolution: Contact technical support
1760	CoolDrive object initialization failed:SV	Resolution: Contact technical support
1761	ESC chip error:SV	Resolution: Contact technical support
1762	ESC chip initialization failed:SV	Resolution: Contact technical support
1763	EtherCAT interrupt initialization failed:SV	Resolution: Contact technical support
1764	EtherCAT PDO mapping initialization failed:SV	Resolution: Contact technical support
1765	Motor voltage level setting failed:SV	Resolution: Contact technical support
1766	Motor pole logarithm setting failed:SV	Resolution: Contact technical support
1767	Motor power rating failed setting	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
	failed:SV	
1768	Motor rated speed setting failed:SV	Resolution: Contact technical support
1769	Motor maximum speed setting failed:SV	Resolution: Contact technical support
1770	Motor rated current setting failed:SV	Resolution: Contact technical support
1771	Motor maximum current setting failed:SV	Resolution: Contact technical support
1772	Motor rated torque setting failed:SV	Resolution: Contact technical support
1773	Motor maximum torque setting failed:SV	Resolution: Contact technical support
1774	Motor line resistance setting failed:SV	Resolution: Contact technical support
1775	Motor line inductance setting failed:SV	Resolution: Contact technical support
1776	Motor back EMF coefficient setting failed:SV	Resolution: Contact technical support
1777	Motor inertia setting failed:SV	Resolution: Contact technical support
1778	Motor type setting failed:SV	Resolution: Contact technical support
1779	Motor torque constant setting failed:SV	Resolution: Contact technical support
1780	Power board matching error:SV	Resolution: Contact technical support
1781	Control board matching error:SV	Resolution: Contact technical support
1782	IO initialization failed:SV	Resolution: Contact technical support
1783	CPU2 initialization failed:SV	Resolution: Contact technical support
1784	Encoder initialization failed:SV	Resolution: Contact technical support
1785	position control initialization failed:SV	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1786	Regulator parameter initialization failed:SV	Resolution: Contact technical support
1787	Overcurrent:SV	Resolution: Check the connection status between the motor and the drive, wiring sequence; reset alarm
1788	Overvoltage:SV	Resolution: Use the correct voltage source, series voltage regulator, series transformer
1789	Low Voltage:SV	Resolution: Confirm voltage wiring, power switch, use the correct voltage source or series transformer

Alarm number	Alarm information	Troubleshooting
1790	Abnormal motor matching:SV	Resolution: Replace with a matched motor; check and reinstall the motor encoder connector
2236	Command pointer wrong row	Resolution: Contact technical support
2237	Instruction is illegal:row	Resolution: delete the specified line number instruction in the background program/macros program
2238	Lack of instruction:row	Resolution: Contact technical support
2239	Lack of backgroud job	Resolution: create the corresponding background job; close the background job
2251	RET input error	Resolution: check background or MACRO job logic
2256	Job memory error	Resolution: check macro program logic
2257	Authorized, please restart	Resolution: no need to handle, just be aware
2258	No macro program	Resolution: edit the corresponding macro program
2299	Robot start in interference	Resolution: disable the enable switch in the interference zone attribute configuration interface
2610	Inst Cache Not Create,Code	Resolution: Contact technical support
2611	Plan Cache Not Create,Code	Resolution: Contact technical support
2612	Intro Cache Not Create,Code:	Resolution: Contact technical support
2613	Cache Clear Err,Code:	Resolution: Contact technical support
2614	Plan Thread Err Err,Code	Resolution: Contact technical support
2615	Intro Thread Err Err,Code:	Resolution: Contact technical support
2616	Transform Thread Err,Code:	Resolution: Contact technical support
2617	Inst Execute Err,Code:	Resolution: Contact technical support
2620	Plan Task Err,SubCode:	Resolution: Contact technical support
2621	Intro Task Err,SubCode:	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
2622	Transform Task Err,SubCode:	Resolution: Contact technical support
2680	Inst Cache Reset Err,Code:	Resolution: Contact technical support
2681	Plan Thread Reset Err,Code:	Resolution: Contact technical support
2682	Intro Thread Reset Err,Code:	Resolution: Contact technical support
2683	Transform Thread Reset Err,Code:	Resolution: Contact technical support
2684	Instruction cache length got Err,Code	Resolution: Contact technical support
2685	Instruction cache member gets Err,Code	Resolution: Contact technical support
2686	Instruction waiting executable state timed out	Resolution: Contact technical support
2687	Wait other thread to destroying timeout, Code	Resolution: Contact technical support
2860	Online must be in local	Resolution: switch to local mode
3093	Modbus TCP slave connection timeout, connection disconnected	Resolution: check slave timeout time setting; check slave network cable connection
3094	Modbus TCP slave received erroneous function code	Resolution: master sends instruction codes supported by slave
3103	Modbus TCP slave received invalid address	Resolution: check slave address setting range; check master sending address
3104	Modbus TCP slave data abnormality detected	Resolution: restart robot, reinitialize
3120	Joint	speed exceed limt. Resolution: check joint speed setting value
3513	system error,pointer is null	Resolution: Contact technical support
3610	on PlayMode,Unreachable Position	Resolution: adjust teaching position points
3624	Robot has been used for	daysContact MFR for maintenance! Resolution: maintenance is required, please contact the manufacturer

Alarm number	Alarm information	Troubleshooting
3626	planning section exceeds the allowed range	Resolution: contact technical personnel
3777	Code plate switch configuration failed	Resolution: Contact technical support
4031	Communication mode set ABN	Resolution: correctly set communication type
4045	the position num is lock, please change	Resolution: unlock operation in position information
4081	No cabinet type file	Resolution: check and confirm that the txt file of the control cabinet model exists in ROBOTTYPE, then restart
4096	System is error	Resolution: check if the path of the body parameter file and control cabinet file is correct
4097	Robot type file data is error	Resolution: Contact technical support to replace parameter file
4099	No robot type file	Resolution: Contact technical support to copy parameter file
8013	Parameter test error	Resolution: Check the setting values of limits and reduction ratios; contact after-sales personnel
8100	Resurrection error:SV	Resolution: Confirm the connection status of the regeneration resistor and reset parameter values

Alarm number	Alarm information	Troubleshooting
8101	Overload:SV	Resolution: Increase motor capacity or reduce load; reset alarm
8102	Excessive speed control error:SV	Resolution: Adjust the rate of change of the input signal or enable filtering; check encoder wiring
8103	Abnormal pulse control command:SV	Resolution: Correctly input pulse frequency; separate the pulse command signal line from other lines
8104	Excessive position control error:SV	Resolution: Adjust the set value according to the actual situation; reduce external load; reset alarm

Alarm number	Alarm information	Troubleshooting
8105	Abnormal voltage in the regeneration state:SV	Resolution: Recalculate the regeneration resistor value, reset correctly; reset alarm
8106	Abnormal position detector:SV	Resolution: Correct wiring; reinstall the connector; reduce encoder communication interference; power on again
8107	Correction anomaly:SV	Resolution: Remove CN1 wiring and perform automatic correction, then clear
8108	Emergency stop:SV	Resolution: Confirm that the emergency switch is in the off state; DI.EMGS reset
8109	Reverse limit anomaly:SV	Resolution: Confirm that the reverse limit switch is in the off state; confirm parameter set values
8110	Forward limit anomaly:SV	Resolution: Confirm that the forward limit switch is in the off state; confirm parameter set values

Alarm number	Alarm information	Troubleshooting
8111	IGBT Abnormal temperature:SV	Resolution: Increase motor capacity or reduce load; check drive output wiring
8112	Memory exception:SV	Resolution: Reset parameters and power on again if it occurs at startup; reset alarm if it occurs during operation

Alarm number	Alarm information	Troubleshooting
8113	Position detector output abnormal:SV	Resolution: Reset alarm; contact the distributor
8114	Outdated serial communication:SV	Resolution: Correctly set the communication timeout related parameter values; ensure correct wiring; reset alarm
8115	Abnormal main circuit power supply:SV	Resolution: Check if the RST power line is loose or not powered; reset alarm
8116	Advance overload warning:SV	Resolution: Change the motor overload output warning level parameter setting; reset alarm

Alarm number	Alarm information	Troubleshooting
8117	Encoder initial magnetic field error:SV	Resolution: Confirm that the motor ground terminal is properly grounded, the encoder signal line is normal
8118	Encoder internal error:SV	Resolution: Confirm that the motor ground terminal is properly grounded, the encoder signal line is normal
8119	Encoder internal data reliability error:S	Resolution: Confirm that the motor ground terminal is properly grounded, the encoder signal line is normal

Alarm number	Alarm information	Troubleshooting
8120	Encoder internal reset error:SV	Resolution: Check the encoder signal line, encoder power, encoder temperature, then power on again
8121	Encoder high voltage error:SV	Resolution: Check if the drive has a charging circuit, if battery installation is abnormal, power on again
8122	Encoder gray code error:SV	Resolution: Power on again, if the anomaly does not clear, replace the encoder
8123	Encoder count error:SV	Resolution: Send the motor back to the distributor or original factory for repair
8124	Motor data anomaly:SV	Resolution: Send the motor back to the original factory for inspection
8125	Motor collision error:SV	Resolution: Power on again, if the anomaly does not clear, replace the encoder

Alarm number	Alarm information	Troubleshooting
8126	Motor power line disconnection:SV	Resolution: Check if the motor power line and ground line are disconnected, wire correctly and ground. Power on again
8127	Abnormal encoder vibration:SV	Resolution: Confirm that the motor vibration range is within 2.5G; reset alarm or power on again
8128	Transfer box 26 PIN terminal disconnection:SV	Resolution: Connect the disconnected part, then power on again to clear. Contact the distributor
8129	Encoder internal communication exception:SV	Resolution: Check if the battery line is reversed, loose, reconnect the battery line and power on again

Alarm number	Alarm information	Troubleshooting
8130	Encoder temperature exceeds protection limit:SV	Resolution: Strengthen cooling to lower the temperature, then power on again to clear
8131	Encoder abnormal alarm status error:SV	Resolution: Confirm motor grounding, encoder signal line normal, reset alarm or power on again
8132	Full closed loop position control error is too large:SV	Resolution: Increase the P1.073 set value, check if the connector is loose, reset alarm
8133	CN5 disconnect:SV	Resolution: Confirm CN5 communication line, reset alarm
8134	Analog speed voltage input is too high:SV	Resolution: Check and confirm if there is a problem with the analog speed voltage source, reset alarm

Alarm number	Alarm information	Troubleshooting
8135	Drive function usage warning:SV	Resolution: Turn off unused filters, set P2.066 to 1 and power on again
8136	Electronic gear ratio setting error:SV	Resolution: Check if the electronic gear ratio setting is within the normal range, power on again
8137	Position detector output abnormal:SV	Resolution: Correctly set parameters, remove encoder interference, reset alarm
8138	Motor parameter identification	Resolution: Power on again to clear

Alarm number	Alarm information	Troubleshooting
	completed:SV	
8139	Automatic motor parameter detection error:SV	Resolution: Check parameter settings, motor normality, reset alarm
8140	Initial magnetic field detection error:SV	Resolution: Check the PM.003.U setting of the motor parameter recognition wizard, reset alarm

Alarm number	Alarm information	Troubleshooting
8141	Motor parameters not confirmed:SV	Resolution: Please execute or re-execute the motor parameter recognition wizard function
8142	Motor type switching causes parameters to be out of range:SV	Resolution: Check if the set values are consistent with the actual motor specifications, power on again
8143	Abnormal motor magnetic field:SV	Resolution: Check Hall components, feedback signal, power on reset
8144	Motor speed is too high:SV	Resolution: Check linear motor parameter settings, abnormal alarm reset

Alarm number	Alarm information	Troubleshooting
8145	Loss of feedback pulse:SV	Resolution: Check motor encoder feedback and parameter settings, eliminate interference sources, power on reset
8146	The position error is too large:SV	Resolution: Prohibit receiving upper-level commands during initial magnetic field detection period
8147	Motor type setting mismatch:SV	Resolution: Check PM.000 setting, power on reset

Alarm number	Alarm information	Troubleshooting
8148	Motor position feedback anomaly:SV	Resolution: Check if the feedback signal is abnormal, power on reset
8149	Position signal transfer box communication failed:SV	Resolution: Confirm whether the wiring of the position signal transfer box is correct, power on reset
8150	Absolute position lost:SV	Resolution: Check battery voltage, re-establish absolute origin coordinate

Alarm number	Alarm information	Troubleshooting
8151	Encoder voltage is too low:SV	Resolution: Replace the battery with the drive controller control power On. Automatic reset
8152	Absolute position circle overflow(encoder):S	Resolution: Check the number of motor revolutions, re-establish absolute origin coordinate. Power on reset
8153	Abnormal signal of optical ruler:SV	Resolution: Check whether the optical scale and read head installation is correct, abnormal alarm reset or power on reset
8154	Encoder vibration warning:SV	Resolution: Please return the motor to the dealer or original factory for repair. Abnormal alarm reset or power on reset
8155	Absolute position circle overflow(drive):SV	Resolution: Check the number of motor revolutions, re-establish absolute origin coordinate
8156	Encoder temperature alarm:SV	Resolution: Strengthen cooling to reduce temperature, then power on reset
8157	Absolute data I/O transmission error:SV	Resolution: Correct the I/O read timing, power on reset

Alarm number	Alarm information	Troubleshooting
8158	Motor type error:SV	Resolution: Check the parameter P2.069 setting value, re-power on
8159	Absolute position lost:SV	Resolution: Confirm whether the absolute origin coordinate has been established, it will be automatically cleared after establishment
8160	The error between inter-coord and encoder-coord too large:SV	Resolution: Re-establish absolute origin coordinate
8161	Encoder type not recognized:SV	Resolution: Immediately replace the motor
8162	Absolute position establishment not completed:SV	Resolution: Power on again and re-establish absolute origin coordinate
8163	Encoder disposal of incomplete alarms:SV	Resolution: Check whether the wiring is correct or whether the connector is loose and connect correctly. Power on reset
8164	Encoder number of turns error:SV	Resolution: If no improvement after abnormal alarm reset, please return the motor to the dealer or original factory for repair

Alarm number	Alarm information	Troubleshooting
8165	Encoder overspeed:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal
8166	Encoder memory error:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal
8167	Encoder single turn absolute position error:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal

Alarm number	Alarm information	Troubleshooting
8168	Encoder absolute number of turns error:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal
8169	Encoder internal error:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal
8170	Encoder parameter setting error:SV	Resolution: Confirm whether the encoder parameters have been written, power on again
8171	Encoder Z phase position lost:SV	Resolution: Please return the motor to the dealer or original factory for repair
8172	Encoder memory is busy:SV	Resolution: Confirm whether the motor ground terminal is indeed grounded, whether the encoder signal line is normal

Alarm number	Alarm information	Troubleshooting
8173	Encoder purge program error:SV	Resolution: Set P0.002 to -80 to check the encoder communication quality, power on
8174	Encoder plate number exception:SV	Resolution: Immediately replace the motor
8175	Excessive driver output current:SV	Resolution: Check the motor power line and power line connector configuration, re-wiring. Abnormal alarm reset
8176	Resurrection setting abnormal:SV	Resolution: Confirm the connection of the regeneration resistor, recalculate the regeneration resistor value and set the correct parameter value
8177	Regenerative resistance overload:SV	Resolution: Replace the regeneration resistor, remove the interference source; abnormal alarm reset

Alarm number	Alarm information	Troubleshooting
8178	Driver function usage alarm:SV	Resolution: Check the currently enabled filters, close the filters that are not in use
8179	Current sensing is disturbed:SV	Resolution: Remove the interference source, if not effective, send the driver back to the dealer or original factory for repair
8180	Automatic gain adjustment command exception:SV	Resolution: Confirm that the command has been issued; confirm the wiring between the upper computer and the driver. Abnormal alarm reset
8181	Automatic gain adjustment stop time is too short:SV	Resolution: Abnormal alarm reset
8182	Automatic gain adjustment inertia estimation anomaly:SV	Resolution: Abnormal alarm reset
8183	Regenerative resistance disconnection:SV	Resolution: Check whether the setting value of P1.053 is correct; abnormal alarm reset

Alarm number	Alarm information	Troubleshooting
8184	DSP firmware upgrade:SV	Resolution: Check whether the firmware upgrade has been done, then power on again
8185	Parameter reset failed:SV	Resolution: Confirm whether there was an abnormal power-off during the reset process, and check the power supply wiring and switch
8186	The absolute position coordinates of the driver and the motor do not match:SV	Resolution: Re-establish absolute origin coordinate
8187	Emergency stop(during deceleration):SV	Resolution: Confirm whether there is a DI(0x47) set in the parameters and triggered by this DI"
8188	Emergency stop:SV	Resolution: Confirm whether there is a DI(0x47) set in the parameters and triggered by this DI"
8189	Control power failure write failure:SV	Resolution: Increase motor capacity or reduce load, power on again, send the driver back for inspection
8190	Flexibe compensation parameter abnormality:SV	Resolution: Re-execute system analysis, set the flexibility compensation parameters
8191	SDO receiving overflow:S	Resolution: Reset NMT, use OD reset or DI.ARST clear

Alarm number	Alarm information	Troubleshooting
8192	PDO receiving overflow:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8193	TxPDO transmission failed:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8194	EEPROM address exceeded limit:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8195	EEPROM CRC calculation error:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8196	Write parameter function is limited:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8197	Bus communication timeout:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8198	Bus communication timeout:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8199	Bus hardware anomaly:SV	Resolution: Reset NMT, use OD reset or DI.ARST clear
8200	Bus data transmission error:SV	Resolution: Reset NMT, reset using OD or clear with DI.ARST

Alarm number	Alarm information	Troubleshooting
8201	Object dictionary data initialization error:SV	Resolution: Reset NMT, reset using OD or clear with DI.ARST
8202	CANopen Synchronization failure:SV	Resolution: Reset NMT or reset using 0x6040
8203	CANopen Sync signal is too fast:SV	Resolution: Reset NMT or reset using 0x6040
8204	CANopen Sync signal timeout:SV	Resolution: Reset NMT or reset using 0x6040
8205	Interpolation mode command failed:SV	Resolution: Turn off USB monitoring function, reset NMT
8206	SYNC Period error:SV	Resolution: Reset NMT or reset using 0x6040

Alarm number	Alarm information	Troubleshooting
8207	Communication synchronization failure:SV	Resolution: Reset NMT or reset using 0x6040
8208	Communication sync signal is too fast:SV	Resolution: Reset NMT or reset using 0x6040
8209	Communication sync signal timeout:SV	Resolution: Reset NMT or reset using 0x6040
8210	Position counter overflow warning:SV	Resolution: Perform homing procedure to establish absolute origin coordinates

Alarm number	Alarm information	Troubleshooting
8211	Indexing coordinates not defined:SV	Resolution: Perform homing, reset alarm
8212	PR Positioning timeout:SV	Resolution: Check PR completion conditions, reset alarm or power cycle
8213	PR Path number out of range:SV	Resolution: Check PR settings, reset alarm or power cycle
8214	Software forward limit:SV	Resolution: Set appropriate deceleration time, clears automatically after leaving the limit
8215	Software reverse limit:SV	Resolution: Set appropriate deceleration time, clears automatically after leaving the limit
8216	Position counter overflow:SV	Resolution: Set appropriate gear ratio, reset alarm
8217	Communication type absolute position command error:SV	Resolution: Establish absolute origin coordinates
8218	Indexing coordinate setting error:SV	Resolution: Check parameter P2.052 setting, reset alarm

Alarm number	Alarm information	Troubleshooting
8219	PR Special filter set too large:SV	Resolution: Check parameter P1.022 setting, reset alarm
8220	Driver internal update parameter program exception:SV	Resolution: Power cycle and re-execute motor parameter identification function
8221	Calculate program timeout:SV	Resolution: Power cycle; if AL520 still occurs, disable flexible compensation function
8222	System failure:SV	Resolution: If AL555 occurs, do not make any changes to the original machine, return it to the factory directly
8223	PR Basic numerical operations:SV	Resolution: Solution: Power cycle
8224	Secondary platform command error:SV	Resolution: Solution: Power cycle
8225	Password matching error:SV	Resolution: Check if the secondary platform project password matches the driver password setting
8226	Write EEPROM timeout failure:SV	Resolution: If the fault persists after multiple power cycles, replace the driver

Alarm number	Alarm information	Troubleshooting
8227	Read EEPROM timeout failure:SV	Resolution: If the fault persists after multiple power cycles, replace the driver
8228	Write EEPROM test error:SV	Resolution: If the fault persists after multiple power cycles, replace the driver
8229	Read EEPROM test error:SV	Resolution: If the fault persists after multiple power cycles, replace the driver
8230	Unrecognized encoder type:SV	Resolution: Check if the encoder model is correct. Replace the encoder
8231	No corresponding model motor:SV	Resolution: Check if the motor model is correct, set the correct motor model
8232	No corresponding model driver:SV	Resolution: Check if the driver model is correct, set the correct driver model
8233	Motor and driver current mismatch:SV	Resolution: Check if the driver model is correct, replace the driver
8234	FPGA does not match motor model:SV	Resolution: Upgrade the program or replace the motor
8235	Multi-turn absolute value encoder setting error:SV	Resolution: Reconfigure according to the motor nameplate or replace a matching motor
8236	DI function repetitive allocation:SV	Resolution: Reassign to different function numbers, then reapply control power

Alarm number	Alarm information	Troubleshooting
8237	Rotation mode upper limit is too larg:SV	Resolution: Change settings so that the mechanical single-turn position upper limit does not exceed 2^{31}
8238	Encoder parameter error:SV	Resolution: Replace the servo driver"
8239	Encoder communication error:SV	Resolution: Check encoder wiring, motor model settings, software version
8240	STO security status:SV	Resolution: Confirm STO function is enabled, STO power supply is normal, replace the driver
8241	STO input exception:SV	Resolution: Confirm STO power supply is normal, STO input is normal, replace the driver
8242	Buffer 5V abnormal voltage detection:SV	Resolution: Test 5V power supply voltage, replace the driver

Alarm number	Alarm information	Troubleshooting
8243	STO input circuit hardware diagnosis failed:SV	Resolution: Replace the driver
8244	PWM Buffer hardware diagnosis failed:SV	Resolution: After multiple power cycles, if the fault code still appears, replace the driver
8245	P-phase overcurrent:SV	Resolution: Collect current feedback from the background, check for abnormalities
8246	U-phase overcurrent:SV	Resolution: Check U phase current H0b38 at the time of failure
8247	V-phase overcurrent:SV	Resolution: Check V phase current H0b39 at the time of failure
8248	N-phase overcurrent:SV	Resolution: Collect current feedback from the background, check for abnormalities
8249	MCU the location command is updated too fast:SV	Resolution: Internal chip damage, replace the servo driver
8250	Encoder communication timeout:SV	Resolution: Check H0B30 function code; if the driver is abnormal, replace the driver

Alarm number	Alarm information	Troubleshooting
8251	Current acquisition fault:SV	Resolution: Eliminate interference sources; replace the driver
8252	FPGA current loop operation timeout:SV	Resolution: Contact electrical technicians
8253	Output short to ground:SV	Resolution: Rewire or replace the drive cable, replace the motor, replace the servo driver
8254	Speed protection:SV	Resolution: Follow UVW phase sequence wiring; power cycle; confirm encoder type and wiring

Alarm number	Alarm information	Troubleshooting
8255	Main circuit overvoltage:SV	Resolution: Measure the main circuit cable driver side input voltage to ensure it meets specifications
8256	Main circuit undervoltage:SV	Resolution: Measure the main circuit cable non-driver side input voltage to ensure it meets specifications

Alarm number	Alarm information	Troubleshooting
8257	Phase failure:SV	Resolution: Replace the cable and correctly connect the main circuit power lines
8258	PL signal error:SV	Resolution: After multiple power cycles, if the fault persists, use standard voltage signals

Alarm number	Alarm information	Troubleshooting
8259	Motor overspeed:SV	Resolution: Follow the correct UVW phase sequence wiring; check overspeed fault threshold
8260	Speed feedback overflow:SV	Resolution: Check if bit 9 of H0B30 is 1
8261	Angle identification:SV	Resolution: Confirm encoder communication is not interfered, check encoder hardware wiring
8262	Not open UVW angle recognition:SV	Resolution: Swap any two phases in the UVW phase sequence, then start recognition

Alarm number	Alarm information	Troubleshooting
8263	Motor overload:SV	Resolution: Confirm that the cable wiring is correct; power cycle the system. If the fault persists, replace the servo drive

Alarm number	Alarm information	Troubleshooting
8264	Motor blocking:SV	Resolution: Rewire according to the correct wiring diagram or replace the cable
8265	High junction temperature:SV	Resolution: Contact a technician
8266	Radiator overheating:SV	Resolution: Improve cooling conditions for the servo drive and reduce ambient temperature
8267	Air-cooled motor temperature is too high:SV	Resolution: Measure the forced air motor temperature to see if it is too high; cool down the motor

Alarm number	Alarm information	Troubleshooting
8268	The gain of one-click adjustment is too low:SV	Resolution: Manually enable vibration suppression to eliminate vibration; check if the positioning threshold is too small
8269	Encoder battery failure:SV	Resolution: Confirm whether the battery is connected during power-off; replace with a new battery of matching voltage
8270	Encoder multi-turn count error:SV	Resolution: Set 200D-15h=2 to clear the fault, power cycle the system, and replace the motor if necessary
8271	Encoder multi-turn count overflow:SV	Resolution: Power cycle the system. If necessary, perform a home return operation again
8272	Absolute encoder error:SV	Resolution: Ensure that function code H0B28 is not set to 0
8273	Absolute value encoder single turn solution error:SV	Resolution: Check if bit 7 of H0B28 is set to 1
8274	Encoder write failure:SV	Resolution: Check if the encoder cable shield and ground wire are properly connected
8275	Encoder overheating:SV	Resolution: Disconnect the enable signal and allow cooling for some time to lower the encoder temperature

Alarm number	Alarm information	Troubleshooting
8276	Excessive position deviation:SV	Resolution: Rewire according to the correct wiring diagram or replace the cable

Alarm number	Alarm information	Troubleshooting
8277	Position deviation overflow:SV	Resolution: Rewire according to the correct wiring diagram or replace the cable
8278	Encoder read-write check exception:SV	Resolution: Check the wiring; replace the servo motor if the encoder fails
8279	Position instruction increment is too large at atime:SV	Resolution: Confirm the maximum motor speed; check the communication timing from the host computer

Alarm number	Alarm information	Troubleshooting
8280	Position command increment persists too large:SV	Resolution: Confirm the maximum motor speed; check the communication timing from the host computer
8281	Command overflow:SV	Resolution: Stop sending limit commands when the host computer recognizes valid servo limit signals
8282	Rotation mode command exceeds maximum lap position:SV	Resolution: Set the target position within the upper and lower limits
8283	Soft limit position setting error:SV	Resolution: Reset and ensure 607D-01 is less than 607D-02
8284	Origin position setting error:SV	Resolution: Set the home offset within the soft limit and mechanical single-turn limits
8285	Gear ratio overrun:SV	Resolution: Set the gear ratio within the specified range
8286	No sync signal:SV	Resolution: Replace the master station and connect the IN and OUT ports in the correct order
8287	PDO mapping overrun:SV	Resolution: Check the number of self-indexed configurations in 1600h or 1A00h

Alarm number	Alarm information	Troubleshooting
8288	Sync lost:SV	Resolution: Check the communication line shielding, grounding of the drive, and network port condition
8289	Network state switching error:SV	Resolution: Check the host computer's network state switching program
8290	IRQ lost:SV	Resolution: Check the communication line shielding, grounding of the drive, and network port condition
8291	ESI test error:SV	Resolution: Check the XML version; reflash the XML file
8292	Bus read EEPROM failed:SV	Resolution: After multiple power cycles, if the error persists, replace the servo drive
8293	Bus update EEPROM failed:SV	Resolution: After multiple power cycles, if the error persists, replace the servo drive
8294	EtherCAT peripheral anomaly:SV	Resolution: Reflash the FPGA firmware; replace the servo drive
8295	Sync period setting error:SV	Resolution: Set the synchronization period to an integer multiple of 125us or 250us

Alarm number	Alarm information	Troubleshooting
8296	Excessive synchronization period error:SV	Resolution: Measure the 'synchronization period' and increase the manufacturer parameter 200E-21h
8297	Abnormal system parameters:SV	Resolution: Rewrite parameters; restore system parameters to default; replace the servo drive
8298	200h/2001h group parameters of the abnormal:SV	Resolution: Set the drive model correctly, then power cycle; rewrite bus-type motor parameters
8299	Logical configuration failure:SV	Resolution: MCU cannot establish communication with FPGA. Replace the servo drive

Alarm number	Alarm information	Troubleshooting
8300	Software version not match:SV	Resolution: Update FPGA or MCU software to match each other
8301	MCU run a timeout:SV	Resolution:. After multiple power cycles, if the fault persists, replace the servo drive
8302	Current loop time out:SV	Resolution: After multiple power cycles, if the fault persists, replace the servo drive
8303	Instruction update timeout:SV	Resolution: Disable unnecessary functions
8304	Abnormal program:SV	Resolution: Restore system parameters to default and power cycle; if still alarming, replace the servo drive
8305	Encryption chip check failure:SV	Resolution: Power off and restart to see if the fault persists; contact the manufacturer to reflash the encryption software
8311	1-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding
8312	2-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding
8313	3-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding
8314	4-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding

Alarm number	Alarm information	Troubleshooting
8315	5-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding
8316	6-Axis encoder communication anomaly:SV	Resolution: Check the encoder connection cable, separate the encoder cable from the power cable, and handle grounding
8317	1-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8318	2-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8319	3-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8320	4-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8321	5-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8322	6-Axis motor overcurrent:SV	Resolution: Correctly connect the motor cable; secure loose connectors; retune gain to eliminate oscillation
8323	1-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive
8324	2-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive

Alarm number	Alarm information	Troubleshooting
8325	3-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive
8326	4-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive
8327	5-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive
8328	6-Axis current acquisition bias anomaly:SV	Resolution: Check the sampling circuit or replace the drive
8329	1-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8330	2-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8331	3-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8332	4-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8333	5-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8334	6-Axis FO:SV	Resolution: Correctly connect the motor cable; tighten any loose connectors; retune gain to eliminate oscillation
8335	Bus overvoltage:SV	Resolution: Adjust the power supply voltage to within the product specification range
8336	Bus undervoltage:SV	Resolution: Increase the power supply

Alarm number	Alarm information	Troubleshooting
		capacity, replace the power supply, ensure that the power supply voltage is stable within the specification
8337	1-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8338	2-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8339	3-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8340	4-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8341	5-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8342	6-Axis excessive position deviation:SV	Resolution: Check encoder wiring and settings; adjust control parameters
8343	1-Axis motor stall:SV	Resolution: Adjust control parameters
8344	2-Axis motor stall:SV	Resolution: Adjust control parameters
8345	3-Axis motor stall:SV	Resolution: Adjust control parameters
8346	4-Axis motor stall:SV	Resolution: Adjust control parameters
8347	5-Axis motor stall:SV	Resolution: Adjust control parameters
8348	6-Axis motor stall:SV	Resolution: Adjust control parameters
8349	1-Axis motor overspeed:SV	Resolution: Optimize motor control parameters
8350	2-Axis motor overspeed:SV	Resolution: Optimize motor control parameters
8351	3-Axis motor overspeed:SV	Resolution: Optimize motor control parameters
8352	4-Axis motor overspeed:SV	Resolution: Optimize motor control parameters
8353	5-Axis motor overspeed:SV	Resolution: Optimize motor control

Alarm number	Alarm information	Troubleshooting
		parameters
8354	6-Axis motor overspeed:SV	Resolution: Optimize motor control parameters
8355	1-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8356	2-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8357	3-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8358	4-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8359	5-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8360	6-Axis motor overload:SV	Resolution: Appropriately reduce acceleration; check for blockage; re-establish initial position
8361	1-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8362	2-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8363	3-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8364	4-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8365	5-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8366	6-Axis encoder battery undervoltage:SV	Resolution: Replace the external battery
8367	EtherCAT abnormal bus	Resolution: Check network port

Alarm number	Alarm information	Troubleshooting
	communication:SV	communication cable connections
8368	STO abnormal terminal wiring:SV	Resolution: Check STO wiring
8369	Abnormal servo control:SV	Resolution: Change control authority to motion controller, power cycle
8370	Input power cord missing phase:SV	Resolution: Reconnect or replace cables; use correct input power with voltage meeting specifications
8371	1-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8372	2-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8373	3-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8374	4-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8375	5-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8376	6-Axis motor fast overload:SV	Resolution: Emergency brake and check if the robot exceeds the safe operating range
8377	1-Axis encoder multi-turn exception:SV	Resolution: Replace the encoder battery; perform a multi-turn reset operation on the encoder
8378	2-Axis encoder multi-turn exception:SV	Resolution: Replace the encoder battery; perform a multi-turn reset operation on the encoder
8379	3-Axis encoder multi-turn exception:SV	Resolution: Replace encoder battery; perform encoder multi-turn zeroing operation
8380	4-Axis encoder multi-turn exception:SV	Resolution: Replace encoder battery; perform encoder multi-turn zeroing operation

Alarm number	Alarm information	Troubleshooting
8381	5-Axis encoder multi-turn exception:SV	Resolution: Replace encoder battery; perform encoder multi-turn zeroing operation
8382	6-Axis encoder multi-turn exception:SV	Resolution: Replace encoder battery; perform encoder multi-turn zeroing operation
8383	Regenerative discharge resistance overload:SV	Resolution: Reduce start-stop frequency; increase acceleration/deceleration time constant; reduce load inertia
8384	1-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8385	2-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8386	3-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8387	4-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8388	5-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8389	6-Axis encoder overspeed failure:SV	Resolution: Reduce motor operating speed; check encoder, wiring, and parameters and motor parameters
8390	EEPROM parameter exception:SV	Resolution: Initialize driver parameters; check driver hardware
8391	1-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold

Alarm number	Alarm information	Troubleshooting
8392	2-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold
8393	3-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold
8394	4-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold
8395	5-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold
8396	6-Axis EtherCAT process data error:SV	Resolution: Check if there is any abnormality in the actual position feedback; reduce acceleration or increase threshold
8397	1-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it
8398	2-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it
8399	3-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it
8400	4-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it
8401	5-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it

Alarm number	Alarm information	Troubleshooting
8402	6-Axis excessive deviation of power-on position:SV	Resolution: Check if the mechanical position has changed, confirm that the mechanical zero point is normal, and then clear it
8403	1-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8404	2-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8405	3-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8406	4-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8407	5-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8408	6-Axis output missing phase:SV	Resolution: Check motor power lines; check power line connections at the driver output terminals
8409	1-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring
8410	2-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring
8411	3-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring
8412	4-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring

Alarm number	Alarm information	Troubleshooting
8413	5-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring
8414	6-Axis encoder status flag bit exception:SV	Resolution: Replace encoder battery; clear encoder status flag; check wiring
8415	24V control power undervoltage:SV	Resolution: Check 24V control power supply wiring, load; replace power supply; replace driver
8416	CPU operation exception:SV	Resolution: Standardize driver peripheral wiring, add anti-interference measures; replace driver
8420	Abnormal system parameters:SV	Resolution: Restore system parameters to initialization, then rewrite parameters
8421	The programmable logic configuration is faulty:SV	Resolution: MCU cannot establish communication connection with FPGA. Replace servo driver

Alarm number	Alarm information	Troubleshooting
8422	FPGA software version too early:SV	Resolution: Contact electrical technicians
8423	Internal program exception:SV	Resolution: Restore system parameters to initialization, then power on; replace servo driver
8424	Programmable logic interrupt failure:SV	Resolution: Replace servo driver
8425	Pulse output overspeed:SV	Resolution: Contact technicians
8426	Parameter storage failure(read write):SV	Resolution: Contact technicians
8427	200h/2001h group parameters of the abnormal:SV	Resolution: Set the wrong driver model, power on, then set correctly, then power on again
8428	Product matching fault:SV	Resolution: Check if the encoder model is correct. Replace encoder
8429	Servo ON command invalid fault:SV	Resolution: Turn off the servo enable signal from the upper computer; turn off redundant servo enable signals
8430	Absolute position function with encoder matching failure:SV	Resolution: Reset 200D-01h according to the motor nameplate or replace with a matching motor

Alarm number	Alarm information	Troubleshooting
8431	DI function duplication:SV	Resolution: Reassign to different function numbers, then power on control
8432	DO distribution of transfinite:SV	Resolution: Reassign to different function numbers
8433	Data check error in motor ROM or parameters not saved:SV	Resolution: Replace with mutually compatible drivers and motors
8434	Over current 1:SV	Resolution: Contact electrical technicians
8435	Over current 2:SV	Resolution: Collect current feedback from the backend, check for abnormalities
8436	D/Q axial current overflow fault:SV	Resolution: Replace the servo driver
8437	FPGA system sampling operation timrd out:SV	Resolution: Internal chip damage, replace the servo driver

Alarm number	Alarm information	Troubleshooting
8438	The output shorted to the ground:SV	Resolution: Re-wire or replace the driver power line; replace the motor; replace the servo driver
8439	UVW phase sequence errors:SV	Resolution: Contact technicians
8440	coaster:SV	Resolution: Wire according to correct UVW phase sequence; power cycle; set appropriate parameters

Alarm number	Alarm information	Troubleshooting
8441	Main circuit is overvoltage:SV	Resolution: Measure the input voltage on the driver side of the main circuit cable to ensure it meets specifications
8442	Main circuit electrical undervoltage:SV	Resolution: Measure the input voltage on both non-driver side and driver side of the main circuit cable

Alarm number	Alarm information	Troubleshooting
8443	The motor speed:SV	Resolution: Wire according to correct UVW phase sequence; check overspeed fault threshold
8444	Angle identification failure:SV	Resolution: Confirm encoder communication is not interfered with, check encoder hardware wiring
8445	Drive overload:SV	Resolution: Contact technicians

Alarm number	Alarm information	Troubleshooting
8446	Motor overload:SV	Resolution: Wire according to correct wiring diagram; replace larger capacity driver and matching motor

Alarm number	Alarm information	Troubleshooting
8447	Motor blocked:SV	Resolution: Rewire according to correct configuration, or replace cables
8448	Radiator overheating:SV	Resolution: Improve cooling conditions for the servo driver, reduce ambient temperature
8449	External encoder ruler is faulty:SV	Resolution: Contact technicians
8450	AD sampling overvoltage:SV	Resolution: Contact technicians
8451	Encoder battery failure:SV	Resolution: Measure battery voltage. Replace with a new battery of matching voltage
8452	Encoder multiple turn count error:SV	Resolution: Set 200D-15h=2 to clear the fault; if still faulty, replace the motor

Alarm number	Alarm information	Troubleshooting
8453	Encoder multiple turn count overflow:SV	Resolution: Execute H0D-20=2, power cycle; re-perform home position return operation
8454	Encoder interference:SV	Resolution: Rewire; replace encoder cable; replace servo motor
8455	Abnormal encoder data:SV	Resolution: Confirm encoder cable connections are correct, or check for broken wires or poor connections
8456	Encoder encoder check is abnormal:SV	Resolution: Rewire; replace encoder cable; replace servo motor
8457	Excessive position deviation:SV	Resolution: Rewire according to correct configuration, or replace cables

Alarm number	Alarm information	Troubleshooting
8458	All closed loop position deviation too large:SV	Resolution: Rewire according to correct configuration, or replace cables
8459	Electronic gear ratio setting exceeds the limit:S	Resolution: Set the gear ratio within the specified range
8460	The upper and lower limits of the software location are incorrect:SV	Resolution: Reset parameters
8461	The origin position is incorrect:SV	Resolution: Reset parameters

Alarm number	Alarm information	Troubleshooting
8462	Synchronization loss:SV	Resolution: Check if the communication line has shielding, if the driver is properly grounded, and if the network port is damaged
8463	Z signal loss:SV	Resolution: Contact technicians
8464	Network initialization failed:SV	Resolution: Burn FPGA firmware; driver fault. Replace the servo driver
8465	Synchronization period is incorrect:SV	Resolution: Modify the synchronization period setting value to be an integer multiple of the command scheduling period
8466	Synchronization period error is too large:SV	Resolution: Increase the manufacturer parameter 200E-21h
8467	Full loop use error:SV	Resolution: Use external encoder feedback mode, i.e., 200F-01h can only be 1
8470	Drive other axes abnormal	Resolution: Replace the encoder; check encoder wiring; calibrate encoder sine/cosine AD coefficients

Alarm number	Alarm information	Troubleshooting
8471	Motor brake wiring broke	Resolution: Replace the encoder; check motor encoder wiring; add ferrite core
8472	Drive short circuit 2	Resolution: Check mechanical load transmission method; reduce acceleration of position target value from the host computer
8473	Dynamic brake overload	Resolution: Increase dynamic braking stop interval time
8474	ESC EEPROM error	Resolution: Replace the driver
8475	2nd Excessive velocity following error	Resolution: Reduce actual mechanical load of the motor; optimize control parameters; check motor brake wiring
8476	Parameter Checksum Error 1	Resolution: Set the power supply voltage within specifications, perform parameter initialization
8477	Parameter Format Error 1	Resolution: Servo unit may be faulty. Replace the servo unit
8478	System Checksum Error 1	Resolution: Measure power supply voltage; confirm power-off time; replace the servo unit

Alarm number	Alarm information	Troubleshooting
8479	Main Circuit Detector Error	Resolution: Servo unit may be faulty. Replace the servo unit
8480	Parameter Setting Error 1	Resolution: Confirm that the servo unit and servo motor capacities match
8481	Encoder Output Pulse Setting Error	Resolution: Confirm Pn212, set Pn212 to an appropriate value
8482	Parameter Combination Error	Resolution: Reduce electronic gear ratio; increase JOG movement speed in the program"
8483	Semi-closed/Fully-closed Loop Control Parameter Setting Error	Resolution: Confirm Pn002.3 setting to match full-closed loop module
8484	Combination Error	Resolution: Ensure that the servo unit and servo motor capacities match
8485	Unsupported Device Alarm	Resolution: Confirm product combination, change to a compatible combination
8486	Abnormal pitch setting of Linear encoder grating ruler	Resolution: Contact technicians
8487	Canceled Servo ON Command Alarm	Resolution: Power cycle the servo unit or perform a software reset

Alarm number	Alarm information	Troubleshooting
8488	Overcurrent or Heat Sink Overheated	Resolution: Modify wiring if incorrect; replace cable if short circuit occurs
8489	Regeneration Error	Resolution: Confirm connection of external regeneration resistor and Pn600 value

Alarm number	Alarm information	Troubleshooting
8490	Regenerative Overload	Resolution: Set power supply voltage within specifications; adjust operating conditions
8491	Main Circuit Power Supply Wiring Error	Resolution: Ensure the power setting matches the used power supply
8492	Overvoltage	Resolution: Measure power supply voltage, adjust AC/DC power supply voltage to within range

Alarm number	Alarm information	Troubleshooting
8493	Undervoltage	Resolution: Measure power supply voltage; adjust power supply voltage to normal range, increase power capacity
8494	Main-Circuit Capacitor Overvoltage	Resolution: Replace the servo unit
8495	Overspeed	Resolution: Confirm servo motor wiring issues; reduce command value or adjust gain
8496	Overspeed of Encoder Output Pulse Rate	Resolution: Reduce encoder frequency division pulse setting; reduce motor speed
8497	Vibration Alarm	Resolution: Reduce motor speed or speed loop gain, correctly set inertia ratio
8498	Autotuning Alarm	Resolution: Reduce load to below allowable inertia ratio
8499	Overload: High Load	Resolution: Confirm wiring; verify motor overload characteristics and running commands, motor speed

Alarm number	Alarm information	Troubleshooting
8500	Overload: Low Load	Resolution: Confirm wiring; verify motor overload characteristics and running commands, motor speed
8501	Dynamic Brake Overload	Resolution: Confirm the operating status; reduce the command speed of the servo motor; reduce the inertia ratio
8502	Dynamic Brake Overload	Resolution: Confirm the operating status; reduce the command speed of the servo motor; reduce the inertia ratio
8503	Overload of Surge Current Limit Resistor	Resolution: Reduce the number of ON/OFF cycles of the main circuit power; replace the servo unit
8504	Heat Sink Overheated	Resolution: Reduce the ambient temperature; change the reset method of the alarm; replace the servo unit
8505	Built-in Fan in SERVOPACK Stopped	Resolution: Confirm if foreign objects are stuck; replace the servo unit

Alarm number	Alarm information	Troubleshooting
8506	Encoder Backup Error	Resolution: Confirm the encoder connection; perform encoder setting operations
8507	Encoder Checksum Error	Resolution: Replace the servo motor; replace the servo unit
8508	Absolute Encoder Battery Error	Resolution: Connect the battery correctly; replace the battery; replace the servo unit
8509	Encoder Data Error	Resolution: Replace the servo motor; correctly wire the encoder peripherals
8510	Encoder Overspeed	Resolution: Replace the servo motor; replace the servo unit

Alarm number	Alarm information	Troubleshooting
8511	Encoder Overheated	Resolution: Adjust the ambient temperature of the servo motor to below 40°C
8512	Encoder grating ruler error	Resolution: Contact technical support
8513	Encoder module error	Resolution: Contact technical support
8514	External Encoder Error	Resolution: Set the origin position without moving the motor; replace the external encoder
8515	External Encoder Error of Module	Resolution: Replace the external encoder; replace the serial conversion unit
8516	External Encoder Error of Sensor	Resolution: Replace the external encoder
8517	External Encoder Error of Position	Resolution: Follow the handling method specified in the manufacturer's user manual
8518	External Encoder Overspeed	Resolution: Use the external encoder below its maximum speed
8519	External Encoder Overheated	Resolution: Replace the external encoder
8520	Current Detection Error 1	Resolution: Replace the servo unit
8521	Current Detection Error 2	Resolution: Replace the servo unit
8522	Current Detection Error 3	Resolution: Replace the servo unit; repair the motor cable
8523	System Alarm 0	Resolution: Replace the servo unit
8524	System Alarm 1	Resolution: Replace the servo unit

Alarm number	Alarm information	Troubleshooting
8525	System Alarm 2	Resolution: Replace the servo unit
8526	System Alarm 3	Resolution: Replace the servo unit
8527	System Alarm 4	Resolution: Replace the servo unit
8528	Servo Overrun Detected	Resolution: Confirm if there is a problem with the motor wiring; replace the servo motor; replace the servo unit
8529	Phase error detection	Resolution: Contact technical support
8530	Magnetic pole sensor failure	Resolution: Contact technical support
8531	Inconsistent Phase information	Resolution: Contact technical support
8532	Magnetic pole detection failed	Resolution: Contact technical support
8533	Overtravel detection during magnetic pole detection	Resolution: Contact technical support
8534	Incomplete magnetic pole detection	Resolution: Contact technical support
8535	Magnetic pole detection exceeds the range of activity	Resolution: Contact technical support
8536	Magnetic pole detection failed 2	Resolution: Contact technical support
8537	Absolute Encoder Clear Error and Multiturn Limit Setting Error	Resolution: Replace the servo motor; replace the servo unit
8538	Encoder Communications Error	Resolution: Confirm the encoder cable; replace the servo unit; correctly wire the encoder peripherals

Alarm number	Alarm information	Troubleshooting
8539	Encoder Communications Position Data Error	Resolution: Confirm if there is a problem with the layout of the encoder cable
8540	Encoder Communications Timer Error	Resolution: Implement anti-interference for encoder wiring; reduce mechanical vibration or correctly install the servo motor
8541	Encoder Parameter Error	Resolution: Replace the servo motor; replace the servo unit
8542	Encoder Echoback Error	Resolution: Confirm the encoder wiring; the maximum wiring distance is 50m
8543	Multiturn Limit Disagreement	Resolution: Correctly set the value of Pn205; replace the servo unit

Alarm number	Alarm information	Troubleshooting
8544	Full closed-loop serial conversion unit communication failure	Resolution: Confirm the external encoder wiring; use the correct specified cable
8545	Feedback on communication failure of the purchasing module(Timer stop)	Resolution: Correctly wire around the serial conversion unit
8546	Position Error Overflow	Resolution: Confirm if there are any poor contacts in the motor cable or encoder cable
8547	Position Error Overflow	Resolution: Confirm the position deviation at servo ON; correct the position deviation too large alarm value
8548	Position Error Overflow	Resolution: Clear the position deviation; set the correct speed limit value
8549	Motor-load Position Error Overflow	Resolution: Reverse install the external encoder or set “external encoder rotation direction reversed

Alarm number	Alarm information	Troubleshooting
8550	Main Circuit Cable Open Phase	Resolution: Confirm if there is a problem with the power wiring; correct the power imbalance
8551	Soft Position Limit Error	Resolution: Set the value of 607D within the correct range
8552	Home Offset Set Error	Resolution: Set the values of 607D and 607C within the correct range
8553	Sync Lost	Resolution: Use shielded twisted pair cable with shielding function; wire according to standards
8554	Bus Interrupt	Resolution: Use shielded twisted pair cable with shielding function; wire according to standards
8555	Ethercat Init Failure	Resolution: Burn the corresponding xml file; replace the servo driver
8556	Encoder Communication Timeout	Resolution: Contact technical support
8557	Encoder Data Abnormal	Resolution: Contact technical support
8558	Drive Soft Start Error	Resolution: Contact technical support
8559	Encoder Overspeed During Battery	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
	Drive	
8560	Encoder Full Absolute Value State	Resolution: Contact technical support
8561	Encoder Counting Error	Resolution: Contact technical support
8562	Encoder Multi-turn Value Overflow	Resolution: Contact technical support
8563	Encoder Overtemperature	Resolution: Contact technical support
8564	Encoder Multi-turn Value Abnormal	Resolution: Contact technical support
8565	Encoder Battery Undervoltage Fault	Resolution: Contact technical support
8566	High SYNC commun Error	Resolution: Contact technical support
8567	Drive Internal Error 2	Resolution: Contact technical support
8568	Drive short circuit 3	Resolution: Contact technical support
8569	Servo Error Code Unknown	Resolution: Contact technical support
8570	Bus Undervoltage Alarm	Resolution: Contact technical support
8571	Ecat Commu Break	Resolution: Contact technical support
8572	Motor Phase A Current Overcurrent	Resolution: Contact technical support
8573	Motor Phase B Current Overcurrent	Resolution: Contact technical support
8574	Motor Phase C Current Overcurrent	Resolution: Contact technical support
8575	Q Axial Current Too Large	Resolution: Contact technical support
8576	D Axial Current Too Large	Resolution: Contact technical support
8577	Hardware Failure(short out)	Resolution: Contact technical support
8578	Current Sensor Zero Drift Value Too Large	Resolution: Contact technical support
8579	Bus Voltage is Overvoltage	Resolution: Contact technical support
8580	Bus Voltage is Undervoltage	Resolution: Contact technical support
8581	Motor Temperature Overheats	Resolution: Contact technical support
8582	Drive Overtemperature	Resolution: Contact technical support
8583	Drive Initialization Error	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
8584	Encoder CRC Checksum Error	Resolution: Contact technical support
8585	Velocity Overshoot	Resolution: Contact technical support
8586	Position Overshoot	Resolution: Contact technical support
8587	Brake Fault Alarm	Resolution: Contact technical support
8588	Discharge Exceeds Maximum Voltage	Resolution: Contact technical support
8589	Motor Phase A Detection Failure	Resolution: Contact technical support
8590	Discharge Resistor Detection Failure	Resolution: Contact technical support
8591	Motor Temperature 2 Overtemperature	Resolution: Contact technical support
8592	Motor Temperature 3 Overtemperature	Resolution: Contact technical support
8593	Discharge Overtemperature	Resolution: Contact technical support
8594	AB Encoder Fault	Resolution: Contact technical support
8595	HALL Sensor Fault	Resolution: Contact technical support
8596	Transmission Fault	Resolution: Contact technical support
8597	Tamagawa Encoder Communication Abnormality	Resolution: Contact technical support
8598	HALL Encoder UVW Abnormality	Resolution: Contact technical support
8599	Incremental Encoder Z Detection Abnormality	Resolution: Contact technical support
8600	Transmission Fault	Resolution: Contact technical support
8601	Fan 0 Fault	Resolution: Contact technical support
8602	Fan 1 Fault	Resolution: Contact technical support
8603	Drive Validation Error	Resolution: Contact technical support
8604	Bleed Fault	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
8605	Hardware Failure	Resolution: Contact technical support
8606	Bus Overvoltage Alarm	Resolution: Contact technical support
8607	Encoder Forward AB Pulse Loss	Resolution: Contact technical support
8608	Encoder Battery Undervoltage Warning	Resolution: Contact technical support
8609	Encoder 2 Overspeed During Battery Operation	Resolution: Contact technical support
8610	Encoder 2 Absolute Position State	Resolution: Contact technical support
8611	Encoder 2 Counting Error	Resolution: Contact technical support
8612	Encoder 2 Multi-turn Value Overflow	Resolution: Contact technical support
8613	Encoder 2 Overheating	Resolution: Contact technical support
8614	Encoder 2 Multi-turn Value Abnormality	Resolution: Contact technical support
8615	Encoder 2 Battery Undervoltage Fault	Resolution: Contact technical support
8616	Encoder 2 Battery Undervoltage Warning	Resolution: Contact technical support
8617	Encoder 2 CRC Checksum Error	Resolution: Contact technical support
8618	Encoder 2 Communication Timeout	Resolution: Contact technical support
8619	Encoder 2 Data Abnormality	Resolution: Contact technical support
8620	Security Selection Module Detection Failure Alarm	Resolution: Contact technical support
8621	Feedback Selection Module Detection Failure Alarm	Resolution: Contact technical support
8622	Security Selection Module Not Supported Alarm	Resolution: Contact technical support
8623	Feedback Selection Module Not	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
	Supported Alarm	
8624	Abnormal Signal Input Time for Safety Function	Resolution: Contact technical support
8625	Abnormal Torque Command and Torque Feedback Difference	Resolution: Contact technical support
8626	Abnormal Position Command	Resolution: Contact technical support

7.2 Warnings

In the case of the warnings listed below, the servo driver will not be powered off, the robot will pause, and the robot is not allowed to be started directly. If it is desired to move the robot, make ensure that the system does not have a warning, and press the Cancel button to reset the warning. The warning is included in the error message.

Alarm number	Alarm information	Troubleshooting
113	Modbus TCP master connection timeout	Resolution: Check network connection
114	Modbus TCP master illegal data address	Resolution: Check the master station configuration
115	External Device Communicate Unnormal	Resolution: Check communication equipment
116	Position record failed	Resolution: Contact technical support
166	the record job name is different	Resolution: Reopen the job to be executed, contact technical support
169	the job of string not exist	Resolution: Check if the job is legal or contact technical support
243	VS Para Init Error	Resolution: Check the damping parameter configuration
341	Automatically clear memory	Resolution: Contact professional technical support
400	Unreachable Position:	Resolution: Reteach the trajectory, change

Alarm number	Alarm information	Troubleshooting
		the motion type to MOVJ
401	Prepare to reach unreachable Position	Resolution: Adjust the teaching position
608	External stop	Resolution: Check safety door related line
616	Job is hold because of SafeGateRequest	Resolution: Handle according to the warning level, check if the safety door needs to be opened
617	Offline external stop	Resolution: Check the authenticity of the external stop signal
722	Adapt configure failure	Resolution: Contact technical support
723	Adapt algorithm error	Resolution: Contact technical support
724	Collision caused a robot to pause:	Resolution: Check the robot position, clear the collision
725	Torsion calculation error	Resolution: Contact technical support
726	GravityCompensation initializaton error	Resolution: Contact technical support
727	CollisionDetection initializaton error	Resolution: Contact technical support
728	Friction data error	Resolution: Contact technical support
729	Load identification execute exception	Resolution: Contact technical support
730	Dynamic Para Configuration Error	Resolution: Check configuration
731	GravityCompensation Error	Resolution: Contact technical support
751	Self-learning collision detection is only supported by Planning Library 2.0"	Resolution: Switch the planning library to 2.0
752	No self-learning has been performed, so cannot be detect	Resolution: Execute the self-learning process for this area
753	Collision detection parameter storage exceeded limit	Resolution: Optimize the collision detection area, reduce collision detection instructions
848	No local boot	Resolution: Verify the situation and switch

Alarm number	Alarm information	Troubleshooting
		the current mode of the robot
849	External pause, no starting	Resolution: Resolve the pause and then start the program
857	Please wait a moment,then cancel alarm	Resolution: Resolve system alarms and reset
864	Driver can not operate	Resolution: Check if the drive has any alarms or abnormalities. Clear the alarm and retry
869	Off-line external pause	Resolution: Reset the external pause on the upper machine
880	Axis motion not enable	Resolution: Switch to local mode to start the program. Enable axis movement, restart the control cabinet
893	LENZE DCF is invalid	Resolution: Contact after-sales personnel
898	MCS cycle is abnormal	Resolution: Contact professional technical support
904	Alarm C registration Error	Resolution: Contact technical support
905	Alarm C array exceed limit	Resolution: Contact technical support
907	User memory exceed limit	Resolution: Contact technical support
911	Input pramaters invalid	Resolution: Contact professional technical support
912	Pnio abnormal,please check!	Resolution: Contact technical support
913	Driver Temperature protect:SV	Resolution: Open the cabinet door to cool down, adjust the ambient temperature
914	Driver Utilisation protect:SV	Resolution: Check if the load is too heavy, excessive load inertia, high temperature, drive parameters
915	BECKHOFF disconnect,check it	Resolution: Check the module connection line
918	NT151 disconnect,check it	Resolution: Contact professional technical support

Alarm number	Alarm information	Troubleshooting
931	the application is invalid during application instruction initialization	Resolution: contact technical support
932	the numbers of application instruction beyond bounds during initialization	Resolution: contact technical support
933	the numbers of instruction formats beyond bounds during initialization	Resolution: Contact technical support
984	ISA feedback: fan abnormal	Resolution: Contact technical support
985	high control cabinet temperature	Resolution: Open the cabinet door to cool down, adjust the ambient temperature
995	Application shortcut key parameter storage CF card failed	Resolution: Contact professional technicians
999	Alarm code	does not exist. Resolution: Contact technical support
1004	Dnet connection break:slave	Resolution: Check communication cables and power
1006	Robot has been used for	daysContact MFR for maintenance! Resolution: Contact technical support
1007	Invalid registration code	Resolution: Confirm the registration code
1022	T6 not quadrature matrix	-
1024	dnet data lose:	Resolution: Troubleshoot network connection issues
1040	The current job cannot be terminated while in an interrupt job	Resolution: Stop the execution of the interrupt job before closing
1111	END instruction cannot follow MOV SPI instruction	Resolution: Check the job content
1791	Control power undervoltage warning:Axis	Resolution: Restore the control power voltage to normal
1792	Ambient temperature too high:Axis	Resolution: Lower the ambient temperature
1793	STO trigger:Axis	Resolution: Restore the emergency stop

Alarm number	Alarm information	Troubleshooting
		switch to normal
1794	Torque monitoring saturation warning:Axis	Resolution: Actual torque current is below the set value of 0x20A7
1795	CPU1 overload warning:Axis	Resolution: Restart or replace the driver
1796	CPU2 overload warning:Axis	Resolution: Restart or replace the driver
1797	Changed the valid parameters for re-energizing:Axis	Resolution: Soft reset or restart the driver
1798	Emergency stop switch triggered:Axis	Resolution: Restore the DI channel switch configured as an emergency stop switch
1799	Mechanical origin not calibrated:Axis	Resolution: Perform homing operation on the driver, exit PP or CSP mode
1900	Motor overload warning:Axis	Resolution: Reduce motor load, increase running acceleration and deceleration time, etc
1901	Speed limit warning:Axis	Resolution: Reduce the speed target value or speed profile value
1902	DC bus undervoltage warning:Axis	Resolution: Adjust the driver input power, replace the driver
1903	Fault history exception:Axis	Resolution: Soft reset or restart the driver
1904	AD invalid correction factor:Axis	Resolution: Soft reset or restart the driver
1905	CoE abnormal communication parameters:Axis	Resolution: Soft reset or restart the driver
1906	Servo parameters restore default values:Axis	Resolution: Ensure that the used parameters are correct, the servo parameter version is not 0
1907	Does not support setting control mode:Axis	Resolution: Set the correct control mode on the upper computer
1908	First encoder battery undervoltage	Resolution: Check the encoder battery wiring

Alarm number	Alarm information	Troubleshooting
	warning:Axis	and ensure reliable connection
1909	Second encoder battery undervoltage warning:Axis	Resolution: Check the encoder battery
1910	Drive internal warning:Axis	Resolution: Replace the driver
1911	First encoder communication anomaly warning:Axis	Resolution: Check the encoder
1912	Second encoder communication anomaly warning:Axis	Resolution: Check the encoder
1913	First encoder communication timeout warning:Axis	Resolution: Replace the encoder or check the wiring
1914	Second encoder communication timeout warning:Axis	Resolution: Replace the encoder or check the wiring
1915	First encoder data anomaly warning:Axis	Resolution: Replace the encoder or perform a soft reset and restart
1916	Second encoder data anomaly warning:Axis	Resolution: Replace the encoder or perform a soft reset and restart
1917	Position limit warning:Axis	Resolution: Make the position value within the soft limit range, check the limit switches
1918	Abnormal alarm of location planning parameters:Axis	Resolution: Set planning acceleration, deceleration, speed, etc. parameters
1919	SDO write failure alert:Axis	Resolution: Execute the fault clearance command
1920	Origin configuration error:Axis	Resolution: Correctly set the homing-related parameters
1921	Encoder internal warning:Axis	Resolution: Restore the encoder battery voltage to normal, perform encoder zeroing
1922	Energy consumption brake resistance	Resolution: Correctly set the servo

Alarm number	Alarm information	Troubleshooting
	overload:Axis	parameters
1923	Frequency divider pulse output setting failure:Axis	Resolution: Modify the value of H05-17
1924	The return to the origin times out:Axis	Resolution: Contact technical support
1925	Encoder battery warning:Axis	Resolution: Replace the battery
1926	DI emergency brake:Axis	Resolution: Resolve the emergency state

Alarm number	Alarm information	Troubleshooting
1927	Motor overload warning:Axis	Resolution: Contact technical support
1928	Motor power line broken:Axis	Resolution: Check the phase current effective value, check the motor power line
1929	The parameters must be powered on again to take effect:Axis	Resolution: Confirm that the effective time is the power-on function code. Power on
1930	Parameter storage channel:Axis	Resolution: Check the upper computer system
1931	Forward overshoot warning:Axis	Resolution: Check whether the DI terminal logic is valid

Alarm number	Alarm information	Troubleshooting
1932	Reverse overrange warning:Axis	Resolution: Check whether the DI terminal logic is valid
1933	Internal failure of encoder:Axis	Resolution: Repeatedly connect the power, replace the servo motor
1934	Error setting back to zero mode:Axis	Resolution: Set the parameter values according to specifications
1935	Parameter identification failure:Axis	Resolution: Contact technical support
1939	Servo ON invalid command failure:Axis	Resolution: Turn off the servo enable signal from the upper computer
1940	Inertia identification failed:Axis	Resolution: Contact electrical technicians
1941	Origin return warning:Axis	Resolution: Contact electrical technicians
1942	Abnormal origin return switch:Axis	Resolution: Contact electrical technicians
1943	Encoder battery warning:Axis	Resolution: Replace with a new voltage-matched battery
1944	Emergency stop:Axis	Resolution: Check the position command sent by the upper computer
1945	DI setting invalid:Axis	Resolution: Set a valid DI logic selection value

Alarm number	Alarm information	Troubleshooting
1946	DO setting invalid:Axis	Resolution: Set a valid DI logic selection value
1947	The model identification check code failed:Axis	Resolution: Set H01-72=1 to shield the machine type recognition function
1948	Motor overload warning:Axis	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1949	Regenerative bleeder resistance overload:Axis	Resolution: Contact technical support
1950	External regeneration discharge resistance is too small:Axis	Resolution: Check the external braking resistor between P and C
1951	Warning of overtemperature of drain pipe:Axis	Resolution: Contact technical support
1952	Change parameters need to be powered on again to take effect:Axis	Resolution: Confirm that the change takes effect after power-on. Power-on
1953	Frequent parameter storage:Axis	Resolution: Check if the upper computer system frequently and rapidly modifies function codes

Alarm number	Alarm information	Troubleshooting
1954	Forward override warning:Axis	Resolution: Check if the DI function is set on the DI terminals of the 2003h group
1955	Reverse override warning:Axis	Resolution: Check if the DI function is set on the DI terminals of the 2003h group
1956	Torque ripple compensation failed:Axis	Resolution: Check the encoder wiring
1957	Origin regression mode setting error:Axis	Resolution: Contact technicians, check if the setting value of 0x6098 is within the normal range
2001	Assignment empty	Resolution: Contact technical support
2002	Assignment absent	Resolution: Open the job to verify if the called job exists
2004	Over 8 interruption opened	Resolution: Check the program, delete unnecessary interrupts, close interrupts in time
2005	Interruption is opened	Resolution: Check the program logic, close interrupts, check the operation process
2007	Interruption not configured	Resolution: Check the program, check interrupt configuration
2008	Forbid call interruption PROG	Resolution: Check the program, check interrupt configuration
2009	Same pause was open	Resolution: Check the logic of the interrupt instruction
2011	External signal error	Resolution: Contact professional technicians
2014	Illegal execution of background task	Resolution: Contact technicians
2020	Position exceed limit	Resolution: Confirm that the robot movement is within the movement range

Alarm number	Alarm information	Troubleshooting
2021	Same Curve Instruction less than 2	Resolution: Contact technical support
2022	Curve Path Need Other Move Point as Start Point	Resolution: Contact technical support
2023	Curve assist point not found	Resolution: Contact technical support
2024	Current point is close to next point	Resolution: Contact technical support
2025	Target point is close to next point	Resolution: Contact technical support
2026	Current point is close to last point	Resolution: Contact technical support
2027	Target point is close to last point	Resolution: Contact technical support
2028	Curve Plan Error	Resolution: Contact technical support
2029	Curve degree more than 180	Resolution: Contact technical support
2030	Degenerate circle	Resolution: Reteach the position points, contact after-sales personnel
2031	Arc three points collinear Err	Resolution: Contact technical support
2032	Welding command behind MOVJ	Resolution: Check the program logic
2033	Current point is close to curve reference point	Resolution: Contact technical support
2034	Target point is close to curve reference point	Resolution: Contact technical support
2040	Stop abnormally	Resolution: Check the system status
2080	Added position variables have different frame	Resolution: Unify the type of the position variable coordinate system
2081	Hcfa clean code SDO map fail	Resolution: Restart the robot
2082	Hcfa clean code SDO data send fail	Resolution: Check the ecat bus configuration
2083	Subtracted position variables have different frame	Resolution: Suitable position variable information

Alarm number	Alarm information	Troubleshooting
2084	Parameter setting error in calling analog file number	Resolution: Check the corresponding parameter configuration
2085	Poweroff recover error, initial complete	Resolution: Contact professional technicians
2086	Position Error Overflow Alarm at Servo ON	Resolution: Perform settings to clear the warning value
2087	Overload	Resolution: Confirm the motor wiring, encoder wiring, etc
2088	Vibration	Resolution: Reduce the motor speed, correctly set the moment of inertia ratio
2089	Regenerative Overload	Resolution: Set the power supply voltage within the specification range
2090	Dynamic Brake Overload	Resolution: Reduce the motor command speed, moment of inertia, DB stop count
2091	Absolute Encoder Battery Error	Resolution: Check the battery and servo unit
2092	Undervoltage	Resolution: Adjust the power supply voltage, increase the power capacity, etc
2093	Overtravel	Resolution: Confirm the wiring of the over-travel signal, take anti-interference measures

Alarm number	Alarm information	Troubleshooting
2094	Position Error Overflow	Resolution: Confirm the problem with the motor or encoder, contact technicians
2212	IO communication abnormal	Resolution: Contact technical support
2214	Group IO scan code error	Resolution: Contact technical support
2220	Invalid delay time	Resolution: Modify the time parameter of the instruction
2230	Label not exist	Resolution: Set an existing label
2233	Axis interference zone alarm	Resolution: Go to the axis interference zone page to disable the enable, and move the robot to the safe area
2234	Location storage type is JT	Resolution: Only support the conversion of position variables stored in T6 method
2235	File number exceeds	Resolution: Check the coordinate system number
2240	Job call itself	Resolution: Set to another existing subroutine
2241	Job does not exist	Resolution: Check for duplicate definitions
2242	Job cann't return	Resolution: Check if the subroutine has a RET instruction
2243	Subjob has no RET	Resolution: Write the subroutine job
2244	Not configure main program	Resolution: Create a main job, re-register the main job
2245	Main program does not exist	Resolution: Create a main job, re-register the main job
2250	Running stack exceed limit	Resolution: Check the job logic
2252	B-Prog loading prohibition	Resolution: Check the job logic

Alarm number	Alarm information	Troubleshooting
2254	Background job call failed	Resolution: Check the job logic
2263	Frame	Don't dispose. Resolution: Change to a supported coordinate system type
2270	Instruction is jump	Resolution: Contact technical support
2272	Node memory assign failure	Resolution: Contact technical support
2274	Insert empty node	Resolution: Contact technical support
2276	Pulse signal NoBeyond 16	Resolution: Check the job logic
2280	BG not start normally	Resolution: Start the background program through the menu
2290	Don't enter interference zone	Resolution: Check the teaching path
2293	playstack is error	Resolution: Contact technical support
2294	No Move after IO Delay Instruction	Resolution:
2295	SpeedError Get	Resolution: Check the job content
2296	Same IO has been scanned	Resolution: Please check the PATH IO instruction parameters
2297	IO Type Error	Resolution: Contact technical support
2298	Non-motion command before IO delay instruction	Resolution: Check the job content
2300	Local position delete error	Resolution: Contact technical support
2301	Team IO input NoNot exist	Resolution: Check if the group IO input judgment instruction's group number exceeds 8
2302	Team IO output NoNot exist	Resolution: Check if the group IO output judgment instruction's group number exceeds 8
2303	Team IO configure error	Resolution: Check group IO configuration
2304	No configure team IO	Resolution: Check group IO configuration
2305	IO Scan over 128	Resolution: Reduce the number of output IOs on the path
2350	Command parameters exceed	Resolution: Re-enter the instruction, contact

Alarm number	Alarm information	Troubleshooting
		technical support
2351	parameters N exceed	Resolution: Check the instruction parameter input
2352	Next isn't motion command	Resolution: Check the job logic
2353	Get position information error	Resolution: Check the job logic
2355	Speed percentage exceeds	Resolution: Modify the speed percentage parameter
2468	Trace condition is not satisfied	Resolution: Check the corresponding IO output
2501	MCURV ARGUMENT ERROR	Resolution: Replace the points and reissue
2502	MCURV WAIT ABORTED	Resolution: Check if the condition has triggered a pause
2690	Joint on trajectory	Resolution: contact technical support
2700	pthread execute error: pthread	Resolution: Contact professional technical support
2701	inst buffer pop error:	Resolution: Contact professional technical support
2702	inst buffer put error	Resolution: Contact technical support
2703	inst foresight error	Resolution: Contact professional technical support
2704	inst unavailabale	Resolution: Do not use this instruction
2707	get inst cache data length error	Resolution: Contact technical support
2708	No lookahead inst exist	Resolution: Contact technical support
2710	Instruction cache error	Resolution: Contact technical support
2712	Lookahead waiting timeout	Resolution: Contact technical support
2714	No motion position information was obtained	Resolution: Check the position variable information, if empty, teach it
2716	—	—
2718	—	—
2720	—	—

Alarm number	Alarm information	Troubleshooting
2722	Internal algorithm data error	Resolution: Contact technical support
2724	No lookahead task	Resolution: Contact technical support
2727	Planning lib 1.0 doesn't support MOVC PP related instructions	Resolution:
2728	Planning lib 2.0 doesn't support MOVC related instructions	Resolution: Confirm the content of the job instruction
2729	Duplicate IO input configuration	Resolution: Check the input IO configuration
2730	Duplicate IO output configuration	Resolution: Check the output IO configuration
2861	Offline motion pose para is illegal	Resolution: Change the parameters of the offline motion instruction
2898	Realtime data Start/Stop Err:errCode:	Resolution: Contact technical support
2899	Realtime data thread error,errCode:	Resolution: Contact technical support
2900	The realtime position point does not update continuously,data channel is:	Resolution: Contact technical support
2901	Please select loop type	Resolution: Try selecting a loop type, or contact technicians
3000	Inexistent job	Resolution: Open a job and perform instruction search
3100	Joint	exceed limit. Resolution: Confirm if the zero position is normal
3101	Union limit value exceed	Resolution: Move axis 2 or axis 3 to make the joint limit not out of bounds
3102	Approaching singularity interval, unable to move forward	Resolution: Modify the path to avoid approaching the singular point
3106	singularity interval segment error	Resolution: Contact technical support
3110	play sem accumulation	Resolution: Contact technical support
3200	Command modification error	Resolution: Re-enter the instruction
3201	Position is a joint coord and is not	Resolution: The position variable is in joint

Alarm number	Alarm information	Troubleshooting
	supported as an offset value	coordinate system and does not support as an offset value, please adjust
3332	Divisor can't be 0	Resolution: Check the job logic
3333	CALC result exceed limit	Resolution: Check the job logic
3334	Square number is negative	Resolution: Contact technical support
3340	Starting is not in situ,number	Resolution: Check the job logic
3350	Frame NoExceed limit	Resolution: Check the job logic
3601	No power on the arm(warn)	Resolution: put arm Power on
3604	Priority interrupt close(warn)	Resolution: Check the configuration
3605	Output port(warn)	does not exist. Resolution: Pause the job for processing, check the configuration of the IO number
3606	Keep output port(warn)	Resolution: Check the configuration
3607	Input port(warn)	does not exist. Resolution: Pause the job for processing, check the configuration of the IO number
3608	Reserved input port(warn)	Resolution: Check the configuration
3628	No ENDIF instruction	Resolution: Verify the job content
3629	Backward Move not supported	Resolution: Mask this instruction or delete it
3630	INTERLOCK area unconfigured	Resolution: Check the INTERLOCK area configuration
3631	INTERLOCK area not requested	Resolution: Check the file configuration
3632	No IF THEN instruction	Resolution: Check the job content
3633	Current Tool/User Frame is different in teach	Resolution: Verify the coordinate system usage
3634	User Frame Calculated Failure	Resolution: Contact technical support
3635	Joint Type Unsupported	Resolution: Contact technical support
4016	Offline motion exe error in opmotion	Resolution: Check the following alarms
4017	Offline motion status flag is error	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
4018	Can't get next P in opmotion	Resolution: Insert the next point
4019	Data abnormal (NaN)	Resolution: Contact technical support
4020	TCP above or below the shaft	Resolution: Adjust the position point
4060	Abnormal external axis,please check	Resolution: Check the external axis configuration
4061	external axis approaching joint limit	Resolution: Adjust the position point
4202	communication is error between rc and teachbox	Resolution: Check the communication line or contact technicians
4203	please move cursor for restore	Job on Pause
5000	CAN init error	Resolution: Try to restart the robot
5002	CAN send buffer full	Resolution: Check hardware communication lines
5003	CAN write data error	Resolution: Check hardware connection, contact technical support
5004	CAN read data error	Resolution: Check hardware connection, contact technical support
5005	Serial open error	Resolution: Try to restart the robot, contact technical support
5006	Serial set baud rate error	Resolution: Contact technical support
5007	Serial get terminal error	Resolution: Contact technical support
5008	Serial set terminal error	Resolution: Contact technical support
5009	Serial not support baudrate	Resolution: Check baud rate settings
5010	Serial not support databit	Resolution: Contact technical support
5011	Serial not support parity	Resolution: Contact technical support
5012	Serial not support databit	Resolution: Contact technical support
5013	Serial not support port	Resolution: Check serial port configuration
5014	Serial send buffer full	Resolution: Contact technical support
5015	Serial receive error	Resolution: Contact technical support
5016	UDP socket creat error	Resolution: Contact technical support
5017	UDP send error:	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
5018	position var coord error	Resolution: Check the position variable coordinate system, contact technical support
5019	Get MOV P data, coord num abnormal	Resolution: Check the position variable coordinate system, contact technical support
5020	USB-CAN open error	Resolution: Contact professional technicians
5021	USB Serial set baud rate error	Resolution: Contact professional technicians
5022	USB Serial get terminal error	Resolution: Contact professional technicians
5023	USB Serial set terminal error	Resolution: Contact professional technicians
5024	USB Serial not support databit	Resolution: Contact professional technicians
5025	USB Serial not support parity	Resolution: Contact professional technicians
5026	USB Serial not support stopbit	Resolution: Contact professional technicians
5027	USB Serial send buffer full	Resolution: Contact professional technicians
5028	USB Serial not support port	Resolution: Check the port number configuration
5029	USB Serial receive error	Resolution: Contact professional technicians
5030	Parity or send format error	Resolution: Contact professional technicians
5031	Command word not support	Resolution: Contact professional technicians
5032	Parameter set incorrect or not support	Resolution: Contact professional technicians
5033	Not read configuration parameters	Resolution: Contact professional technicians
5034	USB-CAN is incorrect	Resolution: Contact professional technicians
8015	DAT file size error, file:	Resolution: Contact after-sales or development personnel
8016	DAT file read failure	Resolution: Contact after-sales or development personnel
8018	FRAM initialization failed	Resolution: Restart, replace the mainboard
8627	Data Setting Warning 1	Resolution: Contact technical support
8628	Data Setting Warning 2	Resolution: Contact technical support
8629	Data Setting Warning 3	Resolution: Contact technical support
8630	Data Setting Warning 4	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
8631	Data Setting Warning 5	Resolution: Contact technical support
8632	Command Warning 1	Resolution: Contact technical support
8633	Command Warning 2	Resolution: Contact technical support
8634	Command Warning 4	Resolution: Contact technical support
8635	Command Warning 5	Resolution: Contact technical support
8636	Command Warning 6	Resolution: Contact technical support
8637	Approaching Undervoltage	Resolution: Contact technical support
8638	Command Warning 7	Resolution: Contact technical support
8639	Data Clamping Outside Range	Resolution: Contact technical support
8640	Positive/Negative Limit Exceeded	Resolution: Contact technical support
8641	Soft Limit Setting Error	Resolution: Contact technical support
8642	Origin Offset Setting Error	Resolution: Contact technical support
8643	Synchronization Frame Data Loss	Resolution: Contact technical support
8644	EtherCAT Communication Interruption	Resolution: Contact technical support
8645	EtherCAT Network Initialization Failure	Resolution: Contact technical support
8646	Torque Limit Error	Resolution: Contact technical support
8647	Unable to Enable When Hardware Base is Not Blocked	Resolution: Contact technical support
8648	Any Safety Interface Input Not Connected to +24V	Resolution: Contact technical support
8649	Emergency stop warning	Resolution: Contact technical support
9033	Axis number changed,restart	Resolution: Restart
9034	Restart and zero calibration	Resolution: Confirm whether to zero the new axes and restart
9138	Path restoration abnormal	Resolution: Contact technical support
9402	Out of breakpoint recovery	Resolution: Move the robot to the teaching point through forward and reverse motion
9403	Execution mode error	Resolution: Execute a job that matches the

Alarm number	Alarm information	Troubleshooting
		mode
9500	System uncharge	—
9536	Activate with order box	Resolution: Contact technical support
9539	Order pausing	—

7.3 Prompt

In the case of prompts listed below, the robot is not affected in any way, and the message is displayed in the prompt line of the teach pendant. The prompt is not included in the error message.

Alarm number	Alarm information	Troubleshooting
100	No power on the arm(error)	Resolution: Power on, enable
102	Energy saving switch on, operation axis/FWD/BWD/start can trigger power on	After selecting SERVEON, click the [Start] button to initiate operation.
117	External device is in shielded state	Resolution: The communication between external devices is blocked
167	After instruction edit, superadd button lost its response	Resolution: Reinsert instructions, the append button function will be restored
168	After job management, the superadd button lost its response	Resolution: Reinsert instructions, the append button function will be restored
170	Reconnect the FTP Server	Reconnect the FTP Server
220	FRAM wrong (assignment)	Resolution: Do not continue to add instructions, Contact technical support
239	Fast stop operation is not supported in continuous motion	Resolution: Do not trigger quick stop operation
608	External stop	Resolution: Check safety door related line
648	Mode Switch,Request Poweroff	—
650	please wait a moment,requesting poweroff	Resolution: Cancel the alarm and then enable
699	Robot pause	Resolution: Resume

Alarm number	Alarm information	Troubleshooting
709	Current Robot not support Singularity Avoid	Resolution: Contact technical support
733	GravityCompensaton reset failure	Resolution: Contact technical support
734	CollisionDetection reset failure	Resolution: Contact technical support
735	Robot not support load identification	No action required.
736	Test run complete	No action required.
737	Trajectory execution complete	No action required.
738	Calculation complete	No action required.
739	Motion break down!Restart	Resolution: Restart
740	Load identification fail,confirm no load trajectory execute or not	No action required.
741	Load identification fail,confirm on load trajectory execute or not	No action required.
742	CD parameter storage failure	Resolution: Contact professional technical personnel
743	Over limits,Axis	Resolution: Adjust robot axis
744	Axis3 range:2~4 degrees	Resolution: Adjust axis amplitude to a reasonable range
745	Axis5 range:20~40 degrees	Resolution: Adjust axis amplitude to a reasonable range
746	Axis6 range:20~60 degrees	Resolution: Adjust axis amplitude to a reasonable range
747	File save load para	Resolution: Contact technical support
748	Recovery load para failure	Resolution: Contact technical support
749	Load type conflict	Resolution: Contact technical support
750	On Load Execute Compltet,Trajectory	No action required.
760	No Load Execute Compltet,Trajectory	No action required.
842	No response after power-on	Resolution: Check power-on response

Alarm number	Alarm information	Troubleshooting
		hardware line
845	No receiving unlock reply	Resolution: Check driver hardware line
846	Emergency stop and reset	Resolution: Check emergency stop safety relay and related hardware line
847	No local servo power on	Resolution: Remote external servo power on, switch to local mode servo power on
854	Local mode is selected	No action required.
855	Remote mode is selected	No action required.
856	Disable teaching box button	Resolution: Switch to local mode
882	Bus parameter abnormal	Resolution: Restart the system and try again
883	Code data has been cleared	No action required.
884	Driver state has been reset	No action required.
886	Bus parm optimized	No action required.
887	Clear encoder value,restart	No action required.
902	Teaching box ES start	Resolution: Rotate the teaching pendant button clockwise to the raised position
1000	Driver is initializing	Resolution: Wait for 2 minutes, if ineffective, Contact technical support
1001	本体解抱闸电流报警	Resolution: Relearn current, check body motor brake
1002	外部轴解抱闸电流报警	Resolution: Relearn current, check external axis motor brake
1003	解抱闸电流学习完成	Resolution:
1005	signals	accumulated over 10. Resolution: If it appears frequently, please Contact technical support
1008	License initialization fail	Resolution: Try temporary authorization and contact after-sales personnel
1009	License key length incorrect	Resolution: Please check if the authorization

Alarm number	Alarm information	Troubleshooting
		password is entered correctly
1011	Repair licence success	Resolution: Try temporary authorization and contact after-sales personnel
1012	Generate temporary licence success	Resolution: Please contact Newron after-sales personnel
1013	Record licence time failed	Resolution: Restart the robot or contact after-sales personnel
1014	Sync CF licence file failed	Resolution: Restart the robot or contact after-sales personnel
1015	Real-time display beyond the boundary	Resolution: Contact technical support
1016	Configuration changed,reboot	No action required.
1017	Configuration unchanged	No action required.
1018	No license key on usb	Resolution:Please confirm that there is an RCllicence.txt file in the CF path
1025	Maintenance is required after	days.Please contact the manufacturer!
1026	Maintenance is required after	hours.Please contact the manufacturer!
1051	Reading CF card fail:	Confirm communication status.
1052	Writing CF card abnormal:	Confirm communication status.
1053	Assignment format error:	Confirm communication status.
1054	Too many pages, please close redundant pages!	No action required.
1055	Assignment transferred	No action required.
1056	CF card failed to open	Resolution: Contact technical support
1057	Assignment transferring,wait	Await completion of the prompt.
1058	Certification file not exist	Resolution: Contact Newron after-sales personnel
1059	Cover all assignments?	Resolution: Press OK or Cancel
1060	CF card failed to open	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
	OperationLogDB.db	
1061	Assignment initializing,wait	Resolution: Wait for initialization to complete
1062	Assignment initialized	No action required.
1063	Assignment database abnormal:	Resolution: Contact technical support
1164	Move inst type error	Resolution: Check the taught job instructions or Contact technical support
1800	High motor load rate:	Resolution: Contact technical support
1801	Large regeneration rate:	Resolution: Contact technical support
1802	Encoder Battery running out	Resolution: Contact technical support
1803	Driver fan fault	Resolution: Contact technical support
1804	Encoder communication alarm	Resolution: Contact technical support
1805	High temperature of motor:	Resolution: Contact technical support
1806	Vibration detected	Resolution: Contact technical support
1807	Driver component aging	Resolution: Contact technical support
1808	Raster abnormal	Resolution: Contact technical support
1809	Raster communication alarm	Resolution: Contact technical support
1810	High motor load rate	Resolution: Contact technical support
1811	Large regeneration rate	Resolution: Contact technical support
1812	High temperature of driver	Resolution: Contact technical support
1813	Main circuit exceeds 105V	Resolution: Contact technical support
1814	Low control voltage	Resolution: Contact technical support
1815	Encoder Battery running out	Resolution: Contact technical support
1816	Limit torque	Resolution: Contact technical support
1817	Limit speed	Resolution: Contact technical support
1818	Excessive position deviation	Resolution: Contact technical support
1819	Position command over limit	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
1820	High motor load rate	Resolution: Contact New Robot after-sales personnel
1821	Hign driver load rate	Resolution: Contact New Robot after-sales personnel
1822	MTS abnormal	Resolution: Contact New Robot after-sales personnel
1823	Parameter setting abnormal	Resolution: Contact New Robot after-sales personnel
2003	Wrong command code	Resolution: The current command code does not match the current configuration application
2006	Priority interrupt close	Resolution: Check and operate the teaching program to ensure interrupts are enabled
2010	Performing wrong	Resolution: Query other alarm information to confirm the cause of the alarm
2016	B-Prog executing, don't edit	Resolution: Close background job to edit
2018	B-prog in an endless loop or exceeds max row numbers:200,BG	Resolution: Check background job
2095	Remote mode user IO input simulation warning detected	Resolution: Contact technical support
2096	Remote mode user IO output simulation warning detected	Resolution: Contact technical support
2200	Output port	does not exist. Resolution: Power off the system and reconfigure the output port number
2201	Keep output port	Resolution: Reset the output port
2202	Invalid output	Resolution: Check the current IO number configuration
2203	Invalid input!	Resolution: Check the current IO number configuration

Alarm number	Alarm information	Troubleshooting
2204	Input port	does not exist Resolution: Check the current IO number configuration
2205	Reserved input port!	Resolution: Contact technical support
2210	Port	waiting fail. Resolution: Check the signal line
2222	Assignment completed	No action required.
2223	Master registration succeed	No action required.
2224	Master registration fail	No action required.
2225	Main assignment not found	Resolution: Please confirm whether the main job has been registered
2226	the designated jobname is not exist	Resolution: Please confirm whether the job name is correct or exists
2227	the designated jobname is not set	Resolution: Please go to the loop page to configure the job
2228	Extern abort, the designated job should not be start	Restart after external stop release
2245	Main program does not exist	Resolution: Create a main job, re-register the main job
2246	Move robot to BP	Resolution: Switch to teaching mode and move the robot forward to the instruction position
2247	Clear assignment stack	Prompt message
2248	External pause	Resolution: Release external pause
2249	Clear assignment stack first	Resolution: Re-select the program at the time of pause to clear the job stack and start the operation
2253	RET is not in Sub-assignment	Resolution: Add RET instruction at the end of the sub-job, adjust the call logic
2255	Can't set position variable	Resolution: Please select P variables with the same coordinate system parameters for

Alarm number	Alarm information	Troubleshooting
		operation
2259	Authorized, please restart	Prompt message
2264	job flag is clear	Prompt message
2278	IO is not reset	Resolution: Add pulse reset
2398	Enter standby state in interference zone	Prompt message
2399	Remove standby state in interference zone	Prompt message
2705	inst passed during pause process, instcode	Resolution: Contact professional technical personnel
2706	Restart failed. Execute standard start procedure.	Resolution: Contact professional technical personnel
2725	Planning lib 1.0 doesn't support MOVC PP related instructions	Resolution: Do not use MOVC PP related instructions
2726	Planning lib 2.0 doesn't support MOVC related instructions and applications	Resolution: Do not use MOVC related instructions
2809	Annotation Data Write failure	Resolution: Contact technical support
2850	Offline mode,don't operate	Prompt message
2851	Offline mode selected	Prompt message
2857	Please turn to offline mode	Prompt message
2858	Offline mode abnormal	Resolution: Contact technical support
2862	Offline set coordinate failed	Resolution: Investigate the above issues
2890	write data in FRAM is failure	Resolution: Contact technical support
2891	write data in file is success	Prompt message
2892	sd file abnormal	Resolution: Contact technical support
2999	Current irq can not run,level:	Resolution: Adjust the timing of the interrupt trigger logic in the job
3001	Interrupt date not update	Resolution: Contact technical support
3019	executing cant operate job	Prompt message

Alarm number	Alarm information	Troubleshooting
3040	shielded state Axes:	Prompt message
3050	Line number not found	Resolution: Contact technical support
3051	Step number not found	Resolution: Contact technical support
3052	label not found	Resolution: Contact technical support
3053	Instruction not found	Resolution: Contact technical support
3090	On play or remote mode,can not adjust speed by teachbox if ExSpeedSwitch is On	Resolution: Turn off the external speed control enable switch, then adjust the speed with the teaching pendant
3091	Modbus TCP slave connections reached 4, unable to accept more connections	Resolution: Contact technical support
3095	speed percent is 0	Prompt message
3096	speed percent is 100	Prompt message
3097	speed step value is error	Resolution: Contact technical support
3098	Speed percent is wrong	Resolution: Adjust the execution speed percentage range to [0,100]
3099	Speed percent to 30%	
3105	Profinet slvae initialize complete	Prompt message
3107	offline data lose	Resolution: Redownload from upper computer
3108	offline data check error	Resolution: Check for errors on the lower computer
3111	job open	Prompt message
3112	ExSpeed is 0, excute as local speed	Prompt message
3310	Interrupt not enable	Resolution: Set interrupt enable to ON, check the teaching program
3311	No interrupt, RESUME invalid	Resolution: Check the teaching program and interrupt configuration
3320	Interrupt unreset	Resolution:The previous interrupt has not been processed yet. Please finish processing

Alarm number	Alarm information	Troubleshooting
		it before triggering another one.
3341	please reopen job manually	Prompt message
3342	Reserved job failed	Manually resume pause mode
3343	this job will be add to phead stack,when other job is running	Prompt message
3345	the job has been on phead before,so this job will be deleted from phead stack	Prompt message
3346	P	position correction successful
3347	PR	position correction successful
3348	please clear job stack when play stack changed	—
3349	this job is running,so it can not be cancel or reserved	—
3356	Remote backup successfully	—
3357	Remote backup failed	—
3358	Remote backup timeout	Resolution:Reset the backup input signal, and then set it again
3359	Remote backup failed,io input signal lost	Resolution:Check whether the backup input signal is normal
3360	Remote backup failed,network exception	Resolution:Check if the network between the PC and the RC can be pinged
3500	Redis memory abnormal, threshold:	Resolution: Contact technical support
3501	Redis clients abnormal, threshold	Resolution: Contact technical support
3502	Redis output buffer abnormal, threshold	Resolution: Contact technical support
3503	Redis input buffer abnormal, threshold:	Resolution: Contact technical support
3504	Redis block clients abnormal, threshold:	Resolution: Contact technical support

Alarm number	Alarm information	Troubleshooting
3505	Redis hit percent abnormal, threshold:	Resolution: Contact technical support
3506	Redis subscribe channels abnormal, threshold:	Resolution: Contact technical support
3507	Redis key space abnormal, threshold:	Resolution: Contact technical support
3602	background assignment empty	Resolution: Contact technical support
3603	background Assignment absent	Resolution: Contact technical support
3609	System uncharge(error)	Resolution: Please power on
3611	Label not exist(error)	Resolution: Check if the current Label number exists in the job
3612	Job call itself(error)	Resolution: Change the job name to call, do not call this job
3613	Job does not exist(error)	Resolution: Contact technical support
3614	Job cann't return(error)	Resolution: Do not directly run the sub-job
3615	Running stack exceed limit(error)	Resolution: Contact technical support
3616	Team IO input NoNot exist(error)	Resolution: Check the group IO instruction port number
3617	Team IO output NoNot exist(error)	Resolution: Check the group IO instruction port number
3618	Team IO configure error(error)	Resolution: The low bits of the group IO need to be configured with IO numbers
3619	No configure team IO(error)	Resolution: Please configure the current group IO on the group IO configuration page
3620	Command parameters exceed(error)	Resolution: Modify the instruction parameters within the range
3621	CALC result exceed limit(error)	Resolution: Modify the input value
3622	IO PulseHead is null	Resolution: Cancel the deletion operation of the pulse signal node
3623	Speed percentage exceeds(error)	Resolution: Keep the set speed percentage consistent with the maximum value

Alarm number	Alarm information	Troubleshooting
		configured in the interface, do not exceed
3625	Frame	Don't dispose
3627	parameters N exceed	Resolution: Reduce parameter N
4012	Total DevNet bytes >62	Resolution: Keep the total sum of input length or output length of 8 files within 62
4013	I args data type error	Resolution: Change the instruction parameter
4015	Line calibration result error	Resolution: Confirm the calibration method and recalibrate
4021	Incorrect input parameter	Resolution: Re-enter correct data
4022	Input parameter exceeds	Resolution: Check offset configuration and offset loop count
4023	Non-configure SPI	Resolution: Re-enter correct data
4024	Non-configure SIPN	Resolution: Re-enter correct data
4025	Non-configure SIS	Resolution: Re-enter correct data
4026	Offset function disabled	Resolution: Check teaching program and execution switch settings
4028	Application changed, restart	Prompt message
4029	Configuration changed,start	—
4030	No modified configuration	Prompt message
4032	Model changed,please restart	Prompt message
4034	No enable RC	Resolution: Go to the execution switch page, turn on the compensation enable switch
4035	Configuration changed,saving	Prompt message
4036	Singularity Avoid function disabled	The singularity enable switch is not activated.
4037	Time changed, restart	Prompt message
4038	Driver has not information	Prompt message
4039	Modification without support	Prompt message

Alarm number	Alarm information	Troubleshooting
4040	Encoder clearing,wait	Prompt message
4041	Vector collinear	Resolution: Please set the correct value
4042	Stuffer data save failed	Resolution: Contact technical support
4045	the position num is lock,please change	Resolution: unlock operation in position information
4046	edit lock state setting failed	Resolution: Please try to reset
4047	Current plan type does not support singularity avoid	—
4062	exas reach the limit when move z direction	Prompt message
4084	Max cabinet type is	Resolution: Check and process the number of ROBOTTYPE files
4085	cabinet type name is error	Resolution: Check the control cabinet file name validation and compliance with specifications
4095	Max robot type is	Resolution: Contact technical support
4098	Robot type name is error	Resolution: Contact technical support
4100	Operation not allowed In current mode	Switch to operational mode
4200	Page request is processing, please try later	Resolution: Reoperate
4201	Page request process failed	Resolution: Reoperate, if failed restart the system
5001	Teaching lock locked	Unlock teach mode
5067	NOT STABLE, PLEASE DO NOT POWER UP	Resolution: Wait until the system is stable before powering on
8003	P3:monitoring task begin	Prompt message
8004	P4:DB is initialized	Prompt message
8005	P5:MI completed	Prompt message

Alarm number	Alarm information	Troubleshooting
8009	P9:SI completed	Prompt message
8010	P10:DTI completed	Prompt message
8011	P11:MTI completed	Prompt message
8014	The user IO output failed, The user IO output protection switch is activated	Disable IO output protection
8017	FRAM synchronous completed	—
8019	Dynamic parameter acquisition debug switch is turned on	Prompt message
8020	Motion parameter debug switch is turned on	Prompt message
8021	The acc limit has been removed	Prompt message
9005	job,job stack cleared	Prompt message
9006	job,job stack,system parameters cleared	Prompt message
9013	Memory initialization enable	Prompt message
9014	Memo initialization disable	Prompt message
9015	Memory check right	Prompt message
9016	Memory check error	Restart and observe
9017	Available memory size:	Prompt message
9022	Zero initialization complete	Prompt message
9023	Zero initialization enable	Prompt message
9024	Zero initialization disable	Prompt message
9025	Digital output enable	Prompt message
9026	Digital Output test	Prompt message
9028	Switch changed,restart	Prompt message
9029	Switch changed,restart	Prompt message
9031	Press any key to exit!	Prompt message
9032	Clear assignment(yes/no)?	Prompt message

Alarm number	Alarm information	Troubleshooting
9040	Verify system data enable	Prompt message
9090	Clear error message	Prompt message
9100	Situs 1 recorded	Prompt message
9102	File open failure	Resolution: Open the file multiple times, if still fails, contact NewRobot after-sales personnel
9103	File upload complete	Prompt message
9104	Cover existed file(yes/no)?	Resolution: Confirm or cancel
9105	System backup succeed	Prompt message
9106	Oi-MB completed, restart	Prompt message
9108	U disk unload succeed	Prompt message
9109	System in unload state	Resolution: Reinsert the USB and perform the operation again
9110	System mounted	Prompt message
9111	System mounted succeed	Prompt message
9112	System halted error	Resolution: Restart the controller and remount; or try a different USB
9113	Usb unloaded failure	Resolution: Restart the controller and remount
9115	Overlay backup files	Please confirm
9116	Redo?	Please confirm
9117	Initialization(yes/no)?	Please confirm
9118	Backup system first	Resolution: Perform system backup
9119	Low version job recovery failed	—
9121	Can't upload null assgnment	Resolution: Abandon the operation
9122	Backup files unable to open	Resolution: If opening the backup file fails multiple times, contact technical support
9123	Parameter recovery failure	Resolution: If parameter recovery fails multiple times, contact technical support
9124	Parameter backup failure	Resolution: Suggest unmounting and

Alarm number	Alarm information	Troubleshooting
		remounting then backup
9125	Files upload complete	Prompt message
9126	Files upload failure	Resolution: Suggest unmounting and remounting then backup
9127	Usb files open failure	Resolution: Insert the USB multiple times, if opening the file still fails, replace the USB or contact technical support
9128	Files download complete	Prompt message
9129	Download file create fail	Resolution: Redownload the file, if it still fails contact professional technical support
9130	jobSave.bak open fail	Resolution: File corrupted, redo the backup
9131	Backup complete	Resolution: Job file backup complete
9132	Backup failure	Resolution: Suggest unmounting and remounting then backup
9133	Recovery failure	Resolution: Retry the operation, if it still fails contact technical support
9134	Recovery complete,restart	Resolution: Follow the prompt information to unmount and restart
9135	No param backup/recover	Resolution: Do not perform parameter backup and parameter recovery
9136	Out of ram,recover failure	Resolution: Clear some job space, then restore the job again
9137	Executing break restoring	Resolution: Wait for the breakpoint recovery complete prompt
9138	Path restoration abnormal	Resolution: Contact technical support
9139	Assignment downloaded null	Prompt message: No action required OR Redownload the job.
9140	Button configured not enable	Resolution: Or configure a program for the button
9141	Database backup complete	Prompt message

Alarm number	Alarm information	Troubleshooting
9142	Button program not configure	Prompt message: No action required OR Redownload the job
9143	AR complete,restart	Resolution: Restart the robot
9144	Button program executing	Prompt message
9145	Mode switch complete,restart	Resolution: Restart the robot
9146	Button program terminated	Resolution: Re-execute the button program
9148	Button program complete	Prompt message
9150	Button program not exist	Resolution: Contact technical support
9151	Breakpoint restoration finished	Prompt message
9152	Program execution error	Resolution: Contact technical support
9154	Unsupport command,linenum:	Delete Line
9160	Execute only teaching mode	Resolution: Switch to teach mode, button programs can be executed through buttons
9170	No rotation in RT system	Resolution: Avoid rotating this axis as prompted
9171	No rotation in tool system	Resolution: Avoid rotating this axis as prompted
9172	No rotation in user system	Resolution: Avoid rotating this axis as prompted
9513	Record backup failure	Resolution: Check USB mount and retry, otherwise contact technical support
9514	Same name macro exist	Change the name of the new macro program
9515	Same name assignment exist	Resolution: Re-enter the job name
9516	Same name offline assignment	Resolution: Re-enter the job name
9517	Same name PLC assignment	Resolution: Re-enter the job name
9518	Back name is exist	Resolution: Give the new job a non-duplicate name
9519	No BA exist	Resolution: Change to the correct job name
9530	Order status clear complete	Prompt message
9531	Switch to order job	Resolution: Please switch back to the first

Alarm number	Alarm information	Troubleshooting
		job on the current reservation state page
9532	Configure order job IO	Resolution: Configure the reservation job IO as prompted
9533	Order set clear complete	Prompt message
9534	Order set incorrect	Resolution: Check reservation function configuration
9535	Order job not exist	Resolution: Check if the current job name job exists in the job list
9537	Variable recover complete	Prompt message
9538	Variable backup complete	Prompt message
9539	Order pausing	Prompt message
9542	Assignment delete failed	Check job status
9560	Param input incomplete	Re-entry required
9561	Param input error"	Resolution: Re-enter the parameters
9562	System axis number not match	Resolution: Confirm the total number of axes
9563	COMMmode not match	Resolution: Re-enter the communication method interface and reconfigure according to the prompt
9564	Application not match	Resolution: Ensure the configuration is consistent, restart the controller, then re-operate
9565	Model NUM not match	Resolution: Ensure the configuration is consistent, restart the controller, then re-operate
9566	Acquireing failed	Resolution: Obtain the parameter recovery restriction conditions multiple times, if still fails, Contact technical support
9567	Save failure	Resolution: Obtain the parameter recovery restriction conditions multiple times, if still

Alarm number	Alarm information	Troubleshooting
		fails, Contact technical support
9568	Job download complete	Prompt message
9569	Job system not match	Resolution: Please re-backup this job
9570	Instruction not exist	Resolution: Please re-backup this job
9571	Axis number unmatched,UR	Resolution: Confirm the system axis number
9572	Backup again	Resolution: Re-backup, if still fails, Contact technical support
9573	Assignment switch enable	Prompt message
9574	Assignment switch not enable	Resolution: Click the job conversion enable switch
9575	New operation failed	Resolution: Delete some crashed jobs. Backup unused jobs and delete them
9576	Assignment recovery failed	Resolution: Contact technical support
9577	Operation unsupported	Resolution: Contact technical support
9578	Assignment locked,no edit:	Resolution: Change write protection to OFF in job properties to modify this job
9579	Assignment memory error	Resolution: Contact after-sales personnel
9580	Param recovery complete	Prompt message
9581	Original 2 recorded	Prompt message
9583	Param input	Resolution: Contact technical support
9800	Modification complete	Prompt message
9801	Axle motion banned	Prompt message
9802	Modify wearing complete	Prompt message
9810	New assignment creat failure	Resolution: Check memory size or contact technical support
9811	Illegal assignment name	Resolution: Recheck and modify the job name, if still unclear, Contact technical support
9812	Instruction Inserted Failure	Resolution: contact technical support
9813	Prohit to configure BA	Resolution: Select to configure other job

Alarm number	Alarm information	Troubleshooting
		programs
9814	Instruction modify failure	Resolution: Modify the instruction multiple times and save again, if still fails, Contact technical support
9815	AGC,new job failed	Resolution: Rebuild the job, select the axis group information
9816	Registration function limit	Resolution: Contact after-sales personnel
9817	please set the PosSys in ExeSys	External axis configuration required.
9818	Please set the PosSys in ExeSys and in job	Verify whether the job's external axis group and external axis configuration are set for the positioner (turntable).
9819	no assignment job	Prompt message
9820	no reservation status	Prompt message
9901	Soft limit enabled	Prompt message
9902	Soft limit released	Prompt message
9903	USB disk upgrade program:	Resolution: Prompt information; or Contact technical support
9990	The shortcut key configuration interface is not open,Please press OP1 to open it	Resolution: The shortcut key configuration interface is not open, please press OP1 to open
9991	The working mode has been switched:	Prompt message
9992	User IO output successful	Prompt message

Revision history of Manual

Version	Date	Changes
01	2025.6	Initial version writing